

SMARTargets™

A Methodology for Grounded and Actionable Climate Targets
Aligned with Global Goals – July 2025 Public Comment Draft



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The Draft SMARTargets Methodology was developed with extensive input from an independent scientific peer review panel and the SMARTargets Stakeholder Advisory Group, which includes institutions recognized for their thought leadership across the finance, environmental, regulatory, industry, trade, and consulting sectors. EPRI is deeply grateful for the time, expertise, and thoughtful engagement provided by these groups throughout the development process.

In addition, the methodology has benefited from detailed comments and insights from utility technical experts, EPRI subject matter experts, and a wide range of stakeholders who have engaged with EPRI on this topic during the methodology's scoping and development. These contributions have significantly informed and strengthened the methodology.

As part of EPRI's commitment to scientific rigor and transparency, the draft methodology and scientific foundations underwent a formal independent scientific peer review. EPRI sincerely thanks the distinguished members of the peer review panel for their generous participation and constructive feedback:

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The peer reviewer's comments are those of the individual and not their institutions, the institution is noted for information purposes only. The responsibility for integrating feedback and finalizing the methodology rests solely with SMARTargets' principal investigators and should not be interpreted as endorsement by the reviewers or Stakeholder Advisory Group.

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ABSTRACT

This document presents the July 2025 public comment draft of the SMARTargets Methodology. Public comments were solicited on the draft methodology and supporting materials from July 8th to September 3rd, 2025. The set of materials available to the public included the following:

Primary materials

- SMARTargets Methodology: A Methodology for Grounded and Actionable Climate Targets Aligned with Global Goals
- SMARTargets Scientific Foundations: The Science Relevant to Corporate Greenhouse Gas Emissions Target Setting Based on Science (updated July 27, 2025 to fix hyperlinks)

Supporting materials

- SMARTargets Reporting Template: For Documenting and Validating SMARTargets
- Appendix B: SMARTargets Methodology Implementation Illustrative Examples
- SMARTargets Methodology Independent Scientific Peer Review Comments and Responses

This document includes the methodology and appendices, including Appendix B. The other documents are available by request (srose@epri.com and coctaviano@epri.com). Prior to the public comment period, the methodology was subjected to a formal, independent scientific peer review. See the acknowledgments section and supporting materials for additional details on the scientific review.

The SMARTargets Methodology is a methodology for grounded, actionable climate targets and strategies aligned with science and international climate goals. The methodology helps companies contribute to the pursuit of international climate aspirations while identifying strategies that account for their actual opportunities and risks.

There is growing interest in companies setting climate targets aligned with the United Nations Framework Convention on Climate Change's Paris Agreement. However, current methodologies are not considering issues that are essential to address for companies to identify actionable strategies and to facilitate progress towards the Paris Agreement goals. This includes considering uncertainty, differences in transition opportunities, and the multiple social priorities relevant to decisions.

This study provides a new approach that addresses these issues. The SMARTargets Methodology consists of eight standardized steps, with transparent company-specific assessment and communications of company circumstances and transition opportunities and risks a hallmark of the approach. The new methodology is grounded in assessment of the science that defines the relationship between a global climate outcome and a company. The methodology aligns targets and strategies with the Paris Agreement, conducts transition risk analysis, evaluates alternative strategies, and identifies actionable targets, enabling conditions, and contingency plans.

The SMARTargets Methodology provides significant value to companies and stakeholders. For companies, it helps them pursue ambitious and actionable targets aligned with science and international goals. It also facilitates enhanced planning, risk management, and stakeholder engagement. For investors, and other stakeholders, SMARTargets not only means a company has science-based targets aligned with international goals, but the standardized process and outputs provide transparency, comparability, and credibility to the targets and strategies. Furthermore, by identifying enabling conditions, the methodology facilitates constructive dialogue and coordination for greater emissions reduction ambition and progress.

The revised methodology in response to the public comment feedback is expected to be released in 2026.

Keywords

Climate change	Climate scenarios
Climate-related risk	Science Based Targets
Climate targets	Emissions targets
Net zero	Paris Agreement

EXECUTIVE SUMMARY

This document presents the SMARTargets Methodology—a methodology for grounded, actionable climate targets and strategies aligned with science and international climate goals. The methodology helps companies contribute to the pursuit of international climate aspirations while identifying strategies that account for their actual opportunities and risks.

In the United Nations Framework Convention on Climate Change’s 2015 Paris Agreement, the international climate policy community set a goal to limit global average temperature to “well below” 2°C, with an additional aspirational goal to “pursue efforts” to limit global warming to 1.5°C. Companies worldwide are wanting and needing to understand their role in the pursuit of the Paris Agreement goals in terms of potential transitions, opportunities, and risks.

There is also growing interest in companies setting greenhouse gas (GHG) emissions targets that are aligned with the Paris Agreement. This has given rise to proposed GHG target setting and assessment methodologies and benchmarks. However, assessment of the relevant science has found that existing approaches can be misleading with important issues not considered that are essential for informed decision-making and credible targets and strategies.

Alignment with Science and International Goals

Implementing SMARTargets, by design, results in corporate GHG targets and strategies aligned with science and the Paris Agreement. The methodology is based on strong scientific foundations. Scientific and operational requirements for a science-based methodology are first derived from assessment of all the relevant science. The resulting methodology is “based on science” in that it provides (1) science-based alignment with a global temperature goal, and (2) facilitates science-based decision-making. The former recognizes the ranges of pathways consistent with international goals, pathway assumptions and uncertainties, the implications of alternative pathways, and the limitations of global pathways as corporate benchmarks, while the latter is achieved by helping companies consider uncertainty, their unique transition opportunities, decarbonization and other societal priorities, the need for flexibility, and the resiliency of strategies.

The methodology also results in targets and strategies that are comprehensively aligned with both the Paris Agreement’s temperature goals and its provisions for achieving those goals, which includes recognition of country differences in opportunities, multiple priorities, and enabling factors.

The Methodology

The SMARTargets Methodology consists of eight standardized steps, with transparent company-specific assessment and communications of company circumstances and transition opportunities and risks a hallmark of the approach. Using tailored transition risk analysis and standardized outputs, companies evaluate, pursue, and communicate aspirational targets based on international goals and qualified targets reflecting the greatest emissions reductions

possible while balancing priorities under alternative future conditions. The result is a valuable set of company- and engagement-relevant information on target and transition opportunities, strategies, and risk management, including the identification of opportunities for cooperation and coordination on enabling conditions for greater emissions reductions.

The initial methodology is designed to support utilities worldwide that are providing electric power, electricity transmission and distribution, and natural gas services to communities. The approach, however, generalizes and can be helpful to other industries and companies as a template for evaluating their transition opportunities and risks and pursuing GHG targets.

The Value of SMARTargets

SMARTargets provides significant value to companies and stakeholders. For companies, in addition to helping them pursue ambitious and actionable GHG targets aligned with science and international goals, the methodology facilitates enhanced planning, risk management, and stakeholder engagement with a risk-based approach that helps companies explore a broader set of futures and transition opportunities and typically considered in company planning. The approach also facilitates cross-functional integration across planning, risk management, sustainability, policy, and corporate strategy teams.

For investors, and other stakeholders, SMARTargets not only means a company has science-based targets aligned with international goals, but the grounded, rigorous, well-defined standardized process and outputs provide transparency, comparability, and credibility to the targets and strategies. Furthermore, by identifying enabling conditions, the methodology facilitates constructive dialogue and coordination for greater ambition and progress.

Implementation

Implementing SMARTargets requires planning and coordination with respect to resources, internal collaboration, modeling capabilities (internal or external), and stakeholder engagement. Completing and documenting the analysis requires 6-9 months, with third-party validation an additional activity.

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1. OVERVIEW OF THE METHODOLOGY

The SMARTargets Methodology helps companies contribute to the global pursuit of international climate aspirations while identifying actionable strategies that account for their actual opportunities and risks. The methodology is a systematic, comparable approach for evaluating transition opportunities and risks and developing credible company-level greenhouse (GHG) targets and strategies aligned with science and international goals.

The initial SMARTargets Methodology is designed to support utilities worldwide that are providing electric power, electricity transmission and distribution, and natural gas services to communities. The approach, however, generalizes and can be helpful to other industries and companies as a template for evaluating their transition opportunities and risks and pursuing GHG targets.

Having SMARTargets means:

1. A company has ambitious and actionable GHG targets that are aligned with science and the international climate goals,
2. A company followed a grounded, rigorous, well-defined standardized process (with standardized outputs) that provides transparency and comparability, and
3. The company has created a wealth of company- and engagement-relevant information on decisions and analysis, transition opportunities, targets, strategy, and risk management, including identifying opportunities for cooperation and coordination on enabling conditions for greater emissions reductions.

SMARTargets Methodology Steps

The steps associated with implementing the SMARTargets Methodology are depicted in Figure 1. See the methodology section for detailed step guidance, including the outputs required from each step, and see the methodology justification section for the scientific support for the steps and overall design. Accounting for company specific circumstances is critical to identifying real and actionable low-carbon transition opportunities for companies, and for making progress on addressing climate change. The steps below provide companies with the opportunity to pursue global climate objectives while considering their company-specific and regional economy-wide transition opportunities and risks. The resulting information allows for informed dialogue regarding real transitions and opportunities. Here we briefly characterize each step of the methodology:

- **Step 1: GHG inventories and base year** – In this step, a company will prepare verified GHG inventories and select a base year. A company will need, as possible given current methods, GHG emissions inventories for each of the categories of emissions relevant to their business. The company will also need to have the inventories verified. This would include verifying the company's base year inventory and most recent year inventory if different from the base year. A company will also need to decide on their emissions reduction base

year and apply it consistently. SMARTargets does not mandate a specific base year because there are practical reasons for companies to have different base years and there is no scientific justification for a prescribed base year. See Table 1 for the emissions categories most relevant to utilities.

- 1 **STEP 1: GHG inventories and base year**
- 2 **STEP 2: Practical issues and climate target groupings**
- 3 **STEP 3: Aspirational climate targets aligned with international goals**
- 4 **STEP 4: Company transition risk analysis of opportunities and risks**
- 5 **STEP 5: Strategy and qualified target alignment with international goals**
- 6 **STEP 6: Documentation and communication**
- 7 **STEP 7: Validation and verification**
- 8 **STEP 8: Monitoring and adjusting**

Figure 1. The SMARTargets Methodology implementation steps

- **Step 2: Practical issues and climate target groupings** – In this step, a company will consider company-specific practical issues and identify GHG target groupings. For each emissions category, a company will need to work through a set of questions that facilitate consideration of practical issues that affect a company’s ability to set and pursue targets. Among other things, this includes consideration of whether the emissions can be adequately characterized, whether the emissions are material, and whether the emissions are regulated by Paris Agreement aligned policy. For each category, a company will decide whether to (1) set a target; (2) not set a target because the emissions are already covered by a Paris Agreement aligned policy or another company’s SMARTargets, or (3) not set a target with justification based on practical considerations. There is a limited set of conditions where not setting a target can be appropriate. In these circumstances, a company will need to document and justify the decision, and, as possible, communicate plans for addressing the issue to facilitate future targets. The company will also communicate how the categories for which they are pursuing targets will be grouped.

Targets can be by emissions category, by GHG across categories (e.g., CO₂), or across GHGs and categories.

- **Step 3: Aspirational climate targets aligned with international goals** – In this step, a company will select generic aspirational targets aligned with the international goals. Based on the categories for which the company will be setting targets, a company will identify the generic aspirational CO₂, non-CO₂, or CO₂e intermediate and 2050 target levels using global total emissions pathway ranges aligned with the international climate goals. The global pathways help identify aspirational goals and coordinate effort, but they do not provide information about actual individual company opportunities and risks (see Scientific Foundations document). In subsequent steps of the methodology, the company will evaluate their transition opportunities and risks and identify enabling conditions for achieving these aspirational levels. Enabling conditions, such as supportive policies, cooperation with others, and carbon dioxide removals, may be necessary for achieving the aspirational targets while managing the challenges and risks. Such information will inform and facilitate coordination and the development of enabling strategies.
- **Step 4: Company transition risk analysis of opportunities and risks** – In this step, a company will perform company-tailored transition risk analysis of aspirational and qualified targets to identify enabling conditions and their transition strategy, including risk management contingency plans. A company will evaluate alternatives for achieving emissions reduction levels— aspirational and other (qualified) levels—based on tailored company transition risk scenario analysis of transition opportunities, enabling conditions, risks, and risk management options. Qualified targets reflect the greatest emissions reductions possible, in terms of managing challenges and risks, under different potential future conditions. The Paris Agreement refers to this as the “highest possible ambition” in reference to considering both emissions reductions and the effort required, which accounts for differences in circumstances and multiple priorities. The transition risk analysis consists of a well-defined process of seven phrases (Figure 2). It starts with defining reference conditions that exist in all plausible futures and itemizing uncertainties for possible evaluation. It also includes an iterative refinement process of risk-based scenario design and evaluation of the potential transition opportunities and risks for potential futures. In the end, a company may have a range of qualified targets, with different qualified reductions consistent with different future conditions.
- **Step 5: Strategy and qualified target alignment with international goals** – In this step, a company will evaluate transition strategy and qualified target alignment with the international goals. A company will evaluate *qualitative* alignment of their transition strategy relative to the types of transitions robustly observed in all global pathways consistent with the international goals. The company will also evaluate *quantitative* alignment of their qualified targets with the international goals relative to the pathway ranges consistent with the international goals, but recognizing that this is not a conclusive test due to the limitations of global pathways for representing company opportunities and assessing company alignment (see Scientific Foundations document).
- **Step 6: Documentation and communication** – In this step, a company will document and communicate their SMARTargets. Using the SMARTargets Reporting Template, a company

will document and communicate their implementation of the SMARTargets Methodology. This includes documenting methodology implementation choices with justification, transition analysis insights, targets, strategies, and milestones.

- **Step 7: Validation and verification** – In this step, a company will validate their SMARTargets and verify their progress. Using the SMARTargets Reporting Template, a company can undertake validation and progress verification activities. SMARTargets facilitates internal and third-party validation according to international best practices for validation and verification procedures based on the principles of impartiality, independence, and objectivity. Companies will need to establish internal controls, auditing practices and consider their climate-related risk disclosure needs and regulations. Note that independent third-party validation services for corporate GHG targets will need to develop to provide the appropriately grounded, trained, objective, and independent assurance that investors and other stakeholders require. These services do not currently exist.
- **Step 8: Monitoring and adjusting** – In this step, a company will monitor their milestones and adjust their strategy as needed. Periodically, a company will want to assess developments as the future unfolds relative to their milestones to determine if their climate strategy needs adjustment according to their contingency plans. The company will also want to evaluate whether conditions have changed such that the set of potential futures are substantively different than what was analyzed. If so, the company would want to consider refreshing their SMARTargets analysis.

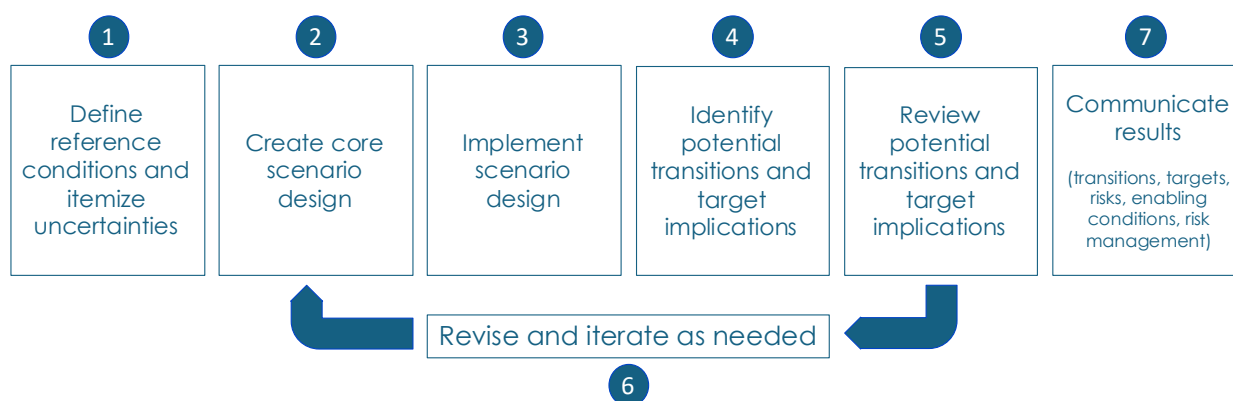


Figure 2. Phases for creating a tailored company-specific low-carbon transition risk scenario analysis and informing the evaluation and pursuit of GHG targets

Emissions Categories and Targets

In applying the methodology, a utility needs to consider each of the emissions categories shown in Table 1. The categories listed in Table 1 are those from the GHG Protocol that are most relevant for utilities. SMARTargets is not dependent on the GHG Protocol categories. Other GHG accounting approaches could be used to define emissions categories.

As noted above, GHG targets can be set by emissions category, by GHG across categories, or across GHGs and categories. For each category, however, a utility will need to decide whether

to (1) set a target; (2) not set a target because the emissions are already covered by a Paris Agreement aligned policy or another company's SMARTargets, or (3) not set a target with justification based on practical considerations.

As noted, there is a limited set of conditions where not setting a target can be appropriate. In these circumstances, a company will need to document and justify the decision and, as possible, communicate plans for addressing the issue(s).

Given resource constraints, a company may also have to set targets progressively, prioritizing targets for the most significant emissions categories and progressively expanding to cover other emissions as resources permit. A company would also document this decision and their plans for future SMARTargets.

Table 1. Emissions categories a utility needs to consider for target setting

Source: EPRI using WRI/WBCSD (2004, 2011).

Scope 1
Direct Emissions from Stationary Combustion
Direct Emissions from Mobile Combustion
Direct Emissions from Process Sources
Direct Emissions from Fugitive Sources
Scope 2
Indirect Emissions from Purchased/Acquired Electricity
Indirect Emissions from Purchased/Acquired Steam
Indirect Emissions from Purchased/Acquired Heating
Indirect Emissions from Purchased/Acquired Cooling
Scope 3
Category 3: Fuel- and energy-related activities (not included in scope 1 or scope 2)
Category 11: Use of sold products

Standardized Comparable Outputs and Insights

Companies will communicate their SMARTargets using the SMARTargets Reporting Template. Specifically, implementing the methodology will produce the following standardized and comparable outputs and insights:

- Aspirational Targets with enabling conditions,
- Qualified Targets and their associated conditions,
- International climate goal alignment of aspirational and qualified targets and company strategies,

- Alternative regional low-carbon transitions and implications insights with respect to multiple priorities, including affordability, reliability, and sustainability,
- Identified risks and strategies (risk management and overall transition strategies), and
- Documentation of the methodology implementation, including GHG inventory for required emissions categories, practical consideration responses and target setting groupings, and transition risk modeling and scenario design details, such as modeling approach, reference conditions and uncertain factors, core scenario design definitions and specifications, and criteria for determining risks and enabling conditions.
- Companies are also encouraged to communicate on the SMARTargets Methodology itself and its scientific foundations to help educate audiences and facilitate dialogue.

SMARTargets Implementation Logistics

Implementing SMARTargets requires coordination and resources—time, human, and analytical resources. To effectively identify and communicate meaningful climate targets and strategies, coordination across corporate business units is recommended. The SMARTargets analysis can be informed by and inform many company business units and teams. Figure 3 illustrates these opportunities for utilities, where there are relationships with planning activities and risk management, as well as other corporate functions. Companies should consider engaging staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on transition scenario analysis and modeling.

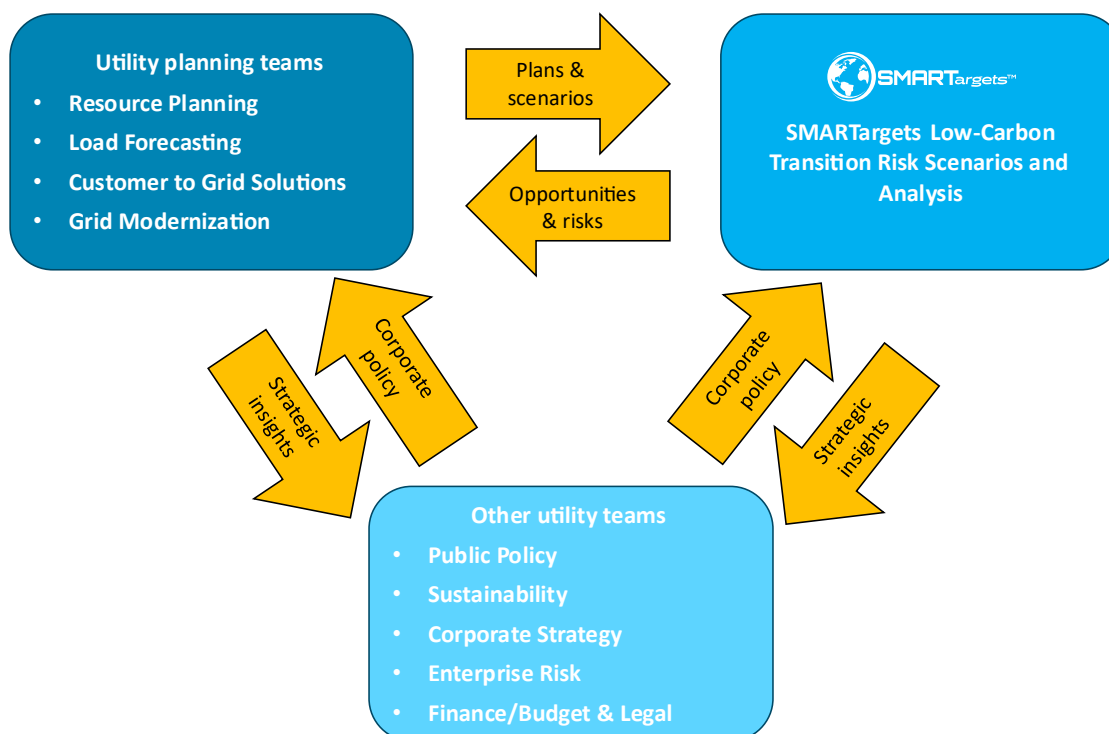


Figure 3. SMARTargets low-carbon transition risk analysis is informed by and informing many utility business units

It is also important to have the appropriate modeling capability (internally or externally) to evaluate the transition opportunities, implications, and risks of alternative regional economy-wide future conditions on company-relevant markets and investment environments. Utilities will need to do either regional economy-wide or enhanced utility planning modeling. Regional (sub-national) economy-wide modeling is recommended but not required. The SMARTargets transition risk analysis does not depend on a particular modeling framework; however, for the insights sought, there are minimum required modeling characteristics (see methodology section). Modeling scope (technology, system, economic, geographic, temporal), dynamics, and even the solution algorithm can affect the types of scenarios and insights produced; thus, communicating the framework used is important. Utility planning modeling that considers alternative economy-wide conditions by varying assumptions can be used. For some utilities, near-term use of current planning modeling capabilities with economy-wide changes considered via exogenous sensitivities may be practical. In the long-run, regional economy-wide modeling is recommended to evaluate potential changes in regional markets, activity, and relationships across sectors and to explicitly identify regional economy-wide transition opportunities and risks.

In addition, there are opportunities along the way for soliciting external input from, for example, a stakeholder advisory group. Implementing the full SMARTargets Methodology—analysis to communications—can take 6-9 months. Third-party validation is an additional activity.

Overall, SMARTargets transition risk analysis is designed to complement and supplement current utility planning analyses and processes—providing broader and longer-term strategic perspectives of potential uncertain factors economy-wide that can affect energy system development, fuel and other commodity markets, and ultimately company transition opportunities and risks. SMARTargets’ risk focus is another differentiating feature, with a scenario design explicitly focused on characterizing and evaluating the transition risk decision space. In general, the SMARTargets analysis should be informed by, informing, and even possibly integrating with, utility planning.

“Based on Science”

In addition to a new methodology, for more informed dialogue and decision-making, the SMARTargets Initiative is providing a scientific foundation for all corporate GHG target setting and evaluation, and transition planning and evaluation, methodologies. The SMARTargets Scientific Foundations document is a resource for guiding and grounding methodologies and an educational resource for informed dialogue.

The SMARTargets Methodology is “based on science,” or more accurately informed by science since the science provides guidance but is not able to tell us exactly what each company should do. The methodology is grounded in and emerges from assessment of all the science relevant to company-level transitions and GHG target setting (see Scientific Foundations document). More specifically, we identify key scientific observations from assessing the science from which we

derive the following scientific requirements for any methodology to be “science-based” (Figure 4):

- **The methodology must have science-based alignment with a global temperature.** This entails recognizing the following: there are broad ranges of global and sub-global pathways consistent with a global temperature goal, assumptions matter and are uncertainties that need to be evaluated, the different pathways aligned with a global temperature vary in their implications, and global pathways in general have severe limitations as company benchmarks but the pathways can provide high-level guidance. The limitations of global pathways as benchmarks include that they do not represent individual companies, their markets, or their transition opportunities and the global pathways are missing company-relevant uncertainties, such as policy design and local factors (e.g., market, customer, resource, policy). Global pathways, however, can help identify aspirational goals and coordinate effort even though they do not provide information about actual company opportunities and risks.
- **The methodology must facilitate science-based decision-making.** Insights from the relevant science highlight that science-based decision-making requires consideration of uncertainty regarding potential future conditions; recognition that companies have different transition opportunities due to differences in, among other things, assets, markets, regulatory structure; accounting for and informing the multiple social priorities that a company is responsible for beyond decarbonization, such as affordability and reliability; providing companies with flexibility to respond as the future unfolds; and, supporting company resilience by helping a company identify robust overall strategies to different futures.

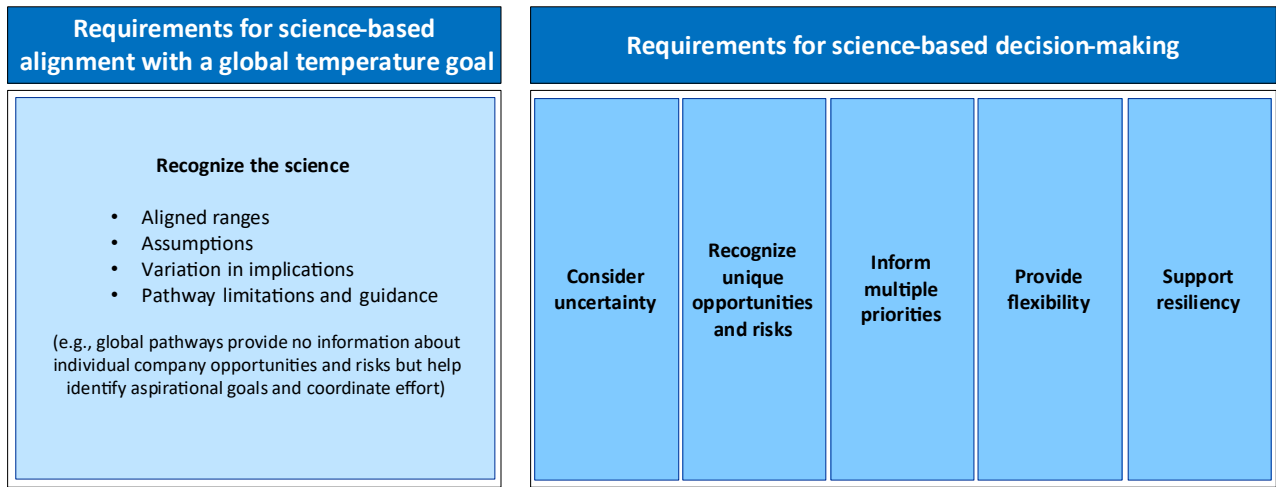


Figure 4. Requirements for a methodology to be “science-based”

Alignment with the Paris Agreement

The SMARTargets Methodology, by design, results in company targets and strategies that align with the Paris Agreement—both its temperature goals and its provisions for achieving those goals. The Paris Agreement calls for “[h]olding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5°C above pre-industrial levels.” The Paris Agreement also includes provisions for achieving the temperature goals, such as recognition of country differences in opportunities, multiple priorities, and enabling factors such as cooperation.

Overall, the SMARTargets Methodology provides alignment with the Paris Agreement in three ways that are consistent with science: quantitative targets aligned with the Paris Agreement temperature goal, qualitative alignment with the Paris Agreement strategies for achieving the temperature goal, and assessment of emission reduction transitions that align with the Paris Agreement and science by recognizing differences in opportunities and enabling conditions. Since alignment with the Paris Agreement and science is by design, a company’s resulting SMARTargets targets and strategies will be automatically aligned by implementing the methodology.

Informing Stakeholders, Serving Investor Needs, and Facilitating Economy-Wide Decarbonization

The SMARTargets Methodology delivers a scientifically grounded, transparent, consistent, and comparable process and outputs that provides users and stakeholders, including investors, with confidence in the approach, results, and insights, as well as, of course, the company’s targets and strategy. The approach is a scientifically derived, systematic, in-depth, comparable target setting and low-carbon strategy development process that each company implements. In addition, SMARTargets provides the following standardized outputs using templates for consistent documentation:

- The company’s aspirational targets and enabling conditions and qualified targets for other conditions,
- The company’s enabling conditions, strategy, risks, risk management, and qualitative Paris Agreement alignment,
- The company’s transition risk analysis (the modeling, scenario design and specification, potential quantitative transitions, and insights), and
- The company’s implementation of the process, including the portions of the company’s GHG inventory that are covered by their targets, regulation, or other company SMARTargets, their practical issues decisions and justifications, and confirmation of their full implementation of the SMARTargets Methodology.

In addition, the SMARTargets transition risk analysis prioritizes economy-wide decarbonization by generating insights regarding interactions and opportunities for cooperation and coordination across the economy that are essential for achieving low-carbon economy-wide transitions.

SMARTargets meets the needs of investors by providing the information sought for credible company low-carbon transition plans, and by transparently informing on company-specific decarbonization opportunities, challenges, enabling conditions, risks, and risk management. Specifically, SMARTargets is aligned with investor needs that have been communicated in various contexts, such as the Net Zero Investment Framework (NZIF), Climate Action 100+, the Transition Plan Taskforce guidance, and TCFD guidance updates.

Methodology Development

The SMARTargets Methodology was developed with broad input from stakeholders, industry technical experts, the scientific community, and the general public. The engagement activities included independent scientific peer review, a stakeholder advisory group, utility technical expert consultations, EPRI scientific expert feedback, a public comment process (ongoing), and public and stakeholder outreach throughout development. Details regarding the peer review feedback and the resulting revisions are available on the SMARTargets public comment process website.

Who is EPRI?

EPRI is a non-advocacy, nonprofit, scientific research organization with a public benefit mandate. EPRI strives to advance knowledge and facilitate informed discussion and decision-making. EPRI's Energy Systems and Climate Analysis research group is a long-standing and well-respected member of the research community relevant to this topic. Specifically, EPRI has over fifty years of recognized expertise in, among other things, climate scenarios, climate-related risk assessment, energy and societal transitions, energy technologies and systems, climate impacts, policy evaluation, and sustainability. EPRI also has recognized research community and scientific leadership, producing peer reviewed publications and contributing to scientific assessments and expert panels, including the National Academies of Sciences, Engineering and Medicine, U.S. Environmental Protection Agency's Science Advisory Board, the Intergovernmental Panel on Climate Change (IPCC), the U.S. National Climate Assessment, the scientific steering committee of the Integrated Assessment Modeling Consortium (IAMC), international and national research community studies, and the Task Force on Climate-Related Financial Disclosures (TCFD) Advisory Group for Scenario Guidance.

Assistance with the Methodology

For questions or assistance with the methodology, or if you would like an introductory briefing, please contact EPRI's scientific leads: Dr. Steven Rose (srose@epri.com) and Dr. Claudia Octaviano (coctaviano@epri.com).

2. WHY SMARTargets?

In the United Nations Framework Convention on Climate Change's 2015 Paris Agreement, the international climate policy community set a goal to limit global average temperature to “well below” 2°C, with an additional aspirational goal to “pursue efforts” to limit global warming to 1.5°C. Companies are eager to understand their role in achieving the Paris Agreement goals in terms of potential low-carbon transitions, opportunities, and risks.

In many countries, decarbonization efforts have begun at national, local, and company levels, and there is growing interest in companies setting greenhouse gas (GHG) emissions targets that are aligned with the Paris Agreement. This has given rise to proposed target setting and assessment methodologies and benchmarks (e.g., SBTi, 2023, 2021; TPI, 2023). However, assessment of the relevant science and additional research has found that there are important issues that are essential to address for companies to identify practical and actionable targets and strategies for achieving outcomes aligned with science and international goals. These include consideration of uncertainty, recognition of differences in decarbonization opportunities and risks, and accounting for multiple societal priorities. While current approaches aim to be “science-based,” they tend not to consider these issues and instead opt for “one-size-fits-all” solutions for all companies within a sector worldwide, which contradicts current climate science and sound decision-making. In addition to being informed by science, companies must also consider practical issues such as fiduciary duty, policy and regulatory environments, public perception, and economic implications. Consideration of these issues is essential for identifying actionable targets and strategies.

Overall, considering these issues when setting targets will facilitate progress by producing strategies that are appropriate for a company and their communities because they reflect a company's transition opportunities, including the company's assets, markets, systems, regulatory environment, barriers, risks, and set of societal objectives, such as sustainability, affordability, reliability, and local economic development. In addition, SMARTargets is an opportunity to elevate transition planning and risk management, as well as dialogue with stakeholders.

What is Novel about SMARTargets?

The SMARTargets Methodology is designed to help companies contribute to the global pursuit of international climate aspirations while identifying viable transition strategies. The methodology is a systematic, comparable approach for evaluating transition opportunities and risks and developing credible company-level GHG targets and strategies. What specifically is novel about the SMARTargets Methodology?

- **Priorities greater than corporate target setting:** SMARTargets prioritizes transition risk management and economy-wide decarbonization.
- **Scientific alignment:** SMARTargets is aligned with all the science relevant to aligning with international goals and grounded company-level decision-making on targets, strategies, and

risk management. Scientific alignment of the methodology was achieved by assessing the relevant science and undertaking a formal, independent scientific peer review.

- **Comprehensive Paris Agreement alignment:** SMARTargets aligns with the Paris Agreement's temperature goals as well as provisions for achieving those goals.
- **Supports good decisions and actionable strategies:** With company-specific transition risk analysis that considers a company's circumstances, opportunities, and uncertainty, SMARTargets generates actionable insights for a company's decarbonization and risk management strategies.
- **Facilitating greater ambition:** SMARTargets identifies enabling conditions for greater emissions reduction ambition that facilitates coordination with customers and policymakers.
- **Facilitating economy-wide decarbonization:** SMARTargets generates insights regarding interactions and opportunities for cooperation and coordination across the economy that are essential for achieving economy-wide decarbonization.
- **Transparency and support for substantive communications and dialogue:** SMARTargets requires companies to be transparent about their transition opportunities and risks, creating critical information for communicating, justifying, and understanding their transition strategies, which is necessary for productive dialogue and progress, as well as cooperation on enabling conditions.
- **Public benefit by enhancing decision-making:** SMARTargets is beneficial to companies, investors, customers, and communities by helping companies develop appropriate viable strategies and consistent targets, properly informing decisions, and avoiding creating risk that can result from imposing abstract and arbitrary benchmarks.
- **Facilitating independent validation:** SMARTargets facilitates internal and third-party validation according to international best practices of impartiality, independence, and objectivity.

SMARTargets Supports Investor Needs

Investors have communicated their needs for credible company low-carbon transition plans through various initiatives. For instance, the [Net Zero Investment Framework](#) (NZIF) has identified five key components: comprehensive net-zero aligned emissions targets, a credible strategy for delivering on the targets, engagement to support target achievement, clear climate solution strategies, and emissions and accounting disclosure. Other initiatives, such as [Climate Action 100+](#), the [Transition Plan Taskforce](#) guidance, and TCFD guidance updates, have also communicated these requirements. SMARTargets meets these needs and more, providing information for all five requirements as well as transparently and consistently informing investors on company-specific decarbonization opportunities, challenges, enabling conditions, risks, and risk management.

What Does it Mean to have SMARTargets?

Implementing the SMARTargets Methodology means a company has:

1. Ambitious and actionable GHG targets that are aligned with science and the international climate goals,
2. Followed a grounded, rigorous, well-defined standardized process with standardized outputs that provide transparency and comparability, and
3. A wealth of company- and engagement-relevant information on decisions and analysis, transition opportunities, targets, strategy, and risk management that inform stakeholders, as well as cooperation and coordination.

SMARTargets Applicability for Utilities and Others

The initial SMARTargets Methodology is designed to support utilities worldwide that are providing electric power, electricity transmission and distribution, and natural gas services to communities. The approach, however, generalizes and can be helpful to other industries and companies as a template for evaluating their transition opportunities and risks and pursuing GHG targets. Given differences across sectors in emissions sources, abatement opportunities, markets, policy environments, and uncertainties, the methodology would need to be customized for other sectors.

3. THE SMARTargets METHODOLOGY

This section presents the details of the SMARTargets Methodology. The section begins by introducing the methodology structure, discussing how SMARTargets complements current company planning and other activities, and describing methodology implementation requirements. The section then walks through each of the methodology implementation steps in detail.

Introduction

Methodology Structure

Implementing the SMARTargets Methodology entails eight steps. Figure 5 provides an overview of the steps and Table 2 provides brief descriptions of each step and its outputs. As shown, each step generates important information that informs the next step. Accounting for company specific circumstances is a key feature of the methodology. Doing so allows a company to identify their transition opportunities and risks, from which they can develop ambitious and actionable climate targets and strategies.

The methodology steps guide utilities through the process required for evaluating, pursuing, communicating, and validating ambitious GHG reduction targets. The initial steps focus on preparing foundational inputs, such as compiling GHG inventories and considering practical issues to determine which emissions to include in company targets. Once emissions categories and target groupings are defined, the company identifies generic aspirational targets aligned with science and the Paris Agreement. These are then evaluated through a company-specific transition risk analysis, which helps identify transition opportunities, risks, and enabling conditions. This analysis also supports the development of qualified targets—actionable goals that reflect the company’s highest level of reductions under different conditions. The strategies and qualified targets identified from the analysis are then evaluated for alignment with international goals and science. Subsequent steps provide guidance for documenting, communicating, and validating the SMARTargets, as well as verifying progress over time. The final step emphasizes the importance of monitoring for milestones regarding future conditions and adjusting strategies as needed, and as possible. Together, the set of steps provide a structured, yet flexible, framework that helps companies align their climate targets and strategies with international goals, identify actionable and ambitious targets and strategies, and manage transition risks.

Overall, the methodology provides companies with the opportunity to pursue international climate objectives while recognizing their company-specific and regional economy-wide possibilities. The resulting information allows for informed dialogue regarding real potential transitions and opportunities for working together to achieve global climate outcomes.

Coordinating and Integrating SMARTargets with Current Company Activities

The SMARTargets transition risk analysis can be informed by and inform many company business units and teams. Figure 6 illustrates these opportunities for utilities, where there are relationships with planning and risk management activities, as well as other corporate functions.

- 1 STEP 1: GHG inventories and base year**
- 2 STEP 2: Practical issues and climate target groupings**
- 3 STEP 3: Aspirational climate targets aligned with international goals**
- 4 STEP 4: Company transition risk analysis of opportunities and risks**
- 5 STEP 5: Strategy and qualified target alignment with international goals**
- 6 STEP 6: Documentation and communication**
- 7 STEP 7: Validation and verification**
- 8 STEP 8: Monitoring and adjusting**

Figure 5. The SMARTargets Methodology implementation steps

Table 2. The SMARTargets Methodology implementation steps, short descriptions, and outputs

Implementation step	Short description	Outputs
STEP 1: GHG inventories and base year	Assemble GHG inventories and select a base year	Verified GHG inventory and emissions reductions base year
STEP 2: Practical issues and climate target groupings	Consider company-specific practical issues and identify target groupings	Decisions on practical issues, including emissions categories covered (by regulation or company targets), the categories for which targets are being set, and how these categories are grouped for target evaluation.
STEP 3: Aspirational climate targets aligned with international goals	Select generic aspirational targets aligned with the international goals	Aspirational targets to evaluate
STEP 4: Company transition risk analysis of opportunities and risks	Perform company-tailored regional transition risk analysis of aspirational and qualified targets to identify enabling conditions and transition strategy, including contingency plans	Company regional transition opportunities and risks, aspirational and qualified targets and their enabling conditions, and transition strategies (low-carbon & risk management)
STEP 5: Strategy and qualified target alignment with international goals	Evaluate transition strategy and qualified target alignment with the international goals	International goal alignment assessments of qualified targets and strategies
STEP 6: Documentation and communication	Document and communicate SMARTargets	SMARTargets documentation
STEP 7: Validation and verification	Validate SMARTargets and verify progress	Validated SMARTargets and verified progress
STEP 8: Monitoring and adjusting	Monitor milestones and adjust strategy as needed	Periodic milestone assessment and, as needed, strategy adjustment and/or SMARTargets refresh

SMARTargets transition risk analysis complements current utility planning analyses and processes, providing broader and longer-term strategic perspectives of potential uncertain factors economy-wide that can affect energy system development, fuel and other commodity markets, and ultimately company transition opportunities and risks. SMARTargets' risk focus is another differentiating feature, with a scenario design explicitly focused on characterizing and evaluating the transition risk decision space. In general, the SMARTargets analysis should be informed by, informing, and even possibly integrating with, utility planning.

For instance, the transition risk analysis should be informed by utility planning such as considering current resource retirement and additions plans, load forecasts, grid modernization plans, customer initiatives, and planning uncertainties. Similarly, the transition risk analysis should be informing planning by elucidating economy-wide opportunities and risks for potential

planning consideration and providing alternative boundary conditions and inputs to resource planning.

SMARTargets evaluates factors and identifies opportunities and risks typically beyond utility planning analysis, such as the potential decarbonization implications for fuel markets (prices, quantities, composition), end use demand levels, responses, and technology adoption, energy demand drivers outside the power sector, inter-regional transmission, and potential policies and uncertainties in other sectors and regions. SMARTargets can also help define alternative exogenous inputs to planning (e.g., fuel prices; load levels and composition; inter-regional power flows). Finally, the SMARTargets transition risk analysis could eventually be integrated with planning analyses. There is the potential for explicit coordination and linking to generate complementary sets of market and company system transition results.

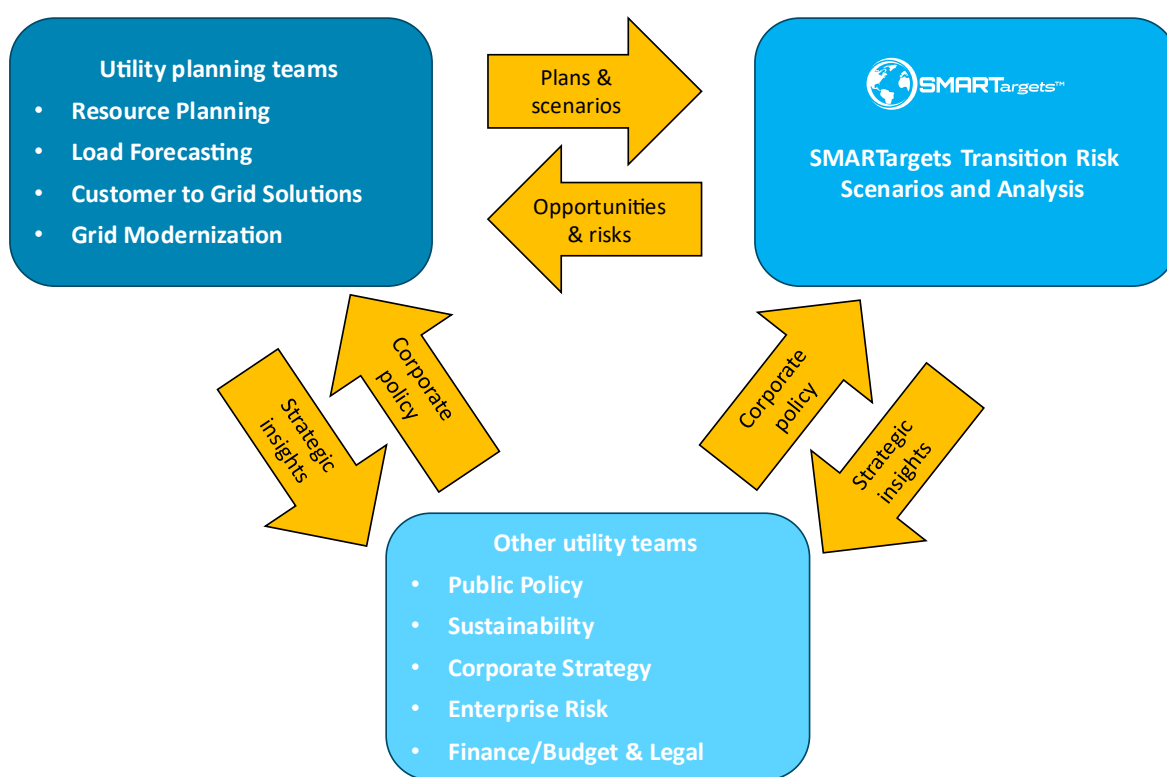


Figure 6. SMARTargets low-carbon transition risk analysis is informed by and informing many utility business units

SMARTargets Coordination and Resource Requirements

Coordination and Resources in General

Implementing SMARTargets requires coordination and resources. As illustrated above, SMARTargets' decisions and analysis can be informed by and inform many existing company business units and teams (Figure 6). To effectively identify and communicate meaningful climate targets and strategies, coordination across corporate business units is recommended.

Resources are also required for implementing the methodology. These include people, time, and modeling resources. Companies should consider engaging staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on transition scenario analysis and modeling.

In addition, there are opportunities along the way for soliciting external input from, for example, a stakeholder advisory group. These are educational opportunities, as well as opportunities for feedback on a company's planned analysis and, eventually, their transition insights and proposed strategy. Finally, implementing the methodology—from analysis to communications—can take 6-9 months, with third-party validation an additional activity beyond that timeframe.

Modeling Resources and Capability

For the company-tailored transition risk analysis (Step 4 in Figure 5), it is important to have the appropriate modeling capability (internally or externally). The SMARTargets Methodology, however, is model agnostic. It does not require use of a specific model, but it does have minimum modeling capability requirements.

The SMARTargets' transition risk analysis is designed to provide high-level strategic risk insights and guidance. It is not designed for individual asset operations planning. The transition risk scenarios discussed in the methodology steps below are crafted to characterize future potential regional markets, looking beyond company systems to economy and sector transitions. Note that a company may need to evaluate multiple regions if they operate in more than one and the markets are unique.

The methodology recommends that utilities use either regional economy-wide or enhanced utility system planning modeling, with regional (sub-national) economy-wide modeling preferable. In the near-term, however, using existing planning modeling may be practical for some companies. If so, the company should consider capturing economy-wide changes via alternative assumption sensitivities (discussed below). In the longer-run, we encourage companies to use regional economy-wide modeling to explicitly account for sector linkages and evaluate economy-wide opportunities, risks, and risk management strategies.

Transition risk analysis for some categories of utility GHG emissions will require the modeling of potential transitions of full systems—sectors, energy, or economy. For instance, evaluating transition opportunities and risks for stationary combustion CO₂ from electricity generation, natural gas related methane (CH₄) fugitive emissions, or downstream gas consumption CO₂ emissions requires supply and demand system modeling to evaluate future potential market conditions—market size and composition. However, some emissions do not require such complex modeling and can be evaluated using simpler abatement cost techniques (e.g., employee commuting emissions, business travel emissions). Methodologically, similar conceptual steps (Figure 5) will still be needed to evaluate uncertainty in emissions reduction opportunities, but the process can be simplified and streamlined. Note that the discussion below focuses on transition risk analysis based on system modeling.

While the transition risk analysis steps do not depend on a particular modeling framework, modeling scope (technology, system, economic, geographic, temporal), dynamics, and even the solution algorithm can affect the types of scenarios and insights that can be produced. Note that creative scenario designs and specifications can be used to circumvent some model scope limitations, such as varying market condition assumptions.

To evaluate and select a modeling framework, a company will need to understand their current modeling resources and capability related to each GHG emissions category and the transition opportunity and risk insights sought. A company may need to collaborate with others who can provide the modeling framework needed. For instance, companies may want to model demand response and/or fuel market size and composition. Also, modeling economy-wide decarbonization transitions and their implications across sectors requires economy-wide modeling. Below we discuss both types of modeling frameworks—economy-wide and planning—and minimum capability requirements for each. Going forward, there will be opportunities for partnerships to advance utility modeling tools to generate additional insights, as well as opportunities for development of tools for SMARTargets applications for other sectors.

Regional economy-wide modeling

SMARTargets recommends scenarios and modeling designed to provide companies with a strategic picture of regional economy-wide transitions and decarbonization opportunities and risks. With economy-wide decarbonization being sought by stakeholders, the methodology is designed to explore what economy-wide decarbonization might look like and take.

Economy-wide modeling tends to have a broader scope (economic, system, and temporal) than company planning modeling, and be less inhibited by policy or process restrictions and, therefore, able to explore a wider set of “what if” transition possibilities. State/region economy-wide modeling elucidates potential supply and demand market changes and provides strategic insights regarding future market and system conditions important to planning and overall corporate strategy.

Valuable economy-wide modeling insights include regional power sector transitions, sector interactions (e.g., end-use electrification, incentives to use low-carbon electricity, electrolytic hydrogen supply and demand), economy-wide renewable fuel opportunities, gas market size and fuel composition, potential fuel price changes, and opportunities for inter-regional transmission and low-carbon power trading (e.g., EPRI, 2023; Blanford et al, 2023; Cheng et al, 2023; Ueckerdt et al, 2021; Luderer et al, 2021).

For a SMARTargets implementation, a *regional economy-wide modeling framework* should have the following minimum characteristics:

- Sub-national regional resolution,
- Economy-wide economic resolution,
- 2050+ modeling horizon,

- 10-year or less time steps (to, at a minimum, model 2030, 2040, and 2050 emissions),
- The ability to evaluate GHG emissions constraints and identify cost-minimizing transitions,
- The ability to derive market equilibrium energy prices and quantities,
- The ability to derive energy supply and demand technology choices,
- The ability to estimate changes in inter-regional energy markets, and
- Dynamic optimization to capture expectation implications for investments. (preferred, not required).

The last item, dynamic optimization, is not required; however, it is the recommended solution algorithm for accounting for market expectation effects on investment choices. Numerous country-specific models with the above capabilities exist. For instance, for the U.S., EPRI's US-REGEN (Blanford et al, 2023), NREL's ReEDS, PNNL's GCAM-USA, and MIT's USREP (Yuan et al, 2019) are examples.¹

Utility planning modeling

If a company is unable to do economy-wide modeling, they should do utility planning modeling and clearly communicate what they are capturing and not capturing with their framework. The company should also use external information to approximate and evaluate alternative broader economy-wide conditions. Specifically, the company should design scenarios considering alternative input assumptions regarding sector and economy-wide conditions that could affect planning modeling solutions. For instance, using external economy-wide results to guide scaling up and down of electrification trends, proxy increases in low-carbon electricity demand and/or the availability of renewable fuels due to end-use decarbonization policies, proxy data center/AI growth implications on electricity load, proxy low-carbon electricity inter-regional export/import opportunities, or proxy variation in the size of the gas market.

EPRI or other experts can support this need with insights and recommendations for data sources and alternative condition specifications from the scientific literature. For example, US economy-wide decarbonization studies have found that end-use decarbonization policies could result in electrification increase of 60-75%. Similarly, recent studies have suggested 6-18% data center/AI load growth nationally from 2023 to 2030 and regional growth up to 150% from 2020 to 2040. In addition, US regional analysis has projected the possibility of significant changes in inter-regional power trading associated with economy-wide decarbonization. For a SMARTargets implementation, a utility planning modeling framework should have the following minimum characteristics:

- Sub-national regional resolution,
- 2050+ modeling horizon,

¹ EPRI US-REGEN: <https://us-regen-docs.epri.com/>. NREL ReEDS: <https://www.nrel.gov/analysis/reeds/>. PNNL's GCAM-USA: <https://im3.pnnl.gov/model?model=GCAM-USA>. MIT's USREP: <https://globalchange.mit.edu/publication/17331>.

- 5-year or less time steps,
- The ability to evaluate system and GHG emissions transitions (i.e., comparing emissions outcomes to constraints or identifying cost-minimizing transitions),
- The ability to inform affordability discussions with price and/or cost results, and
- The ability to derive energy supply technology choices.

Numerous models with the above capabilities exist. For instance, there are common system planning models, such as Plexos, Encompass, PowerSIMM™, and Aurora.² There are also national electric sector models, such as, for the U.S., EPRI's US-REGEN electric sector version (Merrick et al, 2024), NREL's ReEDs, and ICF's IPM.³

Integrated modeling

In the future, a company may want to consider integrated, or coupled, economy-wide and utility planning modeling that provides a complementary and integrated set of market and system specific transition results. For now, the processes should be informing each other, with SMARTargets providing big-picture strategic insights regarding potential economy-wide and sector markets, transitions, and risks, as well as planning analysis sensitivity ideas and inputs. The modeling could eventually be more intimately integrated, or even mathematically coupled. Note, however, that some current utility planning analyses are constrained by regulatory and other processes that could impact what can be done in terms of integrating the modeling. This would be an important consideration.

Technology Agnostic

It is important to note that SMARTargets is *technology agnostic*. The methodology does not favor or preclude any technology *a priori*. Instead, the methodology encourages companies to utilize modeling that allows them to evaluate having and not having options so that they understand the implications and trade-offs for their targets and strategy and can share that information with others and inform dialogue.

Step 1. GHG Inventories and Base Year

Step 1 Outputs: Verified GHG inventory and emissions reductions base year.

The first step towards setting GHG reduction targets is for a company to assemble their GHG inventory and select a base year. A company will need GHG emissions accounts for each of the categories of emissions relevant to their business. A company will also need to decide on their

² Energy Exemplar Plexos: <https://www.energyexemplar.com/plexos>. Yes Energy Encompass: <https://www.yesenergy.com/encompass>. Ascend Analytics PowerSIMM™: <https://www.ascendanalytics.com/solutions/planner>. Energy Exemplar Aurora: <https://www.energyexemplar.com/aurora>.

³ EPRI US-REGEN: <https://us-regen-docs.epri.com/>. NREL ReDS: <https://www.nrel.gov/analysis/reeds/>. ICF IPM (used by USEPA): <https://www.epa.gov/power-sector-modeling>.

emissions reduction base year and apply it consistently. Finally, the company will need to have the inventories verified. This would include verifying the company's base year inventory and most recent year inventory if different from the base year. The inventory, understanding and documentation of methods, and the base year inform subsequent steps in the methodology.

Specifically, Step 1 includes:

- Assembling the GHG inventory, including characterizing trends,
- Selecting an appropriate base year, and
- Documenting inventory methods, inventory verification, and identifying gaps and limitations.

Assembling the GHG Inventory, Including Characterizing Trends

A corporate greenhouse gas (GHG) inventory is a comprehensive and systematic accounting of all GHG emissions and removals associated with a company's operations over a defined period—typically one year. It forms the foundation for understanding a company's emissions profile and is necessary for developing emissions reduction strategies.

There are several methodologies available for developing a corporate greenhouse gas (GHG) inventory, such as the GHG Protocol and ISO 14064. Companies should select and clearly document the methodology or protocol they use to prepare their inventory. The SMARTargets Methodology is designed to be inventory methodology agnostic and does not rely on any specific approach. However, it requires companies to consider all emissions categories listed in Table 3. While these categories reference those defined by the GHG Protocol, using the GHG Protocol itself is not mandatory. Appendix A provides a mapping of key emission sources to each category and scope specifically tailored for electric and gas utilities to assist in identifying relevant sources when using an alternative methodology. For instance, combustion emissions from electricity generation must always be reported, regardless of the chosen protocol.

In addition to identifying current emissions based on the most recent verified inventory, it is important to understand historical trends by analyzing emissions data across as many years as a company has consistent records. This information is useful to communicate and informs base year selection. Visualizing this data through graphs can illustrate trends and highlight emissions increases or reductions.

Table 3. Emissions categories your company needs to consider for target setting

Source: EPRI using WRI/WBCSD (2004, 2011)

Scope 1
Direct Emissions from Stationary Combustion
Direct Emissions from Mobile Combustion
Direct Emissions from Process Sources
Direct Emissions from Fugitive Sources
Scope 2
Indirect Emissions from Purchased/Acquired Electricity
Indirect Emissions from Purchased/Acquired Steam
Indirect Emissions from Purchased/Acquired Heating
Indirect Emissions from Purchased/Acquired Cooling
Scope 3
Category 3: Fuel- and energy-related activities (not included in scope 1 or scope 2)
Category 11: Use of sold products

Selecting a Base Year

Once a company has a clear understanding of their emissions inventory and trends, they can proceed to selecting a base year. A base year is the specific year against which a company's GHG emissions are compared when measuring progress toward emissions reduction targets. It serves as a reference point for tracking emissions reductions over time.

SMARTargets does not mandate a specific base year. Companies should consider the following in selecting their base year:

- **Communications of progress.** A company should consider a base year that would capture their historical emissions changes. In particular, a company should choose a base year that allows the company to communicate progress to date and the additional planned abatement action.
- **Availability of reliable, verifiable, and consistent emissions data.** The selected base year must be supported by high-quality data. A third-party verified emissions inventory with, at a minimum, a limited level of assurance for the selected base year is required to ensure the credibility and the integrity of future comparisons.
- **Local, regional or national policy base years.** Where appropriate, companies may choose to align their base year with relevant policies—such as the base year used in their country's Nationally Determined Contribution (NDC), state decarbonization targets, or regulatory requirements. While not required, this alignment can help clarify how the company's targets relate to broader climate goals and may enhance stakeholder understanding, particularly in policy or investor communications.

Why not standardize the base year across all companies?

SMARTargets does not mandate a specific base year because there is no scientific justification for a particular year. However, there are practical reasons for companies to have different base years.

At the international negotiation level, as well as corporate level, there have been discussions about what base year should be used for computing percent reductions in emissions and communicating targets. This includes discussion about whether the base year should be updated and if all countries or companies should use the same base year. For example, the current U.S. NDC emissions reduction pledge has a 2005 base year. From a company point of view, revising the base year can have the unintended consequence of not acknowledging emissions reductions to date. Updating the base year can affect perceptions regarding effort and progress. Thus, a company should consider a base year that would capture their historical emissions changes.

On its own, there is no scientific argument for revising the base year. It is simply redefining the reference point. As a matter of fact, the current focus on 2015, as seen in IPCC reports and elsewhere, is in large part an artifact of science community data reporting. Databases, such as associated with the IPCC's latest assessment report (Byers et al, 2022; Riahi et al, 2022) or the IEA's latest net-zero scenario (IEA, 2023) no longer include standardized reporting of earlier historical years, such as 1990 and 2005. The shift to 2015 in part simply reflects changes in data availability rather than a scientific rationale for redefining the base year.

The key issue here is that changing global conditions are making limiting warming to the Paris Agreement goals more challenging. Specifically, global emissions continue to rise and global emissions pathways for limiting warming to the Paris Agreement goals are therefore becoming even more ambitious. Together, this implies higher percent reductions as the percent reductions for achieving a similar 2050 outcome are now based on today's higher starting point, compared to a decade ago. However, the global emissions trend may be inconsistent with the company's emissions trend. If a company has declining emissions over the same period that global emissions are increasing there is yet another disconnect between company and global transitions that should be recognized.

Thus, the SMARTargets base year recommendations for a company setting a target are to:

- Use the latest global emissions pathways to inform the aspirational targets the company evaluates (Step 3),
- Choose a base year that allows the company to communicate their progress to date and the additional abatement action planned,
- Ensure that the same base year is used in communicating the company's target, their transition risk pathways, and the global pathways, and
- Quantitatively communicate the global and company emissions trends to help stakeholders understand how each has progressed and to contextualize the company's role within the broader climate transition.

As indicated above, companies should use the same base year in communicating their targets, their transition risk pathways, and the global pathways used when communicating alignment with international goals (see illustrative examples in Appendix B). It is straightforward to re-scale global pathway percent reduction ranges to different base years.⁴

Updating the base year

In some circumstances, it is appropriate for a company to consider updating their base year. In particular, companies that undergo structural changes might need to revise their base year to more clearly communicate their emissions trends and benchmarks. For example, a company might consider revising their base year following the purchase or sale of assets that redefine a company's emissions sources and GHG inventory (GHG Protocol, 2005; EPRI, forthcoming). If this is the case, factors motivating the revision should be clearly communicated.

It is also a good practice to develop a clear corporate policy that sets conditions for considering a base year update. If a company's base year recalculation policy is triggered, the company may also need to revisit their SMARTargets and they will want to inform the public and provide justification.⁵ See the GHG Protocol for guidance on base year recalculation and EPRI's Technical Note on this topic (GHG Protocol, 2005; EPRI, forthcoming).

Documenting Inventory Methods, Inventory Verification, and Identifying Gaps and Limitations

Document Methods

A company needs to document the protocols or methodologies used to prepare its GHG emissions inventory. While SMARTargets does not mandate the use of specific methodologies, it does require them to be documented.

The company will also need to justify the selected base year. It is recommended that companies follow reasonable criteria (e.g., that provided by the GHG Protocol), such as choosing a base year for which verifiable emissions data are available and ensuring it represents typical business operations (avoiding atypical high or low-emission years). Companies should also ensure that the base year selection has properly considered any material structural changes in the company and that it represents an accurate datum for benchmarking the company's emissions reductions. The base year should also relate to the company's business goals and context, such as capturing business strategies or aligning with mandatory reporting requirements in their jurisdiction. Additionally, as discussed, companies should select a base year that facilitates

⁴ Companies can choose the base year but need to ensure consistency with the base year used for deriving the global pathway percent reduction ranges and then determining the midpoints for each year pre-2050 and in 2050 associated with Figure 7 and Table 6. EPRI will provide the data needed so that companies can estimate the required reduction percentages based on their chosen base year.

⁵ This update will trigger the need to revalidate your target, but an expedited process can be designed to facilitate target validation under these specific circumstances.

communication of historical emissions trends, highlighting past achievements and demonstrating planned additional reductions going forward.

The company should also document inventory verification activities, including the inventory years verified and their justification for verification gaps or delays, especially if the inventory is more than five years old.

Finally, as mentioned, the validation of SMARTargets requires a third-party verified GHG inventory. This verification must be conducted by an accredited Validation and Verification Body (VVB) and must meet, at least, a limited level of assurance for accuracy and completeness.

Gaps and Limitations

Companies are required to transparently document gaps or limitations in estimating GHG emissions for the categories listed in Table 3. These gaps may include issues such as incomplete or unreliable activity data, outdated or uncertain emission factors, or missing data for specific sources. In addition to identifying these limitations, companies should assess how they may affect the accuracy of emissions tracking and target-setting and outline any plans or strategies in place to address them over time.

It is important to recognize that GHG accounting is a rapidly evolving field. New methodologies are continually being developed, and existing protocols are regularly updated to reflect advances in science, data availability, and policy needs. For example, the GHG Protocol is currently undergoing a major revision that may influence how emissions are categorized and reported under frameworks like SMARTargets. Companies are encouraged to stay informed about these developments and to understand the limitations of the methodologies they currently use.

As highlighted by Kaplan and Ramanna (2021), one significant limitation of current corporate GHG accounting practices—particularly those based on the GHG Protocol—is the systematic double counting of indirect (Scope 3) emissions across value chains. This can obscure accountability and complicate efforts to manage climate-related risks. In response, alternative approaches such as the e-liability method are being developed to provide more precise, auditable, and transaction-level emissions tracking (Heller, T. and Seiger, A, 2021; Seiger and Roston, 2022; Ronston et al 2023, Ronston et al 2024a, Ronston et al 2024b). These methods aim to improve transparency and better support climate risk management, especially for companies with complex supply chains and potential exposure to climate-related financial risks. Importantly, regardless of whether the GHG Protocol, e-liability, or another GHG accounting approach is used, companies will still want to assess their transition opportunities and risks and identify climate and risk management strategies by following the remaining SMARTargets steps.

Transitioning to a new accounting framework like e-liability could be a significant undertaking. As such, it will be important for companies to understand the potential benefits of emerging alternatives to be able to make informed decisions and to respond to stakeholders.

Step 2. Practical Issues and Climate Target Groupings

Step 2 Outputs: Decisions on practical issues, including emissions categories covered (by regulation or company targets), the categories for which targets are being set, and how these categories are grouped for target evaluation.

In this step, a company will consider company-specific practical issues and identify GHG target groupings. This step helps companies assess practical considerations that may influence their ability to set and achieve GHG emissions targets and inform their evaluation of targets (in Step 4). For each emissions category in Table 3, a utility will need to work through a set of questions that facilitate consideration of practical issues that can affect the company's ability to set and pursue targets.

There is a limited set of conditions where not setting a target can be appropriate. In these circumstances, a company will need to document and justify the decision, and, as possible, communicate plans for addressing the issue to facilitate future targets. The company will also communicate how the categories for which they are pursuing targets will be grouped. Targets can be by emissions category, by GHG across categories, or across GHGs and categories.

For each category, a company will decide whether to:

1. Set a target (using the subsequent SMARTargets steps);
2. Not set a target because the emissions are already covered by a Paris Agreement aligned policy or another company's Paris Agreement aligned targets,
3. Not set a target with justification based on practical considerations that currently preclude target setting, or
4. Not set a target because the category is not relevant to their business.

As indicated by the choices, a company's emissions could be covered by Paris Agreement-aligned strategies in more than one way: their own derived SMARTargets, Paris Agreement-aligned regulation, and other's Paris Agreement aligned targets (SMARTargets or otherwise).

Target Setting Practical Considerations

Drawing on insights from the science (see Scientific Foundations), the following are questions a company should address for each emissions category the company is considering for targets:

1. Is the emissions category adequately estimated and reported?
2. Are the emissions material to the company's overall GHG inventory?
3. Are the emissions already regulated by a Paris Agreement-aligned policy?
4. Has another company in the value chain already set Paris Agreement-aligned targets for this source? (*Applicable only to indirect emission sources*)
5. Does the company have control or influence over the emissions?

Table 4 helps guide the company's responses to the questions. Note that while the table is structured as Yes/No answers, the company is required to provide justification for each of their

responses in Table 5 and the SMARTargets Reporting Template. The questions help identify target setting opportunities and uncertainties, as well as specific situations where it may currently be impractical or inefficient for a company to set targets.

The Scientific Foundations document discusses each of the above practical issues. For instance, a GHG inventory is a pre-requisite for setting a target, but for some emissions categories the level of detail in the inventory might be insufficient to set and track progress for a target. In addition, it might be impractical to set targets for relatively minor emissions categories, such as categories representing less than 5% of your total emissions.⁶ In this case, while target setting may be precluded by the inventory, you should communicate your strategy for addressing the inventory issue as possible. Similarly, setting targets for emissions that are either already managed by Paris Agreement aligned strategies via regulation, or with other company's targets aligned with the Paris Agreement and science (SMARTargets or otherwise), can be economically inefficient and unnecessarily constrain regulatory compliance or affect the other company's targets strategy, respectively. Paris Agreement alignment of regulations or another company's targets can be determined using the Step 5 approach for SMARTargets. A company may still choose to set targets in these cases. If they do, their targets should be consistent with regulatory compliance or the source's targets, respectively.

When to Consider Setting a Target

Given the questions above, a company should consider setting a target for an emissions category if:

- The emissions can be adequately estimated and reported,
- The emissions are material to the company's GHG accounts,
- The emissions are not already regulated to a Paris Agreement-aligned level, **and**
- Another company in the value chain has not already set Paris Agreement-aligned targets (SMARTargets or another type of science-based target aligned with the Paris Agreement) for the same emissions (applies to indirect emissions only).

When Target Setting may be Impractical

Alternatively, there is a narrow set of circumstances under which setting a target may not be practical or necessary:

- The emissions cannot be adequately estimated or reported,
- The emissions are not material to the company's GHG accounts,
- The emissions are already regulated to a Paris Agreement-aligned level, **or**
- Paris Agreement-aligned targets have already been set for these emissions by another company in the value chain (applies to indirect emissions only).

⁶ A Company should use the same materiality threshold used in GHG inventory verification and they should report this threshold in the SMARTargets Reporting Template. If they have not identified a materiality threshold, they should use 5% based on the GHG Protocol Corporate Standard (WRI/WBCSD, 2004, p. 69).

Limited Control or Influence

Limited or partial control or influence over an emissions category is not, by itself, a reason to exclude the category from target setting consideration. However, it can make achieving a target more difficult and less economically efficient than direct management of the emissions at the source (Scientific Foundations). As such, a company should consider their control/influence for each emissions category. When possible, it should be considered in the form of uncertainties that are evaluated in Step 4 when analyzing transition opportunities and challenges for reducing the category's emissions. For instance, the implications of a lack of control/influence can be evaluated by analyzing uncertainties in the drivers that impact things like generation dispatch, load growth, gas market size, or customer energy and technology choices. In addition to evaluating the implications, the analysis can help identify opportunities for influence and collaboration that could facilitate emissions reductions.

Summarize and Document Decisions

After considering each of the practical issues questions above, a company will designate each of the emissions categories as one of the following:

1. Covered by their SMARTarget derived targets,
2. Covered by Paris Agreement-aligned regulation,
3. Covered by Paris Agreement-aligned targets from a company in their value chain,
4. Not setting a target due to practical considerations, or
5. Not setting a target because not relevant to their business.

The company will need to communicate and summarize their designation decisions using Table 5 (which is in the SMARTargets Reporting Template). This is essential for transparency, validation, and informing both internal and external stakeholders. Furthermore, for categories where targets are being pursued, the company will conduct transition risk analysis (Step 4) on the target groupings they have chosen to evaluate the opportunities and risks associated with the targets and to inform the company's climate and risk management strategies. Table 5 also helps a company transparently communicate what emissions are covered and how, as well as help document progressive target setting (discussed below).

Target Groupings

Targets can be set by emissions category, by greenhouse gas (GHG) across categories, or across both GHGs and categories. A company should use Table 5 to communicate how their emissions categories map to their target groupings (category, GHG, or GHG & category). If a company uses a modeling framework capable of integrating different emissions categories and/or GHGs, they should consider setting targets that encompass those categories and gases. This approach is preferable, as it enables the identification of cost-effective abatement opportunities across emissions categories. If the framework accounts for multiple GHGs, targets can be expressed in carbon dioxide equivalents (CO₂e). However, if integration across categories and/or gases is not

feasible within the current modeling framework, it is acceptable to model and evaluate targets separately by category.

Progressive Target Setting

The emissions categories (and groupings) for which a company intends to set targets is also a list of categories (and groupings) for which the company intends to do Step 4 transition analysis to identify opportunities and risks for achieving the targets. These analyses require resources to complete—staff time and financial resources. If completing all the analyses at once is not feasible due to resource constraints, a company is encouraged to adopt a progressive target-setting approach. This involves prioritizing Scope 1 and Scope 2 categories and setting targets now for those categories the company is currently able to evaluate, while clearly communicating their plans to assess and set targets for other categories over time. This information should also be included in Table 5. Even if resource constraints prevent a company from evaluating and setting targets for some categories now, it is valuable to communicate their intentions to set targets for these categories in the future.

Table 4. Practical considerations for target setting

Question	Yes	No
Is the emissions category adequately estimated and reported?	Continue considering setting a target.	Consider not setting a target that includes this emissions category; and, consider and communicate measures to estimate and report this category in the future.
Are the emissions material on the GHG accounts (e.g., above 5% of total emissions or another defined materiality threshold)?	Continue considering setting a target.	Consider not setting a target that includes this emissions category.
Are the category's emissions regulated by a Paris Agreement aligned policy?	Consider not setting a target that includes this category to avoid economic inefficiencies and compliance complications. If you determine that a target is necessary, consider how to mitigate any potential inefficiencies or compliance challenges that may arise. <i>Note: Determine Paris Agreement alignment using the "Well-below 2°C" ranges used in Step 5 of the SMARTargets Methodology.</i>	Continue considering setting a target that includes this emissions category. However, it is important to also consider whether regulatory approval for pursuing greater reductions is needed and will be granted.

Table 4 (continued). Practical considerations for target setting

Question	Yes	No
Does the source have Paris Agreement-aligned targets (SMARTargets or equivalent) set by another company in the value chain?	Follow the steps below to consider not setting a target that includes this category to avoid economic inefficiencies and complications for another company's target strategy; or, set the other company's target as your company target. <u>Steps</u> 1. Communicate the source's target set by another company in your value chain, 2. Determine Paris Agreement alignment of the target using the SMARTargets Step 5, 3. Determine whether the target is aligned with science (i.e., it meets the scientific requirements, such as considering uncertainty, unique opportunities, and multiple objectives), 4. Determine whether to set a target using the guidance below to derive a response. <i>If it is a SMARTarget</i> – no need to set a target; or, if still wanting to consider a target, use the source's SMARTarget as the target <i>If not a SMARTarget but aligned with the Paris Agreement and science</i> – no need to set a target; or, if still wanting to consider a target, use the source's target set by another company in your value chain as the target <i>If not a SMARTarget and not aligned with the Paris Agreement and/or science</i> – consider asking the company in your value chain that controls the source to set a SMARTarget or consider setting a SMARTarget taking into account the source's target	Continue considering setting a target.
Is there control or influence over the emissions?	Continue considering setting a target.	Document the control issues and use the insights from the discussion to inform the transition uncertainties in Step 4.

Table 5. Emissions category targets

Table note: Emissions categories are those from Table 3.

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by <u>future</u> Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type (indicate if target for category, GHG, or CO ₂ e)	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company's Paris Agreement Aligned Targets (MtCO ₂ e)			
Scope 1. Direct Emissions from Stationary Combustion								
Scope 1. Direct Emissions from Mobile Combustion								
Scope 1. Direct Emissions from Process Sources								
Scope 1. Direct Emissions from Fugitive Sources								
Scope 2. Indirect Emissions from Purchased/Acquired Electricity								
Scope 2. Indirect Emissions from Purchased/Acquired Steam								

Table 5 (continued). Emissions category targets

Table note: Emissions categories are those from Table 3.

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by future Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type (indicate if target for category, GHG, or CO ₂ e)	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company's Paris Agreement Aligned Targets (MtCO ₂ e)			
Scope 2. Indirect Emissions from Purchased/Acquired Heating								
Scope 2. Indirect Emissions from Purchased/Acquired Cooling								
Scope 3. Fuel- and energy-related activities (not included in scope 1 or 2)								
Scope 3. Use of sold products								

Step 3: Aspirational Climate Targets Aligned with International Goals

Step 3 Outputs: Aspirational targets that will be evaluated in the Step 4 transition risk analysis.

In this step, a company will select generic aspirational targets aligned with the international goals. Based on the Step 2 emissions categories the company identified for evaluating and pursuing targets, and the target groupings they defined (by category, by GHG, or across GHGs). A company will then need to identify for each target grouping aspirational intermediate and 2050 target levels. These are referred to as the Aspirational Targets (ATs) throughout the methodology.

Aspirational and Qualified Targets

Within the methodology, a company will evaluate and pursue Aspirational and Qualified Targets (QTs). ATs are derived from global emissions pathways aligned with international temperature goals. The ATs are used to encourage ambition and help identify enabling conditions for being more ambitious. Given the limitations of global emissions pathways (see Scientific Foundations); in particular, that they do not represent individual company transition opportunities or differences in opportunities across companies, it is not expected that ATs will be attainable by all companies without enabling conditions, such as carbon dioxide removal (CDR) offsets and/or supportive policies. This is why the ATs are referred to as “aspirational.” The ATs are also aspirational because the international climate goals themselves are aspirational, with global emissions pathways consistent with the goals exhibiting ambitious emissions declines and, given current policy trends and the continued rise in global emissions, optimistic assumptions (e.g., global coverage and cooperation, emissions peaking before today).

Evaluating ATs creates the information needed for informed dialogue regarding challenges, risks, enabling conditions, and opportunities for pursuing these kinds of emissions outcomes. SMARTargets encourages companies to evaluate what achieving these targets would entail, identify the conditions needed to support them, and take proactive steps towards creating those conditions. This approach helps companies pursue greater ambition while remaining grounded in practical, risk-informed planning.

QTs, on the other hand, reflect the greatest emissions reductions possible for a company under different potential future conditions. They reflect the maximum emissions reductions a company can pursue under certain conditions while effectively managing challenges and risks. QTs are identified through the company’s Step 4 transition risk analysis, which evaluates alternative plausible future conditions. Depending on the conditions, QTs may be more or less ambitious than ATs. Furthermore, a company can have multiple QTs and pathways with each

associated with different future conditions. Note that, QTs are optional if a company determines that they can manage the challenges and risks of ATs under the range of future conditions they are required to evaluate in Step 4.

The two types of reductions—ATs and QTs—emerge as a pragmatic approach for pursuing the international temperature goals: evaluating the unique opportunities and challenges for individual companies to pursue ATs, informing stakeholders and their expectations regarding AT levels of reductions, and helping companies identify QT reductions and strategies that make progress under different conditions. Both ATs and QTs provide information that can be readily communicated and pursued in collaboration with customers, policymakers, and others.

Figure 7 and Table 6 describe the annual ATs companies should evaluate and pursue based on their Step 2 target grouping decisions. The specific evaluation requirements are discussed further below. The reductions in Figure 7 and Table 6 are the midpoints of the global emissions pathways ranges consistent with limiting global average warming to 1.5°C. The reduction levels are scientifically consistent with achieving 1.5°C and of interest to stakeholders, as well as accepted as aligned with the Paris Agreement, such as net-zero CO₂ by 2050.

The Figure 7 and Table 6 reductions are ambitious and they will be challenging for many companies, which is why SMARTargets also requires companies to identify enabling conditions and evaluate the highest possible ambition under other conditions (the QTs). Generally, enabling conditions include things like facilitating climate policies, technology availability and lower costs, facilitating infrastructure siting and permitting processes, and favorable market conditions and customer preferences.

For each target grouping, a company is required to evaluate a 2030 or 2035 interim emission reduction AT and a 2050 AT. The ATs should be at least as ambitious as those in Figure 7 and Table 6. Companies are welcome to evaluate additional interim targets, but it is not required.

For utilities, the primary GHGs in their inventories are CO₂, methane (CH₄), and sulfur hexafluoride (SF₆). For each emissions category, the company would set aspirational targets based on the GHG produced from that category using Table 6. For example, for CO₂, the company would set ATs of at least -56% or -82% for 2030 or 2035 respectively and -100% (net-zero) for 2050 relative to 2015. For CH₄ and SF₆, the 2050 ATs are -47% and -52% reductions respectively relative to 2015.

The ATs are by GHG because of the different roles each type of GHG plays in global emission pathways that are limiting global average warming to 1.5°C. The different roles of the GHGs are due to differences in how the climate responds to the different gases, as well as the differences in emissions sources and abatement options. In addition, from a company GHG reduction strategy point of view, the GHGs also differ in their uncertainties, risk management

opportunities, and transition risk assessment methodological needs. Some companies may prefer to set an aspirational target across categories by gas (e.g., across all CO₂ categories) or across GHGs (e.g., across CO₂ and CH₄ categories). If the latter, they should set the target pathway according to the CO₂e reductions in Table 6.

Note that Figure 7 and Table 6 show required reductions relative to 2015. As noted in Step 1, SMARTargets does not mandate a specific base year for emissions reduction rates. However, a company's base year should be clearly communicated and the global emissions pathway ranges generating the required targets in Table 6 should be adjusted to the same base year for consistency. This adjustment is straightforward—using the ratio of historical emissions for the years of interest—and EPRI can assist as needed. For instance, the 2030, 2040, and 2050 net CO₂ reductions from a 2005 base year are -48%, -85%, and -99% respectively (versus the -56%, -88%, and -100% reductions from a 2015 base year shown in Table 6).

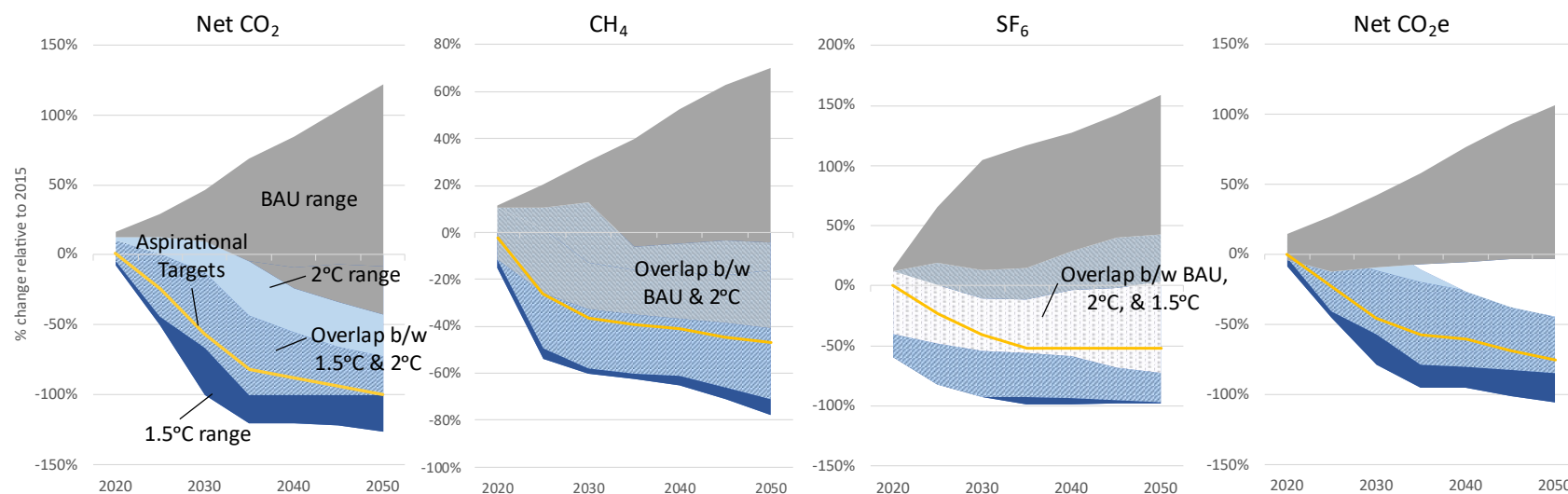


Figure 7. SMARTargets Aspirational Targets and 1.5°C, 2°C, and business as usual (BAU) pathway ranges for global total net CO₂, CH₄, SF₆, and net CO₂e
Source: Developed from the IPCC's Sixth Assessment Report (Byers et al, 2022; Riahi et al, 2022), IEA Net Zero by 2050 (IEA, 2023, 2021), TPI (2023), and PRI (2021) global emissions pathways (EPRI, forthcoming).

Figure note: Each set of targets are the midpoints over time from the respective latest global total emissions percent reduction range consistent with limiting global average temperature to 1.5°C with a 50% likelihood. The 2°C ranges are for global pathways with a 67% likelihood of limiting global average temperature to 2°C. See discussion in the main text for justification and discussion on how the 1.5°C consistent global and sub-global emissions pathway ranges are considered by the methodology. The SF₆ midpoint pathway was modified to be monotonically decreasing. The actual SF₆ midpoints increase post-2030 to -47% in 2050 based on their cost-effectiveness in global pathway solutions. The percent reductions are changes relative to 2015 emissions levels. Companies can choose a different base year that better represents their historical emissions, but they need to adjust the global pathways to derive midpoint target levels consistent with that base year.

Table 6. SMARTargets Aspirational Targets (ATs) for net CO₂, CH₄, SF₆, and net CO₂e and 1.5°C and well-below 2°C global pathway ranges

Source: See Figure 7

Table note: See Figure 7. The well-below 2°C ranges reflects the combined range associated with the latest 1.5°C (50% likelihood) and 2°C (67% likelihood) global pathways.

	Aspirational Targets				1.5°C Ranges				Well-below 2°C Ranges			
	Net CO ₂	CH ₄	SF ₆	Net CO ₂ e	Net CO ₂	CH ₄	SF ₆	Net CO ₂ e	Net CO ₂	CH ₄	SF ₆	Net CO ₂ e
2030	-56%	-37%	-52%	-45%	-13% to -100%	-13% to -60%	-11% to -92%	-11% to -79%	10% to -100%	13% to -60%	13% to -92%	9% to -79%
2035	-82%	-39%	-52%	-57%	-43% to -121%	-16% to -63%	-11% to -93%	-20% to -95%	-4% to -121%	-6% to -63%	15% to -98%	-11% to -95%
2040	-88%	-41%	-52%	-60%	-55% to -121%	-17% to -65%	-3% to -93%	-26% to -95%	-24% to -121%	-5% to -65%	29% to -98%	-26% to -80%
2045	-94%	-45%	-52%	-68%	-66% to -122%	-18% to -71%	-2% to -95%	-35% to -102%	-33% to -122%	-3% to -71%	40% to -98%	-35% to -102%
2050	-100%	-47%	-52%	-75%	-73% to -126%	-16% to -78%	3% to -96%	-45% to -106%	-43% to -126%	-4% to -78%	43% to -98%	-44% to -106%

The ATs are based on global total emissions pathways, instead of sub-global pathways, such as sectoral or regional pathways, like global electric sector emission intensity as is used by some existing methodologies (e.g., SBTi). Total global emissions are used because they have the strongest alignment relationship with global average temperature since they are constrained by the physical dynamics of the climate system. Sub-global results, on the other hand, are contingent on policy and non-policy assumptions that result in an allocation of effort across countries and sectors. However, these assumptions are highly uncertain and country-sector allocation of effort is very sensitive to alternative plausible assumptions. With many sub-global transitions found to be consistent with any global emissions transition, companies need to evaluate these uncertainties, instead of compare themselves to results that are dependent on one alternative. Most importantly, companies need to evaluate their unique transition opportunities and risks. Step 4 of the SMARTargets Methodology requires this essential analysis.

The global emissions pathway ranges are used to derive the ATs because all of the pathways represented within the range are equally credible global transitions and all have been found to be consistent with the global temperature outcome (see Scientific Foundations). The midpoints of the ranges are used for the ATs simply because they are reasonable representative emissions transitions for the set of transitions associated with the ranges.

Finally, to create information on the transition challenges associated with pursuing the Paris Agreement's 1.5°C aspirational goal, the emissions pathway ranges consistent with 1.5°C are used for the ATs instead of broader "well-below 2°C" pathway ranges. The methodology aims at providing insights for the highest ambition goal of the Paris Agreement, and thus ATs are set considering the ranges consistent with 1.5°C. It should be noted, however, that current assessment of global emissions pathways indicates that 1.5°C is likely to be exceeded with important implications for global efforts to pursue this goal. This approach can be revisited as international climate policy evolves. The SMARTargets Methodology conceptual approach itself is not dependent on a particular temperature objective. It can be easily adapted for future developments. See the methodology justification section and Scientific Foundations for additional discussion and insights regarding 1.5°C and "well-below" 2°C global emissions pathways.

Intensity Targets

Intensity targets focus on emissions rates—such as CO₂ per unit of electricity generated or fugitive CH₄/SF₆ emissions per unit of gas or electricity transmitted, respectively. When there is significant uncertainty regarding the activity levels associated with emissions (e.g., electricity load growth, gas market size, or transmission and distribution volumes), and that activity is out of the company's control, intensity targets are frequently considered. This approach allows the company to focus on an aspect of the emissions they can more readily manage—the emissions rate.

However, aligning intensity targets with the Paris Agreement is more challenging than with level (or mass) targets. Doing so requires translating global emissions pathways into sub-global

or sector-specific benchmarks, which depend on uncertain, and frequently unrealistic, assumptions (see Scientific Foundations). Given these limitations, the absolute emissions percent reduction target approach above is used for SMARTargets, along with recommending evaluation of activity level uncertainty. The Step 4 transition risk analysis allows a company to explicitly evaluate and communicate on their activity uncertainty, risk implications, and enabling conditions for achieving targets.

In addition, SMARTargets recommends that a company report emissions intensity changes for the transitions identified in their transition risk analysis. This information helps communicate how the structure of systems might transition with different potential future conditions and activity levels. This information will also help a company currently using intensity targets. The company will be able to connect their current intensity targets to the transitions they are evaluating, which will help identify whether revisions to their intensity targets and strategies might be merited.

Step 4: Company Transition Risk Analysis of Opportunities and Risks

Step 4 Outputs: Company regional transition opportunities and risks, evaluation of aspirational and qualified targets and their enabling conditions, and transition strategies (low-carbon & risk management).

In Step 4, a company will perform tailored transition risk analysis of aspirational and qualified targets (ATs and QTs respectively) to identify enabling conditions and their transition strategy, including risk management contingency plans. If, based on Step 2, a company has decided to evaluate and pursue targets, for an emissions category or across categories, they need to undertake transition risk analysis to identify their opportunities, risks, and risk management strategies for pursuing ATs and QTs.

This approach supports decision-making under uncertainty, which is prudent (see Scientific Foundations). For each target grouping identified in Step 2, the company will undertake transition risk analysis. Note that the same analysis can inform evaluation of target opportunities and risks for more than one grouping. For instance, as shown below, the same core scenarios required for evaluating targets for electricity generation supply emissions provide the uncertainty information needed for evaluating potential gas and T&D targets.

There are seven phases associated with creating the tailored company-specific transition risk scenario analysis (Figure 8):

- *Phase 1: Define reference conditions and itemize uncertainties* – define conditions common to all potential futures and itemize planning uncertainties for consideration in scenarios,
- *Phase 2: Create core scenario design* – combine uncertainties to define a scenario design of plausible extreme sets of future transition conditions,
- *Phase 3: Implement scenario design* – specify assumptions for the alternative future conditions in the scenario design for modeling,

- *Phase 4: Identify potential transitions and target implications* – run the model to identify transition opportunities and risks,
- *Phase 5: Review potential transitions and target implications* – review transition results and insights with company decision-makers,
- *Phase 6: Revise and iterate as needed* – revise scenario design, implementation, and model results as needed until comfortable with results and insights, and
- *Phase 7: Communicate results* – document targets, transitions, risks, enabling conditions, and risk management strategies.

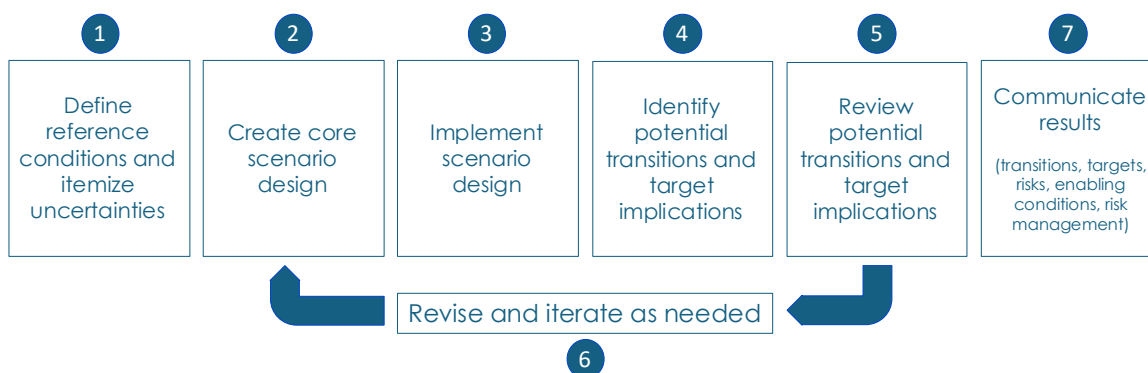


Figure 8. Phases for creating a tailored company-specific transition risk scenario analysis and evaluating GHG targets

This section introduces the SMARTargets scenario design concept and then lays out in detail the required phases, outputs, and decisions associated with implementing the company transition risk analysis. Full analysis implementation examples are provided in Appendix B. In this section, we use electricity supply utility emissions reduction analysis results as our primary examples to illustrate the phases. See Appendix B for implementation examples for target evaluation for utilities providing electricity, gas, and transmission and distribution services.

General Approach

The SMARTargets approach uses scenario analysis to inform decision-making under uncertainty. The scenario results identify potential future states and transitions. The results will also inform decision-maker beliefs about the likelihood of possible transitions, which will help them develop hedging strategies. In the future, companies may want to entertain more advanced computational techniques such as stochastic modeling that explicitly introduces probabilities (e.g., Webster et al, 2022; Zhao et al, 2021).

A well-designed company transition risk scenario analysis can identify the potential future transition and emissions reduction conditions for a company to identify opportunities and risks and to characterize their decision-making space. It requires company-level thinking to identify company-relevant uncertainties and company-relevant future conditions.

Transition risk analysis is a valuable tool for exploring possibilities. It is not a forecast, or prescription. It is intended to explore “what if” possibilities. For instance, what if there is low versus high demand for goods and services? What if there are economy-wide versus sector specific decarbonization transitions? What if carbon dioxide removal (CDR) is allowed versus not allowed as a decarbonization option?

As laid out in Step 3, the SMARTargets Methodology requires evaluation of ATs and QTs for each emissions target grouping. For the ATs, a company will want to evaluate alternative transition strategies and the implications, uncertainties, risks, and risk management options. Identifying the QTs requires exploring different reduction levels to evaluate the incremental implications, challenges, and trade-offs associated with increasing reduction effort. For instance, a company will want to know, and be able to communicate, the incremental implications for economic variables, like investments, prices, and costs, as well as for physical capital and system variables related to new infrastructure and operations.

The details of transition risk assessments will vary with company type, due to differences in emissions sources and abatement strategies. For instance, electric power utilities, gas utilities, and transmission and distribution (T&D) utilities have different sets of emissions sources, and therefore different uncertainties that require variation in approaches for assessing relevant potential emissions, transitions, and abatement opportunities and risks. Furthermore, what variables are most important to decision-making will vary by company.

SMARTargets Scenario Design Concept

A good scenario design can meaningfully explore company transition possibilities, differences, trade-offs, and risk. The challenge, however, is to package the uncertainties into a meaningful but modest number of scenarios to evaluate potential transitions, targets, and risks that can be communicated readily. SMARTargets uses a parsimonious 2-by-2 scenario design logic as a starting point to organize and focus thinking (Table 7). The goals being to define the plausible decarbonization transition space and reasonably bound the risk decision-making space.

Table 7. EPRI 2-by-2 transition risk scenario design concept

		Uncertain Other Conditions (demand, technology, etc.)	
		Facilitating	Challenging
Uncertain Decarbonization Incentives and Options Conditions	Narrower	Narrower & Facilitating	Narrower & Challenging
	Broader	Broader & Facilitating	Broader & Challenging

The basic concept illustrated by Table 7 is that there are two categories of uncertainties—those associated with decarbonization incentives and options and those associated with other conditions unrelated to decarbonization. Decarbonization incentives and options uncertainty includes uncertainties in decarbonization incentives within and across sectors of the economy and uncertainties about the available and eligible decarbonization options. Other uncertainty

includes uncertainties about future goods and services demands, technology costs, new infrastructure constraints, and fuel markets.

This 2-by-2 scenario design concept has been used successfully to evaluate regional low-carbon transition opportunities and risks for utilities (e.g., Alliant Energy, 2023; WEC Energy, 2022). See also EPRI (2022a) for a similar simple scenario approach based on a precursor to the 2-by-2 scenario design. In each case, applying this heuristic method has led to simple, impactful scenario insights with uncertainties determined by the user based on their needs and their unique circumstances that define their decarbonization opportunities.

The simple 2-by-2 construct provides a useful overarching concept, or logic, for conversation, scenario construction, and the grouping of uncertainties, while minimizing the number of scenarios, which facilitates communications and understanding. In transition risk analysis, we are interested in learning about potential risk; therefore, the objective is to define extreme alternative conditions that will elucidate risks. These can be low-likelihood futures, but they should be plausible so that they provide meaningful insights and guidance. The futures should also be meaningfully unique, providing new information, helping define the boundaries of the risk decision space, contributing to a full, and realistic, characterization of potential transitions and risks, and providing substantive insights regarding differences and similarities.

This scenario logic can be a departure from typical company planning processes that tend to focus more on middle-of-road or likely futures. To identify risk, however, exploring the boundaries is needed. Note that, in this plausible extreme scenario design environment, companies are not being asked to pick between extreme decarbonization futures. Instead, they are evaluating the tails of potential future conditions to elucidate risks. If desired, a company can choose an in-between path using the robust strategies and risk management options identified from the analysis.

Broader and Narrower Conditions Related to Decarbonization Incentives and Options

Table 7 is valuable for, among other things, facilitating important initial internal conversations within a company on potential sets of decarbonization incentives and options and other conditions. The Broader and Narrower terms in Table 7 refer to broader or narrower scope and flexibility respectively in terms of decarbonization. For instance, Broader might be defined as an economy-wide decarbonization incentive with a fuller decarbonization options portfolio, while Narrower might be defined as a power-sector-only decarbonization incentive with a more limited decarbonization options portfolio. The specific incentives and options of Broader and Narrower will depend on a company's constraints and conditions, such as whether a company's state, provincial, or regional government prohibits or promotes particular technologies. The steps and examples discussed below illustrate these points.

Facilitating and Challenging Other Conditions

For the uncertain other conditions, the terms Facilitating and Challenging are used in Table 7 to represent conditions that facilitate or challenge the transition to a decarbonized economy. For

example, the Facilitating specification could be defined as slower economic growth, lower low-carbon technology costs, no additional barriers to new infrastructure, and lower input prices, while Challenging could be defined as greater economic growth, higher low-carbon technology costs, constraints on new infrastructure, and higher input prices. Like with Broader and Narrower, the specific elements of Facilitating and Challenging used by a company will depend on their constraints and context, such as local siting and permitting processes and public sentiment. The phases and examples below discuss different types of uncertainties and help illustrate these points.

Expanded Scenario Design

We can also build off the 2-by-2 concept, adding dimensionality to separate categories of uncertainty. Specifically, for SMARTargets, we separate uncertainty regarding decarbonization incentives and options to explore their implications independently. Table 8 is an expanded version of the 2-by-2 table. In Table 8, we have identified the SMARTargets' priority scenarios that are required to characterize the range of possible transitions (in green). They represent "best" and "worst" case transition conditions for each decarbonization incentive environment. The other scenarios in Table 8 are recommended, but not required, to flesh out the space for all the uncertainty combinations. In addition to the required and recommended scenarios, there is also the opportunity for optional sensitivity scenarios to explore within the space, as well as beyond it, by varying individual or subsets of assumptions for more refined insights regarding specific uncertainties or topics. See below for more discussion of optional sensitivity scenarios and examples.

Regional Focus

In general, to capture differences in regional transition opportunities, SMARTargets recommends modeling regional (e.g., sub-national, provincial, state) transitions with regional reductions targets to create insights regarding potential overall system and market transitions. It is important to look beyond company systems to economy-wide transitions to gain insights into potential new markets and investment opportunities and risks. Note that, as discussed in the methodology introduction, if a company is not running a regional economy-wide model, they should run economy-wide-like scenarios by varying exogenous assumptions that proxy economy-wide effects, such as changes in market size and composition.

Table 8. SMARTargets scenario design concept for required (green) and additional recommended scenarios

	Decarbonization Incentive	Decarbonization Options	Uncertain Other Conditions (demand, technology, etc.)	
			Facilitating	Challenging
Uncertain Decarbonization Incentives and Options Conditions	Narrower	Broader	Narrower incentives / Broader options / Facilitating other	Narrower incentives / Broader options / Challenging other
	Narrower	Narrower	Narrower incentives / Narrower options / Facilitating other	Narrower incentives / Narrower options / Challenging other
	Broader	Broader	Broader incentives / Broader options / Facilitating other	Broader incentives / Broader options / Challenging other
	Broader	Narrower	Broader incentives / Narrower options / Facilitating other	Broader incentives / Narrower options / Challenging other

The Seven Phases of Transition Risk Analysis

As indicated in Figure 8, there are seven phases associated with the company-specific transition risk analysis. Each phase and its requirements are discussed below, including who should be involved. In addition, there are opportunities for soliciting external input from, for example, a stakeholder advisory group. These opportunities are identified as well.

Phase 1: Define reference conditions and itemize uncertainties – identify conditions common to all futures and itemize planning uncertainties

- Phase inputs: Previous company planning, strategy, analysis, and risk management experience; and, Step 2 insights regarding activity level uncertainties impacting emissions.
- Phase outputs: Scenario analysis reference conditions and list of uncertain factors to consider in scenario design.
- Who should be involved: Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling.
- Documentation: SMARTargets Reporting Template entries summarizing reference conditions and uncertain transition factors for scenario design consideration.

A critical first conversation is to discuss reference conditions. These are conditions that are not uncertain and that the company sees as a part of all potential futures. For instance, this could be federal or local policies and regulations related to emissions, land use, or economic development that the company expects to be in place in every possible future. It might also be company or regional plans for specific assets that are already in motion, such as projects underway or planned retirements. If, however, there is uncertainty about these, the company

would want to include them in their uncertain factors discussion for potentially inclusion in their scenario design in Phase 2.

A second conversation is to discuss and itemize the uncertain factors that are important to the company. Some will be uncertainties the company already considers in their planning processes, such as uncertainties regarding future demand and input markets. Some will have emerged from Step 2 emissions control discussions of activity level uncertainties and factors, such as uncertainties about the size of the gas market or end-use technology adoption. Others will be newer, and the company may not know whether they are important yet. The company will want to catalogue this latter type of uncertainty as well so that they can consider exploring their importance in the next phase.

Table 9 lists examples of low-carbon transition uncertainties based on EPRI's research, literature assessments, and applied work with companies. These are uncertainties that have been found to be impactful on utility transitions or uncertainties companies have found to be important based on their experience. Uncertain decarbonization incentives and options for evaluation include uncertainty regarding low-carbon transition incentive coverage (e.g., economy-wide, power sector only, technology specific) and the eligible low-carbon options (e.g., CCS, CDR, low-carbon power imports, and/or bioenergy). Other uncertainties include uncertainties regarding other factors, such as load (demand for electricity—clean or otherwise); fuel markets; technology costs; planning, permitting and construction; and not-in-my-backyard constraint uncertainties.

Table 9 is not meant to be exhaustive or prescriptive. Table 9 does, however, identify uncertainty categories that SMARTargets requires be considered (in green). Utilities must consider uncertainty regarding decarbonization incentives, decarbonization options, demand, energy supply and demand technologies, supporting infrastructure, and energy markets. The scientific literature to date has found emissions pathways and decarbonization strategies to be sensitive to these uncertainties; and, therefore, relevant to company risk assessment and planning (see Scientific Foundations).

Keep in mind that some high-level uncertainties can represent a variety of uncertain factors. For example, uncertainty regarding infrastructure additions can represent a combination of uncertainties regarding technology development and commercialization, permitting and siting, local policy, supply chain, not-in-my-backyard (NIMBY) public sentiment, and resource procurement that alone or together could contribute to uncertainty regarding opportunities for new infrastructure. In general, Table 9 is meant to provide ideas to facilitate conversation within a company regarding potential uncertainties to consider in their transition risk scenario design.

Note that, at this point, the objective is to identify relevant uncertainties. Model capability and implementation will be considered later, as we do not want modeling capability to constrain the discussion. In Phase 2, the company will discuss alternatives for representing each uncertainty in their modeling. For models that cannot explicitly represent an uncertainty, there are creative approaches for capturing alternative conditions and sets of conditions.

At the end of Phase 1, the company will document their reference conditions and itemize the identified uncertain factors in the SMARTargets Reporting Template. The template requires consideration of the minimum sets of reference conditions and uncertain factor categories discussed above, and allows for additional conditions and categories to be added. For each of the required uncertainties, the company should document if and how they are being considered. Note that, the company is not required to include all of them, but they are required to consider each and decide on whether to include them.

Table 9. Examples of transition uncertainties

Table note: Required categories in green.

Decarbonization incentives & options uncertainties • Incentive – Economic coverage – e.g., power sector, non-electric sectors, economy-wide – Geographical coverage – e.g., isolated, regional, or national – Timing and stringency • Options – Technology eligibility (nuclear, natural gas, CCS, etc, power imports, allowance markets, offsets, fossil reserve capacity eligibility) • Neighbor's policies (incentives and options)	Other uncertainties • Markets – Load – e.g., population, AI/data centers, EVs, buildings, DERs, changing industry structure, etc. – Inputs – e.g., fuel prices, workforce, other supply chain • Technology – Energy supply – e.g., costs and performance and availability (solar, wind, advanced nuclear, storage, CCS, etc.), water scarcity, additions constraints – Energy demand – e.g., cost, performance, adoption rates, and consumer preferences • Supporting infrastructure – T&D, pipelines, EV chargers, filling stations, etc. – e.g., cost, additions constraints • Non-climate policy – Permitting, clearances, zoning – State and local development plans, resource and transmission planning requirements, DER policy, etc. • Public perception e.g., NIMBY, fossil resource opposition
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Phase 2: Create core scenario design – define plausible extreme alternative future conditions

- Phase inputs: Phase 1 scenario analysis reference conditions and list of uncertain factors to consider in scenario design.
- Phase outputs: The company's core scenario design for the required SMARTargets scenarios.
- Who should be involved: Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling.
- Documentation after iterations: SMARTargets Reporting Template entries summarizing the core scenario design.

Core scenarios

As discussed, the methodology requires companies to evaluate the ATs (Table 6) and QTs and their conditions. Thus, the scenario design needs to facilitate identification of enabling conditions by evaluating the range of possible transition conditions for the company and providing information regarding trade-offs and risks that are relevant to company decision-making.

This phase creates the company's scenario design from their Phase 1 reference conditions and itemized uncertain factors using the SMARTargets' risk-based scenario design logic (Table 7). Specifically, in this phase the company will combine the uncertain factors into plausible, but more extreme, alternative sets of decarbonization conditions for evaluating potential transitions. First, we present the SMARTargets scenario structure for evaluating the ATs and identifying the QTs and then we discuss the details the company needs to define to tailor the scenario design to evaluate their transition possibilities, risks, enabling conditions, and potential risk management strategies.

Scenarios for Electricity Supply Utility Emissions Targets

Table 10 lists the core scenarios required by SMARTargets for evaluating ATs and QTs for companies wanting to set CO₂ targets for electricity supply emissions. Further below we discuss the core scenarios for targets for gas and T&D related emissions. The set of scenarios in Table 10 allow the utility to evaluate the implications of the three priority categories of uncertainties. The scenarios explore the net-zero by 2050 CO₂ AT transition implications for two types of decarbonization incentives (sector specific versus economy-wide net-zero CO₂ by 2050 transitions), two types of decarbonization options (*Fuller* versus *More Limited* sets of options), and two types of other conditions (*Facilitating* versus *Challenging* decarbonization conditions). Discussion of how to define Fuller/More Limited and Facilitating/Challenging is provided below.

Utilities setting targets for electricity supply emissions are required to run reference and AT scenarios. The first two scenarios evaluate reference conditions without emissions constraints to elucidate the implications of the Facilitating and Challenging conditions. The next two AT scenarios evaluate a regional power sector only transition to the net-zero AT in 2050 under alternative combinations of decarbonization options and other conditions. The next two AT scenarios evaluate a regional economy-wide transition to a net-zero AT in 2050 under the alternative decarbonization options and other conditions.

The scenarios in Table 10 may look different from typical utility planning scenarios. This is because the emissions constrained scenarios in Table 10 are intentionally designed to explore risk and evaluate overall decarbonization transitions in the sector and full economy to generate the needed insights regarding opportunities and risks for decarbonizing whole energy systems and economies.

Table 10. Required scenarios for evaluating reductions in electricity supply CO₂ emissions

Target type being evaluated	Target pathway	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Reference	Reference (without CO ₂ constraint)	No	N/A	Facilitating
	Reference (without CO ₂ constraint)	No	N/A	Challenging
Aspirational Targets (net CO ₂ ATs and QTs > ATs)	Power sector transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
	Power sector transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
	Power sector transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
	Power sector transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging
Qualified Targets (net CO ₂ QTs < ATs)	Power sector net CO ₂ transition to X, Y, and Z% in 2050	No	Fuller	Facilitating
	Power sector net CO ₂ transition to X, Y, and Z% in 2050	No	More Limited	Challenging

The economy-wide scenarios will help the company and their stakeholders understand how decarbonization incentives outside the company's sector can impact their transition and decarbonization opportunities. As discussed in the Scientific Foundations document, most low-carbon transition pathways in the literature, and used by stakeholders, assume economy-wide decarbonization incentives, which is not the actual policy context for most companies. However, these policy assumptions have been found to have significant implications for transitions, facilitating more rapid electric sector decarbonization and use of low-carbon electricity and fuels to decarbonize the economy. Different decarbonization incentives will result in different practical decarbonization transitions; and, thus, uncertainty regarding incentives is important to evaluate. The scenarios in Table 10 facilitate this evaluation.

Table 10 also shows the QT scenarios the utility should consider to evaluate the implications of varying levels of emissions reductions for the regional power sector. These scenarios provide insights regarding the potential incremental implications of increasing the level of reductions. To evaluate and identify the companies QTs, a company should run scenarios evaluating a range of reduction targets by 2050 under the Fuller/More Limited and Facilitating/Challenging transition combinations of conditions. For instance, the company could evaluate CO₂ reduction pathways to 30%, 50%, and 80% below 2015 in 2050.

Note that the reference scenarios will help the company identify a reasonable lower bound reduction rate. For instance, if the reference scenarios result in 35% reductions by 2050 without an emission reduction requirement, the company might evaluate a 40% reduction in 2050 with their lowest incremental reduction scenario. Exploring a range of increasingly more ambitious reductions up to the AT levels will create the information needed to understand the incremental implications of pursuing more ambitious reductions. Figure 9 illustrates conceptually how the results across scenarios with increasing reduction levels can be combined to generate incremental insights on costs, prices, investments, new builds, or system variables related to operations. Note that the company may want to revise their incremental reduction levels after seeing the preliminary modeling results to properly identify the reduction levels they can achieve in some conditions while balancing trade-offs and managing risks. Additional scenarios assuming an economy-wide decarbonization are not needed here. The economy-wide

decarbonization scenarios already run for the ATs provide the information needed for evaluating the power sector and energy system implications.

Table 11 lists the additional recommended scenarios for fully fleshing out the possible AT and QT transitions for all the combinations of uncertainties. Table 11 also lists optional sensitivity scenarios for exploring topics of interest further once a company has seen the other results. This is a conversation for Phase 5 of the transition risk analysis.

To implement the scenarios in Table 10 (and Table 11), the company needs to define the Fuller/More Limited decarbonization options and Facilitating/Challenging other conditions. Table 12 provides an example of potential definitions. For the alternative decarbonization option conditions, Fuller/More Limited, the company should define a relatively optimistic set of options for “Fuller” and a relatively pessimistic set of options for “More Limited.” Both sets should be plausible, both should be relevant to policy and stakeholder conversations, and both should be relatively extreme. In Table 12, for instance, Fuller includes a broad portfolio of electric and non-electric sector decarbonization options, including renewables, storage, fossil energy with and without CCS, new nuclear generation, CDR, biopower, hydrogen, other renewable fuel, and low-carbon imported power. Fuller also allows for fossil resources to meet capacity reserve requirements.

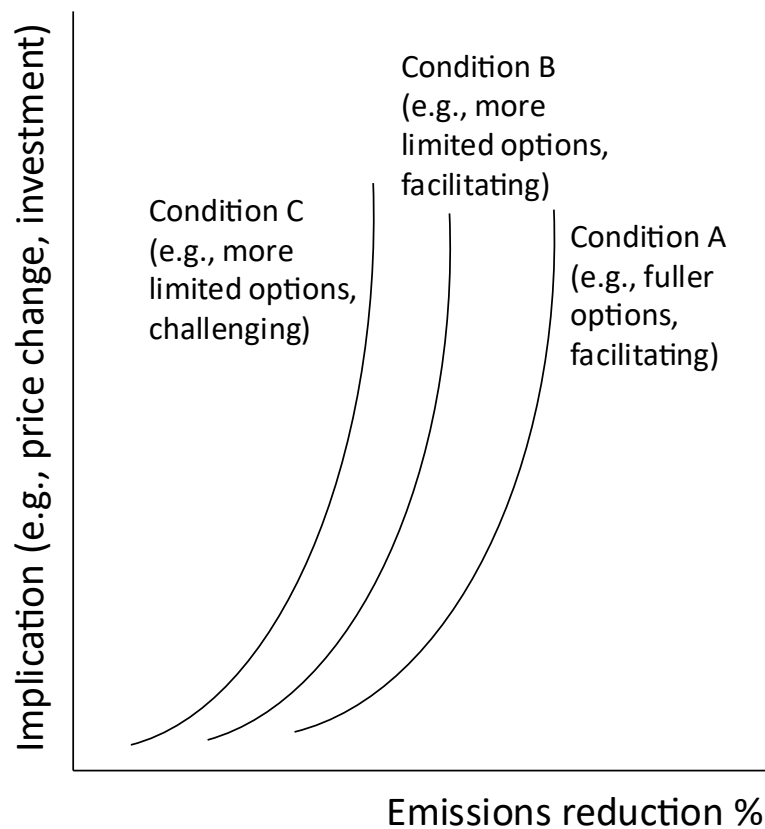


Figure 9. Companies need to develop incremental implications information to evaluate trade-offs and identify their qualified targets

Table 11. Additional recommended and optional sensitivity scenarios for evaluating reductions in electricity supply CO₂ emissions

Required, recommended, optional?	Target type being evaluated	Target pathway	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Additional recommended	Aspirational Targets (net CO ₂ ATs and QTs > ATs)	Power sector transition to net-zero CO ₂ in 2050	No	Fuller	Challenging
		Power sector transition to net-zero CO ₂ in 2050	No	More Limited	Facilitating
		Power sector transition to net-zero CO ₂ in 2050	Yes	Fuller	Challenging
		Power sector transition to net-zero CO ₂ in 2050	Yes	More Limited	Facilitating
	Qualified Targets (net CO ₂ QTs < ATs)	Power sector net CO ₂ transition to X, Y, and Z% in 2050	No	Fuller	Challenging
		Power sector net CO ₂ transition to X, Y, and Z% in 2050	No	More Limited	Facilitating
Optional sensitivities	At company discretion based on insights from other results				

Alternatively, More Limited includes a very narrow portfolio of options, including renewables, existing nuclear generation, storage, electrolytic hydrogen, and declining eligibility of fossil resources for capacity reserves. The company should define Fuller and More Limited as is appropriate to them. Decarbonization technology options will likely vary between companies due to differences in local circumstances, such as local policy. For instance, a company should leave out options that are implausible in all their potential futures. The Fuller/More Limited definitions might also consider other types of emissions offsets (e.g., forestry, agriculture) or a supply of emissions allowances.

It is important to keep in mind that SMARTargets is technology agnostic. The methodology does not favor or preclude any technology *a priori*. Instead, the methodology encourages companies to evaluate having and not having options so that they understand the implications and trade-offs for their targets and strategy and can share that information with others. Thus, companies should default to including options and letting the analysis identify some of the opportunities and risks.

For the alternative other conditions, Facilitating/Challenging, the company should define a set of market, technology, supporting infrastructure, etc. conditions, that could facilitate relatively less and more challenging transitions, respectively. Again, both should be plausible, both should be relevant to policy and stakeholder conversations, and both should be relatively extreme. In Table 12, for instance, the illustrative Facilitating conditions include reference electricity load growth, faster cost improvements for generation technologies, lower barriers for new capacity and transmission additions, lower fuel prices, and faster cost improvements for end-use electric technologies. Challenging is just the opposite of each of these. Note that, at this stage, the alternative conditions are only specified in qualitative terms (e.g., faster/slower, lower/higher). In Phase 3, the company will need to quantitatively specify these conditions so that they can be modeled and analyzed (e.g., x% slower, y% higher).

Table 12. Illustrative Fuller/More Limited and Facilitating/Challenging decarbonization condition definitions

Illustrative FULLER decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions) Advanced nuclear (additions) Fossil Fossil w/CCS CDR (biopower w/ CCS & direct air capture (DAC)) Storage Hydrogen Biopower carbon neutrality Other renewable fuels (w/ and w/o CCS) Low-carbon power imports Fossil capacity reserves permitted (within emissions constraint) Demand conservation and energy efficiency	Illustrative MORE LIMITED decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions only) Storage Electrolytic Hydrogen only Other renewable fuels (w/ and w/o CCS) Fossil capacity reserve eligibility declines to zero in 2050 Demand conservation and energy efficiency
Illustrative FACILITATING other conditions Reference electricity load growth Faster electricity generation cost improvements Economic electricity capacity additions Economic electricity transmission additions Lower fossil fuel prices Faster electric technology cost improvements	Illustrative CHALLENGING other conditions Faster electricity load growth Slower electricity generation cost improvements Constrained electricity capacity additions Constrained electricity transmission additions Higher fossil fuel prices Slower electric technology cost improvements

Scenarios for T&D and gas utility emissions targets

Because T&D and gas utility GHG emissions are a function of the size and composition of the electricity and gas markets, the scenarios required are focused on evaluating the potential future size and composition of these regional markets. This information is an essential input to evaluating opportunities for reducing fugitive SF₆ and gas system CH₄ emissions, CO₂ from line losses, and upstream and downstream CO₂ and fugitive CH₄ gas consumption and development emissions. See Table 13 and Table 14 for the T&D utility and gas utility required scenarios, respectively. From Step 3, the AT for SF₆ in 2050 is a reduction of at least 52% (from 2015); and, for methane, the AT in 2050 is a reduction of at least 47% (from 2015).

Table 13. Required scenarios for evaluating reductions in T&D utility SF₆ and CO₂ emissions

Target pathway type being evaluated	2050 targets for pathways	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Aspirational Targets (SF ₆ & net CO ₂ ATs and QT > ATs)	SF ₆ 52% & net CO ₂ 100%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging
Qualified Targets (SF ₆ & net CO ₂ QTs < ATs)	SF ₆ A, B, C% & net CO ₂ X, Y, Z%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging

Table 14. Required scenarios for evaluating reductions in gas utility CH₄ and CO₂ emissions

Target pathway type being evaluated	2050 targets for pathways	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Aspirational Targets (CH ₄ & net CO ₂ ATs and QTs > ATs)	CH ₄ 47% & net CO ₂ 100%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging
Qualified Targets (CH ₄ & net CO ₂ QTs < ATs)	CH ₄ A, B, C% & net CO ₂ X, Y, Z%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging

For evaluating reductions for T&D and gas utility emissions, the required core scenarios include scenarios to evaluate the electricity and gas markets without and with power sector or economy-wide decarbonization transitions. These scenarios help T&D and gas utilities evaluate GHG reductions when there may or may not be electric sector or economy-wide decarbonization transitions. Both types of futures are part of the plausible transition space for T&D and gas utilities and therefore need to be evaluated. See Appendix B for illustrative T&D and gas utility examples using these scenarios.

Table 15 and Table 16 list the additional recommended scenarios for T&D and gas utilities respectively for more fully fleshing out the possible AT and QT transitions for all the combinations of uncertainties. The tables also note the possibility of optional sensitivity scenarios for exploring individual uncertainties of interest once the company has seen the other results.

Table 15. Additional recommended and optional sensitivity scenarios for evaluating reductions in T&D utility SF₆ and CO₂ emissions

Required, recommended, optional?	Target pathway type being evaluated	2050 targets for pathways	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Additional recommended	Aspirational Targets (SF ₆ & net CO ₂ ATs and QT > ATs)	SF ₆ 52% & net CO ₂ 100%	Transition to net-zero CO ₂ in 2050	No	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	No	More Limited	Facilitating
			Transition to net-zero CO ₂ in 2050	Yes	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	Yes	More Limited	Facilitating
	Qualified Targets SF ₆ & net CO ₂ QTs < ATs)	SF ₆ A, B, C% & net CO ₂ X, Y, Z%	Transition to net-zero CO ₂ in 2050	No	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	No	More Limited	Facilitating
			Transition to net-zero CO ₂ in 2050	Yes	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	Yes	More Limited	Facilitating
Optional sensitivities	At company discretion based on insights from other results					

Table 16. Additional recommended and optional sensitivity scenarios for evaluating reductions in gas utility CH₄ and CO₂ emissions

Required, recommended, optional?	Target pathway type being evaluated	2050 targets for pathways	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Additional recommended	Aspirational Targets (CH ₄ & net CO ₂ ATs and QTs > ATs)	CH ₄ 47% & net CO ₂ 100%	Transition to net-zero CO ₂ in 2050	No	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	No	More Limited	Facilitating
			Transition to net-zero CO ₂ in 2050	Yes	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	Yes	More Limited	Facilitating
	Qualified Targets (CH ₄ & net CO ₂ QTs < ATs)	CH ₄ A, B, C% & net CO ₂ X, Y, 2%	Transition to net-zero CO ₂ in 2050	No	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	No	More Limited	Facilitating
			Transition to net-zero CO ₂ in 2050	Yes	Fuller	Challenging
			Transition to net-zero CO ₂ in 2050	Yes	More Limited	Facilitating
Optional sensitivities	At company discretion based on insights from other results					

Optional Additional Sensitivity Scenarios

The required scenarios for electric supply, T&D, and gas utility emissions reductions will generate valuable insights regarding a wide range of potential future conditions and possible transitions for ATs and QTs; and, the additional recommended scenarios will flesh out the space of potential transitions further. A company may also want to evaluate additional possibilities

between or beyond the required and recommended conditions. Targeted sensitivity scenarios can help a company more concretely characterize a risk or enabling condition. For instance, the company may want to evaluate a decarbonization options set between Fuller and More Limited, or other conditions between Facilitating and Challenging. This might simply entail switching one or two assumptions to define a new potential condition for evaluation. For instance, a company could evaluate a future with More Limited conditions but CDR available, or they might evaluate a future with the Challenging conditions but lower barriers for capacity and transmission additions.

Furthermore, a company might want to consider an “expected” set of conditions as a sensitivity to evaluate what they consider to be a most likely future condition. Caution is merited here from a communications point of view. This scenario should not be regarded as a forecast or risk management strategy and the company does not want to distract from the risk insights provided by the other futures that are, by design, teasing out the range of transition opportunities and risks. As such, the company will want to think carefully about how best to communicate an “expected” scenario to avoid misunderstandings.

At the end of Phase 2, the company should document their definitions of Fuller/More Limited and Facilitating/Challenging using the SMARTargets Reporting Template. The template lists the uncertainties that must be considered and some other common possibilities that the company can indicate as being included or excluded. It also provides space for additional possibilities to be added.

Phase 3: Implement Scenario Design – Specify Assumptions for Scenario Modeling

- Phase inputs: The company’s Phase 2 core scenario design for the required SMARTargets scenarios.
- Phase outputs: Specification of core scenario design for the company’s modeling framework.
- Who should be involved: Staff related to planning, sustainability, policy, and strategy, as well as external experts on scenario analysis, modeling, and alternative assumptions.
- Stakeholder engagement: This is a good time to seek external stakeholder feedback.
- Documentation after iterations: SMARTargets Reporting Template entries summarizing the core scenario design specification.

To evaluate the scenario design created in Phase 2, the company will now need to translate the alternative conditions into modeling assumptions. For instance, what specific decarbonization technology cost and performance specifications does the company want to include for the options in their Fuller and More Limited sets? Similarly, if the company is evaluating technology cost uncertainty, they will need to decide what specific alternative technology cost improvements over time they want to assume and for which technologies. If they are evaluating uncertainty regarding new infrastructure development, how do they want to specify less and more constrained conditions for adding new infrastructure? One option here would be to define two types of futures: one that allows new infrastructure based on costs, and another

that does not allow new infrastructure at all. Alternatively, instead of precluding new infrastructure entirely, the second type of future could take a more nuanced approach and constrain annual future infrastructure additions informed by, but not necessarily limited to, historical rates of additions or increases in deployment costs.

Table 17 and Table 18 provide example specifications for the Fuller/More Limited and Facilitating/Challenging conditions for the scenario designs from Phase 2. Table 17 documents some possible sources for technology cost and performance assumptions. Other reputable sources could also be used (e.g., NREL, 2020). Table 18 documents illustrative specifications for the alternative assumptions for Facilitating and Challenging.

When making specification choices, it is important to keep in mind the overarching objective to identify risk and the scenario design logic of defining plausible extremes. Thus, the two specifications for an individual uncertainty should intentionally be extreme, but plausible, alternatives. It is also important to have grounded specifications with documentation of and justification for the choices. For instance, a company can use studies of future technology cost and performance to inform alternative technology improvement specifications; and, use historical information to ground projections for new infrastructure timing and levels, keeping in mind the need to be intentionally optimistic and pessimistic.

Defining emissions constraints: The company will also need to define the emissions constraint pathways they will be imposing on the model. These constraints will constrain the model to producing no more than the emissions reductions specified (though the model could produce emissions below the constraints, as we show in the examples in Appendix B). The model will identify the least-cost transitions and implications associated with achieving the required reductions in each of the alternative futures.⁷ For evaluating the required ATs, we recommend piece-wise linear constraints between the target years. Specifically, we recommend using emissions constraints that are linear from today to the 2030 or 2035 intermediate target, linear from there to any other intermediate target, and then linear beyond that to the 2050 target. For evaluating the incremental reductions associated with identifying QTs, we recommend a set of piece-wise linear constraints that are incrementally below the ATs pathway (Table 6) but follow its shape and lead to the lower 2050 reduction levels.

The outputs from this phase are the specifications for each of the alternative scenarios in the modeling framework. The company will be ready to run their model at this point, which is Phase 4.

Finally, Phase 3 is a good time for external stakeholder feedback on the company's initial scenario design and specification. It is an opportunity to discuss the preliminary futures being considered as well as SMARTargets in general and the planned transition risk analysis.

⁷ Note that, some models may be simulation models, instead of optimization models that find the cost-minimizing strategy. Simulation models identify emissions and cost outcomes for assumed transition portfolios. In this case, for futures and portfolios where the GHG targets are not met, the company would need to evaluate the implications of the additional abatement required to meet the targets.

At the end of Phase 3, the company should document their specifications for their Fuller and More Limited decarbonization options and their Facilitating and Challenging conditions in the SMARTargets Reporting Template.

Table 17. Illustrative summary of energy technology assumptions

Decarbonization technology cost and performance	Comment/description	Reference
Renewables (wind, solar, bio, hydro), nuclear, advanced nuclear, coal, coal w/ CCS, gas, gas w/ CCS, CDR (biopower w/ CCS, DAC), storage, hydrogen	Engineering-based cost and performance specifications with gradual improvement over time	EPRI (2022b)
Biomass carbon neutrality	Carbon neutral assumption based on analysis of net land use and GHG changes associated with increasing biomass supplies	Blanford et al (2022)
Other renewable fuels	Renewable liquid and gas fuels, including renewable gasoline, diesel, and jet fuel, renewable natural gas, and synthetic natural gas produce with renewable energy	Blanford et al (2022)
Low-carbon power imports	Low-carbon electricity can be exported to other regions via intra-regional transmission	EPRI (2023)
Fossil capacity reserves permitted (within emissions constraint)	Fossil generation capacity can be used to meet capacity reserve requirements but declining emissions constraints over time apply to any generation from that capacity	EPRI (2022b)
Demand conservation and energy efficiency	End-use sectors can reduce emissions through technology and fuel substitution that increases efficiency, as well as reductions in overall use	Blanford et al (2022)

Table 18. Illustrative specifications for Facilitating and Challenging other condition assumptions

Other conditions	Facilitating	Challenging
Electricity load growth	Reference = load growth due to reference economic and population growth assumptions and electrification due to electric technology relative cost improvements	Faster = reference load growth conditions plus additional data center load growth based on recent trends resulting in 12% greater total load by 2030 (EPRI, 2024)
Electricity generation cost improvements	Faster = default cost projections and reference learning curve for battery cost improvements over time	Slower = higher cost projections by technology and slower learning curve for battery cost improvements over time
Electricity capacity additions	Economic = allows for all economically viable electric capacity to be built	Constrained = limits the building of economically viable electric capacity to historical build rates
Electricity transmission additions	Economic = allows for all economically viable interregional transmission to be built	Constrained = limits the building of economically viable interregional transmission based on historical transmission build rates
Fossil fuel prices*	Lower = No additional supply costs for fossil fuel production beyond the equilibrium cost determined by the energy economy model's balance of supply and demand for each fuel	Higher = Additional supply costs added to fossil fuel production modeling low resource availability or pipeline expansion
Electric technology cost improvements	Faster = Default cost reductions in EV & heat pump components, faster EV adoption reflective of less range anxiety and greater availability of new electric technologies	Slower = Higher component costs for EV & heat pumps, slower EV adoption representative of more range anxiety and less availability of new electric technologies

* The fossil fuel price assumptions in the table reflect the US-REGEN modeling structure. Other economy-wide frameworks might have fixed assumed (i.e., exogenous) energy prices. For instance, they might be based on lower/higher U.S. Department of Energy Annual Energy Outlook fuel price projections.

Phase 4: Identify Potential Transitions and Target Implications

- Phase inputs: Phase 3 specification of core scenario design for implementation in the company's modeling framework.
- Phase outputs: Preliminary transition analysis pathways and insights.
- Who should be involved: Staff and external experts running the model and processing outputs.
- Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.

This phase creates preliminary transition results, running the company's modeling framework with the scenario specification implementation and generating initial potential transition pathway model results. The modeling provides information regarding the alternative potential

transitions, implications, trade-offs, risks, enabling conditions, robust strategies, contingencies, and more. As the company attempts to run each of the scenarios in the design, they may have to revise their scenario specification implementation if the model is not solving or it is producing strange or unintuitive results.

In addition to running the scenarios, the company will want to develop preliminary graphics—figures and tables—and identify preliminary insights for review. In particular, the company will want to evaluate differences and similarities in transition results between scenarios. For instance, utilities would likely be interested in the alternative potential transitions for, among other things, electricity capacity and generation mixes, electricity demand, gas consumption and composition, power transmission, emissions, costs, and prices.

A sample of the types of results that can be produced with the core scenarios (Table 10) are provided in the figures and table below. They illustrate potential transitions for electricity generation and load, electricity generation capacity, new capacity investments and retirements, gas consumption, and power sector emissions respectively for a U.S. Midwestern region. They also provide examples of transition uncertainty ranges as well as riskier outcomes for this region. For instance, Figure 10, Figure 11, and Figure 12 illustrate the possibility of very different plausible electricity generation, capacity, and demand (load) transitions due to the different conditions—sector versus economy-wide decarbonization, decarbonization options, and the other conditions.

Figures like Figure 10, Figure 11, and Figure 12 and also help identify robust strategies and investment and operations uncertainty. For instance, load increases, wind, solar, nuclear, and storage are key features in all four AT decarbonization transitions, as well as reference futures. As such, they would be robust strategies. Figure 12, in particular, illustrates that additions of wind, solar, and storage are consistent elements of all four decarbonized futures, as is retirements of coal. However, we also see in Figure 12 notable variation in the additions by 2035 and 2050, especially for solar, storage, and wind; and, mixed results for addition and retirement opportunities for gas, nuclear, and hydrogen, with only some conditions conducive to those investments. Overall, the cumulative capacity changes associated with all four futures are substantial but with large variation. By 2035, we find additions of approximately 100 to 195 GW and retirements of approximately 30 to 40 GW. By 2050, the additions and retirements are approximately 2080 to 380 GW and 65 to 80 GW, respectively. For comparison, in 2024, there was 60 GW of new capacity additions planned across the entire U.S., and 7.5 GW of capacity retirements planned (EIA, 2024). The reference futures alone suggest a good deal of cost-effective additions and retirements, with gas, wind, and hydrogen the primary capacity additions and coal the primary retired capacity (Figure 12). Note that fossil related resources are providing notable generation and/or capacity to 2050 in all the net-zero decarbonization futures, with the role and the magnitude varying with the different conditions.

Figure 13 illustrates transition uncertainty that may exist for gas markets. We observe the possibility of increasing or decreasing gas consumption, with natural gas still being consumed across the economy in 2050 in all the transitions. The key uncertainty for the gas market is whether there is an economy-wide, an electric sector, or no decarbonization transition. We also

see that in the economy-wide decarbonization futures, the gas market in 2050 is largest with Fuller decarbonization options and Facilitating conditions (versus More Limited and Challenging conditions). This is primarily due to the availability of CDR and the lower price of gas in the Fuller-Facilitating future.

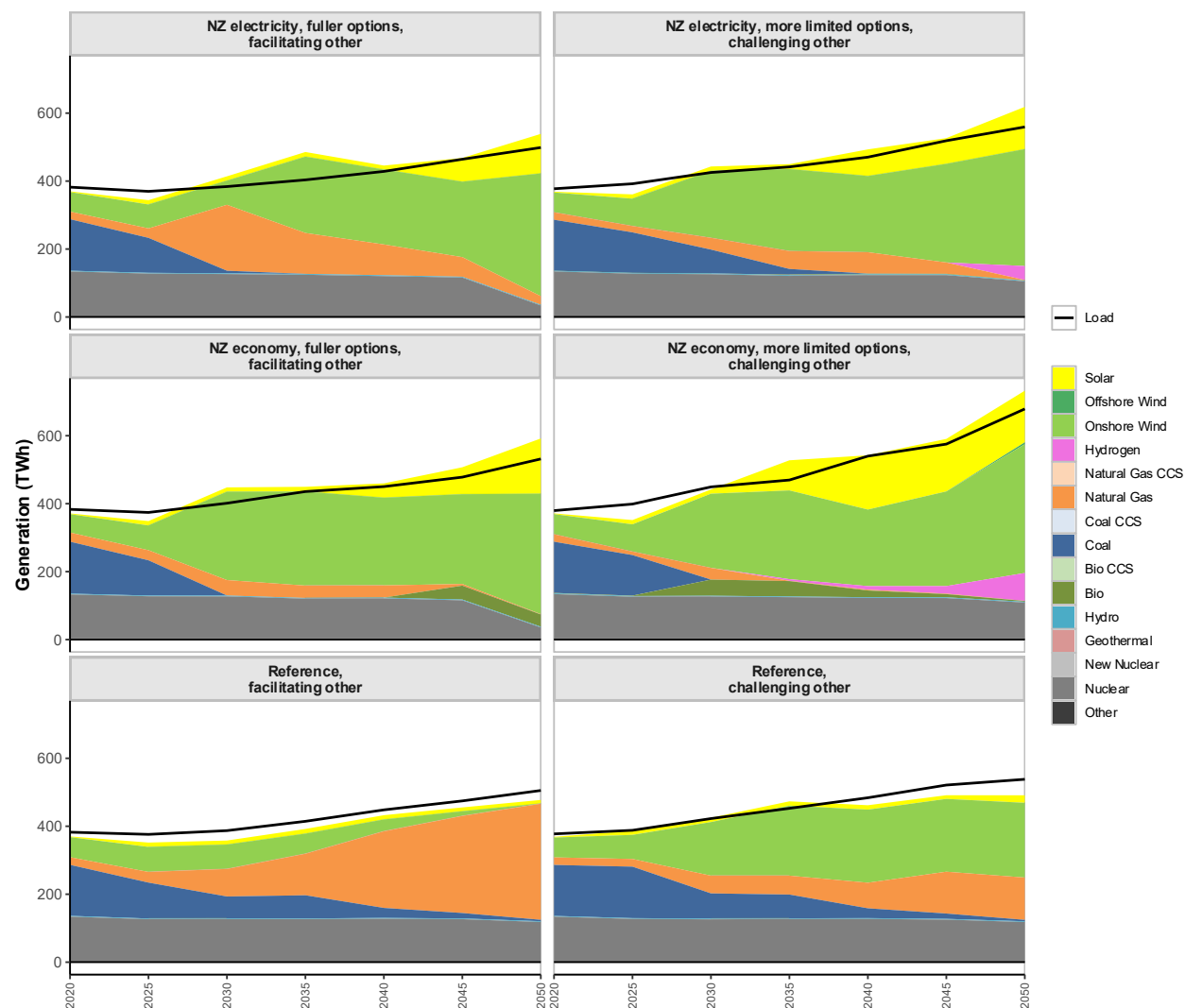


Figure 10. Illustrative regional electricity generation and load transitions for the required electricity supply emissions Aspirational Target scenarios

Figure note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification. Top row results associated with net-zero CO₂ by 2050 power sector transitions with Fuller and More Limited decarbonization options available and Facilitating and Challenging market and technology conditions. Middle row results associated with net-zero CO₂ by 2050 economy-wide transitions with Fuller and More Limited decarbonization options available and Facilitating and Challenging market and technology conditions. Bottom row results associated with reference economy-wide transitions with Facilitating and Challenging market and technology conditions. Natural gas includes a mix of renewable and synthetic natural gas, and dual fuel capacity includes capacity using mixes of fossil and renewable fuels (gas and liquid).

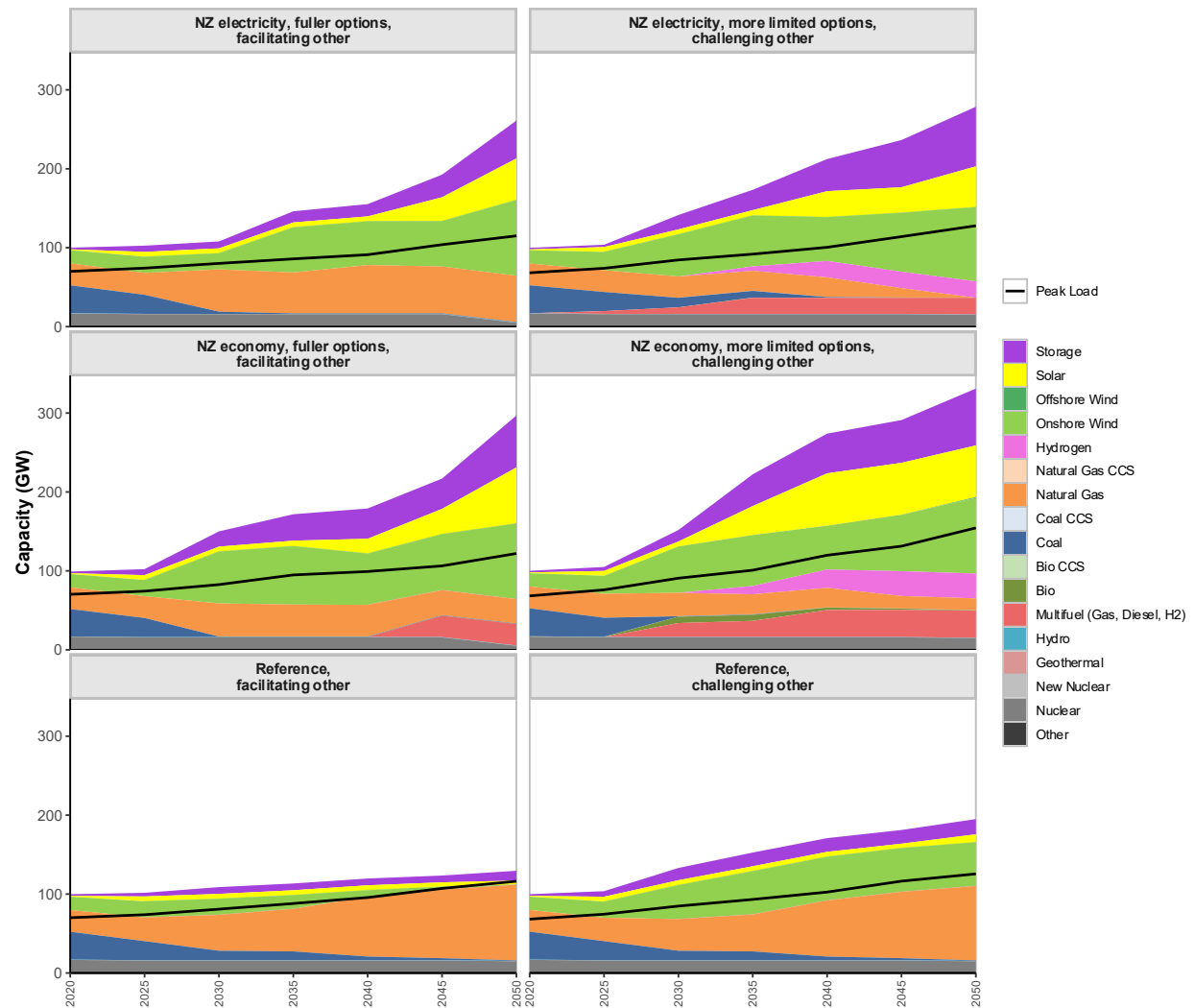


Figure 11. Illustrative regional electricity generation capacity transitions for the required electricity supply emissions Aspirational Target scenarios

Figure note: Figure 11 note.

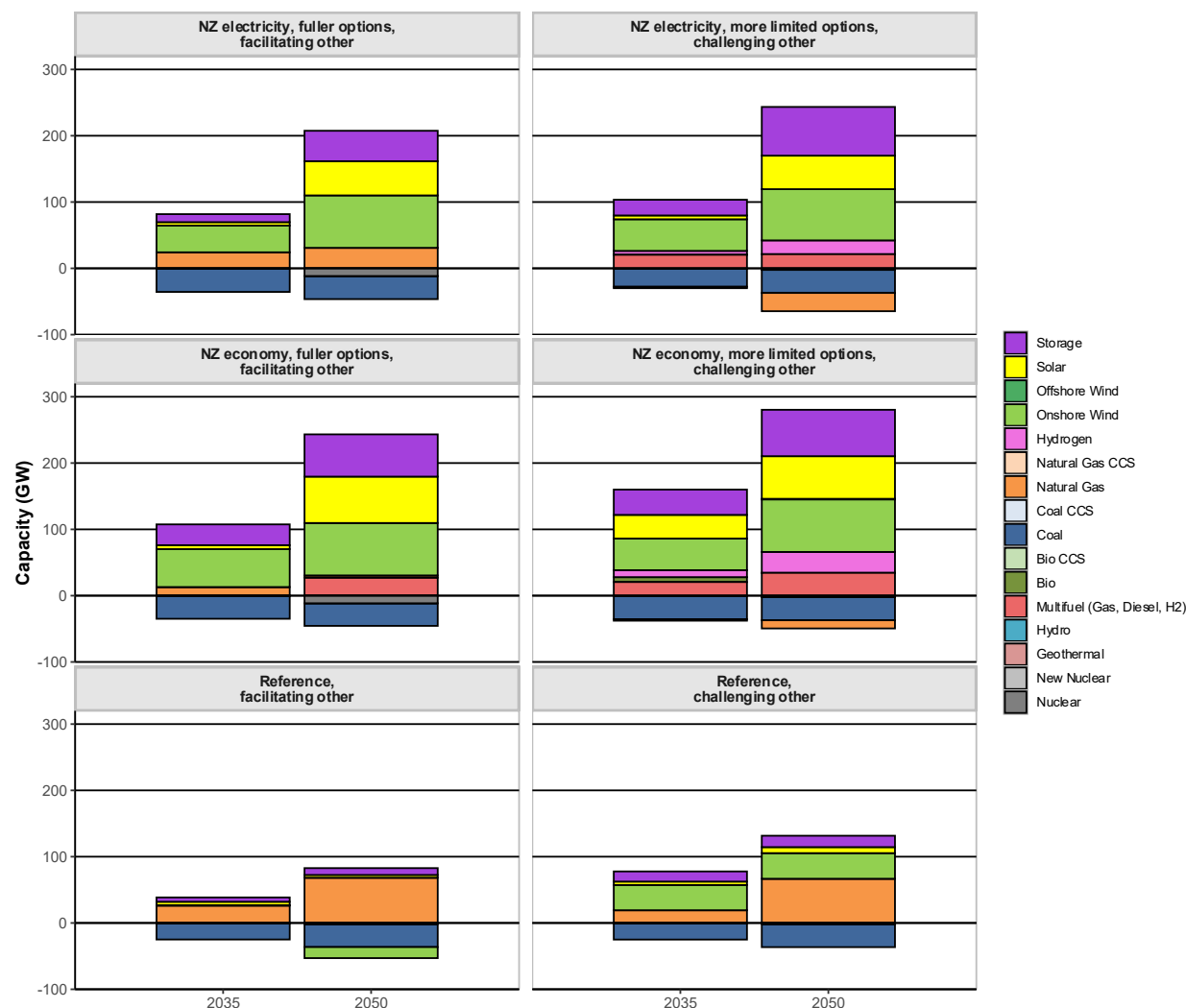


Figure 12. Illustrative regional electricity generation capacity cumulative additions and retirements by 2035 and 2050 for the required electricity supply emissions Aspirational Target scenarios

Figure note: Cumulative capacity additions and retirements are positive and negative, respectively. Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification. Natural gas includes a mix of renewable and synthetic natural gas, and dual fuel capacity includes capacity using mixes of fossil and renewable fuels (gas and liquid).

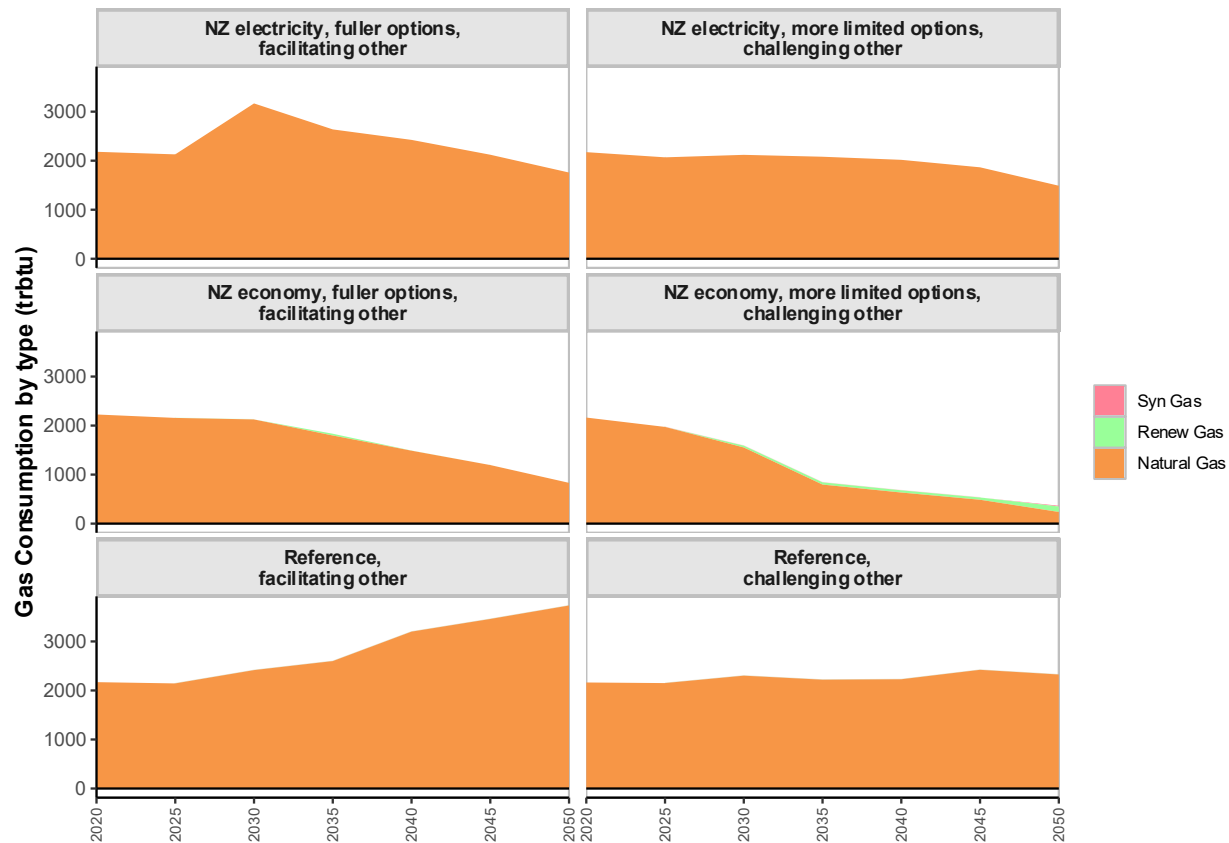


Figure 13. Illustrative regional gas consumption transitions for required gas utility emissions Aspirational Target scenarios

Figure note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification.

Figure 14 illustrates how the power sector decarbonization transition could be much quicker when there is an economy-wide versus power sector only decarbonization incentive. This is an example of an enabling condition and is an important result discussed more in Appendix B. It is also an example of QTs with reductions greater than the ATs. In this context, the economy-wide constraint on emissions requires decarbonization of the end-use sectors as well, such as transportation, buildings, and industry. Forcing decarbonization of end-use activities creates incentives to use low-carbon options, including low-carbon electricity, which incentivizes faster adoption of low-carbon generation capacity in the power sector (Figure 12) and faster reductions in power sector emissions (Figure 14). As reflected in the load results in Figure 11, the economy-wide end-use decarbonization pressure is creating a market for low-carbon electricity and that demand is financially supporting faster decarbonization of the power sector.

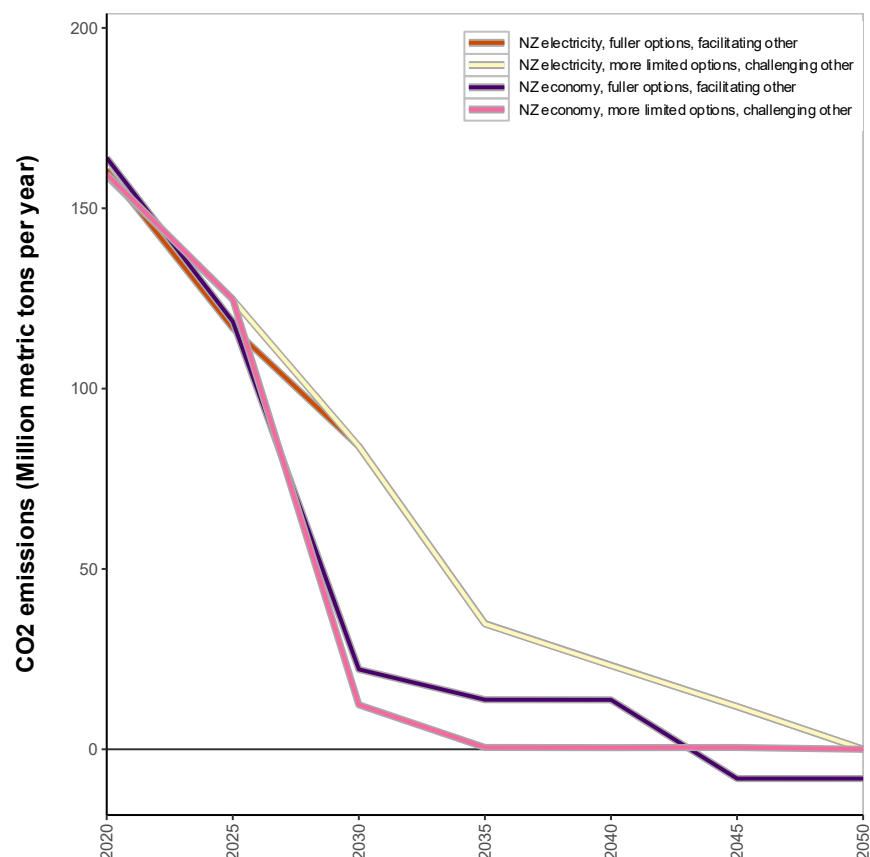


Figure 14. Illustrative regional power sector net CO₂ emissions transitions for the required electricity supply emissions Aspirational Target scenarios

Figure note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification.

Figure 15 provides an example of an incremental implications result from the QT (< AT) scenarios. The incremental reduction scenarios help inform identification of the QTs possible under different conditions, especially when enabling conditions are unavailable. In this case, Figure 15 presents incremental discounted gross electric sector costs through 2035 and 2050 for increasing levels of 2050 power sector emissions reduction ambition (30%, 50%, 80%, and 100%). The 100% results come from the required AT scenarios. Figure 15, we see, in general, increasing discounted costs with larger emissions reductions due to greater generation capacity and transmission additions, O&M costs and power imports. We also see the impact of the Facilitating versus Challenging conditions, where the former is resulting in substantially lower costs of 22-33% and 21-30% by 2035 and 2050 respectively due to the Facilitating environment's lower load growth, lower technology and fuel costs, and reduced barriers for new infrastructure additions.

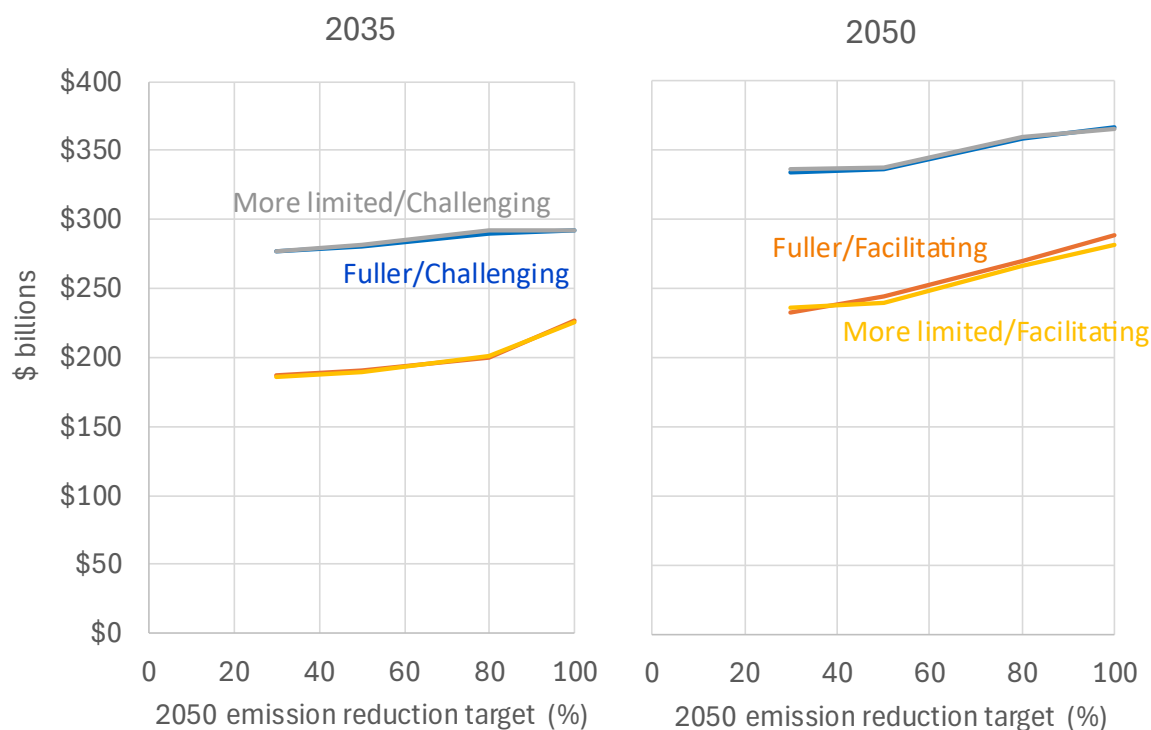


Figure 15. Illustrative regional incremental discounted gross electric sector costs through 2035 and 2050 for the increasing levels of potential 2050 emissions reduction levels

Figure note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification. Gross costs include all fuel, investment, transmission, operations and maintenance, CO₂ pipeline and storage, and power imports costs. Power export revenues not included.

Finally, Table 19 illustrates the type of transition uncertainty insights that can be produced looking across the scenarios, as well as across variables relevant to different priorities that are important to near-term and long-term decisions—affordability, system flexibility and reliability, and sustainability. The ranges help communicate the uncertainty, where wider ranges represent greater uncertainty.

The table also helps identify risky outcomes. For example, there is the possibility of near-term (2035) electricity price increases of almost 370%, with investments of almost \$300 billion and electricity capacity additions of over 100 GWs. In these illustrative results, all the scenarios are producing significant price increases and costs; thus, net-zero is going to be challenging regardless of the conditions. Looking at the individual scenarios behind Table 19, we can also identify the specific conditions contributing to the more and less risky outcomes (see Appendix B for these results). Depending on a company's needs, other metrics and outputs could also be produced, such as individual technology transitions, load growth details, or air pollutant emissions.

The set of figures and tables also provide milestone and contingency insights. For instance, the results suggest the need to evaluate which future is emerging in a few years to determine if the strategy needs to be adjusted. If, for example, the decarbonization options available are turning out to look more like the More Limited set, instead of Fuller, the company, in order to stay on track for their emissions targets, may want to consider alternative strategies for gas, solar, wind, storage, and hydrogen, as well as possibly slowing coal retirements if annual additions of renewables are likely to be constrained. Furthermore, those strategies will depend on whether sector or economy-wide decarbonization incentives are emerging, which is another milestone development the company can monitor. In general, milestones defined in terms of the Fuller/More Limited decarbonization options, the Facilitating/Challenging condition sets, and the state of sector and economy-wide low-carbon transitions, can be used to guide company timing for when to re-evaluate their low-carbon strategy and how they might respond.

Results like those in Table 19 also facilitate more nuanced discussion of risk. For example, utility regulators may not allow recovery of some of the decarbonization costs shown in Table 19, which represents an important consideration and risk. Alternatively, large electricity customers might be willing to pay for low-carbon power, which would help manage the cost risk. It is important to keep in mind that revenues from increased electricity demand in the economy-wide scenarios is not shown; however, those revenues are supporting increased electric sector low-carbon investments and a more rapid decarbonization of the power sector.

Overall, the set of results from Phase 4 shed light on the opportunities, challenges, and enabling conditions for ATs and QTs. See Appendix B for full implementation examples of the methodology and types of insights.

Table 19. Illustrative regional cross-scenario results ranges for multiple variables for the required electricity supply emissions Aspirational Target scenarios

Table note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification.

	Affordability			Stability					Optionality	Sustainability			
	Change in Wholesale Electricity Price	Electric Sector Costs		Total New Generation Builds	Total Generation Retirements	Net Power Imports	Variable Renewable Generation	Electric Load Growth	Dispatchable Resources	CO ₂ Emissions Reductions		CO ₂ Emissions Intensity	In-State Clean Electricity
	% Change vs 2020	\$ Billions NPV		GW from 2020	GW from 2020	% of Load	% of Total MWh	% Change vs. 2020	% of Total Capacity	% Change from 2020		% reduction in tCO ₂ /MWh from 2020	% of MWh
		Total	Capital							Electric	Economy		
Ranges for 2035	+109% to +373%	212 to 305	29 to 72	103 to 196	33 to 41	-23% to -3%	50% to 73%	6% to 24%	47% to 56%	78% to 100%	46% to 95%	79% to 100%	75% to 100%
Ranges for 2050	+21% to +50%	263 to 376	64 to 113	277 to 379	63 to 78	-6% to -2%	74% to 94%	30% to 79%	38% to 45%	100% to 100%	72% to 114%	100% to 104%	95% to 100%

Phase 5: Review Potential Transitions and Target Implications – Based on Decision-Making Needs

- Phase inputs: Phase 4 preliminary transition analysis pathways and insights.
- Phase outputs:
 - a. Preliminary insights regarding potential transitions, targets, and risks,
 - b. Revisions to core scenario design, model implementation, and output reporting, and
 - c. Additional recommended and sensitivity scenarios and specifications the company would like to evaluate.
- Who should be involved: Company decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.
- Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.

This phase consists of critical engagement with the company's decision-makers and their business units to review the preliminary transition analysis results and insights. This phase takes the transition pathway results and insights from Phase 4 as inputs and evaluates them in terms of decision-making needs and considerations. Effective presentation and discussion of the results are important. The phase has two key outputs: agreement on preliminary insights regarding potential transitions, targets, and risks; and, identification of needed revisions to the scenario design, model implementation, and output reporting. In addition to identifying revisions to the core scenario design, this phase also helps identify ideas for additional scenarios.

Given our need to identify enabling conditions, this is the time to think about challenges, trade-offs, risks, and risk management. Are there transition outcomes that the company considers risky and future conditions associated with them that the company would like to avoid? Similarly, are there conditions that allow the company to achieve a target and balance the trade-offs? This phase is also the time to consider the likelihoods of the different futures and potential hedging strategies. If, for instance, the company believes that the likelihood of an economy-wide policy is low, they might invest less in new low-carbon electricity capacity since the demand for the clean electricity is uncertain.

In the Phase 4 illustrative results above, we find the following challenges, trade-offs, risks, and risk management strategy insights:

- *Challenges and trade-offs:* The ATs are being met (or more) in all the transitions, but in each transition, we also observe challenges with affordability and reliability implications, investment implications in capacity build results, and sustainability implications with a range of emissions reductions by 2035.
- *Risks:* The set of results highlight that all three categories of uncertainties create risk for this location, with large variation in the transitions and implications for futures achieving the ATs. Thus, uncertainty regarding the decarbonization incentive (sector versus economy-wide), decarbonization options (Fuller versus More Limited), and other conditions

(Facilitating and Challenging) are all important considerations contributing to uncertainty in the transitions, with some more difficult to manage in terms of trade-offs in prices, costs, load growth, generation and capacity resource choices, traded power, and emissions transitions.

- *Risk management strategies:* Looking across the scenarios, we can identify enabling conditions that allow for transitions with more manageable trade-offs. From careful comparison of scenarios that differ in only one category of uncertainty, we find that encouraging an economy-wide policy with end-use decarbonization incentives would facilitate faster low-carbon generation investments and emissions reductions in the electric sector. We also find that costs are lower when there are the Facilitating (versus Challenging) conditions. Therefore, strategies that accelerate technology cost improvements, reduce capacity additions barriers, or lower fuel costs will be helpful. The lower costs shown in the results help justify pursuing and collaborating on strategies like these. Finally, we find that having Fuller decarbonization options (versus More Limited) has mixed effects on costs. In the power sector only decarbonization transitions the differences are modest, which only highlights the importance of modeling and evaluating how the economics might change. However, having Fuller options is found to help reduce the decarbonization challenge in the economy-wide decarbonization transitions for this region.

As discussed in the overview, and evaluated in the next step (Step 5), companies should keep in mind that Paris Alignment of their transition strategy will be assessed qualitatively relative to the robust transition insights from global emissions pathways and the strategies associated with achieving the Paris Agreement. The company's transition analysis results will be assessed for consistency with these Paris Agreement aligned transition characteristics. The illustrative Phase 4 sample results above, for instance, are qualitatively aligned in that they exhibit increasing reliance on electricity, decreasing reliance on fossil fuels, increasing deployment of low-carbon technologies, and the use of carbon dioxide removal and energy conservation and efficiency (last two not shown). In addition, the transitions and insights align with Paris Agreement recognition of differences in country decarbonization opportunities, multiple societal priorities, and the value of cooperation (in this case across sectors via the economy-wide scenarios).

After developing the preliminary insights, the company will then identify whether revisions are needed to the scenario design, model implementation, and/or output reporting. This conversation should include discussion of whether to run additional scenarios—the recommended additional scenarios above or custom sensitivity scenarios. See the Phase 6 discussion for sensitivity scenario examples.

Phase 6: Revise and Iterate as Needed – Revising Scenario Design, Implementation, and Model Results

- Phase inputs:
 - a. Phase 5 revisions to core scenario design, model implementation, and output reporting, and
 - b. Phase 5 sensitivity scenarios and specifications the company would like to evaluate.

- Phase outputs:
 - a. Revised insights regarding potential transitions, targets, and risks, and
 - b. As needed, additional revisions to scenario design, model implementation, and transition pathways.
- Who should be involved: Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.
- Stakeholder engagement: This is a good time to seek external stakeholder feedback.
- Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.

Iterate as needed to finalize the transition and target results and insights. This entails repeating Phases 2 through 5 until comfortable with the results information and insights. Thus, the company should revise the core scenario design and implementation, add and revise sensitivity scenarios as appropriate, re-run the model with a revised scenario design and implementation, and review results, as necessary. For instance, upon reviewing draft results, the company might revise the scenario design to add a technology to their initial Fuller decarbonization options set (such as agriculture or forestry GHG offsets), remove a technology from the More Limited decarbonization options set, or relax annual new build rate constraints a bit in the Challenging conditions. The company might also wish to run the additional recommended scenarios or sensitivity scenario or explore additional modeling outputs. For example, in the case of sensitivity scenarios, the company might wish to evaluate the benefits of having CDR available in the More Limited and Challenging conditions or, in the case of additional outputs, the company might want to evaluate end-use energy consumption transition details regarding potential transitions in technology adoption and fuel use.

Finally, Phase 6 is another good time for external stakeholder feedback. At this point, after the company's internal iterations, they will have final drafts of their scenario design, results, insights, and strategies to share and discuss. It is an opportunity for education, dialogue, and feedback on the company's potential transitions, challenges, uncertainties, risks, enabling conditions, risk management, and collaboration opportunities.

Phase 7: Communicate Results – Targets, Transitions, Risks, Enabling Conditions, Risk Management

- Phase inputs:
 - a. Phase 6 final scenario design (core and additional scenarios), model implementation, and transition pathways, and
 - b. Phase 6 final insights regarding potential transitions, targets, risks, enabling conditions, and risk management.
- Phase outputs:
 - a. Documented scenario analysis, outputs, and insights,
 - b. Aspirational Targets (ATs), strategies, and risk management, and
 - c. Qualified Targets and their qualifying conditions (QTs) and strategies.

- Who should be involved: The company's decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.
- Documentation: SMARTargets Reporting Template summarizing the following:
 - a. The company's ATs and QTs with corresponding conditions,
 - b. The company's enabling conditions, strategy, risks, and risk management,
 - c. The company's transition risk analysis (modeling, scenario design and specification, potential transitions, insights), and
 - d. The implementation process.

This phase is focused on documenting all of Step 4 for transparency, communications, and validation. The scenario analysis provides a wealth of information that can educate readers on uncertainties, potential futures, opportunities, enabling conditions, and risks, as well as opportunities for collaboration on enabling conditions. This phase also provides justification and transparency for the company's GHG targets, strategies, and risk management.

At the end of Phase 7, the company should use the SMARTargets Reporting Template to finalize the documentation for their Step 4 transition risk analysis (Phases 1 through 7). This includes the summary tables of key output indicators, as well as graphics to help communicate the company's modeling results and insights. Additionally, the company should report the identified enabling conditions for ATs, QTs and corresponding conditions, and the risks and opportunities identified. Finally, the company should indicate their risk management strategies along with their milestones and contingency plans.

Required Outputs

Step 4 has a minimum set of required outputs. Table 20 lists the required outputs related to electric power, gas, and transmission and distribution utility emissions transitions. Additional variables may be added based on company need (e.g., air pollutant emissions, differentiated affordability indicators). Together, the variables in Table 19 provide useful insights related to:

- Transition possibilities,
- Transition uncertainties and risks,
- Transition trade-offs between objectives,
- Robust strategies identified from findings found consistently across transitions,
- Enabling conditions that facilitate outcomes and help manage trade-offs and risks,
- Contingency strategies for responding if different futures emerge, and
- Risk management strategies to reduce the likelihood of riskier outcomes.

Not only are the outputs of transition risk analysis helpful to internal decisions, but they provide valuable material for educating, engaging, and communicating externally, including communicating on enabling conditions that facilitate coordination and collaboration.

Table 20. Required outputs from electric power, gas, and T&D utility transition risk analyses

Table note: The full set of variables and metrics from a specific analysis can vary with decision-maker needs and modeling capabilities.

Electric utility	Gas utility	Transmissions & distribution utility
Affordability		
Electricity prices	Gas prices	Electricity prices and/or T&D rates
Costs	Costs	Costs
Investments	Investments	Investments
System Operation, Flexibility, and Reliability		
Generation levels and mix	Gas consumption (total & by fuel type)	Electricity demand, i.e., load (aggregated, disaggregated)
Generation capacity additions, retirements, and resource mix	Gas consumption (total & by consumption type)	Transmission and distribution flows
Electricity transmission utilization and capacity additions		Inter-regional imported and exported power
Electricity demand, i.e., load (aggregated, disaggregated)		Transmission and distribution capacity expansion
Fuel use		
Sustainability		
GHG emissions	GHG emissions	GHG emissions

Step 5: Strategy and Qualified Target Alignment with International Goals

Step 5 Outputs: International goal alignment assessments of qualified targets and strategies.

In this step, a company will evaluate two types of *qualitative* alignment of their transition strategy. The company will also evaluate *quantitative* alignment of their QTs with the international goals.

Assessing Qualitative Alignment of the Company's Transition Strategy

SMARTargets requires that companies qualitatively align their low-carbon strategies with (a) the types of transitions robustly observed in global emissions pathways consistent with the Paris Agreement temperature goals, and (b) the provisions in the Paris Agreement associated with achieving the temperature goals.

While quantitative energy and economic transition results vary significantly across global emissions pathway models and assumptions (IPCC, IEA, PRI, etc.), they all consistently produce the same set of qualitative low-carbon transition insights (EPRI, forthcoming; EPRI, 2020). These robust characteristics of transitions reliably inform company thinking and strategy, as well as stakeholder expectations. Below is a list of robust insights from global 1.5°C and 2°C emissions transitions. Across models and assumptions, we consistently find:

- Increasing low-carbon technology deployments,
- Increasing reliance on electricity,
- Transitions away from fossil fuels but continued use,
- Use of carbon dioxide removal,
- Improvements in energy conservation and efficiency,
- Differences in regional low-carbon transitions, and
- For any climate goal, substantial uncertainty in the levels, rates of change, and composition (e.g., technologies, energy, activity) of the transition of the global economy.

Furthermore, a company's strategy should qualitatively align with the Paris Agreement's provisions. Paris Agreement provisions, other than the temperature goal, represent important insights and international agreement regarding how to move forward with pursuing the temperature goal. Provisions like recognizing country differences in decarbonization opportunities, multiple societal priorities, and the value of cooperation are as relevant to companies throughout the world, just as they are to countries, in their supporting pursuit of the temperature objectives.

While the opportunities for company decarbonization vary from company to company, the types of robust transitions noted above and the strategies recognized by the Paris Agreement are reliable and aligned guides for company low-carbon strategies. The list above represents a partial list of strategy categories for each company. A company can use the list, along with their Step 4 transition analysis, to guide their strategic discussions to develop specific Paris Agreement aligned strategies for pursuing their targets.

Assessing Alignment of the Company's Qualified Targets

SMARTargets also requires that companies assess quantitative alignment of their QTs. The assessment is based on comparison to the global emissions pathway ranges consistent with the Paris Agreement primary objective of limiting warming to "well-below" 2°C. Comparing to the relevant total CO₂, non-CO₂, or CO₂e global percent reduction emissions pathway ranges aligned with limiting warming to well-below 2°C provides useful high-level insights regarding the consistency of the QTs with pursuing the Paris Agreement temperature goals (Figure 7 and Table 6). For example, a net CO₂ QT of -40% in 2030 would be considered aligned with the Paris Agreement since it falls within the 2030 range of +10% to -100% consistent with limiting global warming to well-below 2°C as defined by the IPCC, which includes both 1.5°C and 2°C (with 67% likelihood) global emissions pathways (see Scientific Foundations). While -40% is less ambitious than the 2030 AT of -56%, it is consistent with global pathways aligned with containing global warming to well-below 2°C.

The high-level insight from this comparison is that QTs within the ranges are aligned with global futures consistent with limiting warming to well-below 2°C. Furthermore, if all companies have QTs within the range, the sum will also be aligned with well-below 2°C. However, it is critical to keep in mind that this comparison is not conclusive. Comparing to the global pathway range is not a conclusive assessment of company alignment, first and foremost, because the global

pathways do not capture the transition opportunities and challenges for individual companies, or differences across companies. In addition, the global pathway ranges do not reflect all the sub-global and local transitions found to be aligned with a global temperature (see Scientific Foundations). For example, broad ranges of US, EU, and China electric sector transitions are found to be consistent with a single global emissions pathway. Furthermore, in well-below 2°C global pathways emissions for some regions can be rising while they are falling for others due to differences in energy demand, population growth, and other drivers. Thus, QTs for some companies may not fall within the global pathway ranges due to their transition challenges. That should not be interpreted negatively. It is simply an indication of their challenges. It is important to keep in mind that the key virtue of, and information gleaned from, the QTs is that they are the highest possible ambition for a company under different conditions while balancing trade-offs and risks. Some companies may face particularly challenging circumstances, such as in developing countries. The Step 4 transition risk analysis will help them identify and communicate their challenges, which will be valuable information for the company and their stakeholders. The QTs represent actionable and substantive contributions to the Paris Agreement goals with additional ambition possible if enabling conditions can be put in place. Policymaker and stakeholder cooperation and coordination to create enabling conditions are especially valuable in these cases.

Alignment with National Goals

Some companies might want to speak to their countries NDC emissions goals as well. The company can simply note the emission goal in percent reduction terms and timing relative to the global total emissions pathways and the company's targets and company low-carbon transition pathways. Visually, the company could overlay the NDC emission goal onto a chart with the relevant global percent reduction pathway range (see illustrative examples in Appendix B). The company can also explain how they see their targets contributing to the national pledge. Comparing to sectoral national goals, however, is not recommended since it can be misleading. As discussed, a company can cost-effectively differ substantially from a sectoral aggregate and from other companies due to their unique opportunities. Furthermore, what is practical for the company will be a function of the specific policy incentives for the sector, as well as other sectors. Finally, if there are sectoral policies regulating emissions, they should be considered within the context of Step 2 where the impact of regulated emissions on target setting is explicitly addressed.

Step 6: Documentation and Communication

Step 6 Output: SMARTargets documentation.

In this step, a company will document and communicate their SMARTargets. Using the SMARTargets Reporting Template, a company will document their implementation of the SMARTargets Methodology. This includes documenting implementation choices with justification, transition analysis insights, targets, strategy, and milestones.

The SMARTargets Methodology provides a scientifically grounded, transparent, consistent, and comparable process and outputs that provide users and stakeholders with confidence in the approach, results, and insights, as well as the resulting company's targets and strategy. The methodology itself is a scientifically derived, standardized, systematic, in-depth process that each company implements and that creates the standardized, comparable information needed to inform stakeholders and validation activities.

Communicating SMARTargets involves presenting a clear and comprehensive package of information about both the process and the outcomes. The SMARTargets Reporting Template helps document a company's implementation of each step of the methodology—naturally guiding the creation of a standardized set of information.

The SMARTargets reporting template consists of a series of structured tables designed to help companies systematically document their implementation of the eight steps of the SMARTargets methodology. By completing these tables, companies will generate standardized outputs that enhance transparency, support effective communication, and enable stakeholders to more easily assess the credibility and alignment of their targets and transition strategies. Each table corresponds to a specific aspect of the implementation process, guiding users through data collection, analysis, and alignment. Specifically, the SMARTargets Reporting Template includes the following sections of information (see Reporting Template for details):

- General Company Information and Stakeholder Engagement
- GHG Inventories and base year
- Practical issues and climate target groupings
- Company Transition Risk Analysis of Opportunities and Risks
- Alignment Assessment
- Documentation and Communication of Targets, Strategies and Indicators
- Implementation and Monitoring

Overall, in applying the methodology, a company will produce the following standardized and comparable information that reflects their process and opportunities:

- a. Paris Agreement-aligned targets and enabling conditions, including Aspirational Targets (AT) and Qualified Targets (QT) and their associated conditions,
- b. Insights into potential low-carbon transitions,
- c. Assessment of risks, risk management strategies, and overall transition strategy, and
- d. Documentation of methodology implementation, including:
 - GHG inventory by required emissions category,
 - Responses to practical considerations and target-setting decisions for each category,
 - Transition risk analysis components:
 - Modeling approach
 - Reference conditions and key uncertainties

- Core scenario definitions and specifications
- Criteria for identifying risks and enabling conditions

A company is encouraged to holistically communicate their strategy. This goes beyond emissions targets and may include the following that together can help create the enabling conditions identified by the company for achieving their targets, increasing ambition, and supporting economy-wide decarbonization: transition plans, research and development (R&D), educational outreach (stakeholder and policy), and collaboration and coordination with customers and supply chain partners.

Implementing SMARTargets creates tremendous leadership opportunities for a company, including educational opportunities to inform and elevate conversation, such as on the science and issues relevant to corporate low-carbon transitions and sound decision-making, and the company's specific decarbonization opportunities, risks, and enabling conditions. The leadership opportunities also include facilitating cost-effective economy-wide decarbonization. For instance, launching a decarbonization initiative with customers and communities for achieving economy-wide decarbonization that is grounded in the Step 4 analysis of transition opportunities, risks, and enablers.

In addition to the standardized outputs, a company should consider communicating on the SMARTargets Methodology itself, including its scientific foundations. This can involve referencing the publicly available methodology, the SMARTargets Scientific Foundations, and other materials and emphasizing how the approach is science-based and aligned with the Paris Agreement (see Justification for the Methodology section).

Step 7: Validation and Verification

Step 7 Outputs: Validated SMARTargets and verified progress.

In this step, a company will validate their SMARTargets and verify their progress. After completing the SMARTargets Reporting Template in Step 6, a company will be able to initiate validation activities. Validation assesses whether the methodology has been properly implemented and whether the target and strategy—by nature forward-looking statements—are supported by sound assumptions, data, and methods. The primary input for validation is the SMARTargets Reporting Template, which documents each step of the methodology’s implementation.

SMARTargets facilitates internal and third-party validation according to international best practices for validation and verification procedures based on the principles of impartiality, independence, and objectivity. Companies will need to establish internal controls, auditing practices and consider their climate-related risk disclosure needs and regulations.

Note that independent third-party validation services for corporate GHG targets will need to develop to provide the appropriately grounded, trained, objective, and independent assurance that investors and other stakeholders require. These services do not exist currently.

In addition to validating, the methodology requires periodic verification of progress. This verification relies on GHG inventories, which are based on observable data and are therefore subject to assurance-level assessments.

Internal and External Validation and Verification of Progress

SMARTargets is designed to be validated using internationally accepted concepts, principles, and best practices. Companies should begin with internal validation, leveraging internal audit processes aligned with recognized standards. This step ensures the methodology has been correctly applied and that the resulting targets and strategies are scientifically grounded and aligned with the Paris Agreement. Internal validation also builds confidence by thoroughly documenting the rationale and process behind the targets—information that is essential for both third-party validation and external communication.

Third-party validation, while optional, is recommended. It provides an independent, authoritative assessment of conformity with the SMARTargets Methodology and validation criteria. To ensure impartiality, third-party validation must be conducted by an accredited organization with no vested interest in the company's targets or strategy. As climate-related risk disclosure becomes more prominent, investing in both internal and external validation can enhance credibility and stakeholder trust.

While internal validation is required to develop a SMARTarget, third-party validation may be expected by certain stakeholders. Companies should evaluate whether external validation is necessary based on their specific context and audience.

Section 5 details the full validation process, including criteria for both internal and external validation, as well as the procedures for verifying progress toward targets. Completing this validation and verification cycle is required to formally claim SMARTargets status.

Step 8: Monitoring and Adjusting

Step 8 Outputs: Periodic milestone assessment and, as needed, strategy adjustment and/or SMARTargets refresh.

In this step, a company will monitor their milestones and adjust their strategy as needed. A company with SMARTargets will not only need to verify progress towards their targets (Step 7), but they will also need to monitor conditions with respect to their milestones to determine if strategy adjustments are needed and/or if updated SMARTargets analysis, targets, and strategies are needed.

The transition risk analysis (Step 4) provides milestone and contingency insights as well as insights regarding transition opportunities and risks. The company should articulate meaningful and observable milestones, and potential responses, based on their analysis, with the milestones clearly communicated in the SMARTargets Reporting Template. In general, milestones defined in terms of the More Limited/Fuller decarbonization options, the Facilitating/Challenging condition sets, and the state of sector and economy-wide incentives and transitions, can be used to guide company timing for when to re-evaluate their low-carbon strategy and how they might respond.

For instance, the Step 4 example results suggest the need to evaluate which future is emerging. If, for example, the decarbonization options available are turning out to look like the More Limited set, instead of the Fuller set, in order to stay on track with their long-run emissions targets, the company may want to consider alternative strategies for gas, solar, wind, storage, and hydrogen, as well as possibly slowing coal retirements if annual additions of renewables are likely to be constrained. Furthermore, those strategies will depend on whether sector or economy-wide decarbonization incentives are emerging, which is another milestone the company can readily monitor. In this case, if economy-wide decarbonization incentives are developing that increase low-carbon electricity demand then a company might consider accelerating clean energy investments (and emissions reductions).

While conditions may not meaningfully change year-to-year, an annual review of conditions relative to the milestones would allow the company to assess and communicate how conditions are evolving and whether their SMARTargets strategies are robust. In addition to elucidating changes in conditions, this review would help identify which futures are consistent with observed changes, determine if shifts in the likelihood of the potential futures are merited, and flag the possibility of new futures that are substantively different than those analyzed. This latter insight would suggest that the company should consider the value of updating their SMARTargets analysis and strategies to account for a revised set of future conditions that are not captured by the existing analysis.

4. COMMUNICATING, EVALUATING AND VALIDATING SMARTargets

This section outlines the SMARTargets validation and verification process, including the institutional ecosystem envisioned to support these services. It provides a clear overview of the process, key actors, and infrastructure needed to deliver state-of-the-art validation and verification tailored to GHG target setting and progress tracking.

This section also serves as guidance for third-party organizations assisting companies with validation and verification and stakeholders evaluating and using SMARTargets.

As already noted, independent, trained, accredited, third-party validation services for corporate GHG targets do not currently exist. As such, companies will need to rely on internal validation in the near-term, as well as the scientific integrity of the SMARTargets approach. SMARTargets is working to facilitate the development of independent, competent, and accredited validation bodies to support robust validation of GHG targets based on best international practice for conformity assessment.⁸

Validating SMARTargets

SMARTargets facilitates internal and third-party validation according to international best practices for internal controls, auditing, and climate-related risk disclosure that follows principles of impartiality, independence, and objectivity. This section describes what is required in terms of the types of activities and different services needed, including accreditation, training, and other support services.

Targets inherently involve forward-looking information related to conditions that may or may not happen. This means that targets can only be validated, not verified.⁹ Establishing robust internal control systems that generate the information supporting emission reduction targets and their validation fosters trust and confidence in the decisions.

Internal and External Validation

SMARTargets provides a methodology that can be validated based on accepted concepts, principles, and best international practices. A company should start with internal validation, leveraging their experience in internal auditing processes that follow best practice.¹⁰

⁸ Companies currently providing GHG Inventories verification and/or Offsets Validation and Verification could be candidates to provide validation of GHG targets. However, they would need to be trained in the methodology and comply with any additional requirement for SMARTargets validation and verification.

⁹ Verification is a process that applies to historical data, for example GHG inventories. Validation, in contrast, is a process that provides an assessment of forward-looking information, considering the reasonableness of the assumptions, methods, etc. that support a given statement.

¹⁰ For instance, by following ISO 9001 internal audit processes.

Additionally, third-party validation by an accredited external body is recommended when such services become available.¹¹

The internal validation process ensures that the company has implemented the SMARTargets Methodology properly; and, by doing so, their targets and strategy are rigorously derived and supported and they have scientific and Paris Agreement alignment. Internal validation provides confidence in the targets and supporting information by documenting the process and basis for targets and strategies, all of which is essential information for third-party validation and external communications. Third-party validation can provide an objective and authoritative opinion on conformity with the SMARTargets Methodology and validation criteria discussed below.

The sections below outline the SMARTargets validation process and criteria for internal and external validation, as well as verification of progress towards targets. Before getting into those details, we discuss validation of alignment with the Paris Agreement. The discussions below are designed to help prepare a company and validators, as well as inform stakeholders.

How is Paris Agreement Alignment Validated?

The SMARTargets Methodology results in corporate climate targets and strategies that are aligned with science and the Paris Agreement. The methodology aligns with science by satisfying the scientific requirements for a science-based methodology: science-based alignment with the Paris Agreement temperature goals and science-based decision-making. Furthermore, the methodology provides alignment with the Paris Agreement in three ways that are consistent with science: quantitative targets aligned with the temperature goals, qualitative alignment of strategies for achieving the temperature goals, and assessment of emission reduction transitions that align with the Paris Agreement and science by recognizing differences in opportunities and enabling conditions. Alignment with science and the Paris Agreement is by design: targets and strategies resulting from SMARTargets will be aligned by implementing the methodology.

External assessments of alignment with the Paris Agreement should similarly strive for the comprehensive science and Paris Agreement alignment of SMARTargets in order to generate scientifically grounded and robust insights. Many existing approaches do not currently achieve this rigorous alignment. Many do not consider global pathway ranges, assumptions, varying implications, or the limitations of those pathways in evaluating corporate alignment with a global temperature goal. Many also do not facilitate science-based decisions by accounting for uncertainty, individual opportunities and risks, or multiple priorities, nor do they consider alignment with the other provisions in the Paris Agreement.

Note that, by conforming with the SMARTargets Methodology and criteria, the targets are also validated as aligned with science and the Paris Agreement.

¹¹ For instance, bodies accredited under ISO 14065:2020 for GHG validation.

SMARTargets Validation and Progress Verification Cycle

Validation and verification is part of a cycle of six stages that companies will undertake and repeat (Figure 16). The following sequence of activities are part of the cycle—from developing SMARTargets to verifying progress:

1. **Developing SMARTargets:** By implementing the SMARTargets Methodology a company will evaluate and pursue targets and identify strategies, including those related to risk management. After working through the steps of the methodology, a company will have completed the Reporting Template and assembled a rich documentation package chronicling the process, results and insights, and decisions.
2. **Internal Validation of Targets:** This stage is the required internal validation of SMARTargets. The validator's job is to confirm the implementation of the methodology based on the documentation and validation criteria below. An independent team within the company, separate from the team implementing the methodology, should validate the implementation using the documentation and validation criteria. The company should leverage internal audit processes in developing the validation procedures. For example, the company should consider establishing internal control procedures to generate and safeguard the documentation resulting from the methodology's implementation, including completing the Reporting Template. The key output from this process is an internal assessment opinion from the validation team on whether the company's implementation of the methodology is reasonable and conforms with the methodology and criteria. The internal validation team might also identify areas for improvement, which could result in implementation and documentation revisions that strengthen communications, increase transparency and the integrity of the decisions, and help prepare the company for third-party validation.
3. **Third-Party Validation:** Third-party validation is recommended but not required. In this stage, companies should decide whether this is needed based on the internal validation, availability of quality (accredited and independent) validation services, and internal and external stakeholders' needs. If the company decides to undertake third-party validation, it will need to work with an accredited validation body through an engagement agreement.
4. **Implementing Emissions Reduction Strategies:** In this stage, the company will be implementing the strategies identified for achieving targets. The company should document emissions changes (via their GHG inventory) and actions taken with respect to emission reduction strategies, including efforts to put enabling conditions in place and collaborate with others.
5. **Internal Verification of Progress Towards the Target Based on the GHG Inventory:** In this stage, the company will internally verify progress towards their targets. The company should have an internal verification process to assess progress on an annual basis that can be communicated (via their sustainability or climate report, regulatory requirements, or by another means). Annually assessing progress is critical to evaluating strategies and identifying opportunities for corrective actions if needed. This assessment can also identify opportunities for being more ambitious and where additional collaboration could be beneficial to facilitate an enabling environment.

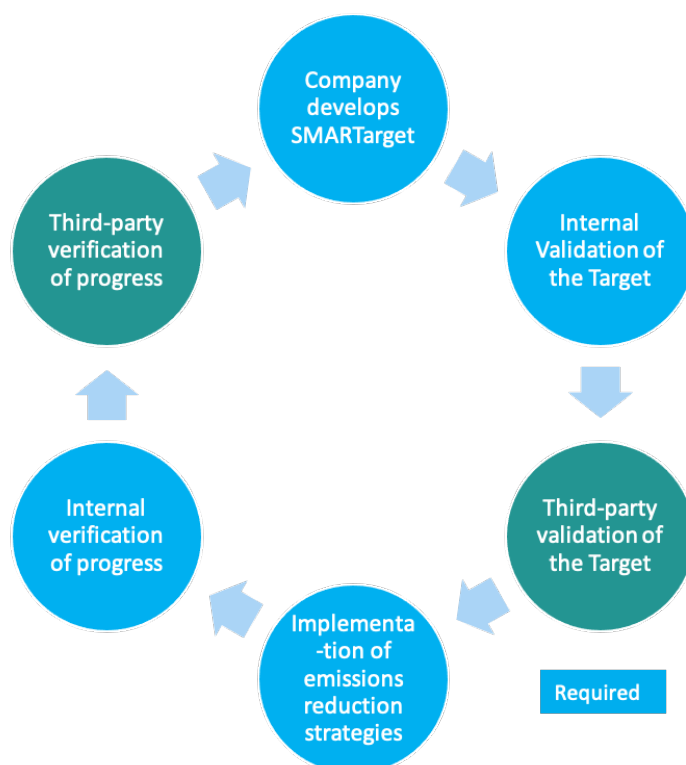


Figure 16. SMARTargets validation and progress verification cycle

6. **Third-Party Verification of Progress Towards a SMARTarget:** Third-party verification of progress is recommended but not required. In this stage, the company should decide whether third-party verification of progress is needed. Among other things, the company should consider the internal verification, GHG inventory verification regulatory requirements, and internal and external stakeholder needs. This type of verification might also be considered when there are special circumstances (e.g., structural changes in the company that affect emissions). This type of verification need not be annual.

The validation and verification process is a cycle because the evaluation of progress might lead a company to revisit their climate targets and strategies based on the conditions emerging, such as making adjustments to improve management of transition risks and/or increase ambition. Figure 17 identifies the key documentation and outputs generated by each stage of the cycle, some of which are common validation and verification products.

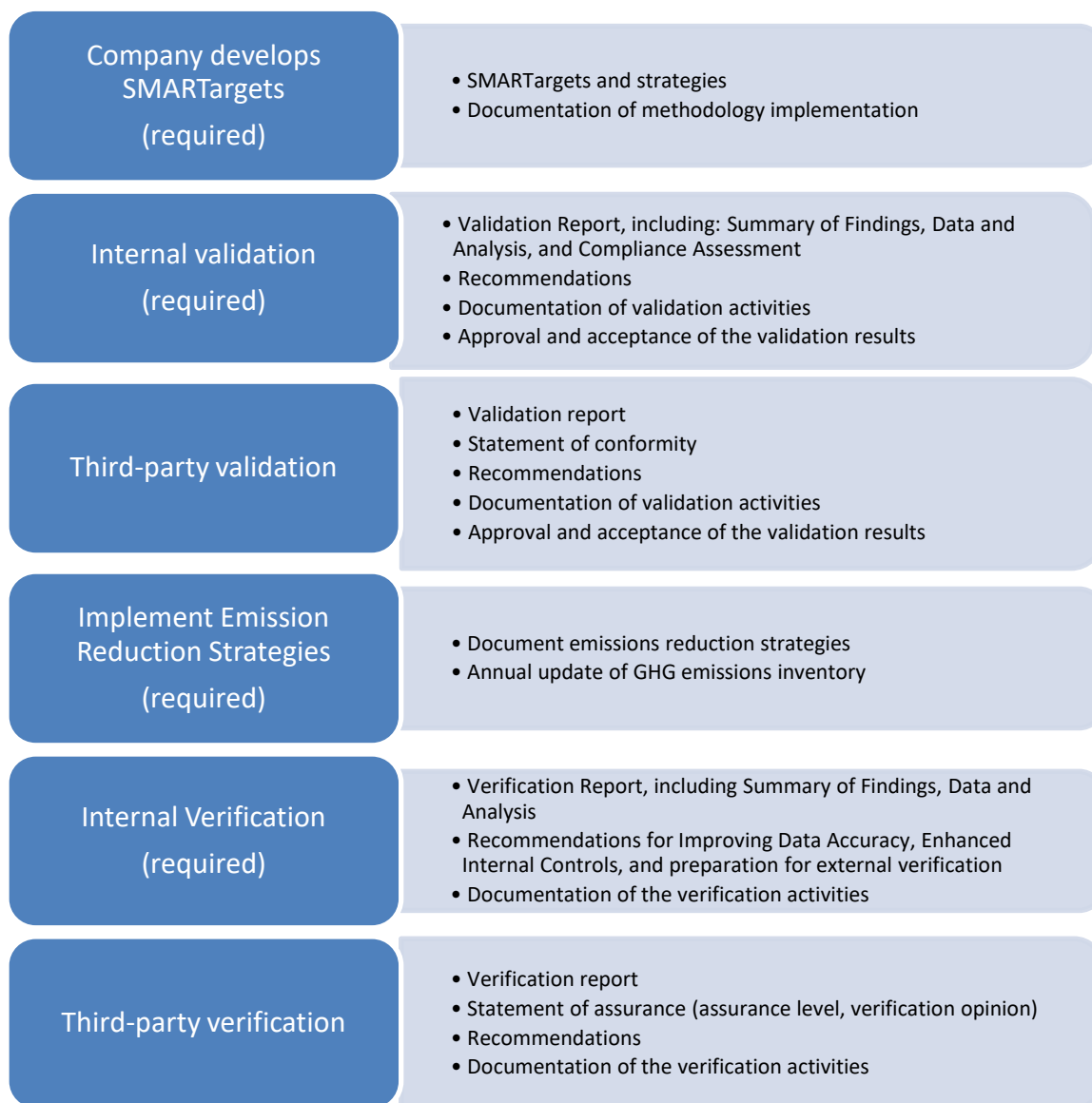


Figure 17. Documentation associated with each stage in the validation and verification cycle

Who can validate and verify?

Internal validation and verification: Companies should establish an Internal Independent Team (IIT) to conduct internal validation. The team should be independent from the team responsible for implementing the SMARTargets Methodology. It is recommended that, when an internal audit team exists, they get involved in the IIT, and that a formal internal control process be established. This includes defining roles and responsibilities within the company and documentation procedures for the methodology implementation.

Third-party validation and verification: SMARTargets third-party validation should be undertaken by an independent validation body with ISO-14065 accreditation and certified training in the SMARTargets Methodology. ISO 14065 is a standard that establishes “General

principles and requirements for bodies validating and verifying environmental information” (ISO, 2020). This process involves an assessment of the Validation and Verification Body’s (VVB) competency, impartiality, and ability to perform validation and verification. In the United States, the American National Standards Institute (ANSI) National Accreditation Board (ANAB) offers accreditation for GHG VVBs according to ISO 14065:2020, (ISO, 2020).¹²

A list of trained and accredited validators and verifiers for SMARTargets will also need to be maintained and made available to the public.

Stakeholder Evaluation of SMARTargets

What does it mean when a company says they have SMARTargets? It means the following:

1. They have ambitious and actionable GHG targets that are aligned with the Paris Agreement global temperature goals and science,
2. They followed a grounded, rigorous, well-defined standardized process that provides transparency and comparability, which includes:
 - a. Using a methodology developed from scientific assessment and requirements, and
 - b. Implementing all the steps of the methodology,
3. They are working to pursue the targets and enabling conditions,
4. They are managing low-carbon transition risk, and
5. They produced a standardized set of outputs transparently documenting the process, decisions, and results, and justifying the strategy.

Overall, it means that the company’s targets and strategies are science-based in their alignment with international temperature goals and in facilitating science-based decision making and that they are aligned with the Paris Agreement’s temperature goals and provisions for achieving those goals.

What should stakeholders be looking for when they evaluate a company’s GHG targets? They should be looking for the following:

- **Alignment with science:** Is the methodology used to develop the targets aligned with all the relevant science associated with global and sub-global low-carbon transitions consistent with limiting global warming to the Paris Agreement temperature goals? Evaluating alignment with science consists of evaluating whether alignment with international temperature goals is science-based in considering ranges, assumptions, varying implications of alternative aligned pathways, and the limitations and guidance of global pathways; in

¹² ISO 14065 accreditation can be provided by various accreditation bodies. A list of ANAB’s Greenhouse Gas VVB Accreditation (Accredited) organizations is available on its [website](#). The company should consult ANAB’s website for the most up to date list of VVBs and review the VVB certificates including the expiry date and scope. The International Accreditation Forum (IAF) accredits National and regional accreditation bodies. For the purposes of this discussion, we refer to an ANAB-accredited body, but this could change depending on your country. For details on ISO 14065: 2020, see <https://www.iso.org/standard/74257.html>.

particular, they do not provide information about actual company opportunities and risks but can help identify aspirational goals and coordinate effort. Evaluating alignment with science also consists of evaluating whether the methodology supports science-based corporate decision-making by considering uncertainty, the unique decarbonization opportunities of individual companies, the set of relevant social priorities, and risk management.

- **Alignment with the Paris Agreement:** Is the methodology producing targets and strategies consistent with the Paris Agreement temperature goals and its provisions for achieving those goals? As noted, SMARTargets are aligned with the Paris agreement in three ways: the targets are quantitatively aligned with the temperature goals, the strategies are qualitatively aligned with the robust transitions consistent with the goals and the strategies within the Paris Agreement for achieving the goals, and the two types of emission reductions (Aspirational and Qualified Targets) align with the Paris Agreement’s recognition of differences in decarbonization opportunities and uncertainty in enabling conditions.
- **Grounded targets:** Are the company’s targets supported by meaningful information? SMARTargets are developed from company-tailored transition risk analysis based on exploring plausible, actionable, and transparent low-carbon transitions and risks relevant to each company.
- **Grounded and viable strategies:** Is the company’s strategy supported and appropriate? The transparent Step 4 company-tailored transition risk analysis provides the information needed for stakeholders to be able to evaluate the viability of the targets and the appropriateness of the company’s strategy.
- **Comparability:** SMARTargets standardized process and outputs provides comparability and facilitates comparison across companies. However, because companies have different decarbonization opportunities and risks, stakeholders should expect the enabling conditions and strategies to vary. The standardized information will help stakeholders understand and interpret the differences and similarities in opportunities, enabling conditions, and risks, and help stakeholders work with companies to facilitate achieving targets.

SMARTargets Validation Template and Criteria

- As noted, implementation of the SMARTargets Methodology includes documenting the process, outputs, and decisions from each of the steps in the SMARTargets Reporting Template. The template collects implementation, justification, and decision details that can be validated according to the criteria listed in this section.

Section 1. Company Information, Implementation Team and Engagements

General Company Identification

- Was the following reported?
 - Company general information identifying the company and points of contact

Company Implementation Team

- Was the following reported?
 - A list of internal organizations involved in the methodology's implementation (e.g., directors, executives, planning, policy, sustainability, risk management, strategy, outreach)?
 - Indication of those responsible for approval of the SMARTargets
 - Indication of those responsible for coordinating the methodology implementation

External Stakeholder Engagement

- Was the following reported?
 - Indication of whether external stakeholders were engaged
 - If engaged, a description of that engagement

Experts, Consultants, and Other Support

- Was the following reported?
 - Indication of whether external experts were involved
 - If involved, a description of that involvement

Step 1. GHG Inventories and Base Year

GHG Emissions Inventory Methodology and Third-party Verification, Target Base Year, Company Emissions Trends

- Was the following reported and, as relevant, reasonably implemented?
 - A verified GHG emissions inventory
 - Indication of the most recent year of the GHG inventory
 - Indication of the base year for the GHG targets
 - Justification for the base year for the GHG targets
 - Consistent use of the base year throughout implementation (e.g., targets, transition pathway analysis, global pathway comparisons)
 - Description of GHG emissions trends related to each category, including the base year and most recent inventory year
 - A summary of the GHG inventory for the base year and most recent year for each emissions category

Step 2. Practical Issues and Climate Target Groupings

- Was the following reported and, as relevant, reasonably implemented?
 - Documentation of the conclusion regarding each practical consideration for each emissions category
 - Documentation of the overall decision and justification regarding pursuing a target for each category (either setting a SMARTarget, not setting a target because the emissions are already covered by Paris Agreement aligned policy or another company's targets, not setting a target with justification based on the practical considerations, or not setting a target because the category is not relevant to business)
 - Summary indication of whether each emissions category is covered by own SMARTargets, Paris aligned regulation, other's Paris aligned targets, or not covered
 - Documentation of target groupings

Step 3. Aspirational Climate Targets Aligned with International Goals

- Was the following reported and, as relevant, reasonably implemented?
 - The Aspirational Targets (ATs) aligned targets that are to be evaluated for each emissions target grouping, which could be by category, by GHG across categories, or across GHGs and categories

Step 4. Company Transition Risk Analysis-of-Opportunities and Risks

Modeling requirements and capability

- Was the following reported and, as relevant, reasonably implemented?
 - The type of modeling approach used (utility planning, economy-wide, or integrated)
 - Summary of the modeling approach and how it meets minimum capability requirements
 - If a utility planning model was used, discussion of how uncertain economy-wide factors were considered
 - If regulations regarding resource planning allow, has the optional use of coordinated (or integrated) economy-wide and planning modeling been considered?

Step 5. Strategy and Qualified Target Alignment with International Goals

- Was the following reported and, as relevant, reasonably implemented?
 - Qualitative assessment of Paris Agreement alignment of the company's transition strategy
 - Quantitative assessment of Paris Agreement alignment of the companies Qualified Targets, including narrative regarding qualifying conditions and challenges to contextualize deviations from the well-below 2°C ranges

Step 6. Documentation and Communication

- Was the following reported and, as relevant, reasonably implemented?
 - The required documentation from implementing the seven phases associated with the transition risk analysis:
 - Phase 1: Summary of reference conditions and uncertain transition factors for scenario design consideration
 - Phase 2: Summary of the core scenario design assumptions
 - Phase 3: Summary of the core scenario design specifications
 - Phase 4-6: No documentation requirement since these are intermediate steps
 - Phase 7: Final full set of scenarios, specifications, modeling results, insights, and conclusions (see below for specific requirements)
 - Final full set of scenario assumptions and modeling specifications (core, recommended, and sensitivity)
 - Have transition risk analysis results, including:
 - Alternative potential transitions and insights
 - Transition risks and robust strategies
 - Aspirational and Qualified Targets and associated conditions, including enabling conditions
 - Criteria used for identifying risks, enabling conditions, and qualified targets
 - Contingency plans for alternative futures
 - Strategies for pursuing targets and managing risks
 - Qualitative assessment of Paris Agreement alignment

Step 7. Validation and Verification

Validation and verification

- Was the following reported and, as relevant, reasonably implemented?
 - Communication to the public of GHG targets
 - Internal validation (required)
 - External validation
 - As relevant, communications on progress and verification activities

Alignment with science and the Paris Agreement

- Was the following reported and, as relevant, reasonably implemented?
 - Alignment with the scientific requirements
 - Confirmation regarding the three types of Paris Agreement alignment (quantitative, qualitative, and two types of emissions reductions evaluated)
 - (Optional) Consideration of alignment with country aggregate emissions pledge (NDC). See Step 5 discussion for guidance.

Step 8: Monitoring and Adjusting

- Was the following reported and, as relevant, reasonably implemented?
 - As relevant, communications on milestones, scenario realization and revisions to strategies

5. JUSTIFICATION FOR THE METHODOLOGY

The SMARTargets Methodology results in corporate climate targets and strategies that are aligned with science and international climate goals. This is by design. A company implementing SMARTargets will achieve both types of alignment automatically. Science and Paris Agreement alignment are objectives sought by stakeholders and companies. This section discusses the justification for the methodology and how alignment with science and the international goals are achieved.

At its core, the SMARTargets Methodology is based on understanding and communicating individual company decarbonization opportunities and risks and identifying aligned and actionable targets and strategies. As such, the following are key company-specific features of the methodology:

- Consideration of practical issues for setting and pursuing targets,
- Evaluation and pursuit of aspirational targets and identifying qualified targets, and
- Assessment of transition opportunities, risks, enabling conditions, and risk management strategies.

Each of these features emerged as a response to the scientific and operational requirements identified for a methodology to be science-based and aligned with the Paris Agreement. Both types of requirements are summarized in this section. See the Scientific Foundations document—*SMARTargets Scientific Foundations: The Science Relevant to Corporate Greenhouse Gas Emissions Target Setting Based on Science*—for detailed discussion of the relevant science, key scientific observations, and the derived scientific and operational requirements for methodologies.

A Science-Based Methodology

Companies and stakeholders want GHG targets and strategies based on science. However, to date, the scientific basis for company-level GHG targets and the criteria for determining whether a target is or is not “science-based” have not been well-defined and transparent. The Scientific Foundations document defines the relevant science and the requirements for a science-based methodology. This section summarizes key insights from the Science Foundations. See the document for additional details, supporting discussion and graphics, and additional references to the literature.

From assessing the relevant science, we have identified scientific and operational requirements for a methodology to be science-based. These requirements apply to climate target setting, climate strategy evaluation, and transition risk assessment methodologies. Below we describe the scientific and operation requirements and the key scientific observations underpinning the requirements.

Scientific Requirements

We identify two types of requirements for a methodology to be “science-based” (Figure 18):

1. The methodology must have science-based alignment with a global temperature goal, and
2. The methodology must facilitate science-based decision-making.

It would be more appropriate to call a methodology that satisfies these requirements “science-informed” since the science is unable to tell us exactly what each company should do. Science only provides guidance regarding what each company needs to consider for scientifically grounded targets and strategies. Given the familiarity of the term “science-based,” however, we will continue to refer to that phrase.

Science-Based Alignment with a Global Temperature Goal

There are many global and sub-global pathways aligned with any global temperature goal. This Key Scientific Observation, and the others discussed below, cannot be ignored. Specifically, alignment with a global temperature goal must recognize and account for the following:

- The ranges of aligned pathways (Key Scientific Observation #1),
- The assumptions in pathways and uncertainty about them (Key Scientific Observation #2),
- The varying implications of different aligned pathways (Key Scientific Observations #3 and #4), and
- The limitations of global pathways, as well as their guidance (Key Scientific Observation #5).

The Key Scientific Observations discussed below emerge from assessment of the science in the Scientific Foundations document. Thus, in seeking or evaluating corporate alignment with any global temperature, we must recognize that there is more than one way to be aligned, conditions could be different than those assumed, and the different conditions imply different aligned pathways and different challenges. The last point is perhaps the most important. It implies that global pathways can help identify aspirational goals and coordinate effort; however, we must recognize what they do not tell us. Specifically, they provide no information at all about an individual company’s actual transition opportunities.

Thus, global pathways can be helpful in providing direction and coordinating effort, but we need to acknowledge the limitations. Given the disconnect between global pathways and company opportunities (Key Scientific Observation #4), we should not expect companies to be able to achieve the global outcomes, nor necessarily want them to for economic efficiency reasons, given differences in opportunities. In addition, a comparison to global pathway results (e.g., global, regional, sectoral, technology) is not a conclusive test of corporate alignment given that the global pathway information does not reflect who the company is now and its transition opportunities. Not only are companies not the global, regional, or sectoral average, the global pathway results are contingent on assumptions that are uncertainties that need evaluation, with many equally credible pathways, and uncertainty and risk being ignored.

In general, public transition expectations for companies should also be aligned with science. As noted, this means we should not be surprised when some companies say that they cannot achieve a global pathway outcome without enabling conditions, such as CDR and supportive

policies. We should expect the challenge of achieving a global pathway outcome to vary across companies, with some finding it harder than others and some finding it impossible without enabling conditions. Most importantly, however, we should expect and want companies to evaluate their individual opportunities, challenges, and risks to be able to identify what they can do under different uncertain conditions and facilitate progress as possible.

Science-Based Decision-Making

Scientific observations also provide guidance regarding what needs to be considered for scientifically grounded decision-making. Specifically, for a methodology to support science-based decision-making it must (Figure 18):

- 1. Consider uncertainty** – A company needs to consider uncertainty about the future when setting a target. Specifically, a company needs to consider the plausible alternative futures because different conditions can imply different practical GHG targets, transitions, and strategies, as well as risks. *Follows from Key Scientific Observations 1, 2, 3, and 4.*
- 2. Recognize uniqueness** – A company needs to consider its transition opportunities and constraints and risks, which can vary by region and company due to differences in endowments, assets, systems, markets, regulatory structure, future options, and more. *Follows from Key Scientific Observations 4, 5, and 6.*
- 3. Account for multiple priorities** – A company needs to consider the set of objectives and social priorities that they are supporting, which could include affordability, reliability, economic development, equity, and sustainability. *Follows from Key Scientific Observations 3 and 4.*
- 4. Provide flexibility** – With the future uncertain, a company needs flexible strategies, with contingency plans and options available, to allow the company to respond and adjust to the different futures that could unfold. *Follows from Key Scientific Observations 1, 2, 3, and 4.*
- 5. Support resilience** – A company’s GHG target should be consistent with overall corporate risk management and support company resilience to potential futures. *Follows from Key Scientific Observations 1, 2, 3, and 4.*

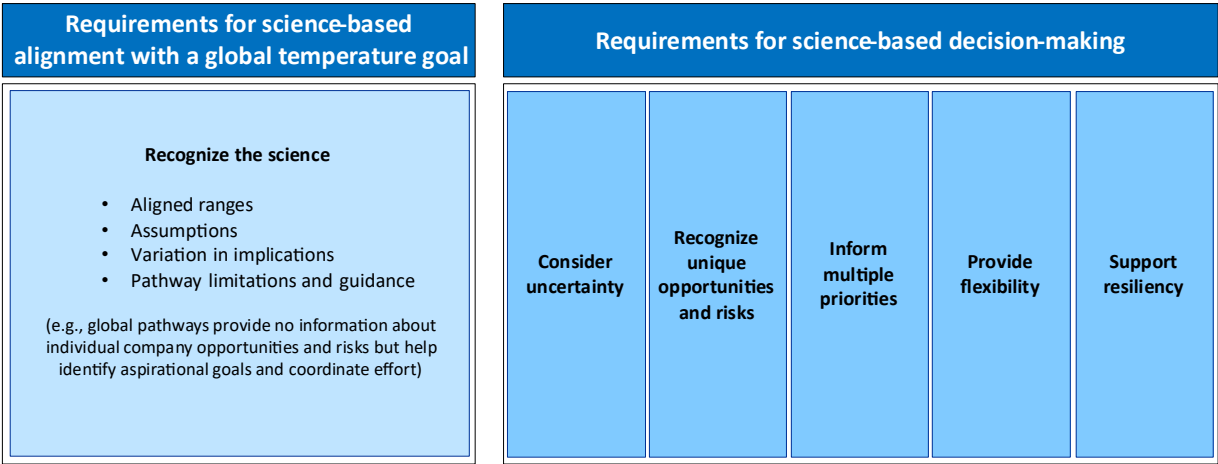


Figure 18. Requirements for a methodology to be “science-based”

Operational Requirements

The scientific observations and requirements, along with the needs of business integration, also provide guidance on the functionality required from a science-based methodology. Specifically, the methodological design needs to achieve the following:

1. Inform the identification of actionable targets and strategies for each company,
2. Help a company assess and manage low-carbon transition risk,
3. Help a company integrate GHG abatement into strategic planning,
4. Help a company communicate the most ambitious level of GHG reductions they can achieve under different conditions and the role of enabling conditions for greater ambition,
5. Assist with climate-related transition risk disclosure, and
6. Facilitate cost-effective economy-wide decarbonization.

The SMARTargets Methodology was designed to meet both the scientific and operational requirements. By doing so, the methodology is well-grounded in all the relevant science and it provides meaningful, actionable results for an individual company based on their opportunities. The methodology also facilitates enhanced planning and risk management and informs internal strategy, investors, and other stakeholders on company opportunities under different conditions, including challenges constraining ambition, and enabling conditions for greater ambition.

Key Scientific Observations and Company Target Setting Insights

The SMARTargets Methodology was built from the ground up, starting with identification and assessment of the relevant science, from which a scientific foundation of Key Scientific Observations and company climate target setting insights were derived, and the scientific and operational requirements above identified.

Assessment of the relevant science began with a fundamental question: what is the relationship between a global average temperature goal and a company (Figure 19)? Answering this question helped identify the science relevant to setting a company target. Specifically, setting a company target requires understanding of climate system dynamics and uncertainties, potential transition pathway dynamics and uncertainties, relationships across scales (global, sub-global, local, and company), and relationships between transition variables (emissions, activity, and policy). From assessing this set of science, we identify the factors that define the relationship between a company and global temperature. These are the factors that define the potential role, opportunities, and risks for a company, which is the information required for informed corporate climate target setting and strategy development.

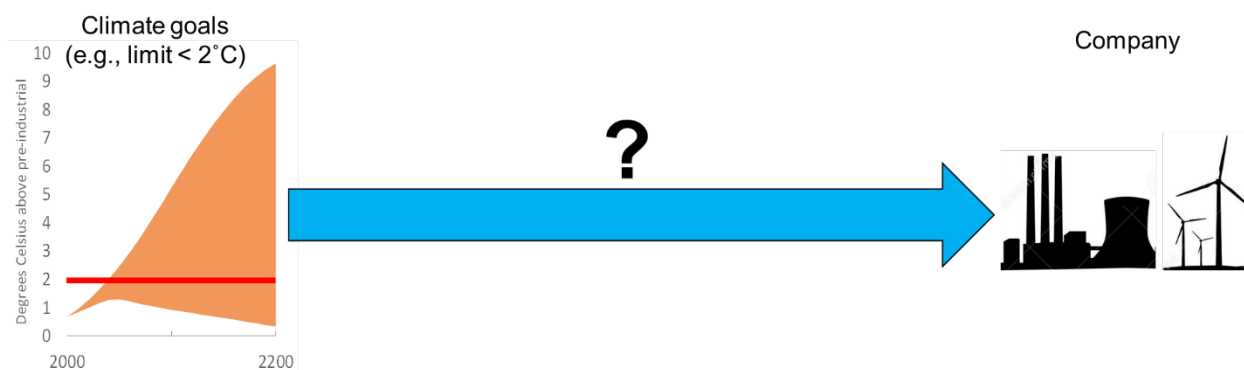


Figure 19. It is essential to understand the relationship between a global climate goal and a company

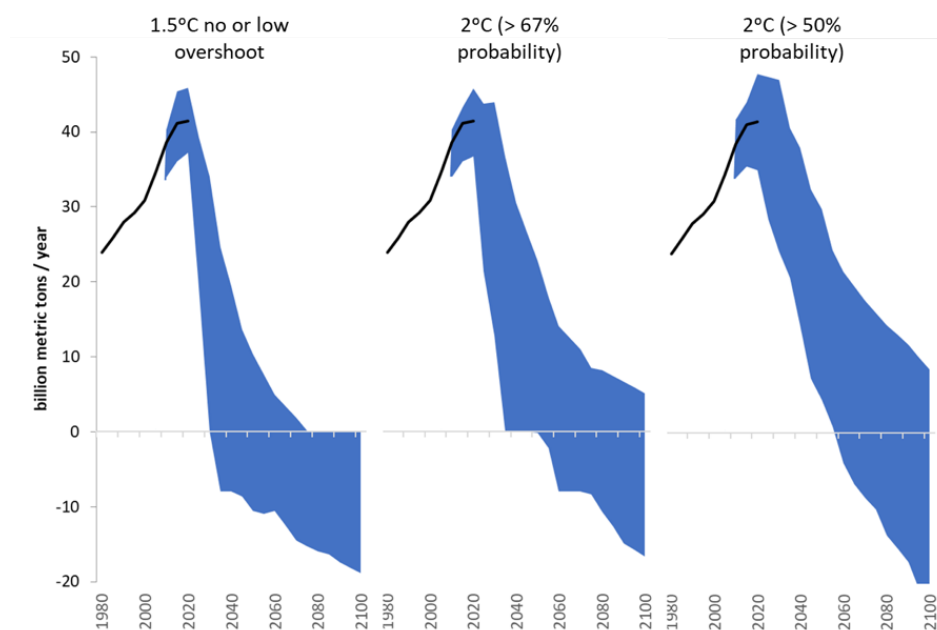


Figure 20. Global net CO₂ emissions pathway ranges consistent with limiting global average temperature to 1.5°C, 2°C with a 67% likelihood, and 2°C with a 50% likelihood

Source: EPRI (forthcoming)

Figure note: Results based on assessment of recent IPCC (Byers et al, 2022; Riahi et al, 2022), IEA (2023, 2021), TPI (2023), and PRI (2021) pathways. The 1.5°C pathway range shown is for pathways categorized by Riahi et al (2022) as “no and low overshoot” of 1.5°C. Full ranges from IPCC data included since all are relevant possibilities.

From assessing the relevant science, we have identified the following Key Scientific Observations (Table 21):

1. Many different transition pathways are consistent (aligned) with achieving a global temperature goal (see, for instance, Figure 20 for global pathway examples),
2. Global pathways aligned with limiting warming to 2°C and 1.5°C have influential but uncertain assumptions,
3. Alternative future conditions can have very different transition implications,

4. Transition opportunities vary by company,
5. Global pathways provide high-level guidance but have significant limitations in informing companies, and
6. Practical issues defining a company's circumstances affect GHG abatement opportunities.

Each Key Scientific Observation has important company GHG target setting implications (Table 21). For instance, in Table 21, the observations that there are many different transition pathways consistent (aligned) with achieving a global temperature goal (Key Scientific Observation #1) yields important company insights. In particular, it elucidates that the science does not provide guidance regarding a specific transition pathway for a company. Table 21 only lists the primary company implication. The Scientific Foundations includes more expansive discussion of company implications. For instance, in the case of Key Scientific Observations #1, there is no one pathway that can fully inform companies on potential transition opportunities. There are many Paris Agreement aligned future conditions and pathways for each company with the relationship to a global pathway weak. It is not a 1-to-1 relationship. Instead, there are many local pathways consistent with a global pathway. As a result, we conclude that the science does not provide guidance regarding a specific transition pathway for a company, which underscores the need for a company to explore their set of transition options.

Table 21. Key Scientific Observations and key company GHG target setting implications

Key Scientific Observation		Key Company Climate Target Setting Implication
1	Many different transition pathways are consistent (aligned) with achieving a global temperature goal	The science does not provide guidance regarding a specific transition pathway for a company
2	Global pathways aligned with limiting warming to 2°C and 1.5°C have influential but uncertain assumptions	Policy design and technology deployment are uncertainties companies cannot ignore
3	Alternative future conditions can have very different transition implications	It is essential to evaluate transition implications uncertainties and risks
4	Transition opportunities vary by company	Each company needs to evaluate its transition opportunities and risks
5	Global pathways provide high-level guidance but have significant limitations in informing companies	Global pathways help identify aspirational levels of effort and facilitate coordination, but it is impractical to expect all individual companies to achieve the aggregate outcomes
6	Practical issues defining a company's circumstances affect GHG abatement opportunities	Assessment of target setting capacity and a company's current situation are necessary

Alignment with the Paris Agreement

Alignment with the Paris Agreement and science-based alignment are different concepts. These are frequently conflated in public dialogue, but it is important to differentiate them. Aligning targets and strategies with science means having a methodology aligned with the scientific and operational requirements discussed above. Aligning with the Paris Agreement, on the other

hand, implies aligning targets and strategies with the provisions of the Paris Agreement, which include the following:

1. Aligning with the Paris Agreement's specific global temperature goals, and
2. Aligning with other provisions of the Paris Agreement associated with achieving the temperature goals.

The Paris Agreement temperature goal is to limit global average temperature increase to “well below” 2°C, and to “pursue” limiting the increase to 1.5°C. As discussed above, the science provides guidance on how to properly align with a global temperature goal for science-based alignment. Aligned global pathways can provide aspirational goals; however, we need to account for the limitations of global pathways in informing or assessing companies. In particular, we need to evaluate company-specific opportunities and risks, which will likely not look like the aggregate global pathway transitions.

Aligning with the Paris Agreement implies also needing to consider the other provisions of the agreement that facilitate achieving the temperature goals. This includes recognition of country differences in opportunities, the pursuit of multiple priorities, and the importance of enabling factors, such as cooperation, for achieving country pledges. In adopting the agreement, the countries recognized key issues like these as being relevant to making progress towards the temperature goals, which the science also highlights as being important. In sum, when discussing alignment with the Paris Agreement, we should be discussing alignment with the temperature goals and the provisions for achieving them.

The SMARTargets Methodology requires both quantitative and qualitative alignment with the Paris Agreement. Quantitative alignment refers to quantitative GHG emissions targets that are scientifically consistent with achieving the agreement's specific temperature goals. Qualitative alignment refers to (1) aligning corporate low-carbon strategies qualitatively with the robust types of transitions found in the global pathways consistent with achieving the Paris Agreement temperature goals, and (2) qualitatively aligning corporate strategies with the strategies represented by the other provisions of the Paris Agreement.

The first type of qualitative alignment leverages the insight that certain types of transitions are found consistently in all Paris Agreement aligned global pathways regardless of the model and assumptions (see Scientific Foundations). These include increased electrification, increased deployment of low-carbon technologies, use of carbon dioxide removal, declining but continued use of fossil energy, increases in energy efficiency, and uncertainty. These robust changes represent reliable guidance for strategies consistent with pursuing the Paris Agreement temperature goals. The second type of qualitative alignment refers to aligning with the types of strategies identified by the international community as important to pursuing the temperature goals. These include recognition of differences in country opportunities, multiple priorities, and enabling conditions. This also provides valuable strategic guidance for company strategies.

In the end, the SMARTargets Methodology aligns with the Paris Agreement in three ways:

1. It produces targets aligned with the temperature goals,
2. It produces company strategies aligned with the types of transitions consistent with the temperature goals and the Paris Agreement's key provisions for achieving the goals, and
3. It requires companies to evaluate two types of emissions reductions—aspirational and qualified targets (ATs and QTs respectively), which aligns with the Paris Agreement's recognition of country differences in decarbonization opportunities, multiple priorities, and uncertainty in enabling conditions.

Alignment of Aspirational and Qualified Targets

As noted, the two-target concept (ATs and QTs) is aligned with the Paris Agreement, as well as science. Recall that, ATs are based on global total emissions pathways consistent with the Paris Agreement temperature goals and they are used to encourage ambition and help identify enabling conditions for greater ambition. QTs, on the other hand, are derived from the transition risk analysis and reflect the greatest emissions reductions possible for a company under different future conditions while successfully balancing other societal priorities, where QTs could be greater than or less than ATs.

The two-target concept recognizes the Paris Agreement temperature goal aspiration while also recognizing differences in country decarbonization opportunities, multiple priorities, and enabling conditions, which, as discussed above, are all elements of the Paris Agreement provisions for achieving the temperature goals. Evaluating the two types of reductions helps companies define and communicate their ambition in terms beyond emissions, creating quantitative information that supports the identification and development of narratives regarding challenges, risks based on multiple priorities, enabling conditions, risk management, and their overall strategy, including pursuing enabling conditions through collaboration, R&D, and other activities.

Quantitatively, the ATs are, by definition, aligned with the Paris Agreement. They are the midpoints of the global total emissions pathways aligned with limiting warming to 1.5°C. Alignment of the QTs is more subtle. Companies are required to compare their QTs to the “well-below” 2°C global pathway ranges. However, this is not a conclusive test of alignment. Comparing to the relevant global emissions pathway ranges aligned with limiting warming to well-below 2°C only provides high-level insights regarding the consistency of the QTs with pursuing the Paris Agreement temperature goals. The high-level insight from this comparison is that QTs within the range are aligned with global futures consistent with limiting warming to well-below 2°C, and if all companies have QTs within the range, the sum will also be within the range and aligned with well-below 2°C.

However, comparing to the global pathway ranges are not conclusive alignment assessments because the global pathway ranges do not capture all the global, sub-global, and local transitions found to be aligned with a global average temperature (see Scientific Foundations). QTs for some companies may not fall within the global pathway ranges. This result should not be interpreted negatively. The QTs, by construction, align with the Paris Agreement. It is

important to keep in mind that the key virtue of the QTs is that they are the highest possible ambition for a company under different conditions while balancing trade-offs and risks. Some companies may face particularly challenging decarbonization circumstances, such as in developing countries. The transition risk analysis helps a company transparently identify and communicate their challenges, and the QTs represent actionable and substantive contributions to the Paris Agreement goals with additional ambition possible if enabling conditions can be put in place. Thus, as discussed above, the QT concept is consistent with the Paris Agreement's recognition of country differences in opportunities, multiple priorities, and enabling conditions. In that regard, it aligns with the Paris Agreement's provisions to pursue the highest possible level of ambition.

Why is the SMARTargets quantitative alignment of ATs and QTs with respect to global total emissions pathway ranges? We use total global emissions, versus sub-global emissions for sectors or countries, because they have the strongest alignment relationship with global average temperature since they are constrained by the physical dynamics of the climate system. Sectoral, or other sub-global results, however, are contingent on policy and non-policy assumptions that are highly uncertain. With many sub-global transitions found to be consistent with any global emissions transition, companies should not be comparing to a global future dependent on an uncertain, and potentially unlikely, assumption. Instead, companies need to evaluate these uncertainties, and most importantly, companies need to evaluate their unique transition opportunities and risks.

The global emissions pathway *ranges* are used to derive the ATs because all of the pathways represented within the ranges are equally credible global transitions and all have been found to be consistent with the global temperature outcome (see Scientific Foundations). The *midpoints* of the ranges are used for the ATs simply because they are reasonable representative emissions transitions for the set of transitions associated with the ranges. Note that the ranges themselves are still very much relevant to the rest of the SMARTargets Methodology. They are a key justification for evaluating uncertainty about future decarbonization conditions and opportunities.

Many current approaches use sectoral information from a global pathway or two to guide or evaluate alignment, such as a global electricity emissions intensity pathway from an IEA, TPI, IPCC, or custom global pathway (e.g., SBTi, TPI, Moody's). This approach is appealing because the sectoral information is believed to be the most relevant to companies within a sector. However, it ignores the other information relevant to science-based alignment with a temperature (Figure 18). We must recognize the ranges of aligned pathways, the assumptions in pathways and uncertainty about them, the varying implications of different aligned pathways, and the limitations of global pathways in informing company strategies.

Despite the many virtues of the SMARTargets' target setting approach, we must acknowledge the limitation that it is asking companies to consider the same CO₂ or non-CO₂ AT levels (e.g., net-zero in 2050 for CO₂). Science has shown us that uniform targets across companies are not cost-effective (Key Scientific Observation #4). However, it has also shown us that there is cost savings through coordination. Thus, coordination across companies, sectors, and countries can

help manage the overall societal costs of limiting warming to well-below 2°C. This is the reason why SMARTargets encourages evaluation or regional economy-wide transitions, the identification of enabling conditions, and company collaboration with stakeholders, including policymakers, suppliers, customers and environmental groups, to create conditions that facilitate achieving the ATs economy-wide. Coordination of this type is a critical role for public policy, which can put in place policies and programs that recognize differences in opportunities and facilitate coordination and the realization of more cost-effective transitions.

Practical Target Setting Considerations for Real Solutions

The practical considerations step in the methodology (Step 2) helps companies consider and communicate issues that can affect a company's ability to set or achieve targets and can inform the company's analysis evaluating targets (Key Scientific Observation #6).

The Scientific Foundations document provides the technical and scientific justification for considering these issues. For instance, a GHG inventory is a pre-requisite for GHG target setting. Furthermore, inventory approaches and capabilities can facilitate, bound, and/or hinder target setting. Similarly, a company should consider whether an emissions category is significant enough ("material") to justify the resources needed for developing a management strategy and managing the emissions. In addition, a company should consider whether an emissions category is already regulated or constrained by company targets at the source, and to what level. This is important for economic efficiency reasons to manage the decarbonization costs to society, and it is important for the company to consider whether targets could facilitate or complicate regulatory compliance. Lastly, setting and pursuing targets for emissions a company does not control is more challenging. Recognizing this situation helps a company identify uncertainties that should be evaluated in their transition risk analysis. Note that, in SMARTargets, a lack of control is a reason for evaluating the related uncertainty. It is not a reason for choosing not to set a target for an emissions category.

Other Approaches

The discussion above defines what it takes for a methodology to be science-based, Paris Agreement aligned, and practical. It provides guidance for all corporate target setting, target evaluation, and transition risk assessment methodologies.

Many current methodologies, however, are not aligned with the scientific requirements above for being science-based: both in terms of science-based alignment with a global temperature goal and science-based decision making. Furthermore, most do not consider the comprehensive alignment with the Paris Agreement defined above, which includes alignment with the temperature goals, as well as the agreement's provisions for achieving those goals.

In terms of scientific requirements (Figure 18), temperature alignment for most methodologies does not recognize the ranges of aligned pathways, the assumptions in pathways and uncertainty about them, implications of different aligned pathways, and the limitations of global pathways with respect to companies, as well as the type of guidance they provide.

Furthermore, many do not help companies make science-based decisions by considering uncertainty, differences in opportunities, multiple priorities, flexibility, and risk management.

As for Paris Agreement alignment, many methodologies do not consider qualitative alignment with the provisions associated with achieving the temperature goals or the robust strategies found in global pathways. In addition, their quantitative alignment is problematic. Many guide or assess company alignment with the Paris Agreement based on quantitative sectoral benchmark information from global pathways, such as a global or OECD electric sector emissions intensity pathway (e.g., SBTi, TPI). This information is not a meaningful benchmark for companies and can be very misleading. Not only is the aggregate information not representative of who a company is today or its transition opportunities, but the sectoral results are dependent upon modeling assumptions that are key uncertainties that need to be evaluated, such as immediate global climate policy with comprehensive coverage and global cooperation. Furthermore, the pathway being used is but one possibility with many equally credible Paris Agreement alternative pathways. Finally, by using one pathway, the approach ignores the other possible conditions and the risk it represents for companies.

Other methodologies also frequently do not acknowledge the practical issues, such as whether emissions are regulated, material, and can be adequately characterized.

By considering the scientific and operational requirements, broader alignment with the Paris Agreement, and practical issues, SMARTargets are aligned with science and international goals, and actionable in identifying actual company opportunities under different conditions, while encouraging ambition, and facilitating constructive dialogue and progress in managing the climate.

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A. GHG ACCOUNTING

	Scope 1 Emission Sources	Scope 2 Emission Sources	Scope 3 Emission Sources
Electric utilities	Stationary combustion at electricity generating facilities: - Boilers and turbines used in the production of electricity, heat, or steam Mobile combustion: - Mobile combustion (trucks, barges and trains for transportation of fuels) Fugitive emissions: - SF₆ emissions from transmission and distribution equipment	Stationary combustion: - Emissions associated with electricity, heat or steam purchased for use by an entity. - Consumption of electricity during transmission and distribution	Category 3: Fuel- and energy-related activities (not included in scope 1 or scope 2) - Emissions from the generation of purchased energy - Emissions from T&D losses*: emissions from generation of electricity that is consumed (i.e., lost) in a T&D system – reported by end user
Natural Gas Utilities	Stationary combustion: - Process heaters - Engines - Turbines - Flares - Incinerators - Oxidizers - Production of electricity, heat, and steam Process emissions: - Process and equipment vents - Maintenance/turnaround activities - Non routine activities Mobile combustion: - Transportation of raw materials/products/waste by company owned vehicles Fugitive emissions: - Leaks from pressurized equipment - Wastewater treatment - Surface impoundments	Stationary combustion: Consumption of purchased electricity, heat, or steam	Category 3: Fuel- and energy-related activities (not included in scope 1 or scope 2) - For upstream emissions of purchased fuels: All upstream (cradle-to-gate) emissions of purchased fuels (from raw material extraction up to the point of, but excluding combustion) Category 11: Use of sold products - The direct use emissions of sold products over their expected lifetime (i.e., emissions from the combustion of natural gas)

B. METHODOLOGY IMPLEMENTATION ILLUSTRATIVE EXAMPLES

Introduction

This appendix provides examples of applying the SMARTargets Methodology to emissions categories relevant for electricity generation, transmission and distribution, and natural gas utilities. The scenario designs and analyses provided in this appendix are illustrative examples of how the methodology can be used to inform target setting, strategy development, risk management, and communications. The examples are intended as proofs of concept for the methodology, with results that are not intended to be applied to any specific company. Actual implementation by an individual company would entail company-specific consideration of practical issues, scenario design (e.g., reference and uncertainty conditions considered), modeling implementation, analysis, and decisions.

In each illustrative example, a transition risk scenario design is used to evaluate transition uncertainty relevant to identifying possible emissions futures, infrastructure and activity transitions, implications, and risks for targets. The target evaluation and setting approaches taken vary between the examples due to differences in uncertainties, greenhouse gas (GHG) sources, abatement options, and analytical needs.

For each example, we provide a brief overview and then work through the eight SMARTargets Methodology steps (Figure 21). For Steps 2 and 4, we walk through the process, as well as provide illustrative results, for considering practical issues and developing a transition risk analysis respectively.

- 1 **STEP 1: GHG inventories and base year**
- 2 **STEP 2: Practical issues and climate target groupings**
- 3 **STEP 3: Aspirational climate targets aligned with international goals**
- 4 **STEP 4: Company transition risk analysis of opportunities and risks**
- 5 **STEP 5: Strategy and qualified target alignment with international goals**
- 6 **STEP 6: Documentation and communication**
- 7 **STEP 7: Validation and verification**
- 8 **STEP 8: Monitoring and adjusting**

Figure 21. The SMARTargets Methodology implementation steps

Recall that a utility needs to apply the methodology to each of the emissions categories listed in Table 22. However, the company is free to set targets by emissions category, by GHG across categories, or across GHGs and categories. The illustrative examples in this appendix focus on target setting for emissions associated with the services provided by each type of utility—electricity generation, transmission and distribution, and gas supply. A utility that provides multiple services, would need to consider analyses similar to that shown for the emissions targets for the individual services.

Table 22. Emissions categories a utility needs to consider for target setting

Source: EPRI using WRI/WBCSD (2004, 2011).

Scope 1
Direct Emissions from Stationary Combustion
Direct Emissions from Mobile Combustion
Direct Emissions from Process Sources
Direct Emissions from Fugitive Sources
Scope 2
Indirect Emissions from Purchased/Acquired Electricity
Indirect Emissions from Purchased/Acquired Steam
Indirect Emissions from Purchased/Acquired Heating
Indirect Emissions from Purchased/Acquired Cooling
Scope 3
Category 3: Fuel- and energy-related activities (not included in scope 1 or scope 2)
Category 11: Use of sold products

Modeling Description

For the examples in this appendix, we implement regional economy-wide modeling for a single region of the United States (U.S.). The regional modeling provides strategic perspectives regarding potential economy-wide and sector-wide decarbonization transitions and risks. Evaluating economy-wide transitions sub-nationally at the regional level allows us to capture and evaluate the implications of the unique regional endowments, markets, and low-carbon options that define decarbonization transition opportunities (EPRI, 2023). Specifically, for the examples, we run EPRI’s U.S. Regional Economy, Greenhouse Gas, and Energy Model (US-REGEN) and generate transition results for a variety of energy system, economic, and environmental variables. In each example, we share a sample of results to illustrate the approach and types of information produced. US-REGEN is a framework for evaluating the economic and environmental impacts of national and regional policies. It combines a detailed dispatch and capacity expansion model of the United States and regional electric sectors with technologically detailed consumer choice models of end-use services, technology choices, and energy demands and markets, including fuel markets.¹³ As needed, for some GHG categories, we also show how GHG abatement supply resources can be incorporated into the analysis to provide target setting insights.

¹³ For US-REGEN model documentation, see <https://us-regen-docs.epri.com/>. Numerous analyses have been published recently using the US-REGEN model, including Bistline et al. (2023), EPRI (2023), Blanford et al (2022), Bistline et al. (2022), and EPRI (2022b).

Illustrative Results for Electricity Generation Emissions Targets

This example illustrates the derivation of SMARTargets for electricity generation CO₂ emissions for a hypothetical company that generates electricity and purchases electricity to serve load. GHG emissions associated with generating electric power are CO₂ emissions due primarily to the stationary combustion of fossil fuels. As such, the company is most concerned with the following two emissions categories (Table 22):

1. Direct own stationary combustion CO₂ and
2. Indirect purchased electricity CO₂.

The example is based on a U.S. Midwestern region and the hypothetical company sits within the region and is interested in understanding how their sector and economy might evolve with decarbonization by 2050 and what might be the implications and risks. Using US-REGEN, we can constrain emissions and evaluate the implications of alternative potential regional low-carbon transitions for electricity supply and investments, electricity demand, energy demand, fuel markets, and emissions.

Below, we describe how this electricity generation utility would apply each step of the SMARTargets Methodology.

Step 1: GHG Inventories and Base Year

Step 1 Outputs: Verified GHG inventory and emissions reductions base year.

The first step is for a company to assemble their GHG inventory and select a base year. A company will need GHG emissions accounts for each of the categories of emissions relevant to their business. A company will also need to decide on their emissions reduction base year and apply it consistently. Note that the company will need to have the inventories verified.

For this example, Table 23 summarizes the emissions for our illustrative utility. The primary source of emissions is own generation, however, indirect emissions from purchased power are also noteworthy. The utility could have other emission sources but, for this example, we assume they represent only a small fraction of the overall inventory.

The emissions trend is illustrated in Figure 22. As shown, emissions over the last decade have been declining. An analysis of the emissions drivers reveals that this is primarily due to shifts away from fossil fuel electricity generation over the period shown.

Table 23. GHG inventory for key categories in electricity generation illustrative example

Source: EPRI using EPA (2024) and illustrative GHG corporate inventories.

Key emissions categories	2015 (base year)	2021 (most recent)
	MMtonCO _{2e}	
Direct stationary combustion CO ₂	66.0	50.2
Indirect purchased electricity CO ₂	16.0	12.2

Given the information available, we assume the utility decides to select 2015 as its base year. This decision is based on considering the accuracy of their inventories, the inventory years verified, and choosing a year that facilitates communications regarding the utility's emissions trends.

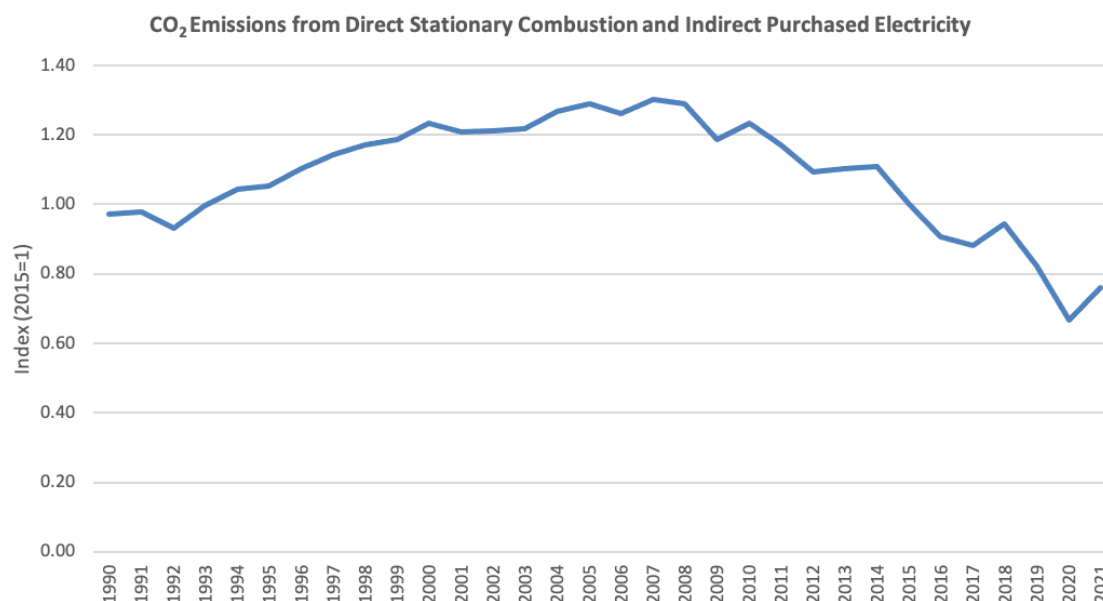


Figure 22. Emissions trend for electricity generation illustrative example

Step 2. Practical Issues and Climate Target Groupings

Step 2 Outputs: Decisions on practical issues, including emissions categories covered (by regulation or company targets), the categories for which targets are being set, and how these categories are grouped for target evaluation.

For this step, our illustrative utility works through the company target setting practical issues, as well as identifies their target groupings. Table 24 presents example responses to the set of practical issues questions for both emissions categories, and below we discuss the implications. This step helps companies assess practical considerations that may influence their ability to set and achieve targets. The company's responses also inform their evaluation of targets (Step 4).

For each emissions category, the company decides whether to:

1. Set a target (using the subsequent SMARTargets steps),
2. Not set a target because the emissions are already covered by a Paris Agreement aligned policy or another company's Paris Agreement aligned targets,
3. Not set a target with justification based on practical considerations that currently preclude target setting, or
4. Not set a target because the category is not relevant to their business.

Table 24. Illustrative practical issues responses for the primary electricity generation emissions categories

Company target setting practical considerations question	Response for own electricity generation emissions (Scope 1 Direct Emissions from Stationary Combustion)	Response for purchased electricity emissions (Scope 3 Category 3 Fuel and Energy Related Activities not in scopes 1 & 2)
Is the emissions category adequately estimated and reported?	Yes, the emissions can be adequately characterized	Yes, the emissions can be adequately characterized
Are the emissions material to the company's overall GHG inventory? (e.g., above 5% of total emissions or a threshold defined by the company)	Yes, the emissions from this source category represent the majority of our emissions	Yes, the emissions from this source category are greater than 5% of our emissions
Are the category's emissions regulated by a Paris Agreement aligned policy?	No, the emissions from this source category are not currently regulated	No, the emissions from this source category are not currently regulated
Does another company in the value chain already set Paris Agreement-aligned targets for this source? (applicable only to indirect emission sources)	n/a – direct emissions category	Yes, the source category does currently have targets, which include 50% reductions by 2030 relative to 2005 and net-zero by 2050. These targets are aligned with the global pathways consistent with the Paris Agreement goals (see Step 5).
Does the company have control or influence over the emissions?	Yes, for the most part, we control the emissions, including dispatch decisions	Partially, we can decide from whom we purchase power but we are also constrained by clean power availability

Based on the responses in Table 24, the company should consider pursuing targets for both their own and purchased power generation CO₂ emissions. Another electricity generating company might have different responses to some of the questions or reach a different conclusion based on the responses in the table. For instance, some utilities do not have controlling ownership of the generation, or do not control the dispatch of their generating units; and, therefore, they do not control the operations and the emissions. In this case, they would want to consider the drivers of this uncertainty in their Step 4 transition risk analysis. Similarly, the generating emissions for some utilities are covered by current state regulations, such as in California or Minnesota in the U.S. In this case, the utility might choose not to set targets because setting a company target could complicate compliance with state regulations. Similarly, for purchased power, some companies lack control because the purchased electricity comes from the market and the emissions are determined by the system operator's dispatch of system units. The lack of control of the market mix of energy would be an important uncertainty to consider in the Step 4 analysis.

For this example, we assume the company decides to proceed with evaluating and pursuing targets for both categories and will therefore undertake subsequent steps, including the Step 4 analysis to evaluate the targets, transition opportunities and risks, and enabling conditions.

In this step, the company also needs to decide on its target groupings. For this example, we assume the company pursues a single CO₂ target that includes both of the CO₂ emissions categories. The companies overall target setting and target grouping decisions are documented in Table 25.

Table 25. Emissions category targets

Table note: Emissions categories are those from Table 22* Emissions not relevant to business or no material to inventory.

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by <u>future</u> Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company's Paris Agreement Aligned Targets (MtCO ₂ e)			
			(indicate if target for category, GHG, or CO ₂ e)					
Scope 1. Direct Emissions from Stationary Combustion	50.2	50.2	CO ₂					CO ₂ emissions from own generation resources are material
Scope 1. Direct Emissions from Mobile Combustion	0.1							Not material*
Scope 1. Direct Emissions from Process Sources	n/a							n/a
Scope 1. Direct Emissions from Fugitive Sources	0.2							Not material*
Scope 2. Indirect Emissions from Purchased/Acquired Electricity	0.3							Not material*
Scope 2. Indirect Emissions from Purchased/Acquired Steam	n/a							n/a
Scope 2. Indirect Emissions from Purchased/Acquired Heating	n/a							n/a
Scope 2. Indirect Emissions from Purchased/Acquired Cooling	n/a							n/a
Scope 3. Fuel- and energy-related activities (not included in scope 1 or 2)	12.2	9.3	CO ₂					CO ₂ emissions from purchased power are material, the rest of the emissions are related to upstream fuels activities and not material
Scope 3. Use of sold products	n/a							n/a

Step 3. Aspirational Targets Aligned with International Goals

Step 3 Outputs: Aspirational targets that will be evaluated in the Step 4 transition risk analysis.

For this step, the utility will select generic Aspirational Targets (ATs) aligned with the international goals for their net CO₂ target covering owned and purchased electricity generation CO₂ emissions. The utility identifies intermediate and 2050 ATs using Table 26, which shows the net CO₂ ATs from Table 6 of the methodology document. The utility must select a 2030 or 2035 intermediate AT, as well as a 2050 AT. They are also welcome to select additional pre-2050 intermediate targets using the table. For this example, we assume the utility chooses to select 2030 and 2040 intermediate ATs and therefore their set of ATs are -56%, -88%, and net-zero (-100%) from 2015 for 2030, 2040, and 2050.

Table 26. Generic Aspirational Targets (% reduction from 2015)

Year	Net CO ₂
2030	-56%
2035	-82%
2040	-88%
2045	-94%
2050	-100%

Step 4. Company Transition Risk Analysis of Opportunities and Risks

Step 4 Outputs: Company regional transition opportunities and risks, evaluation of aspirational and qualified targets and their enabling conditions, and transition strategies (low-carbon & risk management).

Given that the utility has decided to evaluate and pursue targets (Step 2), they will need to undertake transition risk analysis to identify their opportunities, risks, and risk management strategies for pursuing the Step 3 ATs and identifying qualified targets (QTs) and their associated conditions. The utility works through the seven phases associated with the transition risk analysis. The example analysis below informs aggregate net CO₂ targets across both own generation and purchased power CO₂ emissions.

Phase 1: Define Reference Conditions and Itemize Uncertainties

- *Phase inputs:* Previous company planning, strategy, analysis, and risk management experience
- *Phase outputs:* Scenario analysis reference conditions and list of uncertain factors to consider in scenario design
- *Who should be involved:* Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling

- *Documentation:* SMARTargets template summarizing reference conditions and uncertain transition factors for scenario design consideration

For this phase, the company works with their planning, policy, sustainability, risk management, strategy, and outreach staff, as well as any external experts on scenario analysis and modeling with which they choose to collaborate. Together, the team defines their scenario analysis reference conditions and the uncertain factors that they want to consider in their scenario design in Phase 2.

For reference conditions, the company develops lists for announced retirements, new builds underway for new capacity, and new builds underway for new transmission. All will be implemented as constraints in the modeling analysis. The model can retire or build more, but it will not be able to do less. The company might also include local or federal policy drivers that they want to include in every transition future they evaluate. For instance, there might be a state renewables goal of increasing wind and solar generation, or federal incentives for certain technologies. These would be incorporated as either modeling constraints such that the model can do more, but it cannot do less, or economic factors that change the relative and overall costs.

The company staff, and any experts involved, will also develop a list of uncertain factors that they believe are or could be important and that they want to evaluate in some way. In this case, we assume the company identifies the following uncertain factors:

- Electricity supply decarbonization options,
- Electricity supply technology cost and performance improvements,
- Future electricity demand (load),
- Low-carbon generation technology development and commercialization,
- Low-carbon generation technology permitting,
- Supply chain constraints related to low-carbon generation technologies,
- Not in my backyard (NIMBY) issues,
- Resource procurement,
- Support for new transmission infrastructure investments,
- New transmission infrastructure costs,
- New transmission siting and permitting,
- Fuel availability,
- Fuel pipeline capacity/augmentation, and
- End-use technology cost and performance improvements.

Note that the SMARTargets scenario design already accounts for uncertainty about the type of decarbonization—power sector only versus economy-wide. At the end of Phase 1, the utility will document their reference conditions and itemize the identified uncertain factors in the SMARTargets Reporting Template.

Phase 2: Create Core Scenario Design – Define Plausible Extreme Alternative Future Conditions

- *Phase inputs:* Phase 1 scenario analysis reference conditions and list of uncertain factors to consider in scenario design.
- *Phase outputs:* The company's core scenario design for the required SMARTargets scenarios.
- *Who should be involved:* Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling.
- *Documentation after iterations:* SMARTargets Reporting Template entries summarizing the core scenario design.

For this phase, the company continues working with the same team as in Phase 1 to define their core scenario design for the required SMARTargets scenarios. Table 27 presents the SMARTargets Methodology required AT and QT scenarios for electricity supply CO₂ emissions. For the Qualified Targets, the company will define increasing reduction levels for evaluation—X, Y, and Z%. The incremental reductions will help the company identify their qualified CO₂ reduction targets without enabling conditions. The specific X, Y, and Z% incremental reduction levels will be defined after the company reviews the results from the AT scenarios and after the company has identified the AT enabling conditions.

To implement the Table 27 AT scenarios, the company needs to define the future conditions—Fuller/More Limited decarbonization option conditions and Facilitating/Challenging other conditions. Table 27 provides example definitions for each of these conditions. Recall, that the conditions should be intentionally more extreme. Thus, the Fuller and More Limited decarbonization options sets differ notably, with differences in the availability of advanced nuclear, fossil generation with and without CCS, carbon dioxide removal, bioenergy, low-carbon imported power, and the opportunity to use fossil resources to satisfy capacity reserve requirements.

Defining the Facilitating and Challenging other conditions consists of three intermediate steps: (1) combining uncertain factors that have similar qualitative implications into uncertainty categories, (2) combining categories into sets of conditions, and (3) defining the qualitative differences in the relative conditions. The first, combining uncertain factors that have similar qualitative implications into uncertainty categories, entails identifying uncertain factors with similar implications. Below are three examples of sets of uncertain factors identified in Phase 1 that all affect the same condition:

- *Generation resource availability:* affected by technology development and commercialization; permitting, supply chain, and not in my backyard (NIMBY) issues; and resource procurement.
- *Interregional transmission:* affected by support for infrastructure investments, costs, and siting and permitting.
- *Fuel supply:* affected by resource availability and pipeline capacity/augmentation.

This grouping of uncertain factors into uncertainty categories allows the company to compactly incorporate and capture many factors that can affect the same condition from the company’s point of view.

The outcome of the second phase, combining uncertainty categories into sets of conditions, is what is shown in Table 28. The set of uncertainties considered in both the Facilitating and Challenging conditions includes uncertainty regarding electricity load growth, electricity generation cost improvements, electricity capacity additions, electricity transmission additions, fossil fuel prices, and electric technology cost improvements. Finally, the third step, defining the qualitative differences in the relative conditions, requires specifying the relative condition differences for Facilitating and Challenging. For example, in Table 28, the alternative electricity generation cost improvement conditions are either faster or slower cost improvements in Facilitating and Challenging respectively. Similarly, the alternative electricity capacity additions conditions are economic versus constrained, where “constrained” refers to there being non-economic constraints that slow the addition of new generation capacity more than what “economic” conditions alone would allow.

With the scenario conditions defined, the company is ready for Phase 3 and quantification of the future conditions. At the end of Phase 2, the company should document their definitions of Fuller/More Limited and Facilitating/Challenging using the SMARTargets Reporting Template.

Table 27. Required scenarios for evaluating reductions in electricity supply CO₂ emissions

Target type being evaluated	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Reference	Reference (without CO ₂ constraint)	No	N/A	Facilitating
	Reference (without CO ₂ constraint)	No	N/A	Challenging
Aspirational Targets (net CO ₂ ATs and QTs > ATs)	Power sector transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
	Power sector transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
	Power sector transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
	Power sector transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging
Qualified Targets (net CO ₂ QTs < ATs)	Power sector net CO ₂ transition to X, Y, and Z% in 2050	No	Fuller	Facilitating
	Power sector net CO ₂ transition to X, Y, and Z% in 2050	No	More Limited	Challenging

Table 28. Illustrative Fuller/More Limited and Facilitating/Challenging decarbonization condition definitions

Illustrative FULLER decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions) Advanced nuclear (additions) Fossil Fossil w/CCS CDR (biopower w/ CCS & direct air capture (DAC)) Storage Hydrogen Biopower carbon neutrality Other renewable fuels (w/ and w/o CCS) Low-carbon power imports Fossil capacity reserves permitted (within emissions constraint) Demand conservation and energy efficiency	Illustrative MORE LIMITED decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions only) Storage Electrolytic Hydrogen only Other renewable fuels (w/ and w/o CCS) Fossil capacity reserve eligibility declines to zero in 2050 Demand conservation and energy efficiency
Illustrative FACILITATING other conditions Reference electricity load growth Faster electricity generation cost improvements Economic electricity capacity additions Economic electricity transmission additions Lower fossil fuel prices Faster electric technology cost improvements	Illustrative CHALLENGING other conditions Faster electricity load growth Slower electricity generation cost improvements Constrained electricity capacity additions Constrained electricity transmission additions Higher fossil fuel prices Slower electric technology cost improvements

Phase 3: Implement Scenario Design – Specify Assumptions for Scenario Modeling

- *Phase inputs:* The company’s Phase 2 core scenario design for the required SMARTargets scenarios.
- *Phase outputs:* Specification of core scenario design for the company’s modeling framework.
- *Who should be involved:* Staff related to planning, sustainability, policy, and strategy, as well as external experts on scenario analysis, modeling, and alternative assumptions.
- *Stakeholder engagement:* This is a good time to seek external stakeholder feedback.
- *Documentation after iterations:* SMARTargets Reporting Template entries summarizing the core scenario design specification.

For this phase, the company continues working with the same team as in Phases 1 and 2 to specify the core scenario design for implementation in their modeling framework. Phase 2 provides the company with everything they need conceptually and qualitatively in terms of the scenarios they will run. Actually running the scenarios requires quantification of the scenario design. This includes specifying decarbonization technology cost and performance assumptions, as well as what exactly is meant by “faster” and “slower” cost improvements, and “constrained” capacity and transmission additions.

Table 29a and Table 29b present an illustrative specification for each of the assumptions associated with the Fuller/More Limited and Facilitating/Challenging conditions. For the decarbonization technology assumptions, Table 29a notes the technical resource. For the Facilitating and Challenging conditions, Table 29b notes each specification. Remember that the specification definitions are intentionally extreme, but plausible, alternative conditions. They also need to be grounded, and justification should be provided for the specifications. For example, the higher load growth assumption for Challenging is based on recent data center growth trends and new analysis evaluating a potential higher growth future (EPRI, 2024).

The utility will also need to define the emissions constraint pathways they will be imposing on the model. These constraints will constrain the model to producing no more than the emissions reductions specified (though the model could produce emissions below the constraints).

After completing Phase 3 is a good time for external stakeholder feedback on the scenario design and specifications. It is an opportunity to discuss setting SMARTargets and the planned transition risk analysis and preliminary future conditions the company is considering. At the end of Phase 3, the company should document their specifications for their Fuller and More Limited decarbonization options, their Facilitating and Challenging conditions, and their emissions constraints in the SMARTargets Reporting Template.

Table 29. Illustrative summary of energy technology assumptions

Decarbonization technology cost and performance	Comment/description	Reference
Renewables (wind, solar, bio, hydro), nuclear, advanced nuclear, coal, coal w/ CCS, gas, gas w/ CCS, CDR (biopower w/ CCS, DAC), storage, hydrogen	Engineering-based cost and performance specifications with gradual improvement over time	EPRI (2022b)
Biomass carbon neutrality	Carbon neutral assumption based on analysis of net land use and GHG changes associated with increasing biomass supplies	Blanford et al (2022)
Other renewable fuels	Renewable liquid and gas fuels, including renewable gasoline, diesel, and jet fuel, renewable natural gas, and synthetic natural gas produce with renewable energy	Blanford et al (2022)
Low-carbon power imports	Low-carbon electricity can be exported to other regions via intra-regional transmission	EPRI (2023)
Fossil capacity reserves permitted (within emissions constraint)	Fossil generation capacity can be used to meet capacity reserve requirements but declining emissions constraints over time apply to any generation from that capacity	EPRI (2022b)
Demand conservation and energy efficiency	End-use sectors can reduce emissions through technology and fuel substitution that increases efficiency, as well as reductions in overall use	Blanford et al (2022)

Table 30. Illustrative specifications for Facilitating and Challenging other condition assumptions

Other conditions	Facilitating	Challenging
Electricity load growth	Reference = load growth due to reference economic and population growth assumptions and electrification due to electric technology relative cost improvements	Faster = reference load growth conditions plus additional data center load growth based on recent trends resulting in 12% greater total load by 2030 (EPRI, 2024)
Electricity generation cost improvements	Faster = default cost projections and reference learning curve for battery cost improvements over time	Slower = higher cost projections by technology and slower learning curve for battery cost improvements over time
Electricity capacity additions	Economic = allows for all economically viable electric capacity to be built	Constrained = limits the building of economically viable electric capacity to historical build rates
Electricity transmission additions	Economic = allows for all economically viable interregional transmission to be built	Constrained = limits the building of economically viable interregional transmission based on historical transmission build rates
Fossil fuel prices*	Lower = No additional supply costs for fossil fuel production beyond the equilibrium cost determined by the energy economy model's balance of supply and demand for each fuel	Higher = Additional supply costs added to fossil fuel production modeling low resource availability or pipeline expansion
Electric technology cost improvements	Faster = Default cost reductions in EV & heat pump components, faster EV adoption reflective of less range anxiety and greater availability of new electric technologies	Slower = Higher component costs for EV & heat pumps, slower EV adoption representative of more range anxiety and less availability of new electric technologies

* The fossil fuel price assumptions in the table reflect the US-REGEN modeling structure. Other economy-wide frameworks might have fixed assumed (i.e., exogenous) energy prices. For instance, they might be based on lower/higher U.S. Department of Energy Annual Energy Outlook fuel price projections.

Phase 4: Identify Potential Transitions and Target Implications

- *Phase inputs:* Phase 3 specification of core scenario design for implementation in the company's modeling framework.
- *Phase outputs:* Preliminary transition analysis pathways and insights.
- *Who should be involved:* Staff and external experts running the model and processing outputs.
- *Documentation: None.* This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.

For this phase, the company's staff, and any external experts, running the model and processing outputs will work together to develop preliminary transition analysis pathway results and insights. The team will be running the model with the scenario specification

implementation developed in Phase 3, and they will have to debug and likely revise the scenario specification implementation as they attempt to run each of the scenarios in the design. They will also want to review the results for non-intuitive, or otherwise problematic, outcomes that could be an artifact of an incorrect scenario specification or implementation. The team will want to revise and rerun scenarios in these cases.

In addition to running the scenarios, the team will develop preliminary graphics—figures and tables—and identify preliminary insights for review. In particular, the team will want to evaluate differences and similarities in transition results between scenarios. The differences between the scenarios are the most meaningful insights, not the specific values from individual scenarios. The differences will provide insights regarding the direction and order of magnitude of the potential implications of the different conditions. For instance, electricity supply utilities would be interested in the alternative potential transitions for electricity capacity and generation mixes, electricity demand, power transmission, emissions, costs, and prices. The modeling provides information regarding the alternative potential transitions, trade-offs, uncertainty, risks, enabling conditions, robust strategies, contingencies, and more. Below we present sample results for the company’s scenario design. For this example, recall that we are evaluating potential market transitions for a company in the midwestern U.S. These results represent potential alternative market and operational conditions for the company’s system and assets.

Figure 23, for example, presents the electricity sector net CO₂ emissions transitions for the net-zero AT scenarios. The figure includes the results for both the power only (“NZ electricity”) and economy-wide (“NZ economy”) net-zero target scenarios. The power sector only targets reach 100% reductions (net-zero) by 2050, with only slight variations in their cost-effective emissions transitions to 2050 due to the different conditions.

Most importantly, we observe that the economy-wide net-zero policy facilitates more rapid emissions reductions in the power sector than the sector only net-zero in 2050 target. Specifically, the economy-wide net-zero scenarios are resulting in faster emissions reductions and possibly net-zero power by 2035. This result is consistent with previous findings that an economy-wide policy incentivizes end-use decarbonization and creates demand for low-carbon electricity that facilitates faster decarbonization of the power sector (EPRI, 2022b; EPRI, 2022a). The faster decarbonization is enabled by the economy-wide policy and the low-carbon electricity market it creates.

For a given decarbonization transition, power sector or economy-wide, the variation in transitions due to the other conditions is most prominent in the physical energy infrastructure and economic implications. Figure 24, Figure 25, and Figure 26 present results for the four AT net-zero scenarios, and the two reference scenarios, for electricity generation, load (demand), and generation capacity. First and foremost, the figures immediately illustrate that there is significant transition uncertainty. This is shown by the large variation across scenarios in the evolution of the generation and capacity portfolios over time. Thus, there are very different plausible electricity generation, capacity, and load transitions consistent with a net-zero by 2050 power system due to the different conditions. We also see significantly more capacity

additions in the AT futures, compared to the Reference futures (Figure 25 and Figure 26). Thus, significant capacity additions are needed to meet the emissions constraints. The fastest and largest capacity additions are with the economy-wide decarbonization transition due to the revenue provided by the demand for low-carbon electricity across the economy. The higher electricity demand (load) in these futures is evident in Figure 24.

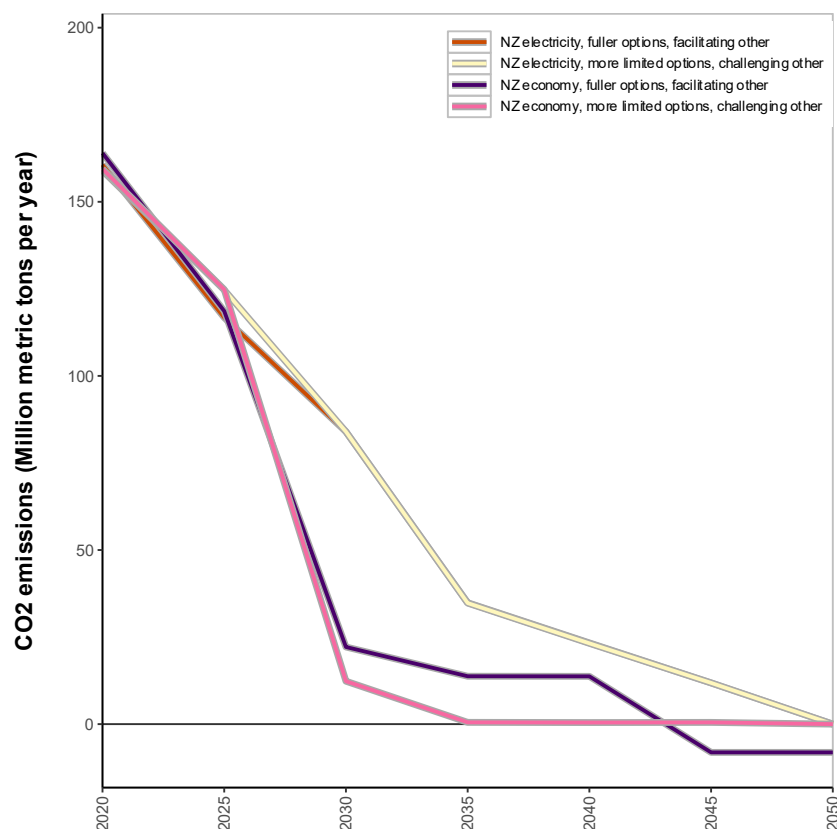


Figure 23. Regional power sector net CO₂ emissions transitions

Source: EPRI analysis

We also observe important technology specific insights that help identify robust strategies and investment and operations uncertainty. Load increases, wind, solar, nuclear, and storage are key features in all four AT decarbonization transitions, as well as reference futures. As such, they would be robust strategies Figure 26, in particular, illustrates that additions of wind, solar, and storage are consistent elements of all four AT futures, as is retirements of coal. However, we also see in Figure 26 notable variation in the additions by 2035 and 2050, especially for solar, storage, and wind; and, mixed results for addition and retirement opportunities for gas, nuclear, and hydrogen, with only some conditions conducive to those investments. Overall, the cumulative capacity changes associated with all four futures are substantial but with large variation. By 2035, we find additions of approximately 100 to 195 GW and retirements of approximately 30 to 40 GW. By 2050, the additions and retirements are approximately 280 to 380 GW and 65 to 80 GW respectively. For comparison, in 2024, there was 60 GW of new capacity additions planned across the entire U.S., and 7.5 GW of capacity retirements planned

(EIA, 2024). The reference futures alone suggest a good deal of cost-effective additions and retirements, with gas, wind, and hydrogen the primary capacity additions and coal the primary retired capacity (Figure 26). Note that natural gas related resources are providing notable generation and/or capacity in all the net-zero decarbonization futures, with coal resources also in use through 2045 under some conditions, and the role and the magnitude of gas and coal resources over time varying substantially with the different conditions. In the four decarbonization futures, in 2035 and 2050, gas capacity could be 25 to 52 GW and 0 to 59 GW respectively, while coal capacity could be 1 to 8 GW and 0 to 1 GW respectively.

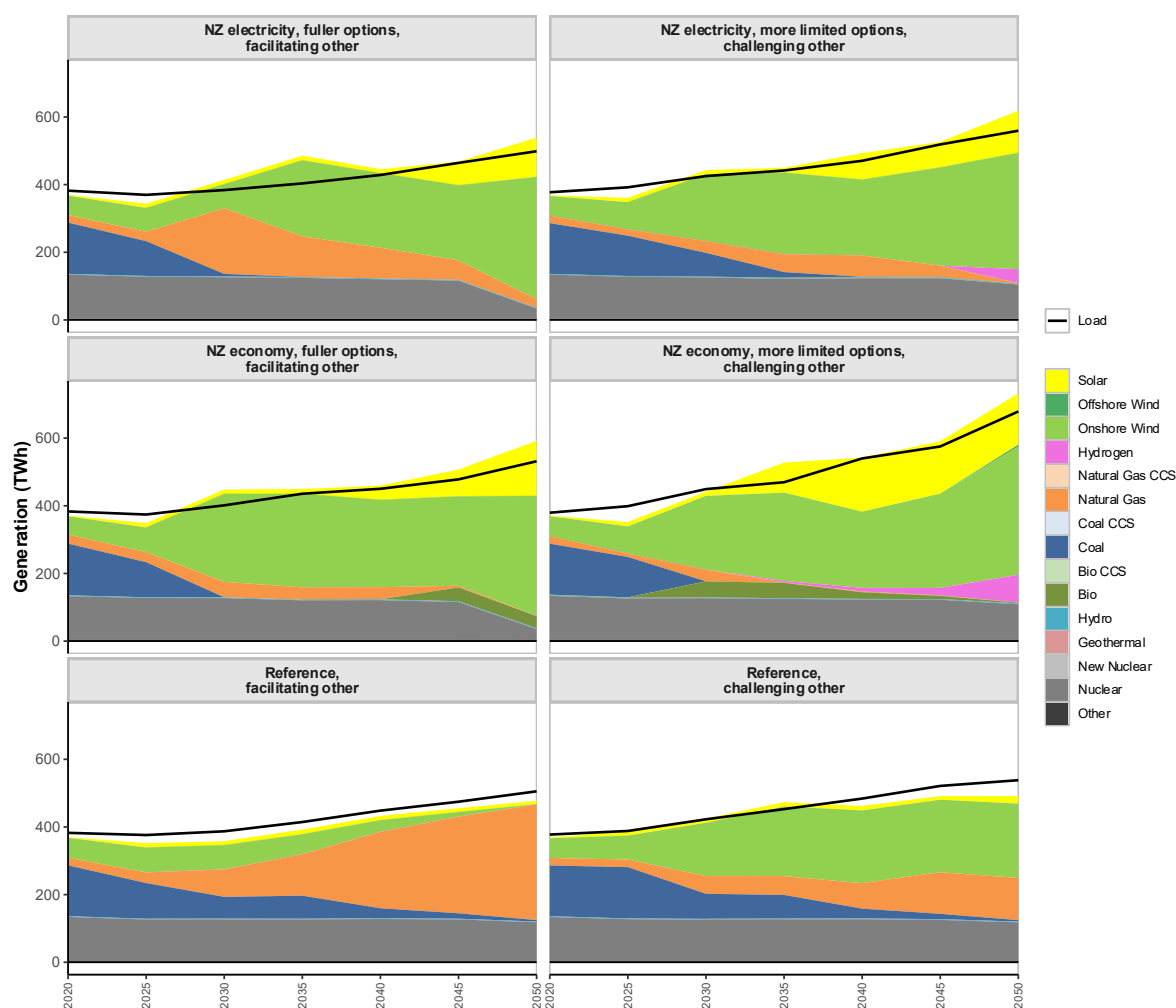


Figure 24. Regional electricity generation and load transitions

Source: EPRI analysis

Figure note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification. Top row results associated with net-zero CO₂ by 2050 power sector transitions with Fuller and More Limited decarbonization options available and Facilitating and Challenging market and technology conditions. Middle row results associated with net-zero CO₂ by 2050 economy-wide transitions with Fuller and More Limited decarbonization options available and Facilitating and Challenging market and technology conditions. Bottom row results associated with reference economy-wide transitions with Facilitating and Challenging market and technology conditions. Natural

gas includes a mix of renewable and synthetic natural gas, and dual fuel capacity includes capacity using mixes of fossil and renewable fuels (gas and liquid).

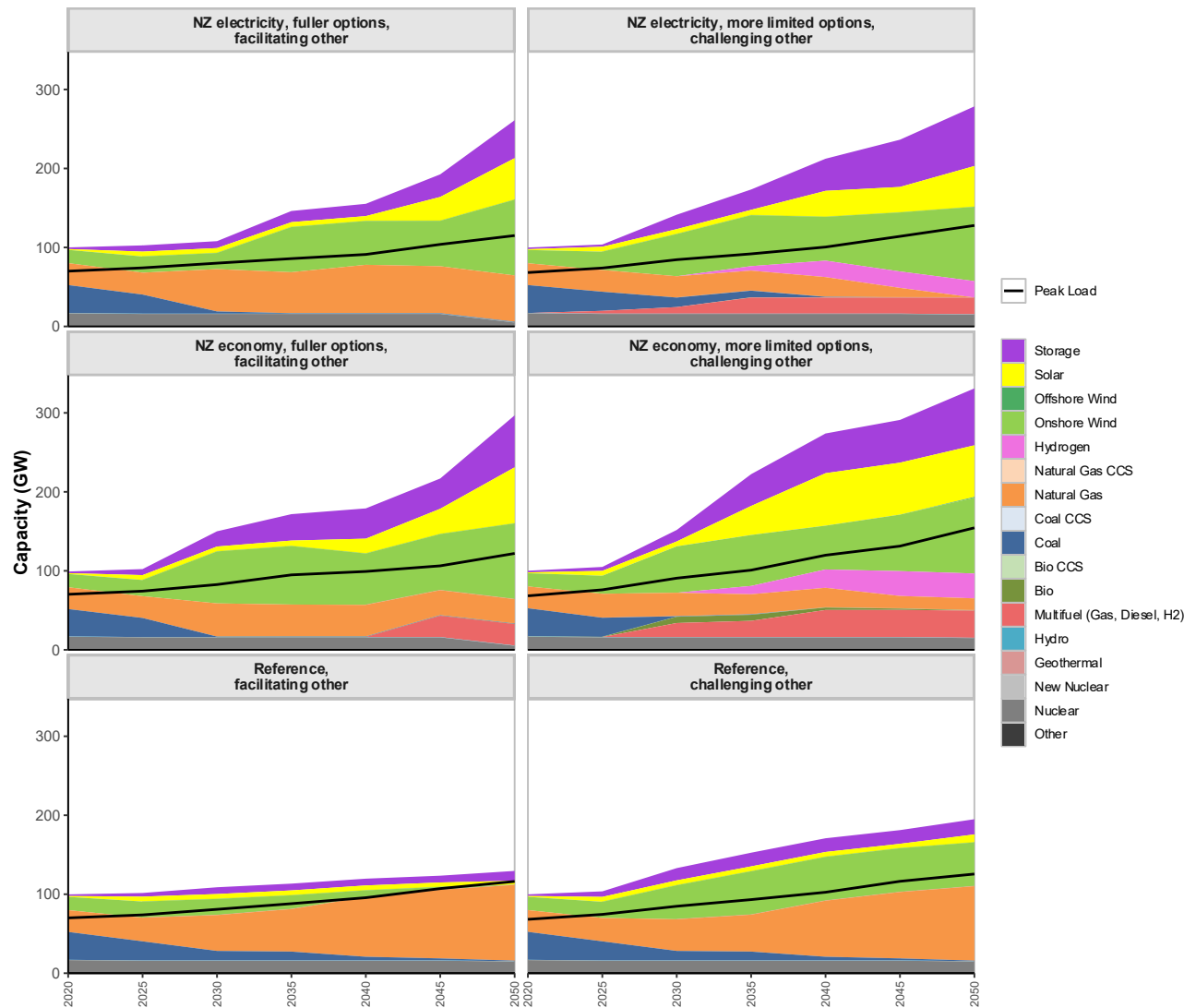


Figure 25. Regional electricity generation capacity transitions

Source: EPRI analysis

Figure note: Figure 24 note.

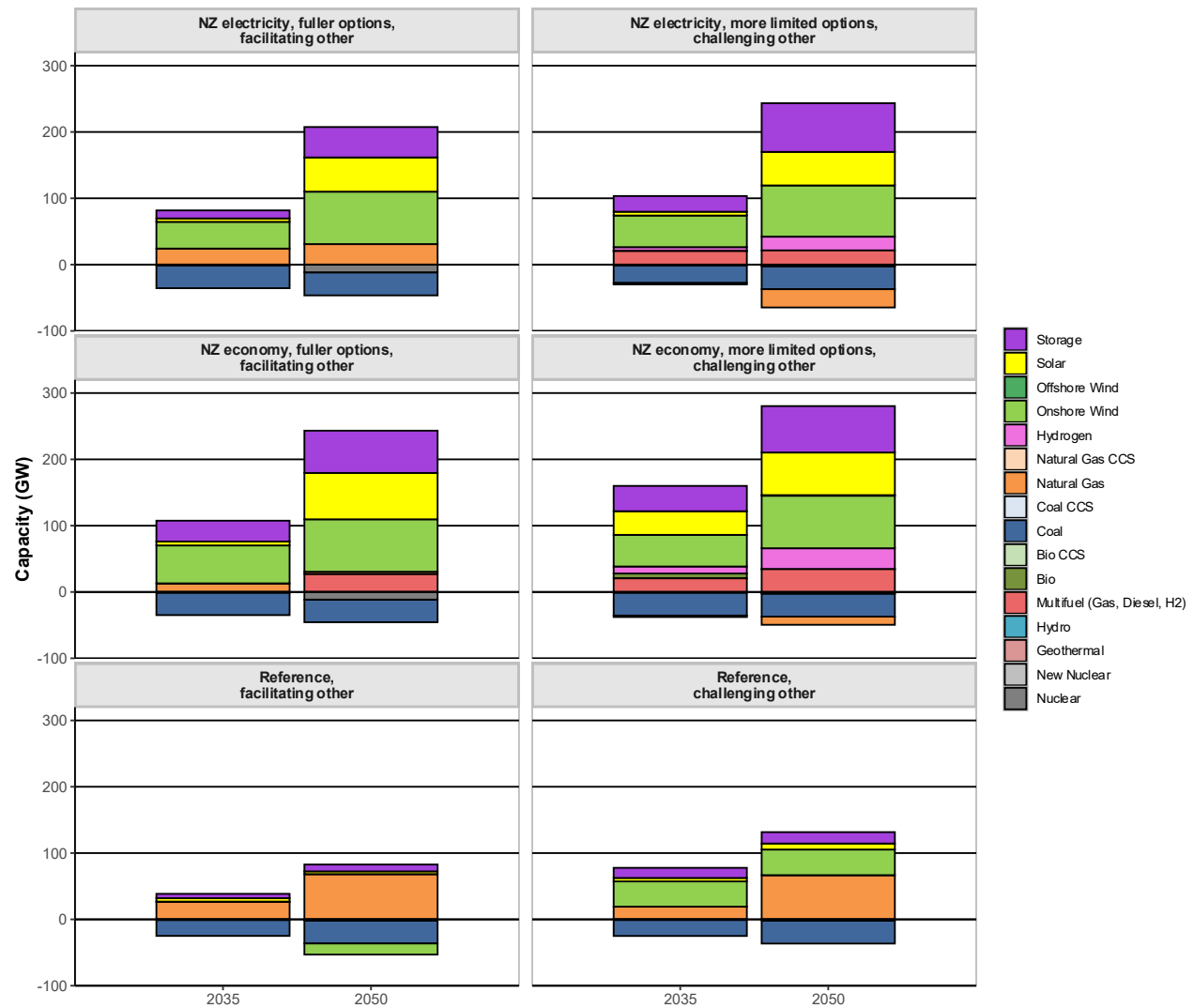


Figure 26. Regional electricity generation capacity cumulative additions and retirements by 2035 and 2050

Source: EPRI analysis

Figure note: Cumulative capacity additions and retirements are positive and negative respectively. Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification. Natural gas includes a mix of renewable and synthetic natural gas, and dual fuel capacity includes capacity using mixes of fossil and renewable fuels (gas and liquid).

The Figure 24 and Figure 25 results also illustrate that the roles of fossil, hydrogen, and imported power resources (net power imports are the space between the load line and the top of the generation portfolio in Figure 24) are highly variable as the different conditions change the relative cost-effectiveness of these options. Note, however, while the fossil generation resource (coal and gas) role is variable across the alternative future conditions, fossil resources are found to be cost-effective parts of the generation and/or capacity portfolios up until 2050 under all conditions. Capacity reserve requirements are met over the period in different ways depending on the conditions, with some combination of coal, gas, nuclear, hydrogen, dual fuel, and storage resources needed. Among other things, gas continues to serve as a cost-effective capacity resource serving reserve requirements. Even in the economy-wide decarbonization transitions with gas generation declining to zero, the capacity value of gas resources for reliability is important. Hydrogen and dual fuel capacity resources are also helpful, especially if the decarbonization options are More Limited and the other conditions are Challenging. The More Limited conditions have fewer decarbonization options, in particular carbon dioxide removal (CDR) is unavailable, and the Challenging conditions are more expensive and constrained in terms of new additions.

Figure 24 also provides insights regarding potential future electricity demand (load) and the system size. In all the scenarios we find load increasing over time. However, as mentioned, the economy-wide decarbonization incentive results in the most rapid increases in load, generation, and capacity additions. As noted, the economy-wide policy is creating greater demand for low-carbon electricity that is supporting the additional capacity investments. Furthermore, the additional load growth from data center demand in the Challenging context is further contributing to projected load growth (see the Challenging versus Facilitating load growth with power sector or economy-wide decarbonization).

The sensitivity of the transitions to the alternative decarbonization options and other conditions depends on whether there is a power sector only or economy-wide net-zero transition. For instance, the generation dispatch observed in the power sector only net-zero transitions are sensitive to the Facilitating versus Challenging conditions but are not very sensitive to the Fuller versus More Limited decarbonization conditions. Meanwhile, the economy-wide net zero transitions are found to be sensitive to both the decarbonization options and the other conditions.

Figure 27 presents net-zero transition results for wholesale electricity power price changes by 2035 and 2050 relative to 2020. Electricity prices are rising in all four AT transitions, with the price increases greater in 2035 (ranging from 109% to 373% in 2035 versus 21% to 50% in 2050). The greater price increases in 2035 indicate that there is an early adjustment challenge as the electric system and economy incur the initial costs of shifting towards decarbonization. The 2035 electricity price increases are greatest in the economy-wide decarbonization transitions due to the greater demand for low-carbon electricity. However, the decarbonization options and other conditions affect the magnitude of the 2035 price increases as well. The highest price increases with economy-wide decarbonization result from the More Limited options and Challenging conditions due to the increased cost of emissions reductions.

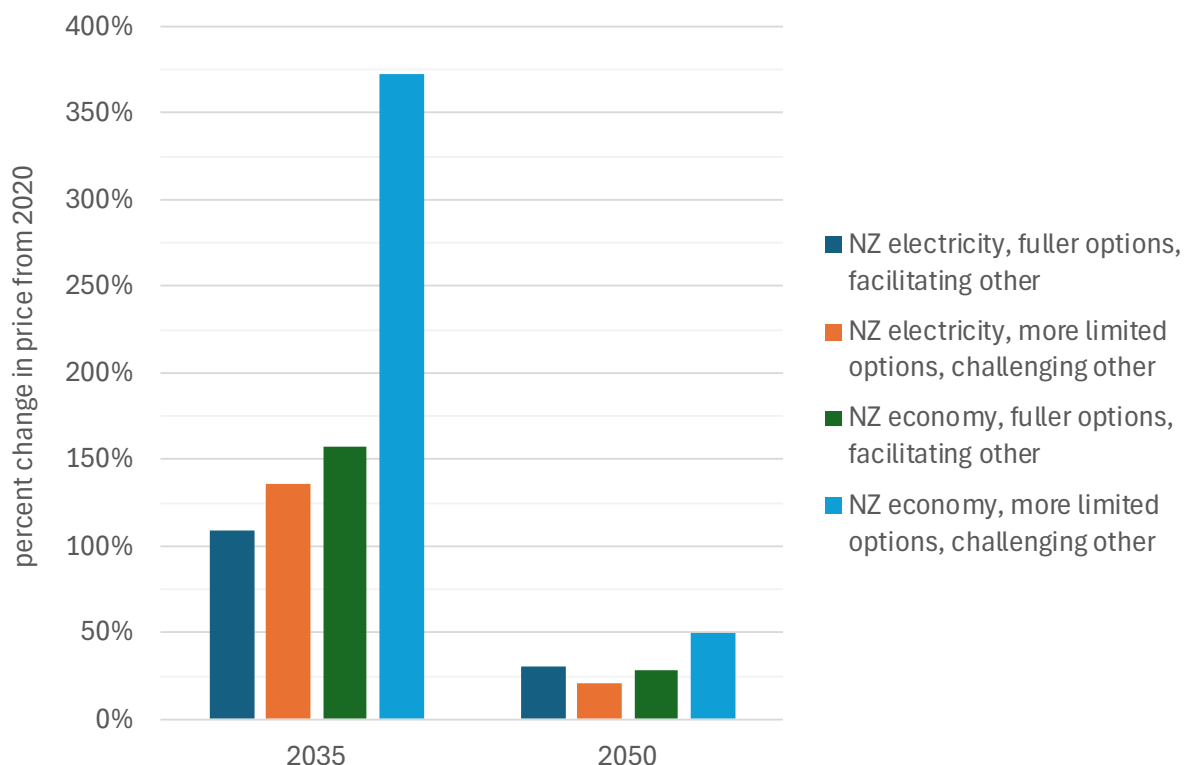


Figure 27. Wholesale electricity price changes from 2020

Source: EPRI analysis

Figure 28 provides net present value electricity system costs through 2050 for the net-zero scenarios, with the costs ranging from \$263 to \$376 billion. For comparison, the reference scenario costs are also shown with costs of \$230 and \$285 billion for the facilitating and challenging conditions respectively. The net-zero transitions result in an increase in projected net present value system costs with the economy-wide net-zero transition the most expensive. The majority of the costs are incurred by 2035 (Table 30). The cost of achieving net-zero—power sector or economy-wide—is higher with the More Limited and Challenging conditions. Fuller-Facilitating is the least expensive and More Limited-Challenging the most expensive.

It is very important to keep in mind that the higher expenses with the economy-wide policy are offset by the revenues from the increased demand for low-carbon electricity. These results are not shown, but it is what is driving the cost-effective more rapid decarbonization of the power sector in the economy-wide scenarios (Figure 23). Thus, investing in low-carbon electricity can be facilitated by policies that incentivize end-use decarbonization and increase the demand for and revenue from low-carbon electricity.

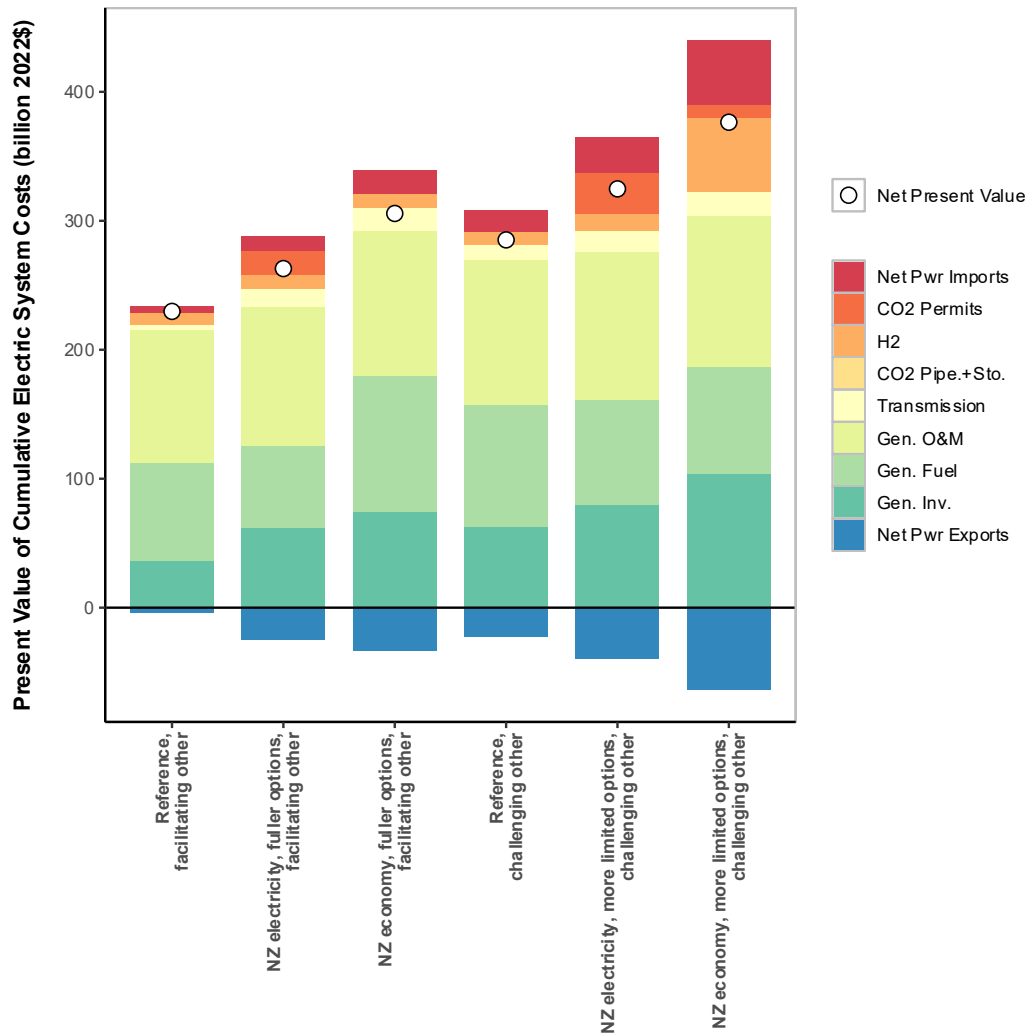


Figure 28. Net present value electricity system costs through 2050

Source: EPRI analysis

In addition to providing important insights on potential transitions and robust strategies, the scenario analysis provides insights regarding key risks, risk factors, and risk management strategies. For instance, Figure 24 and Figure 25 illustrate generation and capacity investment and operations risks, while Figure 27 and Figure 28 highlight affordability risk, with the More Limited-Challenging outcome a risk a company might want to manage. The analysis also provides risk management suggestions, highlighting the risk management value of having more decarbonization options in the future, as well as pursuing Facilitating-like conditions such as lower technology costs and reducing barriers to capacity and transmission additions.

Table 30 provides a valuable summary of results across variables and net-zero scenarios. The variables in the table can be organized according to their relevance to different company and societal priorities. In this case, the results in Table 30 are presented in terms of variables relevant to electricity supply affordability, reliability (system stability and optionality), and sustainability. The table is particularly valuable for summarizing uncertainty and risks. The wide ranges found illustrate that there is significant transition uncertainty relevant to affordability, reliability, and sustainability. We also see trade-offs and riskier outcomes related to affordability and reliability. For example, riskier outcomes are found in the near-term (2035) for electricity prices (potential increases of 370%), costs (possibly as much as \$300 billion), capacity additions (up to almost 200 GWs), system operations (the possibility of about 70% variable renewable resources), and load growth (possibly as much as 25%). Since the results in Table 30 are for the region, the company would scale them to their system, utilizing the implied growth rates in prices, costs, and system changes.

We can also look within the individual scenarios and 2035 and 2050 results (Table 31 and Table 32 respectively) to identify the conditions associated with the riskier outcomes. Table 31 and Table 32 allow the company to define risk criteria that they can then apply to identify “risky” outcomes and scenarios from their perspective. For instance, if the company views a 50% electricity price increase by 2035 as a difficult trade-off and risky, then all of the AT scenarios would be considered risky given their 2035 estimated price increases of about 110% to 370%. This implies that net-zero itself will be challenging under all of the conditions from an electricity price perspective and that the options for managing this risk are limited and the company may need to consider lower emissions reductions in 2035 and possibly 2050 than associated with the net-zero pathway. Similarly, the company might define a threshold for new builds based on historical experience for new investments. Such a threshold would help the company evaluate the viability of 100 to 200 GW of new regional generation capacity by 2035 to support the decarbonization transition futures. In addition to being able to define risky outcomes, the company can use the less risky outcomes to identify enabling conditions that facilitate achieving the net-zero goal without exceeding risk thresholds. For instance, for both power and economy-wide decarbonization, Fuller decarbonization options and Facilitating conditions imply decarbonization with lower electricity price increases and fewer capacity additions (Table 31 and Table 32).

The set of figures and tables also provide milestone and contingency insights. For instance, the results suggest the need to evaluate in a few years which future is emerging to determine if the strategy needs to be adjusted. If, for example, the decarbonization options available are turning out to look more like More Limited conditions, the company would want to evaluate increasing solar and wind investments substantially, as well as possibly slowing retirements as part of a revised cost-effective portfolio, in order to stay on track for their long-run emissions target. In general, milestones defined in terms of the More Limited/Fuller decarbonization options and the Facilitating/Challenging set of uncertainties, as well as in terms of the state of economy-wide transitions, can be used to guide company timing for when to re-evaluate their low-carbon strategy and how they might respond.

The evaluation of the ATs helps the company identify risks and the conditions that would enable the company to achieve the net-zero targets. The company then needs to evaluate the reductions it can achieve without the enabling conditions using the scenarios in Table 27. For this example, we ran scenarios with 30%, 50%, and 80% power sector net CO₂ reduction targets by 2050. These results can be combined with the 100% (net-zero) results to construct incremental response insights regarding the implications of increasing reductions. The insights will help the company identify the best they can do while still effectively managing the trade-offs and risks. These will be the company's QTs, and the conditions associated with them will be the qualifications.

Figure 29 through Figure 34 present the kinds of results needed to help the company identify their QTs. In some charts, the AT results are also shown for comparison. Figure 27 to Figure 34 illustrate how the results across the range of targets can be assembled to provide incremental implication insights. Note, after reviewing the initial results, we assumed this company decided to run the recommended additional scenarios in Table 27 to explore all the decarbonization option and other condition combinations, such as power sector and economy-wide net-zero transitions with Fuller-Challenging conditions. Those additional results are shown as well in Figure 29 to Figure 34.

Table 31. Regional cross-scenario result ranges for multiple variables for the AT scenarios

Source: EPRI analysis

Table note: Negative net power import results indicate net power exports from the region and that generation exceeds load in that year. For example, -23% means that in 2035 generation exceeded load by 23%.

	Affordability			Stability					Optionality	Sustainability			
	Change in Wholesale Electricity Price	Electric Sector Costs		Total New Generation Builds	Total Generation Retirements	Net Power Imports	Variable Renewable Generation	Electric Load Growth	Dispatchable Resources	CO ₂ Emissions Reductions		CO ₂ Emissions Intensity	In-State Clean Electricity
	% Change vs 2020	\$ Billions NPV		GW from 2020	GW from 2020	% of Load	% of Total MWh	% Change vs. 2020	% of Total Capacity	% Change vs. 2020		% reduction in tCO ₂ /MWh from 2020	% of MWh
		Total	Capital							Electric	Economy		
Ranges for 2035	+109% to +373%	212 to 305	29 to 72	103 to 196	33 to 41	-23% to -3%	50% to 67%	6% to 24%	52% to 59%	78% to 100%	46% to 95%	79% to 100%	75% to 100%
Ranges for 2050	+21% to +50%	263 to 376	64 to 113	277 to 379	63 to 78	-6% to -2%	74% to 90%	30% to 79%	43% to 51%	100% to 105%	72% to 114%	100% to 104%	95% to 100%

Table 32. Regional cross-scenario 2035 results for multiple variables for individual AT scenarios

Source: EPRI analysis

Table note: Negative net power import results indicate net power exports from the region and that generation exceeds load in that year. For example, -23% means that in 2035 generation exceeded load by 23%. Pie chart key: wind (green), solar (yellow), gas (orange), coal (dark blue), nuclear (grey), hydrogen (pink), biopower (green), storage (purple), dual fuel (red), and hydropower (light blue).

Scenario	Affordability			Stability						Optionality		Sustainability			
	Change in Wholesale Electricity Price	Electric Sector Costs		Total New Generation Builds	Total Generation Retirements	Net Power Imports	Generation Diversity	Variable Renewable Generation	Electric Load Growth	Capacity Diversity	Dispatchable Resources	CO ₂ Emissions Reductions		CO ₂ Emissions Intensity	In-State Clean Electricity
	% Changevs 2020	\$ Billions NPV		GW from 2020	GW from 2020	% of Load in 2035	Share of Generation on in 2035	% of Total MWh in 2035	% Changevs 2020	Share of Fleet in 2035	% of Total Capacity in 2035	% Change 2035 vs. 2020		% reduction in tCO ₂ /MWh from 2020	% of MWh in 2035
		Total	Capital									Electric	Economy		
NZ electricity, fuller options, facilitating other	+109%	212	29	103	41	-23%		50%	6%		56%	78%	46%	79%	77%
NZ electricity, more limited options, challenging other	+136%	266	36	130	33	-4%		58%	17%		59%	78%	47%	81%	84%
NZ economy, fuller options, facilitating other	+157%	246	35	137	40	-3%		66%	14%		52%	92%	84%	93%	92%
NZ economy, more limited options, challenging other	+373%	305	72	196	41	-11%		67%	24%		54%	100%	95%	100%	100%

Table 33. Regional cross-scenario 2050 results for multiple variables for individual AT scenarios

Source: EPRI analysis

Table note: See Table 32 note.

Scenario	Affordability			Stability						Optionality		Sustainability			
	Change in Wholesale Electricity Price	Electric Sector Costs		Total New Generation Builds	Total Generation Retirements	Net Power Imports	Generation Diversity	Variable Renewable Generation	Electric Load Growth	Capacity Diversity	Dispatchable Resources	CO ₂ Emissions Reductions		CO ₂ Emissions Intensity	In-State Clean Electricity
	% Change vs 2020	\$ Billions NPV		GW from 2020	GW from 2020	% of Load in 2050	Share of Generation in 2050	% of Total MWh in 2050	% Change vs 2020	Share of Fleet in 2050	% of Total Capacity in 2050	% Change 2050 vs. 2020		% reduction in tCO ₂ /MWh from 2020	% of MWh in 2050
		Total	Capital									Electric	Economy		
NZ electricity, fuller options, facilitating other	+31%	263	64	277	71	-2%		90%	30%		43%	100%	72%	100%	95%
NZ electricity, more limited options, challenging other	+21%	325	76	316	78	-5%		77%	48%		47%	100%	73%	100%	100%
NZ economy, fuller options, facilitating other	+28%	306	69	319	70	-6%		88%	39%		44%	105%	102%	104%	100%
NZ economy, more limited options, challenging other	+50%	376	113	379	63	-3%		74%	79%		51%	100%	114%	100%	100%

Figure 29 and Figure 30, for instance, show the generation and capacity portfolios over time for the different 2050 target levels. Figure 31 to Figure 34 show the incremental implications for NPV electric sector costs, wholesale electricity prices, and fossil, variable renewable, and dispatchable resources. For example, Figure 31 presents incremental net present value electric sector costs through 2035 and 2050 for the increasing levels of 2050 emissions reductions.

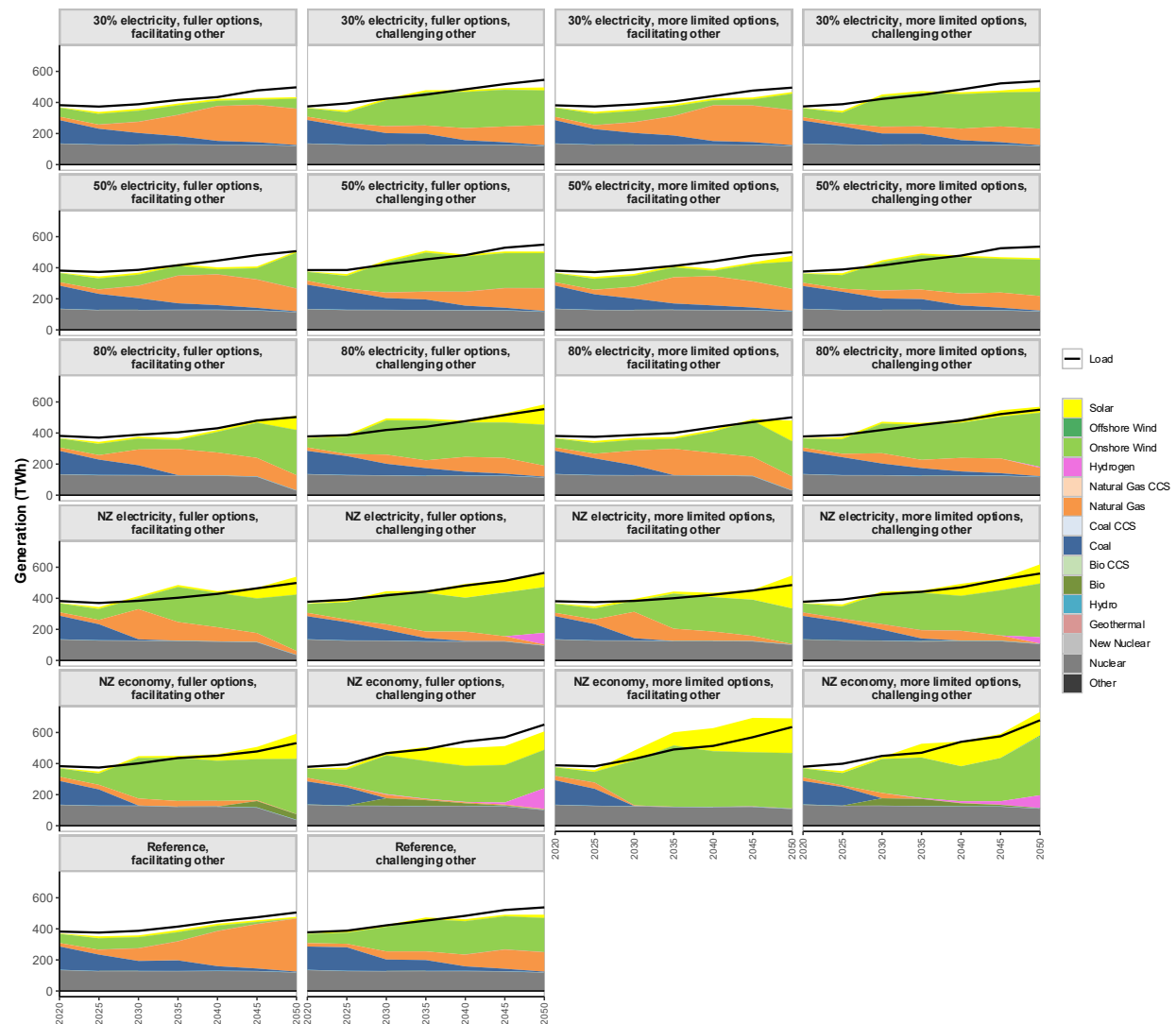


Figure 29. Electricity generation for the increasing levels of emissions reduction ambition, Aspirational Target, and reference scenarios

Source: EPRI analysis

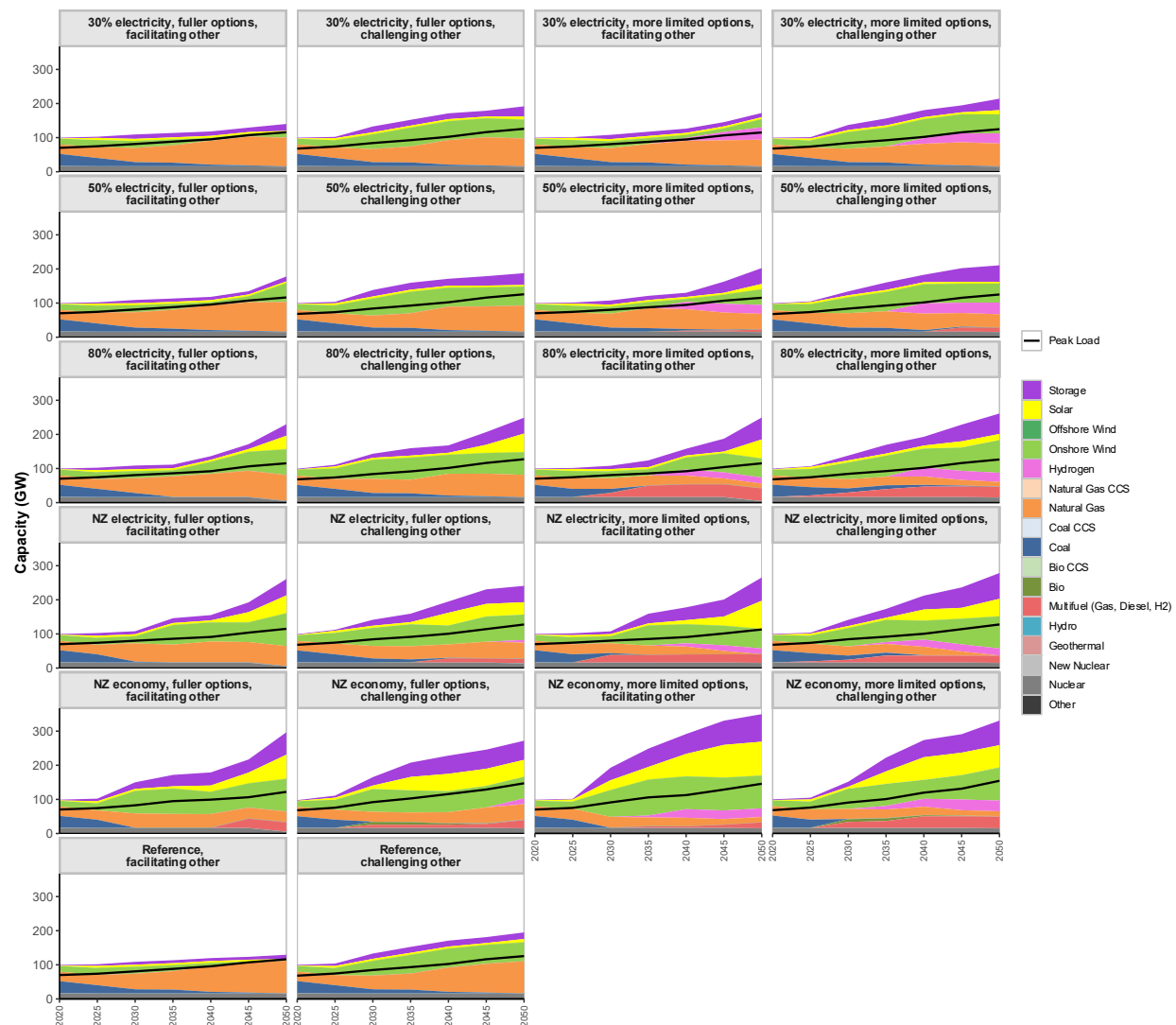


Figure 30. Electricity generation capacity for the increasing levels of emissions reduction ambition, Aspirational Target, and reference scenarios

Source: EPRI analysis

In Figure 31, we see increasing discounted costs with larger reduction levels due to greater generation capacity and transmission additions, O&M costs and power imports. We also see the impact of the Facilitating versus Challenging conditions, where the former is resulting in substantially lower costs (20-30%) in 2035 and 2050 due to the Facilitating environment's lower load growth, lower technology and fuel costs, and reduced barriers for new infrastructure additions. Furthermore, we observe that the costs under Fuller and More Limited conditions are similar in the power sector only decarbonization transitions. Interestingly, we find that the Fuller conditions, with low-carbon power trading available, incentivize additional investments driven by the opportunity for power exports. This sort of result highlights the value of economy-wide modeling of the energy system and inter-regional power markets. A chart like Figure 31 also provides critical enabling condition insights. For example, for less than the costs of 50% reductions with Challenging conditions, 100% reductions could be reached with

Facilitating conditions. Thus, efforts to reduce technology costs and lower barriers for new generation and transmission additions could enable greater ambition without additional costs. Of course, the costs and benefits associated with facilitating these conditions would need to be considered.

Electricity price increases are expected as well in 2035 and 2050 (Figure 32). Electricity price increases are projected for every level of emissions reduction ambition and decarbonization option and other conditions. Larger price increases are projected with the Challenging conditions, and the price increases are similar with Fuller and More Limited decarbonization options. Price increases in 2035 increase with emissions reduction ambition and are expected to be larger than in 2050, with pursuing 30% and 50% reductions resulting in about a doubling of 2035 prices (100% higher than 2020), and pursuing 80% and 100% reductions resulting in even larger 2035 price increases (150% higher than 2020). By 2050, there are still price increases but they decline with emissions reduction level, instead of increase. This is because pursuing larger emissions reductions is resulting in greater new electricity capacity additions by 2035 than would have resulted from pursuing lower emissions reductions. The additional capacity additions in 2035 imply greater electricity supply availability by 2050 and lower price increases in 2050.

Figure 33 and Figure 34 provide additional examples of incremental results and insights, showing changes in incremental fossil generation capacity (coal and gas), and variable renewable energy and dispatchable resource capacity by 2035 and 2050 for each of the target levels and under the alternative decarbonization options and other conditions. In Figure 33, we observe that, by 2050, coal capacity is nearly gone with emissions targets of 30% or higher. By 2035, coal capacity declines significantly (at least 60%) in all conditions and for all levels of reductions. The level of decline is relatively unaffected by the ambition level when there are Challenging conditions. Facilitating conditions, on the other hand, allow coal to be replaced more easily. Fuller option conditions also support a slightly faster decline in coal capacity. On the other hand, there is the possibility of cost-effective natural gas capacity growth through 2035 and 2050 while decarbonizing, where increasing the level of reduction ambition could result in greater gas capacity or less. By 2050, gas capacity could be significantly higher than today or eliminated. Gas growth is found to be cost-effective when there are Fuller decarbonization options, with the level of growth declining with ambition but potentially positive through 2050. Challenging conditions, with higher gas fuel prices, imply lower levels of gas capacity, but also greater costs and price increases (Figure 31 and Figure 32).

Figure 34 provides results relevant to reliability and operations. With the Challenging decarbonization conditions, there is need for significantly greater wind and solar variable renewable energy (VRE) capacity growth by 2035 than with the Facilitating conditions. If the company views 250% VRE growth by 2035 as an upper bound (3.5 times current VRE capacity), that would suggest that a target of no more than 30% might be the best they can do in Challenging conditions. However, with Facilitating conditions, the VRE capacity growth (and investment) needs are substantially lower and the company would be able to pursue 100% net CO₂ reductions by 2050. In 2050, we find potentially greater VRE growth than increases with emissions reduction ambition in all conditions.

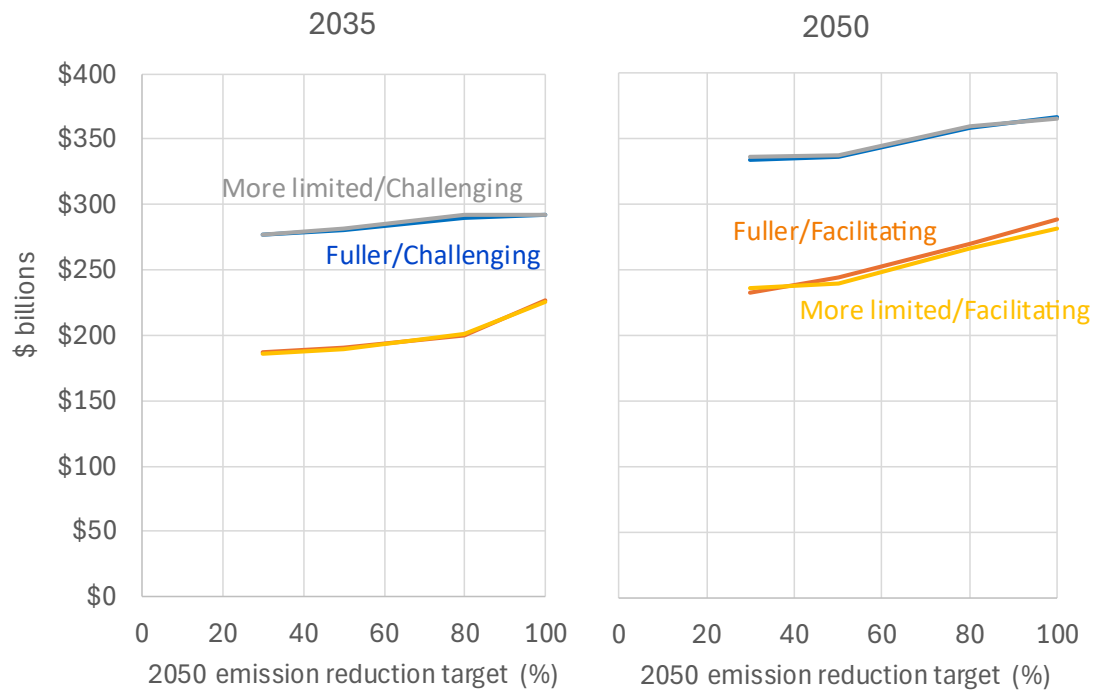


Figure 31. Incremental discounted gross electric sector costs through 2035 and 2050 for the increasing levels of 2050 emissions reductions

Source: EPRI analysis

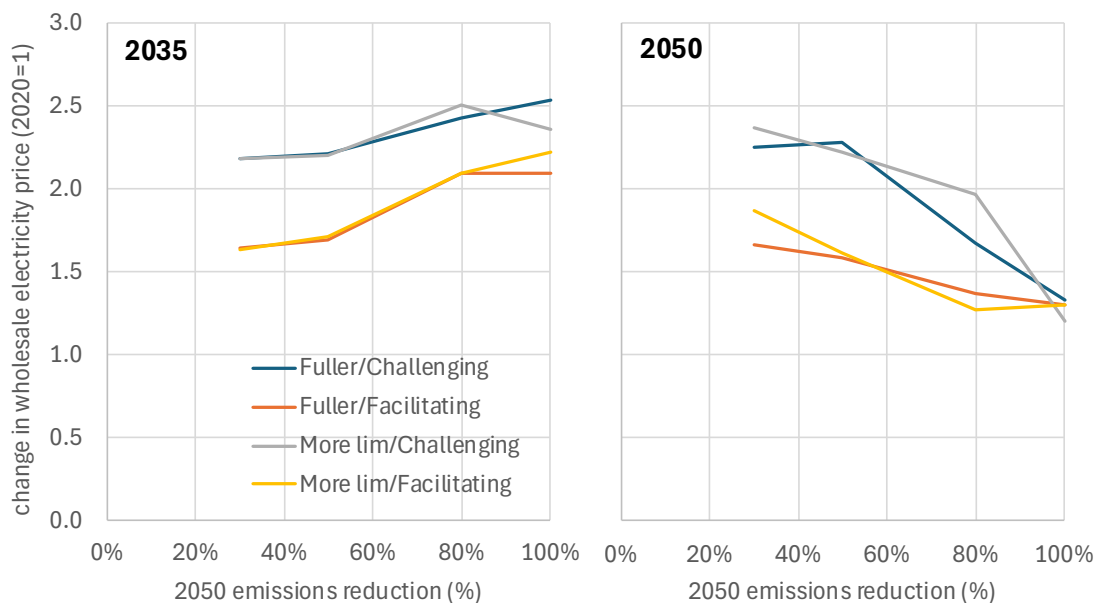
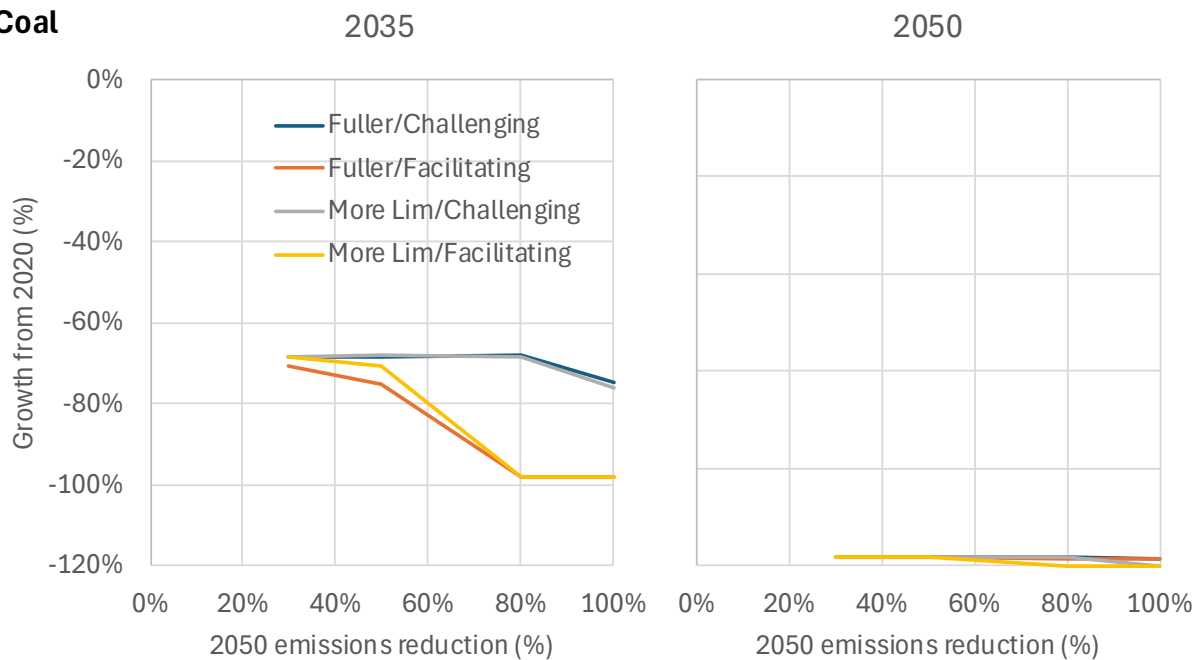


Figure 32. Wholesale electricity price changes in 2035 and 2050 for increasing levels of emissions reduction ambition

Source: EPRI analysis

Coal



Natural gas

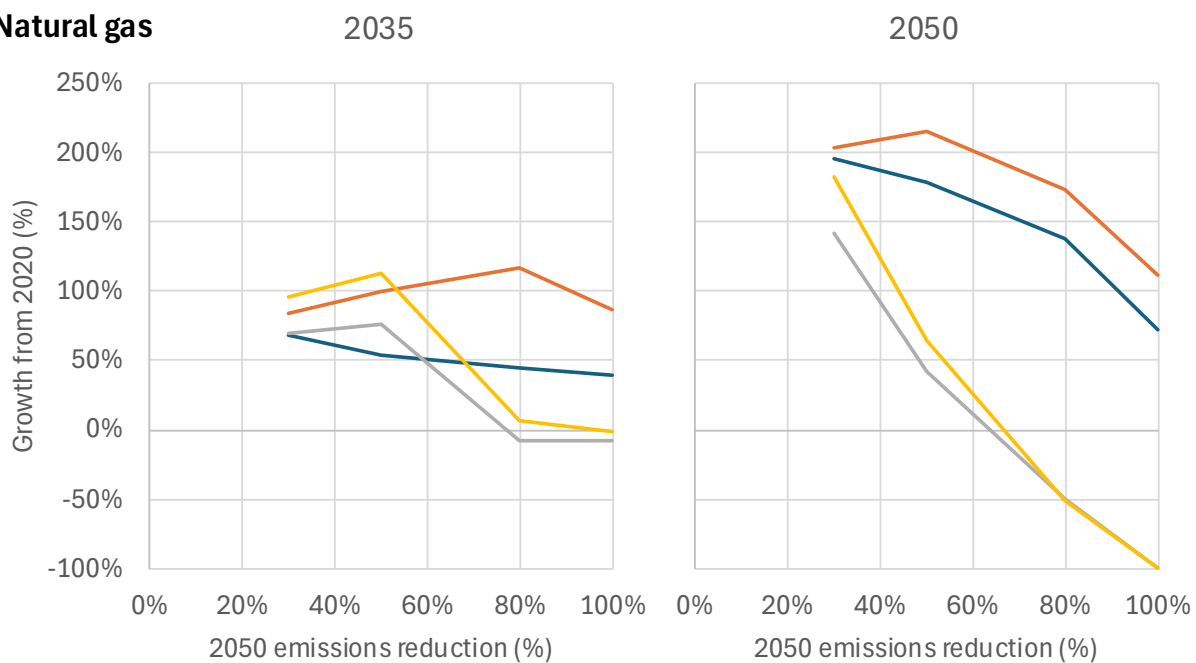
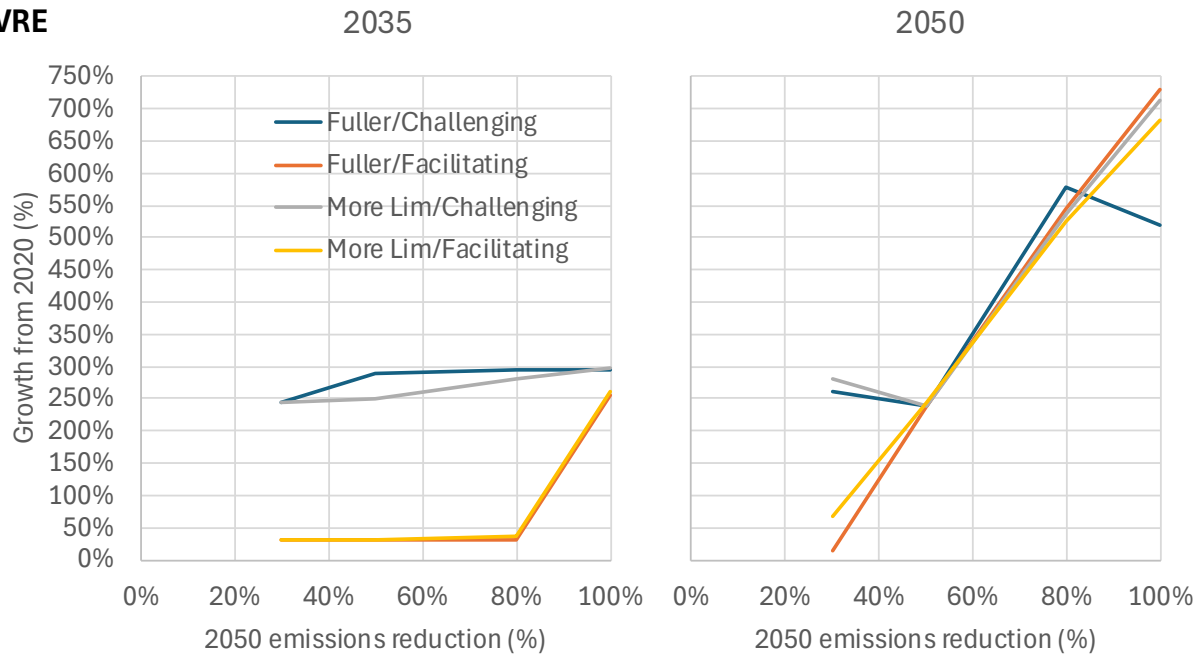


Figure 33. Coal and natural gas generation capacity growth through 2035 and 2050 for increasing levels of emissions reduction ambition

Source: EPRI analysis

VRE



Dispatchable

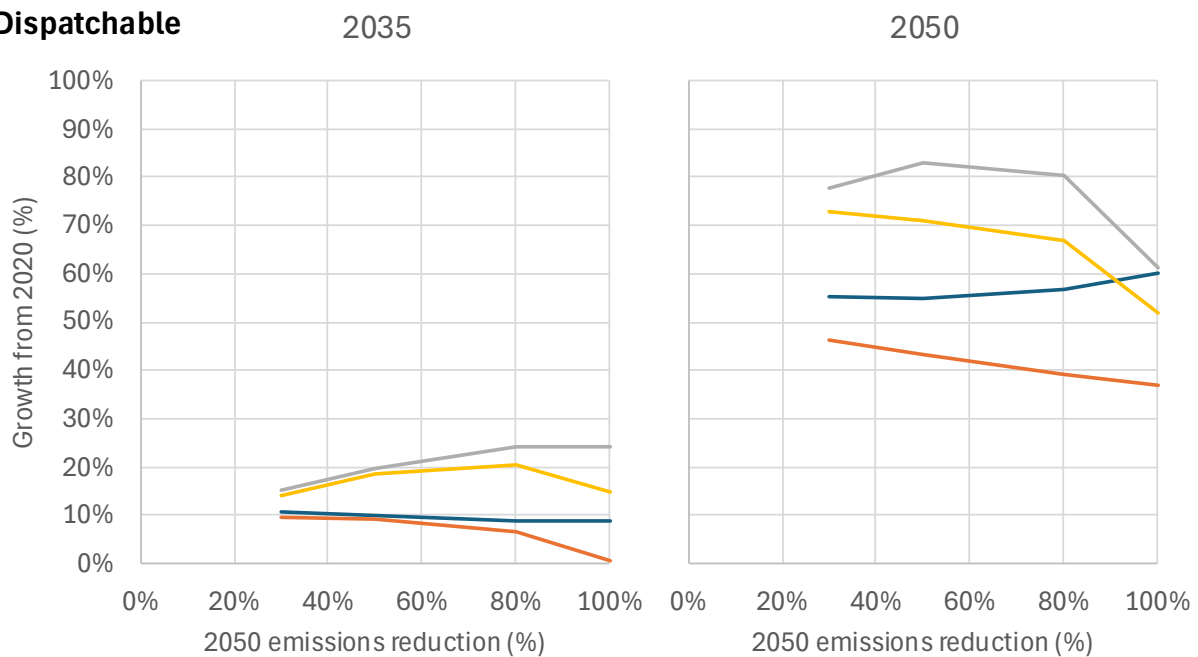


Figure 34. Variable renewable energy (VRE) and dispatchable generation resource capacity growth through 2035 and 2050 for increasing levels of emissions reduction ambition

Source: EPRI analysis. Dispatchable resources include biomass, coal, gas, geothermal, hydropower, hydrogen, nuclear, storage, and mixed fuel generation capacity.

Compared to the VRE capacity growth, there is significantly less variation in the growth of dispatchable resources by 2035 and 2050 across CO₂ reduction ambition levels (Figure 34). Dispatchable resources, which includes natural gas, coal, nuclear, biopower, hydro, hydrogen, storage, and dual fuels, are growing under all conditions for all reduction levels. Growth in dispatchable resources is positive under all conditions and ambition levels, but growth can increase or decrease with ambition depending on the conditions. By 2050, growth in dispatchable resources is significantly higher than through 2035. The largest growth is with More Limited-Challenging conditions, while the smallest growth is with Fuller-Facilitating conditions. The More Limited options decarbonization futures are leaning harder on dispatchable resources. Growth in dispatchable capacity is a robust strategy, but the exact amount is uncertain. Natural gas, nuclear, and storage are key dispatchable resources in all the futures, while coal, hydrogen, and dual fuels could also be important dispatchable resources depending on the future conditions. Regardless of the conditions, growth in dispatchable resources should be part of the company's strategy. While growing VRE is also a robust strategy, the level of growth varies substantially with reduction ambition, ranging from 10% to over 700% by 2050. Note that, in 2035, the VRE growth requirements could increase substantially beyond 80% reductions with the Facilitating conditions. Thus, getting to net-zero by 2050, even in favorable conditions, requires over five times the growth in VRE resources by 2035 than the pathway to 80% reductions.

Incremental figures like Figure 31 to Figure 34 provide helpful target setting and risk management information, as well as educational information for stakeholders. For instance, the Challenging conditions are significantly more expensive. Even for a 30% by 2050 reduction strategy, the company may wish to avoid the Challenging conditions and work with others to put in place the enabling conditions associated with Facilitating. Similarly, as discussed above, Figure 31 provides information regarding opportunities and contingencies for increasing ambition. There is an opportunity for the company to pursue more ambitious reductions; however, it would be contingent on whether Facilitating conditions are available in the future. Similarly, a 100% target with Challenging conditions would imply additional investment costs and would be information a company would want to communicate to stakeholders with an explanation supported by the analysis. Furthermore, the economy-wide results from earlier highlight that end-use decarbonization incentives are an important enabling condition for greater investments in low-carbon generation capacity and faster electric sector and economy-wide decarbonization.

Of course, companies may want to consider additional incremental metrics related to costs, infrastructure, and operations to inform their target setting and communications. They would also want to consider over-time transition results like those in Figure 29 and Figure 30.

Overall, the results from this step provide relevant, actionable, and transparent transition perspectives that are a useful complement to company system planning. The results provide perspectives on potential transitions in overall energy demand, energy markets, and economies that are relevant to the company's potential system and service territory transitions and decarbonization strategies.

It is important to keep in mind that transition opportunities and risks can vary dramatically between regions, with different transition options, different sensitivity to alternative conditions, and different risks. See EPRI (2023) for examples of differences in decarbonization opportunities and risks between regions within the U.S.

Outside of noting that they completed this phase, there is no documentation requirement for the company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete. The company will, of course, be generating draft results and insights from this phase. However, those results are deliberative and documenting them can be confusing and misleading to audiences that are non-technical and that have not been intimately engaged in the development process.

Phase 5: Review Potential Transitions and Target Implications – Based on Decision-Making Needs

- *Phase inputs: Phase 4 preliminary transition analysis pathways and insights.*
- *Phase outputs:*
 1. *Preliminary insights regarding potential transitions, targets, and risks,*
 2. *Revisions to core scenario design, model implementation, and output reporting, and*
 3. *Additional recommended and sensitivity scenarios and specifications the company would like to evaluate.*
- *Who should be involved: Company decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.*

For this phase, the company's planning, policy, sustainability, risk management, strategy, and outreach, staff, as well as external experts, will want to engage the company's decision-makers—directors and executives. The main objective here is to critically evaluate and discuss the preliminary results and insights to identify those they find reliable and meaningful. The additional objectives here are to identify any revisions to the core scenario design, model implementation, and output reporting they might want to make, and to identify additional scenarios and specifications that the company might like to evaluate. The last two items emerge from the company's critique and discussion of the draft results and insights.

Given the need to identify enabling conditions, this is the time for the company to think concretely about trade-offs, risks, and risk management. The Phase 4 discussion above highlighted relevant insights from the perspective of our example company and analysis, noting specific affordability and reliability risks and enabling conditions that identify and justify the company's QTs. Using the results from Phase 4, the company identifies challenges, trade-offs, risks, and risk management strategies. For instance:

- *Challenges and trade-offs:* The ATs are being met (or more) in all the transitions, but in each transition, we also observe challenges with affordability and reliability implications, investment implications in capacity build results, and sustainability implications with a range of emissions reductions by 2035. The set of incremental results highlight that there is no clear reduction level that is fully without consequential challenges. The Challenging conditions even make pursuing 30% reductions by 2050 difficult in terms of price increases, costs, and new capacity additions. Meanwhile, the company also found that some conditions could facilitate even greater reductions than the ATs (see risk management strategies below).
- *Risks:* The set of results highlight that all three categories of uncertainties create risk for this location, with large variation in the transitions and implications for futures achieving the ATs. Thus, uncertainty regarding the decarbonization incentive (sector versus economy-wide), decarbonization options (Fuller versus More Limited), and other conditions (Facilitating and Challenging) are all important considerations contributing to uncertainty in the transitions, with some more difficult to manage in terms of trade-offs in prices, costs, load growth, generation and capacity resource choices, traded power, and emissions transitions.
- *Risk management strategies:* Looking across the scenarios, we can identify enabling conditions that allow for transitions with more manageable trade-offs. From careful comparison of scenarios that differ in only one category of uncertainty, we find that encouraging an economy-wide policy with end-use decarbonization incentives would facilitate faster low-carbon generation investments and emissions reductions in the electric sector. We also find that costs are lower when there are the Facilitating (versus Challenging) conditions. Challenging conditions would result in greater investment costs and could preclude achieving net-zero by 2050. The Facilitating conditions, however, would allow a company in this region to achieve net-zero (100%) for less than the cost to achieve 30% reductions by 2050 under Challenging conditions. Therefore, strategies that accelerate technology cost improvements, reduce capacity additions barriers, or lower fuel costs will be helpful. The lower costs shown in the results help justify pursuing and collaborating on strategies like these. Finally, we find that having Fuller decarbonization options (versus More Limited) has mixed effects on costs. In the power sector only decarbonization transitions the differences are modest, but in the economy-wide decarbonization transitions the differences are notable, which only highlights the importance of modeling and evaluating how the regional economy might change. Specifically, having Fuller options is found to help reduce the decarbonization challenge in the economy-wide decarbonization transitions for this region.

The key strategy for this company for enabling greater emissions reductions is therefore to pursue Facilitating conditions, policies that create demand for low-carbon electricity, and the availability of Fuller options to facilitate economy-wide decarbonization. That said, the company will also want to communicate their QTs. In this case, a transition pursuing 50% reductions by 2050 (with 17% and 33% reductions in 2030 and 2040) would be consistent with a company affordability threshold of 100% price increase by 2035. If the company cannot tolerate a 100% price increase, then they would not be able to support the 50% by

2050 reduction target. The 50% by 2050 target would also be associated with as much as \$280 billion dollars in expenditures and up to a 300% increase in VRE generation capacity by 2035. Note that these 2035 levels of expenditures and VRE capacity additions are consistent with the long-run pursuit of 100% reductions by 2050 with Facilitating conditions. As such, expenditures and VRE additions of these magnitudes would keep the company on track for 50% by 2050 reductions, as well as give the company the opportunity for greater reductions while the company works with others to put Facilitating conditions in place for achieving net-zero emissions by 2050.

This phase is also the time when the company would want to consider the likelihoods of the different futures and potential hedging strategies they might pursue. For example, if the company believes that the likelihood of an economy-wide policy is low, they might invest *less* in new low-carbon electricity capacity since the demand for clean electricity is uncertain. Furthermore, if the company believes that the Challenging conditions are more likely than Facilitating conditions, they may want to communicate the likely additional challenge for achieving the ATs and consider pursuing their QTs.

After developing preliminary results and insights, the company would want to identify whether revisions are needed to the scenario design, model implementation, and/or output reporting. In terms of identifying potential revisions to the scenario design and implementation, the group discussion should consider whether they want to tweak anything. This could include revising what is included in the Fuller and More Limited decarbonization options sets, making Facilitating/Challenging more extreme to further differentiate them, or simply adjusting a specification such as the annual additions constraints for new generating capacity or transmission. This conversation should also include discussion of whether to run additional scenarios—the recommended additional scenarios or custom sensitivity scenarios. As noted above, we assumed the company’s assessment of their preliminary results led them to decide to run the recommended additional scenarios in Table 27. See the Phase 6 discussion for sensitivity scenario examples.

The company discussion on potential sensitivity scenarios should look for meaningful adjustments to one or two assumptions. For example, based on the results above, this company might want to consider running More Limited with CDR scenarios (power sector and economy-wide net-zero) that add CDR to the More Limited decarbonization options set. They might also want to consider running the Challenging scenarios with economic, instead of constrained, capacity additions. This scenario would run the Challenging conditions but with the annual capacity addition constraint removed.

Like Phase 4, outside of noting that they completed this step, there is no documentation requirement for the company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete.

Phase 6: Revise and Iterate as Needed – Revising Scenario Design, Implementation, and Model Results

- *Phase inputs:*
 1. *Phase 5 revisions to core scenario design, model implementation, and output reporting, and*
 2. *Phase 5 sensitivity scenarios and specifications the company would like to evaluate.*
- *Phase outputs:*
 1. *Revised insights regarding potential transitions, targets, and risks, and*
 2. *As needed, additional revisions to scenario design, model implementation, and transition pathways.*
- *Who should be involved: Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Stakeholder engagement: This is a good time to seek external stakeholder feedback.*
- *Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.*

For this phase, the company's planning, policy, sustainability, risk management, strategy, and outreach, staff, as well as external experts, will work together to revise the insights regarding potential transitions, targets, and risks; and, as needed, implement any needed revisions to the scenario design, model implementation, and transition pathways.

Phase 6 is also a good time for external stakeholder feedback. After internal iterations, the company will have a final draft scenario design, results, insights, and strategies to share and discuss. This gives the company an opportunity for education and dialogue, as well as to solicit feedback on their potential transitions, challenges, uncertainties, risks, enabling conditions, risk management, and collaboration opportunities.

For this phase, the company continues to iterate on the scenario design, specification, results, and insights until they have agreement with their decision-makers. Like Phase 5, outside of noting that they completed this phase, there is no documentation requirement for the company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete.

Phase 7: Communicate Results – Targets, Transitions, risks, Enabling Conditions, Risk Management

- *Phase inputs:*
 1. *Phase 6 final scenario design (core and additional scenarios), model implementation, and transition pathways, and*
 2. *Phase 6 final insights regarding potential transitions, targets, risks, enabling conditions, and risk management.*

- *Phase outputs:*
 1. *Documented scenario analysis, outputs, and insights,*
 2. *Aspirational Targets (ATs), strategies, and risk management, and*
 3. *Qualified Targets and their qualifying conditions (QTs) and strategies.*
- *Who should be involved: The company's decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Documentation: SMARTargets Reporting Template summarizing the following:*
 1. *The company's ATs and QTs with corresponding conditions,*
 2. *The company's enabling conditions, strategy, risks, and risk management,*
 3. *The company's transition risk analysis (modeling, scenario design and specification, potential transitions, insights), and*
 4. *The implementation process.*

This phase is the critical communications of all the company has done for the Step 4 analysis. For this phase, the company will want to work with their decision-makers and company planning, policy, sustainability, risk management, strategy, and outreach teams. They will also want to engage their external experts to help with documenting the methodology and effectively, rigorously, and defensibly communicating the results. The experts can also help the company communicate the relevant science to educate readers and ground the methods.

At the end of Phase 7, the company will use the SMARTargets Reporting Template to finalize the documentation for their Step 4 transition risk analysis (Phases 1 through 7). This includes the summary tables of key output indicators, as well as graphics to help communicate the company's modeling results and insights. Additionally, the company will report the identified enabling conditions for ATs, QTs and corresponding conditions, and the risks and opportunities identified. Finally, the company will indicate their risk management strategies along with their milestones and contingency plans.

The company's strategy will consist of more than targets. It will also include transition plans as well as strategies for pursuing enabling conditions and managing risks, where the latter might entail research and development plans, collaboration with customers or other sectors, and policy dialogue. Step 4 has a minimum set of required outputs (Table 20 in the methodology document). Additional variables may be added based on company need (e.g., air pollutant emissions, differentiated affordability indicators).

Step 5. Strategy and Qualified Target Alignment with International Goals

Step 5 Outputs: International goal alignment assessments of qualified targets and strategies.

In this step, the company will evaluate *qualitative* alignment of their transition strategy with the international goals, as well as *quantitative* alignment of their QTs.

Assessing Qualitative Alignment of the Transition Strategy

The company will assess two types of qualitative alignment of their strategy with international goals: (1) alignment with the types of transitions robustly observed in global emissions pathways consistent with the Paris Agreement temperature goals, and (2) alignment with the provisions in the Paris Agreement associated with achieving the temperature goals.

As discussed in the methodology document, the following are robust types of transitions found in all global 1.5°C and 2°C emissions pathways. The following are changes found consistently in well-below 2°C pathways regardless of the model and assumptions:

- Increasing low-carbon technology deployments,
- Increasing reliance on electricity,
- Transitions away from fossil fuels but continued use,
- Use of carbon dioxide removal,
- Improvements in energy conservation and efficiency,
- Differences in regional low-carbon transitions, and
- For any climate goal, substantial uncertainty in the levels, rates of change, and composition (e.g., technologies, energy, activity) of the transition of the global economy.

The Step 4-Phase 4 analysis results above do not cover all these perspectives. However, what they do provide is qualitatively aligned in that they exhibit increasing reliance on electricity, decreasing reliance on fossil fuels, increasing deployment of low-carbon technologies, and use of carbon dioxide removal and energy conservation and efficiency (last two not shown). They also exhibit significant uncertainty in the transitions and, by design, recognize the unique low-carbon transition opportunities.

Furthermore, the transitions and insights align with Paris Agreement's provisions recognizing differences in national decarbonization opportunities, multiple societal priorities, transparency, and the value of innovation and cooperation (in this case across sectors via the economy-wide scenarios).

Assessing Alignment of the Qualified Targets

The company will also assess quantitative alignment of their QTs through comparison to the relevant global emissions pathway ranges consistent with the Paris Agreement objective of limiting global average temperature to "well-below" 2°C. Table 33 shows this comparison from which this utility would conclude that their QTs are quantitatively Paris Agreement aligned. Based on the Step 4 transition risk analysis, we assume this utility identified two sets of QTs:

1. 50% reductions by 2050 (with 17% and 33% reductions in 2030 and 2040)
2. 89% reductions by 2035 (with 93% and 100% reductions in 2040 and 2050)

The conditions associated with the first set of QTs include the following: electric sector only decarbonization transition and Challenging other conditions. The conditions associated with the

second set of QTs include the following: economy-wide decarbonization transition, Fuller options, and Facilitating other conditions. The sets of QTs were identified by considering the transition implications for affordability (prices, costs, investments), new infrastructure requirements, and operational reliability, as well as the likelihood of future conditions.

As discussed in the methodology document, the high-level insight from the Table 34 comparison is that QTs within the ranges are aligned with global futures consistent with limiting warming to well-below 2°C. Furthermore, if all companies have QTs within the range, the sum will also be aligned with well-below 2°C. However, it is critical to keep in mind that this comparison is not conclusive because the global pathways do not capture the transition opportunities and challenges for individual companies, or differences across companies. In addition, the global pathway ranges do not reflect all the sub-global and local transitions found to be aligned with a global temperature. Thus, QTs for some companies may not fall within the global pathway ranges due to their transition challenges. The QTs should not be judged negatively based on this comparison. This outcome is simply an indication of the company's challenges. The QTs should be evaluated in terms of their challenges and the strategies for pursuing them and enabling conditions for additional ambition. The QTs represent the highest possible ambition for a company under different conditions while balancing trade-offs and risks that the company would be transparently identifying and communicating. Specifically, the company would communicate what conditions are determining their QTs, as well as how additional reductions could be possible under other conditions.

Table 34. Qualified Target Paris Agreement alignment assessment

Qualified Target set	Key qualifying conditions	GHG		Qualified Target 2050	Well-below 2°C ranges	Within range
1	Electric sector only decarbonization transition and Challenging other conditions	Net CO ₂	2030	-17%	10% to -100%	Yes
			2040	-33%	-24% to -121%	Yes
			2050	-50%	-43% to -126%	Yes
2	Economy-wide decarbonization transition, Fuller options, and Facilitating other conditions	Net CO ₂	2030	-89%	10% to -100%	Yes
			2040	-93%	-24% to -121%	Yes
			2050	-100%	-43% to -126%	Yes

In addition to the above documentation, the company might also want to communicate quantitative Paris Agreement alignment visually. A graphic like Figure 35 can be created and is a grounded, effective means to quantitatively communicating alignment of the company targets and transition risk analysis pathways with the Paris Agreement temperature goals.

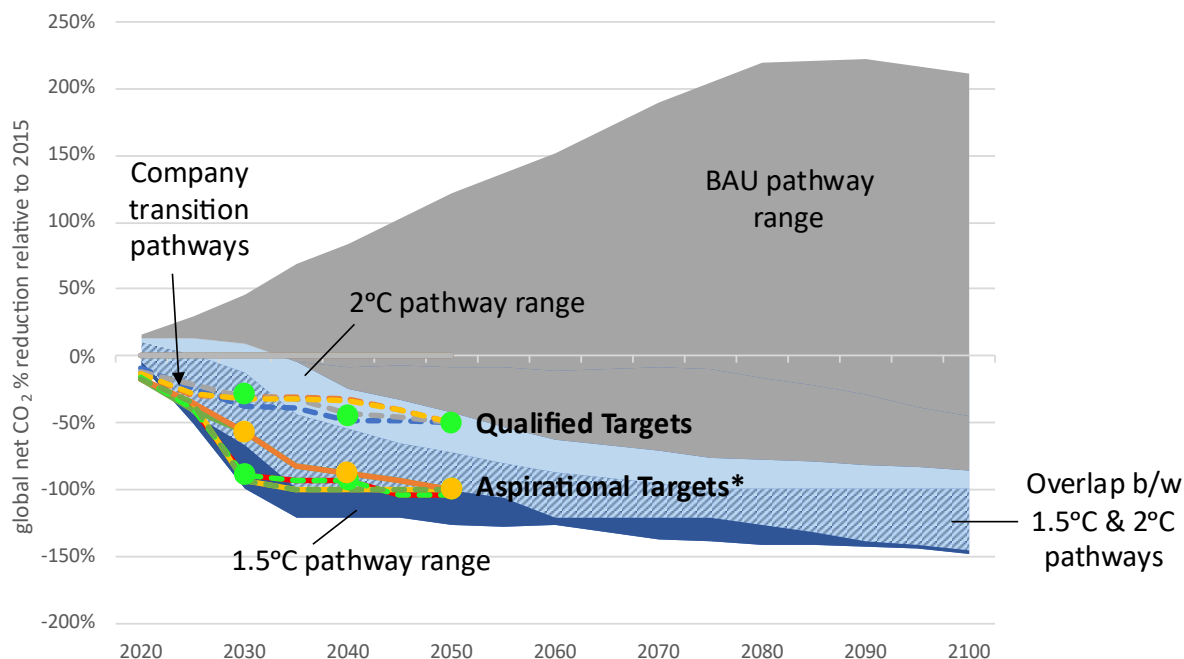


Figure 35. Alignment of illustrative company Aspirational and Qualified net CO₂ Targets and transition risk pathways with the global emissions pathways consistent with the Paris Agreement temperature goals

Source: EPRI

Figure notes: Aspirational Targets denoted with yellow dots with solid line transition pathways. Qualified Targets denoted with green dots with dashed line transition pathways. Figure developed from EPRI (forthcoming) analysis of the IPCC's Sixth Assessment Report, IEA Net Zero by 2050, and other recent global emissions pathways, and the EPRI illustrative US-REGEN analysis completed for this example. See EPRI (forthcoming) and EPRI (2020 and 2018) for detailed assessments of recent and previous global emissions pathway modeling and results, including discussion of global pathway limitations and the importance of using full ranges due to the nature of the IPCC data.

* Enabling conditions may be needed to achieve the Aspirational Targets. See the company transition risk analysis and narrative that identifies the enabling conditions and Qualified Targets and their conditions.

Step 6. Documentation and Communication

Step 6 Output: SMARTargets documentation.

In this step, the utility will document and communicate their SMARTargets implementation using the SMARTargets Reporting Template. This includes documenting implementation choices with justification, transition analysis insights, targets, strategy, and milestones. Through the template, the following information will be assembled:

- General company information and stakeholder engagement,
- GHG Inventories and base year,
- Practical issues and climate target groupings,
- Company transition risk analysis of opportunities and risks,
- Alignment assessment,

- Documentation and communication of targets, strategies and indicators, and
- Implementation and monitoring.

The set of information represents a clear, comprehensive, and transparent package of information about the process and the outcomes. Overall, in applying the methodology, the utility will produce the following standardized and comparable results that reflect their implementation and climate strategy:

- Paris Agreement-aligned targets and enabling conditions, including Aspirational Targets (AT) and Qualified Targets (QT) and their associated conditions,
- Insights into potential low-carbon transitions,
- Assessment of risks, risk management strategies, and overall transition strategy, and
- Documentation of methodology implementation, including:
 - GHG inventory by required emissions category,
 - Responses to practical considerations and target-setting decisions for each category,
 - Transition risk analysis components:
 - Modeling approach
 - Reference conditions and key uncertainties
 - Core scenario definitions and specifications
 - Criteria for identifying risks and enabling conditions

In addition, the company is encouraged to holistically communicate their strategy. This goes beyond emissions targets and includes leadership strategies to help create the enabling conditions for achieving greater emissions reductions and economy-wide decarbonization, such as research and development (R&D), educational outreach (stakeholder and policy), and collaboration and coordination with customers and supply chain partners. For this illustrative utility, the analysis has identified that, among other things, strategies related to the following would be helpful and worth considering: encouraging end-use decarbonization incentives that create a market for low-carbon electricity, having access to carbon dioxide removal technologies, accelerating technology cost improvements, and reducing capacity additions barriers.

In addition, the company might consider communicating on the SMARTargets Methodology itself, including its scientific foundations. This can involve referencing the publicly available methodology, the SMARTargets Scientific Foundations, and other materials and emphasizing how the approach is science-based and aligned with the Paris Agreement (see Justification for the Methodology section).

Step 7. Validation and Verification

Step 7 Outputs: Validated SMARTargets and verified progress.

In this step, the utility will validate their SMARTargets and verify progress. After completing the SMARTargets Reporting Template in Step 6, the utility will be able to initiate validation activities. Validation assesses whether the methodology has been properly implemented and whether the target and strategy—by nature forward-looking statements—are supported by sound assumptions, data, and methods. The primary input for validation is the SMARTargets Reporting Template.

The utility should begin with internal validation, leveraging internal audit processes aligned with recognized standards. This step ensures the methodology has been correctly applied and that the resulting targets and strategies are scientifically grounded and aligned with the Paris Agreement. Internal validation also builds confidence by thoroughly documenting the rationale and process behind the targets—information that is essential for both third-party validation and external communications.

Internal validation is all that is possible in the near-term. Accredited, independent third-party validation services for corporate GHG targets do not currently exist. Third-party validation is recommended once it becomes available. It provides an independent, authoritative assessment of conformity with the SMARTargets Methodology and validation criteria.

Following validation, the utility will communicate the results with the public, including regulators, investors, and other interested stakeholders.

In addition to validating, the utility is required to verify progress periodically. The utility will evaluate actual emissions reductions against its targets using the most recent GHG emissions inventory. The same third-party provider that verifies the GHG inventory could also perform this progress verification, thereby ensuring consistency and credibility with reporting. This verification relies on GHG inventories, which are based on observable data and are therefore subject to assurance-level assessments.

Step 8. Monitoring and Adjusting

Step 8 Outputs: Periodic milestone assessment and, as needed, strategy adjustment and/or SMARTargets refresh.

Under this step, the company will annually monitor their milestones and adjust their strategy as needed. The utility will need to monitor conditions with respect to their milestones to determine if strategy adjustments are needed and/or if updated SMARTargets analysis, targets, and strategies are needed.

The transition risk analysis (Step 4) provides milestone and contingency insights as well as insights regarding transition opportunities and risks. The company should articulate meaningful and observable milestones, and potential responses, based on their analysis, with the milestones clearly communicated in the Reporting Template. For instance, the Step 4 results

above suggest the need to evaluate which future is emerging. If, for example, the decarbonization options available are turning out to look like the More Limited set, instead of the Fuller set, in order to stay on track with their long-run emissions targets, the utility would want to consider alternative strategies for gas, solar, wind, storage, and hydrogen, as well as possibly slowing retirements if annual additions are likely to be constrained. Furthermore, those strategies will depend on whether sector or economy-wide decarbonization incentives are emerging, which is another milestone the utility can readily monitor. If economy-wide decarbonization incentives are developing that increase low-carbon electricity demand then the utility should consider accelerating clean energy investments (and emissions reductions).

Illustrative Results for T&D Utility Emissions Targets

This example illustrates the derivation of SMARTargets for transmission and distribution (T&D) utility emissions for a hypothetical company that provides T&D services supporting the delivery of electricity. T&D infrastructure is critical to the provision of electricity to customers, including supporting the potential increased use of electricity to help decarbonize economies.

The primary categories of GHG emissions for this utility are the following:

1. CO₂ emissions from electricity that is sold to end users (scope 3),¹⁴
2. CO₂ emissions from T&D line losses in the process of transmitting electricity to points of distribution and in the final distribution of that electricity to customers (scope 1, 2, or 3),¹⁵ and
3. SF₆ (sulfur hexafluoride) fugitive emissions from T&D equipment (scope 1),¹⁶

Uncertainty regarding electricity supply and demand creates uncertainty for T&D emissions, targets, and strategies, with uncertainty in total power flows, T&D infrastructure needs, and the mix of electricity generation resources (low-emitting and emitting). T&D utilities need to evaluate these uncertainties to identify opportunities and risks and set actionable targets and practical strategies.

The regulatory environment is also relevant. In some jurisdictions, T&D utilities have the ability to influence generation decarbonization through green procurement strategies, while in other jurisdictions, utilities may not have control or influence over the generation resources. For instance, some T&D utilities may be subject to regulations requiring least-cost electricity

¹⁴ These emissions are classified as scope 3 emissions for companies that do not own generating assets according to the GHG Protocol.

¹⁵ Transmission losses are classified in the GHG Protocol in different scopes depending on company structure. For vertically integrated utilities, T&D transmission losses will be already considered in stationary combustion of electricity generation scope 1. For “wires-only” T&D utilities, transmission losses will be considered scope 2. And for independent power producers, T&D transmission losses that occurred in a separate “wires only company” are scope 3.

¹⁶ According to the GHG Protocol, SF₆ fugitive emissions are classified as scope 1 emissions directly related to the operation of T&D assets.

procurement that constrains their ability to recover potential premiums for low-emitting resources.

In this example, we consider an illustrative T&D utility in a U.S. Midwestern region. The company sits within the region and is interested in understanding how the regional electric power market and economy might evolve with decarbonization by 2050 and what might be the implications and risks. Using US-REGEN, we can constrain emissions and evaluate alternative potential low-carbon transitions for the regional electric power market, the three categories of GHG emissions, and GHG abatement opportunities.

Below, we describe how this T&D utility would apply each step of the SMARTargets Methodology.

Step 1: GHG Inventories and Base Year

Step 1 Outputs: Verified GHG inventory and emissions reductions base year.

The first step is for the company to assemble their GHG inventory and select a base year. The company will need GHG emissions accounts for each of the categories of emissions relevant to their business. The company will also need to decide on their emissions reduction base year and apply it consistently. Note that the company will need to have the inventories verified.

For this example, Table 35 summarizes the emissions for our illustrative utility. The primary source of emissions is CO₂, emissions from purchased power to serve load. The utility could have other emission sources but, for this example, we assume they represent only a small fraction of the overall inventory and are immaterial. Table 35 presents the inventories for the utility's base year and their most recent inventory.

The GHG emissions trends for each category are shown in Figure 36. T&D related CO₂ emissions have been declining over the last decade. An analysis of drivers, reveals that this is due primarily to reductions in fossil fuel-based electricity generation and reductions of energy losses on the T&D network. SF₆ emissions have declined significantly since 1990, primarily due to enhanced controls targeting fugitive emissions. However, since 2015, SF₆ emissions have increased due to an expanding system and additional T&D infrastructure.

Given the information available, we assume the utility decides to select 2015 as its base year. This decision is based on considering the accuracy of their inventories, the inventory years verified, and choosing a year that facilitates communications regarding the utility's emissions trends. Given the relevance of emissions from electricity sold to customers, the utility notes that accounting improvements are needed to better reflect CO₂ emissions from this category (scope 3).

Table 35. GHG Inventory for key categories in T&D utility illustrative example

Source: EPRI with data from US-REGEN, DOE (2024), EPA (2024), and an illustrative T&D utility inventory.

Key emissions categories	2015 (base year)	2021 (most recent)
	MtonCO ₂ e	
CO ₂ from electricity sold to end users	4.5	3.4
CO ₂ from T&D line losses	9.5	6.9
SF ₆ fugitive emissions from T&D equipment	42.1	48.4

Step 2. Practical Issues and Climate Target Groupings

Step 2 Outputs: Decisions on practical issues, including emissions categories covered (by regulation or company targets), the categories for which targets are being set, and how these categories are grouped for target evaluation.

For this step, our illustrative T&D utility works through the company target setting practical issues, as well as identifies their target groupings. Table 36 presents example responses to the set of practical issues questions for all three emissions categories, and below we discuss the implications. This step helps the company assess practical considerations that may influence their ability to set and achieve targets. The company’s responses also inform their evaluation of targets (Step 4). Note that, in our example, we assume the company owns and operates T&D equipment only. It is a “wires-only” utility. They do not own generating assets.

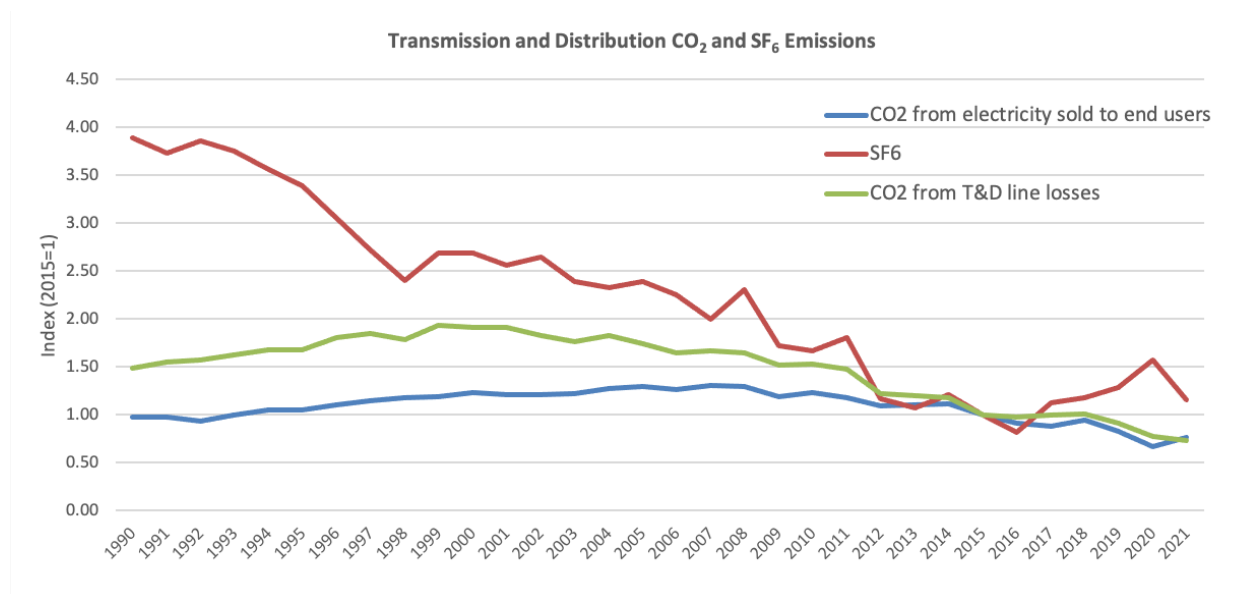


Figure 36. Emissions trends for key emissions categories in T&D illustrative example

Based on the responses in Table 35 above, this company should consider evaluating and pursuing targets for all three categories. The local jurisdiction has Paris Agreement aligned regulations for electricity generation emissions. Thus, the utility should consider whether or not to set targets for CO₂ from electricity sold to end-users since the emissions are already covered and managed to a Paris Agreement aligned level. It is possible that another T&D utility would have different answers. For instance, another company could be in a state with regulation mandating a phase out of SF₆ in gas-insulated equipment. In that circumstance, setting a target for SF₆ could be unnecessary or even complicate/conflict with compliance. The company would want to evaluate its context and decide whether target setting is the best strategy to manage these emissions.

Recall that, for each emissions category, the company decides whether to:

1. Set a target (using the subsequent SMARTTargets steps),
2. Not set a target because the emissions are already covered by a Paris Agreement aligned policy or another company's Paris Agreement aligned targets,
3. Not set a target with justification based on practical considerations that currently preclude target setting, or
4. Not set a target because the category is not relevant to their business.

For the sake of this illustration, we assume the T&D utility decides to pursue targets for all three categories. The targets will have risk since the company does not control the size of the power market or the generation mix. The company has some control over the procurement process and could influence the market by demanding low-carbon electricity. The bottom line is that the utility should be sure to evaluate and communicate the implications of power market uncertainty.

In this step, the company also needs to decide on its target groupings. For this example, we assume the company pursues separate targets for each of the three emissions categories. The company has decided to have separate CO₂ and SF₆ targets due to specific reporting requirements for SF₆, as well as the modeling requirements and options for evaluating abatement of each gas and category. The company's overall target setting and target grouping decisions are documented in Table 36.

Table 36. Practical issues responses for the illustrative T&D utility

Company target setting practical considerations question	Response for CO ₂ from electricity sold to end-users (scope 3 Category 3 Fuel- and energy-related activities not includes in scopes 1 and 2)	Response for CO ₂ from T&D line losses (scope 1, 2, or 3)	Response for fugitive SF ₆ from T&D equipment (scope 1 Direct emissions from fugitive sources)
Is the emissions category adequately estimated and reported?	Yes, emissions reported using location-based and market-based methods	Yes, the emissions are estimated and reported using the location-based method	Yes, emissions are estimated in the inventory since 2000 with improvements over time. Among other things, the company has implemented a monitoring system that has improved the inventory and management ability. However, emissions from the distribution system are estimated using the IPCC's Tier I emissions factors, which are known to be imprecise.
Are the emissions material to the company's overall GHG inventory? (e.g., above 5% of total emissions or a threshold defined by the company)	Yes, the emissions from this source category are greater than 5% of emissions	Yes, the emissions from this source category are greater than 5% of emissions	Yes, the emissions from this source category are greater than 5% of emissions
Are the category's emissions regulated by a Paris Agreement aligned policy?	Yes, the utility's local jurisdiction has Pairs Aligned CO ₂ targets (intermediate and net zero by 2050)	No, the emissions from this source category are not currently regulated	No, the emissions from this source category are not currently regulated
Does another company in the value chain already set Paris Agreement-aligned targets for this source? (applicable only to indirect emission sources)	No, although some utilities in the area are working with the SMARTargets framework to implement targets and design strategies	No, the source does not currently have a Paris Agreement-aligned target	n/a, these are direct emissions
Does the company have control or influence over the emissions?	Limited, the utility can influence the purchase of electricity but it depends on the availability and timeliness of clean energy resources	No, the utility does not control the size of the electricity market, and has only limited control over the energy mix	No, the utility does not control the size of the electricity market

Table 37. Emissions category targets

Table note: Emissions categories are those from Table 22. * Emissions not relevant to business or not material to inventory.

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by <u>future</u> Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type (indicate if target for category, GHG, or CO ₂ e)	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company's Paris Agreement Aligned Targets (MtCO ₂ e)			
Scope 1. Direct Emissions from Stationary Combustion	0.0 (< 0.1)							Not material*
Scope 1. Direct Emissions from Mobile Combustion	0.2							Not material*
Scope 1. Direct Emissions from Process Sources	n/a							n/a
Scope 1. Direct Emissions from Fugitive Sources	3.4	3.4	SF ₆					Fugitive SF ₆ emissions are material
Scope 2. Indirect Emissions from Purchased/Acquired Electricity	7.1	6.9	CO ₂					The target covers CO ₂ emissions from T&D losses which are material, other scope 2 emissions (e.g., electricity purchases for buildings) are not material
Scope 2. Indirect Emissions from Purchased/Acquired Steam	n/a							n/a

Table 37 (continued). Emissions category targets

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by future Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type (indicate if target for category, GHG, or CO ₂ e)	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company's Paris Agreement Aligned Targets (MtCO ₂ e)			
Scope 2. Indirect Emissions from Purchased/Acquired Heating	n/a							n/a
Scope 2. Indirect Emissions from Purchased/Acquired Cooling	n/a							n/a
Scope 3. Fuel- and energy-related activities (not included in scope 1 or 2)	48.4	48.4	CO ₂					CO ₂ emissions from purchased power for resale are material
Scope 3. Use of sold products	n/a							Not material*

Step 3. Aspirational Targets Aligned with International Goals

Step 3 Outputs: Aspirational targets that will be evaluated in the Step 4 transition risk analysis.

For this step, the utility will select generic Aspirational Targets (ATs) aligned with the international goals for their SF₆ and net CO₂ targets. The utility identifies intermediate and 2050 ATs using Table 38, which shows the SF₆ and net CO₂ ATs from Table 6 of the methodology document. The utility must select a 2030 or 2035 intermediate AT, as well as a 2050 AT. They are also welcome to select additional pre-2050 intermediate targets using the table. For this example, we assume the utility chooses to select 2030 and 2040 intermediate ATs and therefore their set of SF₆ ATs are -37%, -41%, and -47% from 2015 for 2030, 2040, and 2050 respectively; and, their set of net CO₂ ATs are -56%, -88%, and net-zero (-100%) from 2015 for 2030, 2040, and 2050 respectively.

Table 38. Generic Aspirational Targets (% reduction from 2015)

	SF ₆	Net CO ₂
2030	-52%	-56%
2035	-52%	-82%
2040	-52%	-88%
2045	-52%	-94%
2050	-52%	-100%

Step 4. Company Transition Risk Analysis of Opportunities and Risks

Step 4 Outputs: Company regional transition opportunities and risks, evaluation of aspirational and qualified targets and their enabling conditions, and transition strategies (low-carbon & risk management).

Given that the utility has decided to evaluate and pursue targets (Step 2), they will need to undertake transition risk analysis to identify their opportunities, risks, and risk management strategies for pursuing the ATs and identifying qualified targets (QTs) and their associated conditions.

The utility works through the seven phases associated with the transition risk analysis. The example analysis below informs all of the company's target types. While the company has to analyze targets for the two different GHGs, they can leverage the same scenarios to evaluate the potential size of the future power market, transmission and distribution needs, and the potential SF₆ and CO₂ emissions. From this information, they can then evaluate abatement opportunities and costs for reducing SF₆ and CO₂.

Phase 1: Define Reference Conditions and Itemize Uncertainties

- *Phase inputs:* Previous company planning, strategy, analysis, and risk management experience
- *Phase outputs:* Scenario analysis reference conditions and list of uncertain factors to consider in scenario design
- *Who should be involved:* Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling
- *Documentation:* SMARTargets template summarizing reference conditions and uncertain transition factors for scenario design consideration

For this phase, the illustrative company works with their planning, policy, sustainability, risk management, strategy, and outreach staff, as well as any external experts on scenario analysis and modeling with which they choose to collaborate to define their scenario analysis reference conditions and uncertain factors that they wish to consider in their scenario design in Phase 2.

The key uncertainty for the T&D utility SF₆ and CO₂ emissions categories is the size and composition of the future regional power market. A power market three times the size of today in 2050 can have three times the level of emissions, depending on changes in the generation mix, and a power market half the size of today in 2050 can have approximately half the level of emissions. Similarly, greater use of low-carbon generation, such as renewables, nuclear, and generation with carbon capture, could lower the CO₂ emissions intensity of the electricity moving through the T&D system. Thus, the company will want to discuss potential reference conditions and identify uncertain factors that can influence the size and composition of the power market.

For reference conditions, our illustrative company considers planned builds and retirements. Like the electricity generation utility example, the T&D utility could also include local and federal policy drivers that would be present in all scenarios, such as state renewables goals or clean energy incentives. They might also consider transmission expansion plans in their jurisdiction, grid hardening strategies, retirements of aging lines, upgrades for T&D infrastructure, generator interconnection needs, and any standards they must meet.

The company staff, and any experts involved will develop a list of uncertain factors that they believe are or could be important and that they want to evaluate in some way. For this example, we assume our company identifies the same uncertain factors that were identified in the electricity generation utility example. They could also group their uncertain factors into sets relevant to generation and T&D infrastructure and operations. In addition, the company identified SF₆ abatement technology costs and performance and within region T&D costs, siting, and permitting as additional uncertainties. The list of uncertainties identified are summarized below:

Uncertainties related to the evolution of the electric system (generation)

- Electricity supply decarbonization options,
- Electricity supply technology cost and performance improvements,
- Future electricity demand (load),
- Low-carbon generation technology development and commercialization,
- Low-carbon generation technology permitting,
- Supply chain constraints related to low-carbon generation technologies,
- Not in my backyard (NIMBY) issues,
- Resource procurement,
- Fuel availability,
- Fuel pipeline capacity/augmentation, and
- End-use technology cost and performance improvements.

Uncertainties related to T&D

- Support for new transmission and distribution infrastructure investments,
- New transmission and distribution infrastructure costs,
- New transmission and distribution siting and permitting, and
- Not in my backyard (NIMBY) issues,
- SF₆ abatement costs and performance, and

Note that the SMARTargets Methodology scenario design for T&D utility emissions already accounts for uncertainty about decarbonization transitions—reference (without decarbonization ambition), power sector only, and economy-wide. Like the electricity generating utility, these are relevant to the T&D utility as significant factors influencing potential power market transitions.

At the end of Phase 1, the utility will document their reference conditions and itemize the identified uncertain factors in the SMARTargets Reporting Template.

Phase 2: Create Core Scenario Design – Define Plausible Extreme Alternative Future Conditions

- *Phase inputs:* Phase 1 scenario analysis reference conditions and list of uncertain factors to consider in scenario design.
- *Phase outputs:* The company's core scenario design for the required SMARTargets scenarios.
- *Who should be involved:* Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling.

- *Documentation after iterations:* SMARTargets Reporting Template entries summarizing the core scenario design.

For this phase, the company continues working with the same team as in Phase 1 to define their core scenario design for the required SMARTargets scenarios. Table 39 presents the required core scenarios for T&D utility emissions from the methodology. For evaluating the ATs for T&D utility emissions, the required scenarios are the net-zero scenarios for electricity generation CO₂ targets and the reference scenarios. The reference scenarios help the T&D utility evaluate the power market and SF₆ and CO₂ targets when electricity or economy-wide decarbonization are not pursued. This type of future is a part of the plausible transition space for a T&D utility and therefore should be evaluated. The company will then want to evaluate a range of target levels for SF₆ and CO₂ for identifying their QTs after they review the results for the ATs and reference conditions.

To implement the Table 39 scenarios, the company needs to define the conditions—Fuller/More Limited decarbonization options and Facilitating/Challenging other conditions. Recall that the conditions should be intentionally more extreme, but plausible. In this example, we use the same condition definitions that were used for evaluating electricity generation targets (Table 40). See the electricity generation target example for discussion of the three intermediate steps associated with grouping and defining the Facilitating and Challenging non-climate-policy conditions.

The outcome of the second phase, combining uncertainty categories into sets of conditions and defining the relative conditions, is what is shown in Table 40. Table 40 is identical to Table 12 that was used for the electricity generation targets example. The set of uncertainties considered in both the Facilitating and Challenging conditions includes uncertainty regarding electricity generation cost improvements, electricity capacity additions, electricity transmission additions, fossil fuel prices, and electric technology cost improvements. Note that, Table 40 does not include fugitive SF₆ abatement technology cost and performance uncertainty. Note, however, that in the subsequent phases analyzing T&D needs, we consider within region T&D cost, siting, and permitting uncertainty.

With the scenario conditions defined, the company is ready for Phase 3 and the quantification of the alternate conditions. At the end of Phase 2, the company should document their definitions of Fuller/More Limited and Facilitating/Challenging using the SMARTargets Reporting Template.

Table 39. Required scenarios for evaluating targets for T&D utility emissions

Target pathway type being evaluated	2050 targets for pathways	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Aspirational Targets (SF ₆ & net CO ₂ ATs and QT > ATs)	SF ₆ 52% & net CO ₂ 100%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging
Qualified Targets (SF ₆ & net CO ₂ QTs < ATs)	SF ₆ A, B, C% & net CO ₂ X, Y, Z%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging

Table 40. Illustrative Fuller/More Limited and Facilitating/Challenging decarbonization condition definitions

Illustrative FULLER decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions) Advanced nuclear (additions) Fossil Fossil w/CCS CDR (biopower w/ CCS & direct air capture (DAC)) Storage Hydrogen Biopower carbon neutrality Other renewable fuels (w/ and w/o CCS) Low-carbon power imports Fossil capacity reserves permitted (within emissions constraint) Demand conservation and energy efficiency	Illustrative MORE LIMITED decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions only) Storage Electrolytic Hydrogen only Other renewable fuels (w/ and w/o CCS) Fossil capacity reserve eligibility declines to zero in 2050 Demand conservation and energy efficiency
Illustrative FACILITATING other conditions Reference electricity load growth Faster electricity generation cost improvements Economic electricity capacity additions Economic electricity transmission additions Lower fossil fuel prices Faster electric technology cost improvements	Illustrative CHALLENGING other conditions Faster electricity load growth Slower electricity generation cost improvements Constrained electricity capacity additions Constrained electricity transmission additions Higher fossil fuel prices Slower electric technology cost improvements

Phase 3: Implement Scenario Design – Specify Assumptions for Scenario Modeling

- *Phase inputs:* The company's Phase 2 core scenario design for the required SMARTargets scenarios.
- *Phase outputs:* Specification of core scenario design for the company's modeling framework.
- *Who should be involved:* Staff related to planning, sustainability, policy, and strategy, as well as external experts on scenario analysis, modeling, and alternative assumptions.
- *Stakeholder engagement:* This is a good time to seek external stakeholder feedback.
- *Documentation after iterations:* SMARTargets Reporting Template entries summarizing the core scenario design specification.

For this phase, the company continues working with the same team as in Phases 1 and 2 to specify the core scenario design for their modeling framework. Phase 2 provides the company with everything they need conceptually and qualitatively in terms of the scenarios they want to run. Running the scenarios, however, requires quantification of the scenario design. This includes specifying decarbonization technology cost and performance assumptions, as well as what exactly is meant by “faster” and “slower” cost improvements, and “constrained” capacity and transmission additions. Since we are using the same condition definitions as used for the electricity generation target example, we also use the same specifications (Table 41a and Table 41b are the same as Table 17a and Table 17b).

Table 41a and Table 41b present an illustrative specification for each of the assumptions associated with the Fuller/More Limited and Facilitating/Challenging conditions. Remember that the specification definitions are intentionally extreme, but plausible, alternative conditions. They are also grounded with justification provided for the specifications.

The utility will also need to define the emissions constraint pathways they will be imposing on the model. These constraints will constrain the model to producing no more than the emissions reductions specified (though the model could produce emissions below the constraints).

After completing Phase 3 is a good time for external stakeholder feedback on the scenario design and specifications. It is an opportunity to discuss setting SMARTargets and the planned transition risk analysis and preliminary future conditions the company is considering.

At the end of Phase 3, the company should document their specifications for their Fuller and More Limited decarbonization options, their Facilitating and Challenging conditions, and their emissions constraints in the SMARTargets Reporting Template.

Table 41. Illustrative summary of energy technology assumptions

Decarbonization technology cost and performance	Comment/description	Reference
Renewables (wind, solar, bio, hydro), nuclear, advanced nuclear, coal, coal w/ CCS, gas, gas w/ CCS, CDR (biopower w/ CCS, DAC), storage, hydrogen	Engineering-based cost and performance specifications with gradual improvement over time	EPRI (2022b)
Biomass carbon neutrality	Carbon neutral assumption based on analysis of net land use and GHG changes associated with increasing biomass supplies	Blanford et al (2022)
Other renewable fuels	Renewable liquid and gas fuels, including renewable gasoline, diesel, and jet fuel, renewable natural gas, and synthetic natural gas produce with renewable energy	Blanford et al (2022)
Low-carbon power imports	Low-carbon electricity can be exported to other regions via intra-regional transmission	EPRI (2023)
Fossil capacity reserves permitted (within emissions constraint)	Fossil generation capacity can be used to meet capacity reserve requirements but declining emissions constraints over time apply to any generation from that capacity	EPRI (2022b)
Demand conservation and energy efficiency	End-use sectors can reduce emissions through technology and fuel substitution that increases efficiency, as well as reductions in overall use	Blanford et al (2022)

Table 42. Illustrative specifications for Facilitating and Challenging other condition assumptions

Other conditions	Facilitating	Challenging
Electricity load growth	Reference = load growth due to reference economic and population growth assumptions and electrification due to electric technology relative cost improvements	Faster = reference load growth conditions plus additional data center load growth based on recent trends resulting in 12% greater total load by 2030 (EPRI, 2024)
Electricity generation cost improvements	Faster = default cost projections and reference learning curve for battery cost improvements over time	Slower = higher cost projections by technology and slower learning curve for battery cost improvements over time
Electricity capacity additions	Economic = allows for all economically viable electric capacity to be built	Constrained = limits the building of economically viable electric capacity to historical build rates
Electricity transmission additions	Economic = allows for all economically viable interregional transmission to be built	Constrained = limits the building of economically viable interregional transmission based on historical transmission build rates
Fossil fuel prices*	Lower = No additional supply costs for fossil fuel production beyond the equilibrium cost determined by the energy economy model's balance of supply and demand for each fuel	Higher = Additional supply costs added to fossil fuel production modeling low resource availability or pipeline expansion
Electric technology cost improvements	Faster = Default cost reductions in EV & heat pump components, faster EV adoption reflective of less range anxiety and greater availability of new electric technologies	Slower = Higher component costs for EV & heat pumps, slower EV adoption representative of more range anxiety and less availability of new electric technologies

* The fossil fuel price assumptions in the table reflect the US-REGEN modeling structure. Other economy-wide frameworks might have fixed assumed (i.e., exogenous) energy prices. For instance, they might be based on lower/higher U.S. Department of Energy Annual Energy Outlook fuel price projections.

Phase 4: Identify Potential Transitions and Target Implications

- **Phase inputs:** Phase 3 specification of core scenario design for implementation in the company's modeling framework.
- **Phase outputs:** Preliminary transition analysis pathways and insights.
- **Who should be involved:** Staff and external experts running the model and processing outputs.
- **Documentation:** *None.* This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.

For this phase, the company's staff and external experts running the model and processing outputs will work together to develop preliminary transition analysis pathway results and insights. The team will be running the model with the scenario specification implementation

developed in phase 3, and they will have to debug and likely revise the scenario specification implementation as they attempt to run each of the scenarios in the design. They will also want to review the results for non-intuitive, or otherwise problematic, outcomes that could be an artifact of an incorrect scenario specification or implementation. The team would want to revise and rerun scenarios in these cases as well.

In addition to running the scenarios, the team will develop preliminary graphics—figures and tables—and identify preliminary insights for review. In particular, the team will want to evaluate differences and similarities in transition results between scenarios. The differences between the scenarios are the most meaningful insights, not the specific values from individual scenarios, as you will learn something about the direction and order of magnitude of implications for the different conditions. For instance, T&D utilities would be interested in the alternative potential transitions for the electricity market size and generation composition, transmission and distribution needs, emissions, and abatement costs for each emissions category. The modeling provides information regarding the alternative potential transitions, trade-offs, uncertainty, risks, enabling conditions, robust strategies, contingencies, and more. Below we present sample results for the company’s scenario design. For this example, we are evaluating potential market transitions for a company in the midwestern U.S. These results represent potential alternative market and operational conditions for the company’s system and assets.

The overall approach is, for each emissions category, to first estimate the future T&D needs under the different scenario conditions and then the associated emissions, after which the company evaluates target setting. For fugitive SF₆ emissions, we evaluate targets by applying marginal abatement cost curves to estimate costs for a range of emissions reduction levels for a given power market. For both CO₂ emissions categories, purchased energy and transmission and T&D line losses, we evaluate targets using the scenario modeling emissions results from which costs can be estimated using the resulting CO₂ prices.

Future T&D needs

The utility begins by assessing the T&D transition needs for the different futures. The T&D utility is primarily interested in the implications of decarbonization on current T&D assets and the need for new infrastructure. The utility can assess needs based on the projected electricity generation transitions. Figure 37 shows the electricity generation size and mix across the required scenarios. The figure is the same as Figure 22 from the electricity generation utility example. In 2035 and 2050, there is significant variation in the generation from variable renewables (wind and solar) across the scenarios, with variable renewable resources potentially as high as 70% and 90% of annual generation respectively. The results also show that, by 2035, load growth could be as high as 25%, and by 2050, as high as 80%. The results indicate that, as early as 2030, grid enhancements are required to accommodate growing load and largescale deployment of renewables. The T&D utility, therefore, would want to consider how they can

manage growth in load and renewables while managing the risk associated with the uncertainty evident in Figure 37.¹⁷

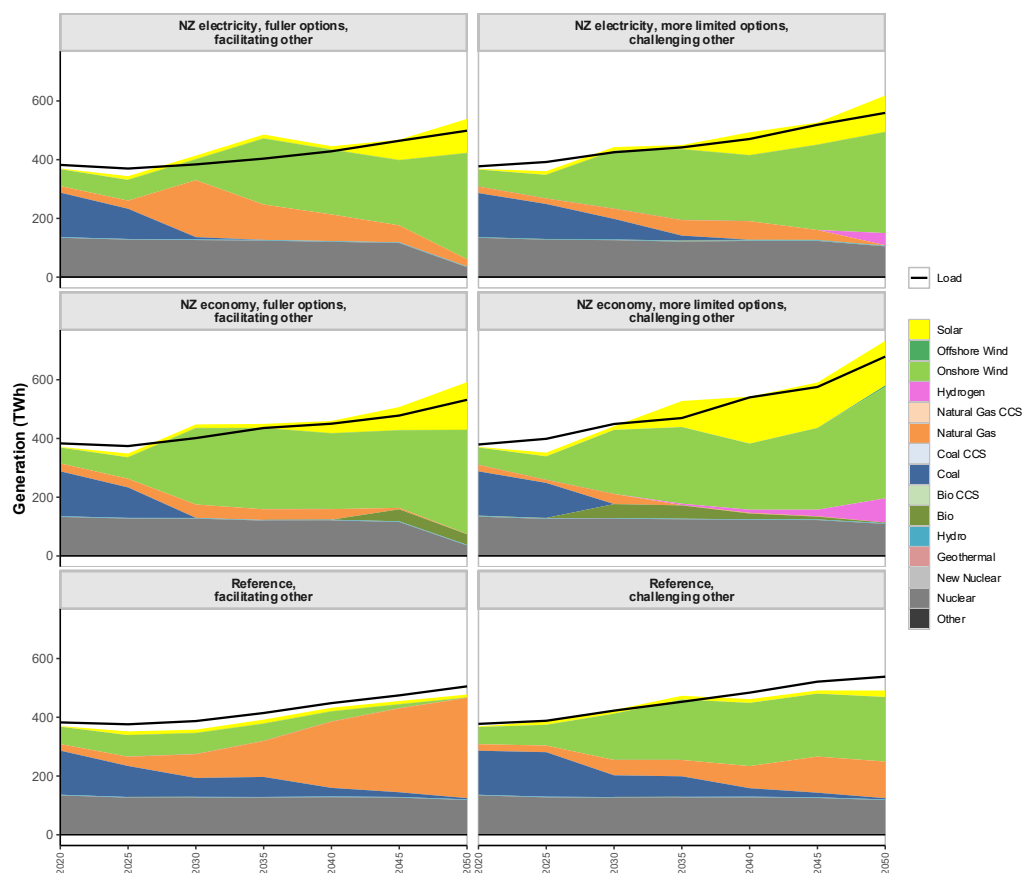


Figure 37. Regional electricity generation and load transitions

Source: EPRI analysis

Figure note: See Figure 22 note.

Future potential inter-regional power trade is also important to understand given the use of transmission assets. Figure 38 shows the power imports into and exports from the region. As indicated in the figure, electricity trade varies across scenarios and over time. Electricity imports help serve load, while exports provide additional revenues to generation resources. As shown in the figure, power trade volumes increase notably in the decarbonization transitions versus Reference futures. We also see that in the decarbonization transition futures, imported power is, in most cases, larger than exported power, suggesting that the region is leaning harder on imports than exports when decarbonizing.

In addition, within region T&D needs also need to be evaluated. Grid enhancements and expansions are needed to support the growth in renewables and load, as well as to alleviate congestion and improve the reliability and resilience of the existing system. In this example, the

¹⁷ Recall that these scenarios evaluate potential load increases due to growing data center demand.

US Midwest utility can utilize the scenario analysis results as well as other transmission expansion information to assess their within region T&D needs.¹⁸ Most transmission expansion plans do not currently project out to 2050 and/or do not consider economy-wide and cross-region deep decarbonization. Thus, the SMARTargets transition analysis provides the utility with a bigger and broader strategic picture of potential transitions and T&D needs.

Figure 39 shows the intra-regional transmission needs assessed by a multi-model scenario study that considered possible decarbonization futures (DOE, 2023).¹⁹ The variable renewable penetration and load growth results from our transition analysis are similar to the levels considered in the DOE (2013) Moderate/Moderate scenario.²⁰ According to the DOE study, the Midwest region had 11.92 Terawatt-miles (TW-mil) of transmission capacity in 2020. By 2035, the study projects an increase in within-region transmission needs of 19% to 174% depending on whether there is moderate load growth and moderate renewable penetration or high load growth and high renewable penetration. By 2040, the study projects that transmission needs could increase by 29% to 196%.

Rough estimates of the investment requirements for this kind of transmission capacity expansion ranges from \$2.3 to \$6.8 billion in 2035 and from \$3.4 to \$10.6 billion in 2050 if using high voltage lines (HV), and from 4.5 to 11.3 billion in 2035 and 6.8 to 17.0 billion in 2050 if done with ultra-high voltage lines (UHV). The latter have several benefits specially for long-range transmission that could help connect renewable resources to distant load centers, as well as help reduce transmission losses.²¹

¹⁸ For example, [MISO Transmission Expansion Plan](#).

¹⁹ The U.S. Department of Energy (DOE) is required to update this analysis of transmission needs every three years and could serve as a basis for estimating grid expansion needs.

²⁰ The moderate/moderate scenarios consider a moderate penetration of renewable energy of less than 80% and moderate load growth less than 71% from 2021 levels (3974 TWh in 2021 to 7000 TWh in 2040). The DOE study also provides transmission needs estimates for higher renewables penetration and load growth.

²¹ High-voltage transmission lines (230 and above) typically ranges from \$1 million to \$3 million per MW-mile, while ultra-high voltage lines (500 kV and above) ranges from \$2 million to \$5 million per mile (De Santis et al 2021).

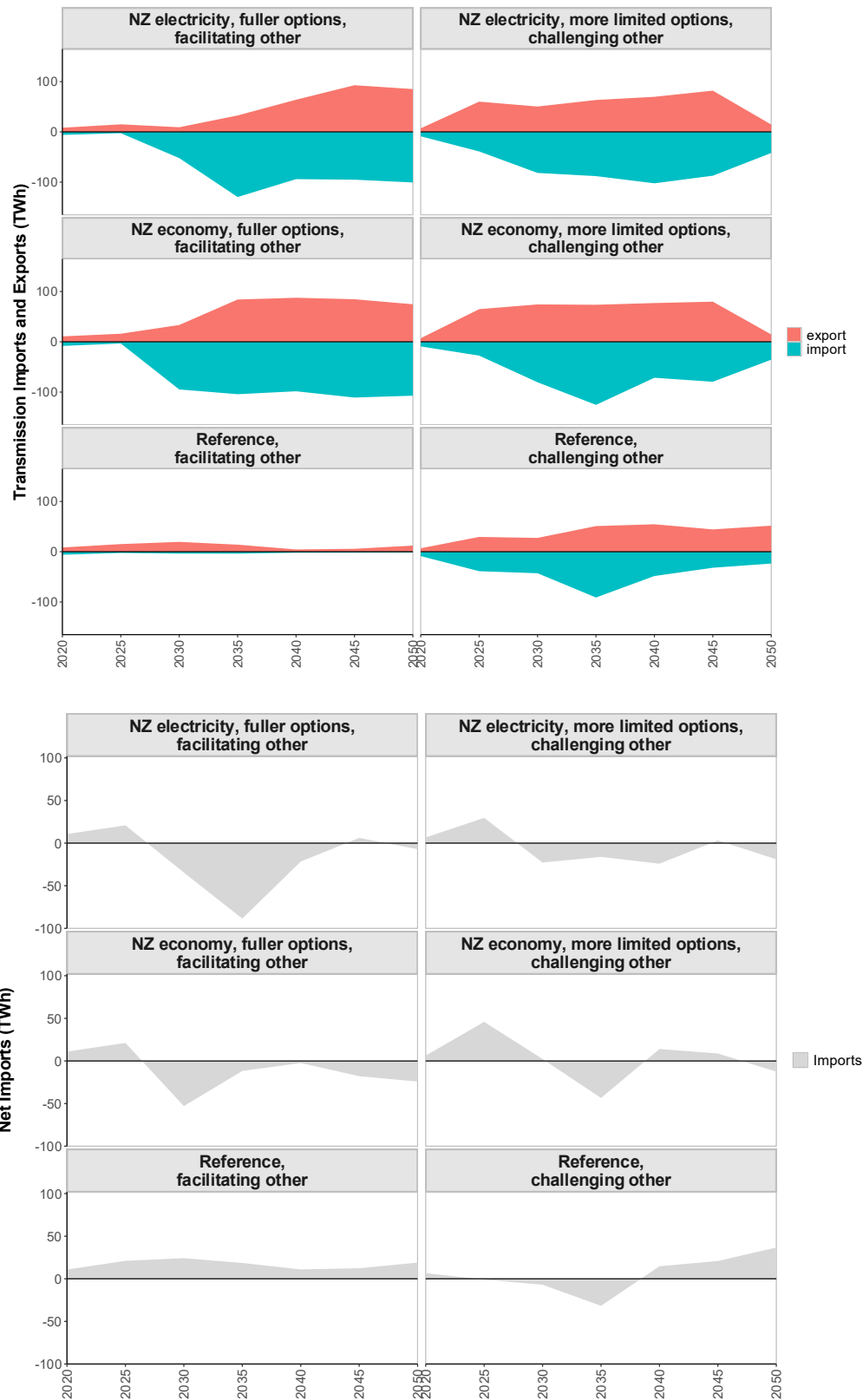


Figure 38. Potential inter-regional electricity imports and exports (top) and net imports (bottom) in the required scenarios

In our example, the utility should estimate the within region infrastructure expansion needs to support the projected transitions. Expansions of at least 20% by 2035 and 30% by 2050 are supported by the DOE (2023) study in combination with the transition scenario modeling results for renewables and load. Investment in HV versus UHV lines could be determined by further exploring the long-range transmission needs and the reduced transmission loss benefits. Even larger expansions of T&D infrastructure are suggested by the extreme DOE (2023) scenarios with 80% load growth and 94% renewables integration. Transmission infrastructure expansion in this case could be as much as 57% by 2035 and 115% by 2050.

Enhancements of distribution infrastructure should also be evaluated, particularly in scenarios where electrification of energy end-use increases and when distributed energy resources (DER) are a larger share of the energy mix. Investments in smart grid technology that enables bi-directional flows of energy and demand response would be important for an efficient grid and to help integrate DER. US-REGEN does not model the details of the distribution grid; however, insights for assessing distribution needs can be derived from the electrification results for transport, buildings and industry. Figure 40 shows potential electricity consumption by sector and transition scenario. Electrification consumption increases in all scenarios particularly for transportation but varies across the potential futures, with the economy-wide decarbonization futures resulting in the largest increases in electricity consumption. Detailed studies of the need for distribution infrastructure could be conducted based on this information, with DER considered as an estimated fraction of solar resources.

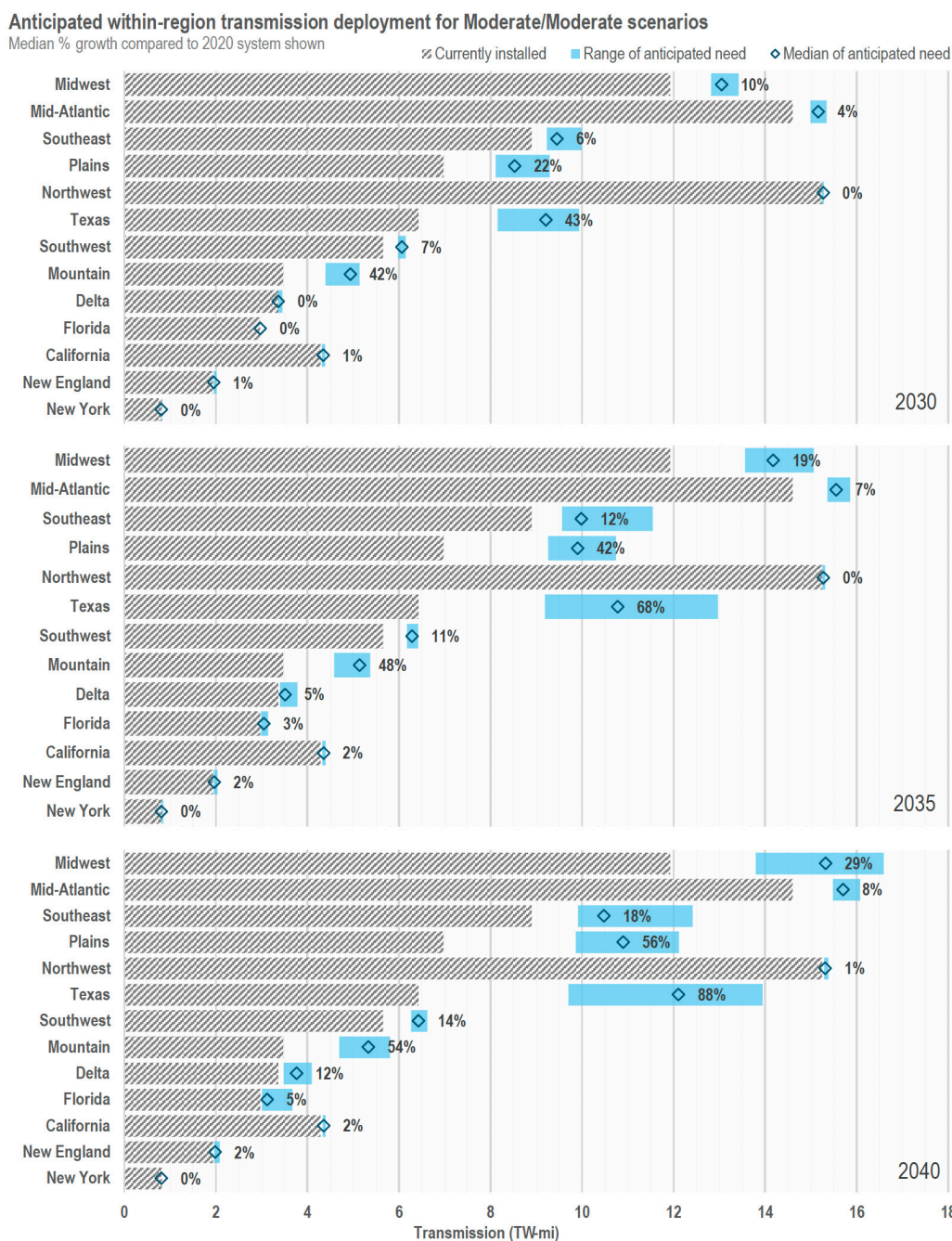


Figure 39. U.S. within-region transmission installed and needed in DOE (2013) decarbonization futures

Source: DOE (2023)

Figure note:

- New transmission relative to 2020 system is shown for all regions for 2030 (top), 2035 (middle), and 2040 (bottom). Median and interquartile range (IQR) of new transmission results shown. Currently installed transmission shown from Denholm et al (2022a).
- Moderate/Moderate scenario is defined as moderate load growth from the 2021 baseline (3,974 TWh) to 7,000 TWh in 2040 and moderate clean energy penetration from the 2021 baseline (38.6%) to 80% in 2040. 2021 values from EIA (2022a).

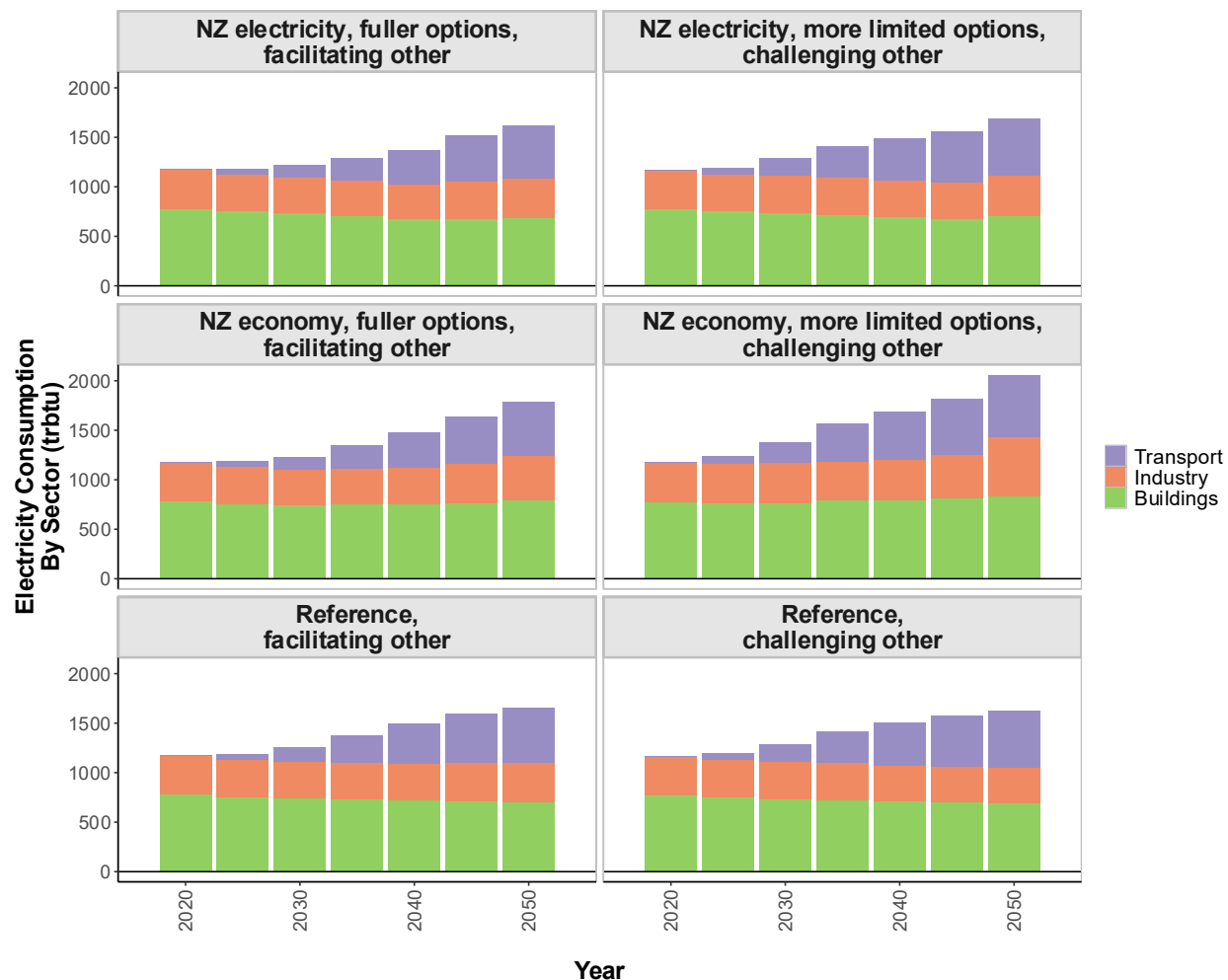


Figure 40. End use electricity consumption by sector for the required scenarios

Future emissions without targets

Once the utility team has a good understanding of potential T&D needs for each scenario, they can translate those to future emissions. The power to serve load CO₂ emissions pathways emerge directly from modeling the power sector transition and emissions with and without decarbonization,²² but it is necessary to do additional calculations to compute the transmission loss CO₂ and fugitive SF₆ emissions. For the former, we use state data on average grid power losses from the states in the region (IEA, 2024), which is an average of 4% for this Midwest region. For the latter, we use US-REGEN data of expected load growth to project T&D requirements and SF₆ use.

Figure 41 presents the 2030 and 2050 CO₂ and SF₆ emissions in percentage change terms (relative to 2015). CO₂ emissions decrease as the grid decarbonizes. While transmission losses

²² We assume the utility is a 'wires-only' company that does not own generation assets, and that all electricity consumed in the service area is delivered using the company's infrastructure.

cannot be fully eliminated, losses will not translate into emissions if the power fed into the grid is clean. If generators decarbonize to net zero, the T&D utility has no additional action since the net-zero CO₂ targets would already be reached. However, if the generators do not fully decarbonize, the T&D company would have to pursue additional reductions with the generators and by reducing line losses. In the meantime, all the scenarios project increasing SF₆ emissions. This result is due to load growth and the expansion of renewable energy, both of which require more T&D infrastructure regardless of the future. SF₆ emissions are projected to increase significantly across all scenarios, with the most pronounced growth occurring in the 'Net Zero Economy-wide, More Limited Options, Challenging Other' scenario. This scenario has the highest demand for electricity due to the demand for low-carbon electricity created by the economy-wide emissions constraint, as well as data center load growth and the More Limited Options resulting in greater dependence on electricity in decarbonizing the whole economy. The scenario also has the largest growth rate in renewable generation capacity greatest of all the futures. These factors together translate into greater T&D requirements and more SF₆ emissions than the other scenarios.

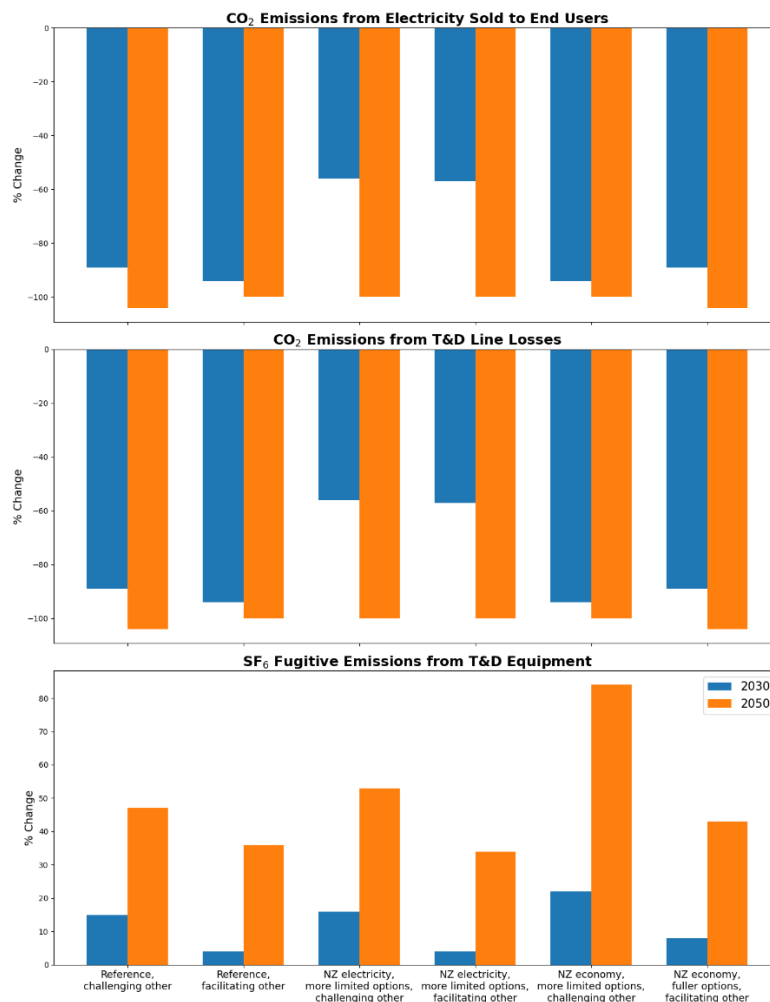


Figure 41. Percent reductions in regional emissions in 2030 and 2050 from power to serve load, T&D losses, and Fugitive SF₆

Target evaluation

With the emissions projections by scenario, the company can now evaluate target setting. For fugitive SF₆ emissions, the company can evaluate targets by applying marginal abatement cost curves to estimate costs for a range of emissions reduction levels. For CO₂ from energy to serve load and T&D losses, the company can evaluate targets using the scenario modeling results for emissions from which costs can be estimated using the implied CO₂ prices.²³

There are a variety of abatement measures that can be applied to fugitive SF₆ emissions from T&D systems (EPA, 2019, Table BT8). For reducing SF₆ emissions the utility can target four streams of emissions:

- Handling: Emissions can result from improper handling of SF₆ during the maintenance of gas-insulated circuit breakers. Abating these emissions can be low cost and implies training technicians to properly handle the gas and purchasing adapter kits to ensure proper fittings are available for connecting hoses to all gas-insulated equipment throughout the system.
- Venting: Emissions result from venting during equipment maintenance and disposal. These emissions can be abated by increasing SF₆ recycling and reuse. It requires investment in SF₆ reclamation and equipment for purifying for reuse.
- Periodic leakage: Emissions result from periodic leakage from equipment. These emissions can be abated by leak detection or repair options (thermal imaging camera and hand-held gas detectors).
- Chronic leakage: Emissions result from chronic leakage from equipment. These emissions can be abated by equipment replacement or use of SF₆ free gas insulated equipment.

²³ In 2050, the implied CO₂ prices range from \$130 to \$1,330/tCO₂. The lower price is associated with Fuller-Facilitating conditions, while the higher price is associated with More Limited-Challenging conditions.

Table 43. SF₆ abatement options and example costs

Source: EPA (2019)

Table note: The first row of each option refers to an uncontrolled system (e.g. in developing countries), and the second row refers to a controlled system (e.g. Europe and North America).

Abatement Option	Project Lifetime (years)	Capital Cost (2010 USD)	Annual Revenue (2010 USD)	Annual O&M Costs (2010 USD)	Abatement Amount (tCO ₂ e)
SF ₆ recycling	15	\$520,156	\$101,919	\$7,592	44,678
		\$78,023	\$1,716	\$20,097	2,048
LDAR (Thermal Imaging Camera)	5	\$20,700	\$1,887	\$151	827
		\$49,615	\$238	\$3,468	284
LDAR (Handheld Leak Detector)	5	\$1,387	\$7,550	\$1,557	3,309
		\$693	\$238	\$7,458	284
Equipment Replacement	40	\$2,886,599	\$17,930	-	7,860
		\$2,844,205	\$1,811	-	2,161
SF ₆ -free Gas Insulated Equipment	50	\$4,328,671	\$17,930	-	7,860
		\$4,373,246	\$1,811	-	2,161
Improved SF ₆ handling	1	\$4,369,910	\$1,029	-	1,228
		\$14,671	\$135,893	\$358	59,570

Estimates of abatement costs for the set of measures can be used to evaluate emissions reduction opportunities; however, the company will need to consider which technologies apply given their system. For illustrative purposes, we use a representative curve (Figure 42) of marginal abatement costs for reducing SF₆ emissions in the U.S., in 2030 and 2050.

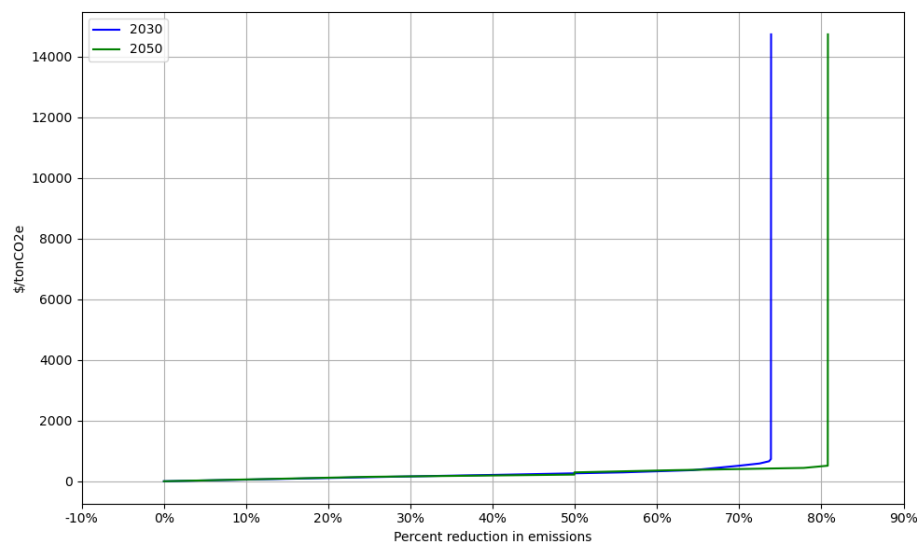


Figure 42. Example marginal abatement cost curves for SF₆ emissions in the U.S. in 2030 and 2050

Source: Harmsen et al 2019.

Figure note: Data for 2030 and 2050 plotted by EPRI and converted to \$/tCO₂e. The original dataset provided by Harmsen et al. (2019) covers 2000 to 2100 in 10-year increments for several countries.

The marginal abatement cost estimates are applied to the alternative SF₆ emissions transition results (Figure 41) to assess the costs of achieving the ATs (Figure 43). Figure 43 presents the net present value abatement costs for meeting the ATs under each scenario for all three emissions categories. These costs reflect the incremental abatement required given the emissions associated with each future (Figure 41).

Because of the increasing SF₆ emissions in all the scenarios, the utility will have to abate SF₆ in each case and incur costs. Abating the CO₂ categories, on the other hand, is a different story since CO₂ reductions occur in most of the scenarios, thereby reducing, if not eliminating, the utility's abatement requirements for meeting the CO₂ ATs. The CO₂ abatement costs in Figure 43 are derived using the implied CO₂ prices needed to achieve decarbonization in the various futures. As shown in Figure 41, only the Reference scenarios require additional abatement. This suggests that the CO₂ targets are achievable if electricity generation decarbonizes at the implied pace. However, without power sector or economy-wide decarbonization, the T&D utility will have to implement abatement measures for the ATs to be met.

Uncertainty in the electricity market (Figure 43) leads to considerable variation in the cost of meeting emissions targets for T&D utilities (Figure 43). For CO₂, the net present value (NPV) cost of achieving net-zero emissions by 2050 from electricity sold to end users ranges from \$0 to \$11,661 million. Similarly, the cost of eliminating emissions from T&D line losses ranges from \$0 to \$466 million. These differences stem from the large variation in CO₂ emissions in the different futures, which the decarbonization transitions reducing emissions to AT levels or beyond and the reference transitions resulting in significant emissions that need to be abated. The Reference-Challenging conditions are the most expensive due to the electricity load growth from data centers and slower improvements in generation costs.

In the case of SF₆, the NPV costs of achieving a 52% reduction by 2050 range from \$562 million to \$5,886 million. Lower-cost outcomes are associated with scenarios featuring smaller electricity markets, such as Reference–Facilitating and Net-Zero Electricity–More Limited–Facilitating. In both cases, the Facilitating slower load without data center growth is a key factor. In contrast, the highest costs are observed in scenarios with larger electricity markets and greater infrastructure needs, such as the Economy-Wide Net Zero–More Limited–Challenging scenario. In these futures, rising electricity demand for end use decarbonization drives T&D infrastructure expansion and increased SF₆ use. Without viable SF₆-free alternatives, emissions from this category are likely to grow—even if handling and monitoring practices improve.

These findings underscore the importance of enabling conditions for SF₆ abatement. Abatement technological improvements or new abatement options can help manage the risk of higher-cost outcomes. While reducing transmission expansion also lowers abatement costs, it would also affect broader economy-wide decarbonization since T&D infrastructure is essential for integrating low-carbon resources into the grid. Facilitating conditions, such as innovation and reduced infrastructure additions barriers, can help reduce costs for both CO₂ and SF₆: supporting a more affordable transition to lower emissions.

As shown in Figure 44, some futures are challenging for achieving the CO₂ and SF₆ ATs. As a result, the company will want to assess the level of emissions abatement achievable under the less favorable conditions. Figure 44 presents incremental net present value abatement costs for each emissions category. For SF₆, reduction pathways leading to 10%, 20%, and 30% reductions in 2050 are evaluated, along with the 52% in 2050 AT. For CO₂, reduction pathways leading to 30%, 50%, and 80% reductions in 2050 are evaluated, along with the 100% in 2050 AT. For all three emissions categories, costs increase with the level of reduction. However, the magnitude of the costs and rate of increase with reduction ambition vary depending on the emissions category and the conditions.

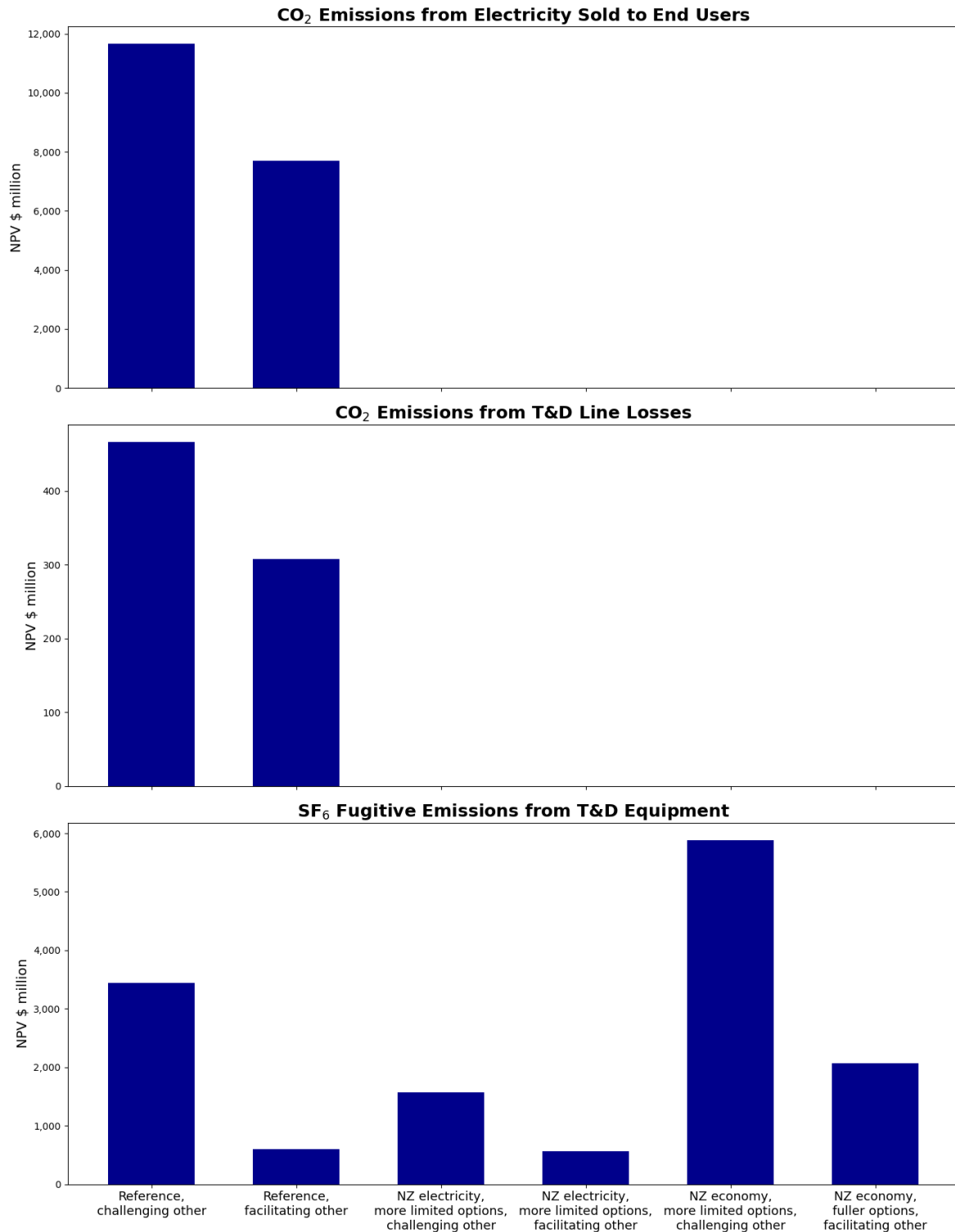


Figure 43. Net present value abatement costs for reaching the CO₂ and SF₆ Aspirational Targets

Figure note: Y-axis' units vary by chart.

The company can use the analysis of incremental costs (Figure 44), to identify their QTs and associated conditions. The utility can identify a cost threshold for reductions based on their regulatory environment and use that information to identify QTs and conditions. For example, for SF₆, a \$2 billion threshold would imply that 52% reductions would be challenging in reference or economy-wide decarbonization futures with Challenging other conditions. If these conditions emerge, Figure 44 identifies QTs of approximately 45% and 30% in 2050 respectively. Since the results in Figure 44 are for the region, the company would scale them for their system, utilizing the implied growth rates and/or their market share, to help them consider their company-specific cost thresholds. Not knowing which future will emerge, the utility could specify both as sets of QTs with pre-2050 and 2050 targets. The figure also highlights that the Facilitating conditions are enabling conditions, lowering the cost of SF₆ abatement in every context versus its Challenging counterpart.

Similar insights can be derived for the CO₂ categories. For instance, if there is \$2 billion threshold here as well, the figure highlights that a net-zero target in 2050 for CO₂ from power to serve load is likely very challenging in both reference futures. However, QTs of approximately 65% and 85% 2050 targets could be pursued in the Challenging and Facilitating reference futures respectively. The utility could use this information to consider company-specific cost thresholds and potentially specify two set of QTs for the CO₂ categories or, given that the lower QT is associated with Challenging conditions, they might simply choose 65% in 2050 and note that Facilitating conditions would enable them to pursue greater reductions.

The regional costs for abating CO₂ from T&D line losses are modest in comparison to the other categories. Nonetheless, the utility can utilize the figure to consider QTs for the reference futures. In this case, we assume that the regional costs of \$450 million for near 100% reductions by 2050 is not prohibitive, and the utility does not need QTs for this category. This of course recognizes that line losses cannot be eliminated entirely.

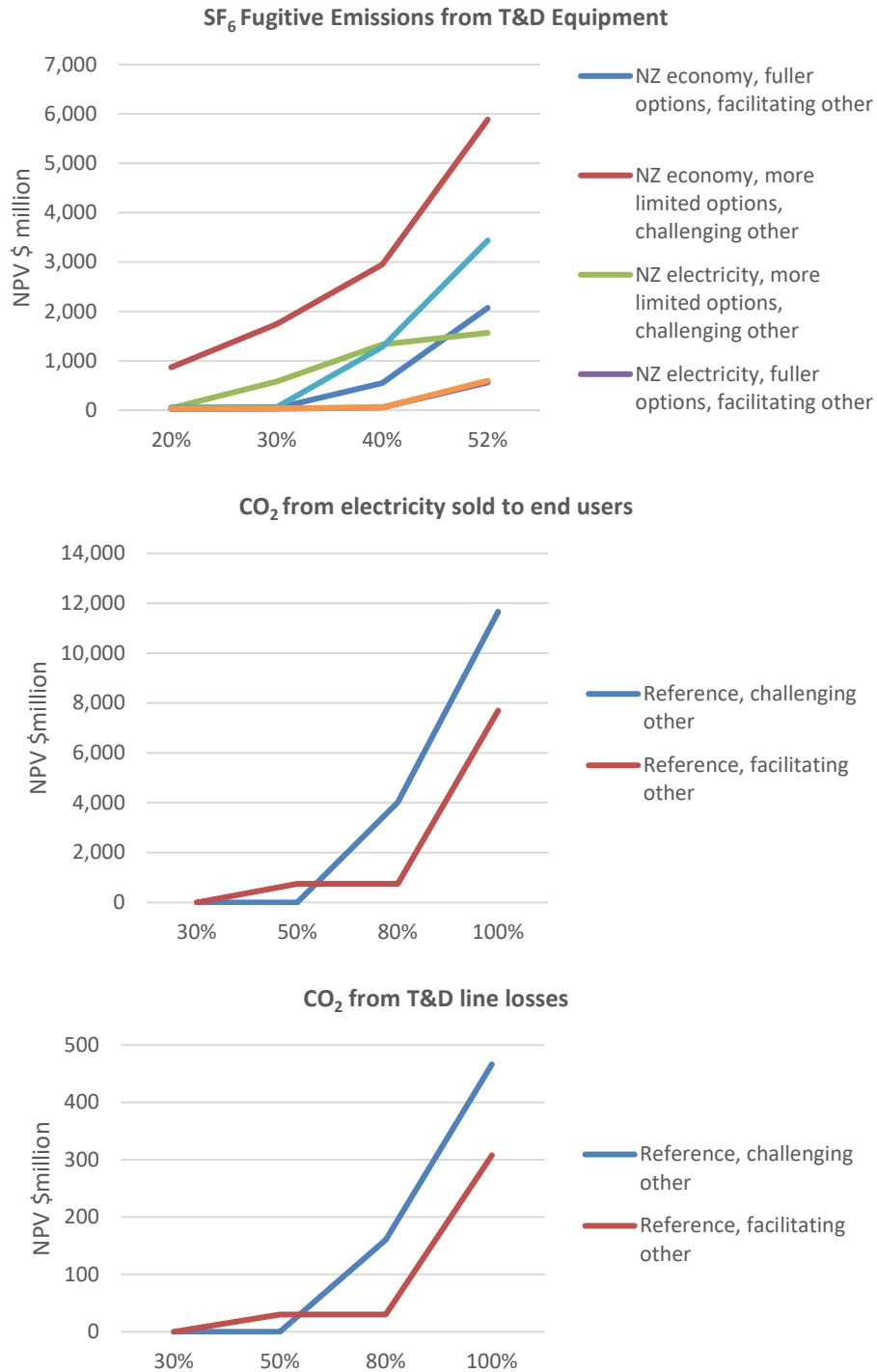


Figure 44. Incremental net present value costs for increasing emission reduction targets in 2050 for T&D fugitive SF₆ emissions (top row), CO₂ from power to serve load (middle row) and CO₂ from T&D losses (bottom)

Figure note: Y-axis' units vary by chart. In the SF₆ chart, the NZ electricity-Fuller-Facilitating cost curve coincides with the Reference-Facilitating curve.

Phase 5: Review Potential Transitions and Target Implications – Based on Decision-Making Needs

- *Phase inputs: Phase 4 preliminary transition analysis pathways and insights.*
- *Phase outputs:*
 1. *Preliminary insights regarding potential transitions, targets, and risks,*
 2. *Revisions to core scenario design, model implementation, and output reporting, and*
 3. *Additional recommended and sensitivity scenarios and specifications the company would like to evaluate.*
- *Who should be involved: Company decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.*

For this phase, the company's planning, policy, sustainability, risk management, strategy, and outreach, staff, as well as external experts, will want to engage the company's decision-makers—directors and executives. The main objective here is to critically evaluate and discuss the preliminary results and insights and identify those they find reliable and meaningful. The additional objectives here are to identify any revisions to core scenario design, model implementation, and output reporting they might want to make, and to identify sensitivity scenarios and specifications that the company would like to evaluate. The last two items emerge from the company's critique and discussion of the draft results and insights.

Given the need to identify enabling conditions, this is the time for the company to think concretely about trade-offs, risks, and risk management from their perspective. The Phase 4 discussion above highlighted related insights from the perspective of our example company and analysis, noting specific risks and enabling conditions that identify and justify the company's QTs. Using the results from Phase 4, the company identifies trade-offs, risks, and risk management strategies. For instance:

- *Risks identified:* The primary risk is the power market (size and generation resources), which is not only a risk in terms of the emissions abatement challenge, but also an overall investment, revenue, and operations risk for a T&D utility. Related to power market risk, there is abatement cost and target attainability risk, where we find that there is the possibility of significantly higher costs associated with achieving the ATs in some plausible future conditions, especially for 52% reductions in fugitive SF₆ emissions by 2050 and 100% reductions in CO₂ from power to serve load.
- *Risk sources:* All categories of uncertainties create risk for the company. Thus, uncertainty regarding the decarbonization incentive (none versus sector versus economy-wide), decarbonization options (Fuller versus More Limited), and Other conditions (Facilitating versus Challenging) are all important considerations contributing to uncertain transitions with some more difficult to manage trade-offs in load growth, traded power, and emissions from purchased power to serve load. Uncertainty about the decarbonization transition

alone creates risk for the CO₂ targets because the company does not know if they will have no or significant abatement costs. Similarly, for SF₆, while all the futures imply abatement costs, the costs for achieving the ATs ranges from \$500 million to \$6 billion. And, Challenging conditions make the cost of abatement higher for all the emissions categories. It is important to recognize that revenue recovery for T&D infrastructure is essential, and tariff schemes should reflect the critical role of T&D in supporting decarbonization efforts.

- *Risk management strategies:* Looking across the scenarios, we can identify enabling conditions and contingencies. We find that economy-wide decarbonization incentives, Fuller decarbonization portfolios, and Facilitating Other conditions result in greater and faster decarbonization of power generation and therefore help manage the risk for CO₂ targets for T&D line-loss and electricity sold to serve load. R&D lowering the costs of line efficiency improvement measures are also practical strategies for helping lower line-loss CO₂ abatement costs, while R&D lowering SF₆ abatement technology costs will help manage SF₆ abatement costs. Pursuing Facilitating Other conditions is also found to be a risk management option for SF₆ targets. As for contingencies, if the company finds themselves in an economy-wide decarbonization future with More Limited and Challenging conditions, they may have to reconsider a 52% SF₆ target, unless lower cost SF₆ abatement technologies have emerged or offsets are available. For CO₂ targets related to line losses and sold power, the company depends on external electricity generators to achieve compliance. In the Reference futures—where no emissions targets are imposed on generators—setting lower reduction targets may be all that is possible given the cost challenge, especially under Challenging conditions. For all three emissions categories, pursuing Facilitating conditions, such as faster technology cost improvements and reduced barriers to new additions, will support greater emissions reductions and help manage abatement costs.

This phase is also the time when the company would want to consider the likelihoods of the different futures and potential hedging strategies they might pursue. For example, if the company believes that the likelihood of an economy-wide policy is low and Challenging conditions are more likely than Facilitating, they might want to invest more in R&D for reducing T&D line losses and abating fugitive SF₆ emissions, as well as explore and communicate what emissions reductions are possible if they cannot bring the abatement costs down.

After developing preliminary results and insights, the company would want to identify whether revisions to the scenario design, model implementation, and/or output reporting. In terms of identifying potential revisions to the scenario design and implementation in particular, the group discussion should consider whether they want to tweak anything. This could include revising what is included in each of the Fuller and More Limited decarbonization options sets, making Facilitating/Challenging more extreme to further differentiate them, or simply adjusting a specification such as the annual additions constraints for new transmission.

The company discussion on potential sensitivity scenarios should look for meaningful adjustments to one or two assumptions. For example, based on the results above, this company might want to consider running more sensitivities to load growth and transmission constraints

and/or evaluating potential improved grid enhancement or improvements in SF₆ abatement technology costs.

Like Phase 4, outside of noting that they completed this phase, there is no documentation requirement for the company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete.

Phase 6: Revise and Iterate as Needed – Revising Scenario Design, Implementation, and Model results

- *Phase inputs:*
 1. *Phase 5 revisions to core scenario design, model implementation, and output reporting, and*
 2. *Phase 5 sensitivity scenarios and specifications the company would like to evaluate.*
- *Phase outputs:*
 1. *Revised insights regarding potential transitions, targets, and risks, and*
 2. *As needed, additional revisions to scenario design, model implementation, and transition pathways.*
- *Who should be involved: Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Stakeholder engagement: This is a good time to seek external stakeholder feedback.*
- *Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.*

For this phase, the company's planning, policy, sustainability, risk management, strategy, and outreach, staff, as well as external experts, will work together to revise the insights regarding potential transitions, targets, and risks, and, as needed, implement any needed revisions to the scenario design (core and sensitivity scenarios), model implementation, and transition pathways.

For this phase, our company continues to iterate on the scenario design, specification, results, and insights until they have agreement with their decision-makers.

Phase 6 is also a good time for external stakeholder feedback, especially given the important role of stakeholder consultation in T&D planning. After internal iterations are complete, the company will have final drafts of their scenario design, results, insights, and strategies to share and discuss. This gives the company an opportunity for education and dialogue, as well as an opportunity to solicit feedback on their potential transitions, challenges, uncertainties, risks, enabling conditions, and collaboration opportunities.

For this phase, the company continues to iterate on the scenario design, specification, results, and insights until they have agreement with their decision-makers. Like Phase 5, outside of noting that they completed this phase, there is no documentation requirement for the

company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete.

Phase 7: Communicate Results – Targets, Transitions, Risks, Enabling Conditions, Risk Management

- *Phase inputs:*
 1. *Phase 6 final scenario design (core and additional scenarios), model implementation, and transition pathways, and*
 2. *Phase 6 final insights regarding potential transitions, targets, risks, enabling conditions, and risk management.*
- *Phase outputs:*
 1. *Documented scenario analysis, outputs, and insights,*
 2. *Aspirational Targets (ATs), strategies, and risk management, and*
 3. *Qualified Targets and their qualifying conditions (QTs) and strategies.*
- *Who should be involved: The company's decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Documentation: SMARTargets Reporting Template summarizing the following:*
 1. *The company's ATs and QTs with corresponding conditions,*
 2. *The company's enabling conditions, strategy, risks, and risk management,*
 3. *The company's transition risk analysis (modeling, scenario design and specification, potential transitions, insights), and*
 4. *The implementation process.*

This phase is the critical communications of all the company has done for the Step 4 analysis. For this phase, the company will want to work with their decision-makers and company planning, policy, sustainability, risk management, strategy, and outreach teams. They will also want to engage their external experts to help with documenting the methodology and effectively, rigorously, and defensibly communicating the results. The experts can also help the company communicate the relevant science to educate readers and ground the methods.

At the end of Phase 7, the company will use the SMARTargets Reporting Template to finalize the documentation for their Step 4 transition risk analysis (Phases 1 through 7). This includes the summary tables of key output indicators, as well as graphics to help communicate the company's modeling results and insights. Additionally, the company will report the identified enabling conditions for ATs, QTs and corresponding conditions, and the risks and opportunities identified. Finally, the company will indicate their risk management strategies along with their milestones and contingency plans.

The company's strategy will consist of more than targets. It will also include transition plans as well as strategies for pursuing enabling conditions and managing risks, where the latter might entail research and development plans, collaboration with customers or other sectors, and

policy dialogue. Step 4 has a minimum set of required outputs (Table 20 in the methodology document). Additional variables may be added based on company need.

Step 5. Strategy and Qualified Target Alignment with International Goals

Step 5 Outputs: International goal alignment assessments of qualified targets and strategies.

In this step, the company will evaluate *qualitative* alignment of their transition strategy with the international goals, as well as *quantitative* alignment of their QTs.

Assessing Qualitative Alignment of the Transition Strategy

The company will assess two types of qualitative alignment of their strategy with international goals: (1) alignment with the types of transitions robustly observed in global emissions pathways consistent with the Paris Agreement temperature goals, and (2) alignment with the provisions in the Paris Agreement associated with achieving the temperature goals.

As discussed in the methodology document, the following are robust types of transitions found in all global 1.5°C and 2°C emissions pathways. The following are changes found consistently in well-below 2°C pathways regardless of the model and assumptions:

- Increasing low-carbon technology deployments,
- Increasing reliance on electricity,
- Transitions away from fossil fuels but continued use,
- Use of carbon dioxide removal,
- Improvements in energy conservation and efficiency,
- Differences in regional low-carbon transitions, and
- For any climate goal, substantial uncertainty in the levels, rates of change, and composition (e.g., technologies, energy, activity) of the transition of the global economy.

SF₆ reductions are also a robust characteristic of the global pathways, with reductions in SF₆ producing activity levels and the utilization of SF₆ abatement measures. The transition analysis results can be assessed for consistency with these robust Paris Agreement aligned transition characteristics. As shown in the modeling results, the T&D company strategies will need to rely on both the deployment of low-carbon technologies and increased electrification—particularly under economy-wide net-zero scenarios. The scenarios also indicate a transition away from fossil fuels, though continued use persists in many cases. The set of strategies also includes increasing deployment of low-carbon technologies and energy conservation and efficiency and the use of carbon dioxide removal. They also exhibit significant uncertainty in the transitions and of course, by design, recognize the unique regional transition opportunities.

The assessment, however, is different from that for electricity generation emissions strategies since the utility is needing to consider alternative power markets conditions that are primarily out of their control. The assessment here should therefore focus on the strategy elements the

utility can control and influence. For example, in all cases the company strategy is reducing fugitive SF₆ and CO₂ emissions by utilizing fugitive SF₆ and CO₂ line-loss abatement technologies.

Furthermore, the transitions and insights align with Paris Agreement's provisions recognizing differences in national decarbonization opportunities, multiple societal priorities, transparency, and the value of innovation and cooperation (in this case across sectors via the economy-wide scenarios).

Assessing Alignment of the Qualified Targets

The company will also assess quantitative alignment of their QTs through comparison to the relevant global emissions pathway ranges consistent with the Paris Agreement objective of limiting global average temperature to "well-below" 2°C. Table 44 shows this comparison. Based on the Step 4 transition risk analysis, the utility identified the following fugitive SF₆ and power sold to end use CO₂ sets of QTs:

1. Fugitive SF₆ QTs #1: 45% reductions by 2050 (with 45% in 2030 and 2040),
2. Fugitive SF₆ QTs #2: 30% reductions by 2050 (with 30% in 2030 and 2040), and
3. CO₂ from power sold to end users QTs: 65% reductions by 2050 (with 36% and 57% reductions in 2040 and 2050).

The conditions associated with the first set of fugitive SF₆ QTs include a growing reference regional power market with a decarbonization transition and Challenging other conditions with high load growth, high gas prices, high generation technology costs, and constraints on new generation and transmission additions. The conditions associated with the second set of fugitive SF₆ QTs include a growing regional power market due to an economy-wide decarbonization transition, More Limited decarbonization options including CDR being unavailable, and emissions constraints on capacity reserves, as well as the Challenging other conditions. The conditions associated with the set of power sold to end users CO₂ QTs include a more slowly growing regional power market due to no regional decarbonization transition and the Challenging other conditions. The QTs were identified by primarily considering the transition implications for affordability. Other factors could also be considered, such as new infrastructure constraints, operational reliability, and the likelihood of future conditions.

Table 44. Qualified Target Paris Agreement alignment assessment

Qualified Target set	Key qualifying conditions	GHG	Qualified Target 2050		Well-below 2°C ranges	Within range
Fugitive SF ₆ #1	Growing regional power market due to reference conditions without a decarbonization transition and Challenging other conditions	SF ₆	2030	-45%	13% to -92%	Yes
			2040	-45%	29% to -98%	Yes
			2050	-45%	43% to -98%	Yes
Fugitive SF ₆ #2	Growing regional power market due to an economy-wide decarbonization transition, More Limited decarbonization options including CDR being unavailable and emissions constraints on capacity reserves, as well as the Challenging other conditions	SF ₆	2030	-30%	13% to -92%	Yes
			2040	-30%	29% to -98%	Yes
			2050	-30%	43% to -98%	Yes
Sold power to end users CO ₂	Flat or growing regional gas market due to no regional decarbonization transition in power sector or economy-wide and non-CDR offsets unavailable	Net CO ₂	2030	-36%	10% to -100%	Yes
			2040	-57%	-24% to -121%	Yes
			2050	-65%	-43% to -126%	Yes

Based on the Table 44 comparison, this utility would conclude that all their QTs are quantitatively Paris Agreement aligned. As discussed in the methodology document, the high-level insight from the Table 44 comparison is that QTs within the ranges are aligned with global futures consistent with limiting warming to well-below 2°C. Furthermore, if all companies have QTs within the range, the sum will also be aligned with well-below 2°C. However, it is critical to keep in mind that this comparison is not conclusive because the global pathways do not capture the transition opportunities and challenges for individual companies, or differences across companies. In addition, the global pathway ranges do not reflect all the sub-global and local transitions found to be aligned with a global temperature. Thus, QTs for some companies may not fall within the global pathway ranges due to their transition challenges. As discussed in the methodology document, the QTs should not be judged negatively based on this comparison. This outcome is simply an indication of the company's challenges. The QTs should be evaluated in terms of their challenges and the strategies for pursuing them and enabling conditions for additional ambition. The QTs represent the highest possible ambition for a company under different conditions while balancing trade-offs and risks that the company would be transparently identifying and communicating. Specifically, the company would communicate what conditions are determining their QTs, as well as how additional reductions could be possible under other conditions.

The company can also visually communicate overall alignment with the Paris Agreement (PA). A graphic like Figure 45 can be created and is an effective means to defensibly and quantitatively communicate ambition, as well as alignment of the company targets and transition risk analysis

pathways with the Paris Agreement temperature goal. The company will want to include a narrative with Figure 45 discussing the information shown and describing SMARTargets' comprehensive Paris Agreement alignment—quantitative, qualitative, and in the company-specific evaluation of two reduction types (ATs and QTs) that recognizes differences in transition opportunities, challenges, risks, and enabling conditions.

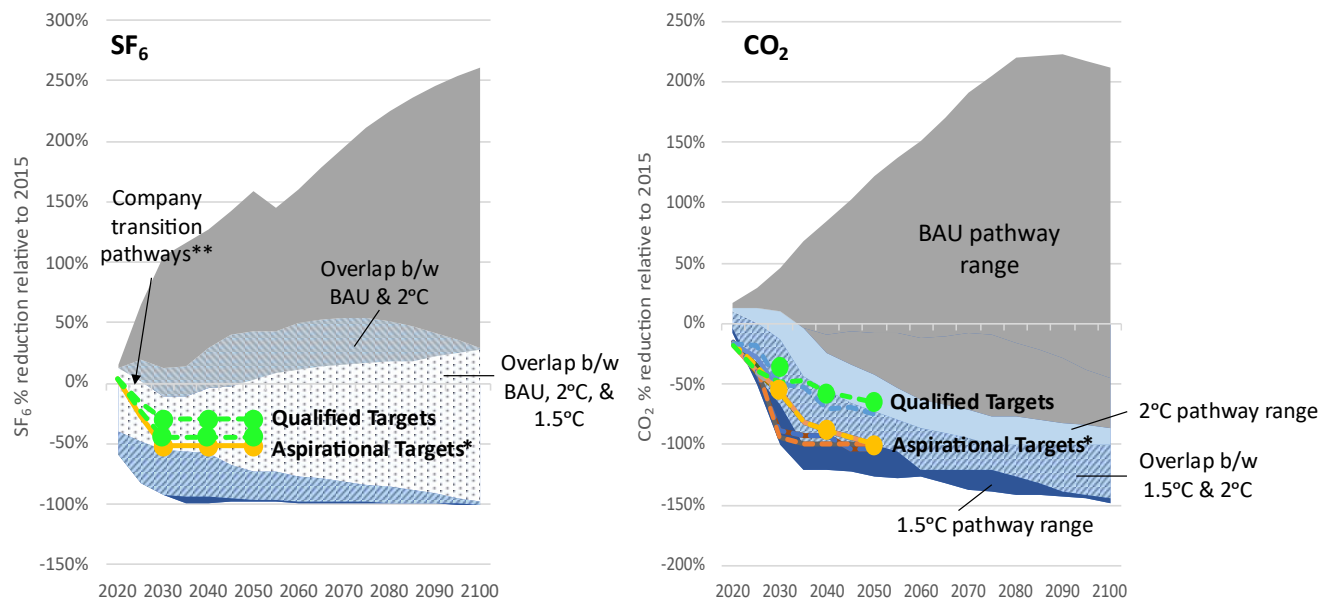


Figure 45. Alignment of illustrative company fugitive gas CH₄ and gas consumption CO₂ Aspirational and Qualified Targets with the global emissions pathways consistent with the Paris Agreement temperature goals

Source: EPRI

Figure notes: Aspirational Targets denoted with yellow dots with solid line transition pathways. Qualified Targets denoted with green dots with dashed line transition pathways. Developed from EPRI (forthcoming) analysis of the IPCC's Sixth Assessment Report, IEA Net Zero by 2050, and other recent global emissions pathways, and the EPRI illustrative US-REGEN analysis completed for this example. See EPRI (forthcoming) and EPRI (2020 and 2018) for detailed assessments of recent and previous global emissions pathway modeling and results, including discussion of global pathway limitations and the importance of using full ranges due to the nature of the IPCC data. Solid and dotted lines indicate AT and QT transition pathways respectively.

* Enabling conditions may be needed to achieve the Aspirational Targets. See the company transition risk analysis and narrative that identifies the enabling conditions and Qualified Targets reductions and their qualifying conditions.

** Shown are the CH₄ and CO₂ transitions after the additional abatement required for achieving the Aspirational and Qualified Targets identified from the Step 4 transition risk analysis.

Step 6. Documentation and Communication

Step 6 Output: SMARTargets documentation.

In this step, the utility will document and communicate their SMARTargets implementation using the SMARTargets Reporting Template. This includes documenting implementation choices with justification, transition analysis insights, targets, strategy, and milestones. Through the template, the following information will be assembled:

- General company information and stakeholder engagement,
- GHG Inventories and base year,
- Practical issues and climate target groupings,
- Company transition risk analysis of opportunities and risks,
- Alignment assessment,
- Documentation and communication of targets, strategies and indicators, and
- Implementation and monitoring.

The set of information represents a clear, comprehensive, and transparent package of information about the process and the outcomes. Overall, in applying the methodology, the utility will produce the following standardized and comparable results that reflect their implementation and climate strategy:

- Paris Agreement-aligned targets and enabling conditions, including Aspirational Targets (AT) and Qualified Targets (QT) and their associated conditions,
- Insights into potential low-carbon transitions,
- Assessment of risks, risk management strategies, and overall transition strategy, and
- Documentation of methodology implementation, including:
 - GHG inventory by required emissions category,
 - Responses to practical considerations and target-setting decisions for each category,
 - Transition risk analysis components:
 - Modeling approach
 - Reference conditions and key uncertainties
 - Core scenario definitions and specifications
 - Criteria for identifying risks and enabling conditions

In addition, the company is encouraged to holistically communicate their strategy. This goes beyond emissions targets and includes leadership strategies to help create the enabling conditions for achieving greater emissions reductions and economy-wide decarbonization, such as research and development (R&D), educational outreach (stakeholder and policy), and collaboration and coordination with customers and supply chain partners. For this illustrative utility, the analysis has identified that risk management is important given the significant uncertainty and lack of control over the size of the power market and the generation resources.

Among other things, for fugitive SF₆, this could include research and development that lowers the cost and increases the availability of technologies for reducing the fugitive SF₆ emissions, such as SF₆-free gas insulation alternatives. For their largest source of emissions, CO₂ associated with sold power to end users, abatement cost risk management might include encouraging the decarbonization of power and the availability of decarbonization options. For all sources, abatement cost risk management might include pursuing the Facilitating conditions, such as generation technology cost improvements and reducing barriers to new generation and transmission.

Finally, the company would also want to consider communicating on the SMARTargets Methodology itself, including its scientific foundations. This can involve referencing the publicly available methodology, the SMARTargets Scientific Foundations, and other materials and emphasizing how the approach is science-based and aligned with the Paris Agreement (see Justification for the Methodology section).

Step 7. Validation and Verification

Step 7 Outputs: Validated SMARTargets and verified progress.

In this step, the utility will validate their SMARTargets and verify progress. After completing the SMARTargets Reporting Template in Step 6, the utility will be able to initiate validation activities. Validation assesses whether the methodology has been properly implemented and whether the target and strategy—by nature forward-looking statements—are supported by sound assumptions, data, and methods. The primary input for validation is the SMARTargets Reporting Template.

The utility will begin with internal validation, leveraging internal audit processes aligned with recognized standards. This step ensures the methodology has been correctly applied and that the resulting targets and strategies are scientifically grounded and aligned with the Paris Agreement. Internal validation also builds confidence by thoroughly documenting the rationale and process behind the targets—information that is essential for both third-party validation and external communications.

Internal validation is all that is possible in the near-term. Accredited, independent third-party validation services for corporate GHG targets do not currently exist. Third-party validation is recommended once it becomes available. It provides an independent, authoritative assessment of conformity with the SMARTargets Methodology and validation criteria.

Following validation, the utility will communicate the results to the public, including regulators, investors, and other interested stakeholders.

In addition to validating, the utility is required to verify progress periodically. The utility will evaluate actual emissions reductions against its targets using the most recent GHG emissions inventory. The same third-party provider that verifies the GHG inventory could also perform this progress verification, thereby ensuring consistency and credibility with reporting. This verification relies on GHG inventories, which are based on observable data and are therefore subject to assurance-level assessments.

Step 8. Monitoring and Adjusting

Step 8 Outputs: Periodic milestone assessment and, as needed, strategy adjustment and/or SMARTargets refresh.

Under this step, the company will annually monitor their milestones and adjust their strategy as needed. The utility will need to monitor conditions with respect to their milestones to determine if strategy adjustments are needed and/or if updated SMARTargets analysis, targets, and strategies are needed.

The transition risk analysis (Step 4) provides milestone and contingency insights as well as insights regarding transition opportunities and risks. The company should articulate meaningful and observable milestones, and potential responses, based on their analysis, with the milestones clearly communicated in the Reporting Template. For instance, the Step 4 results above suggest the need to evaluate which future is emerging. If, for example, Facilitating conditions are emerging lowering generation technology costs, barriers to additions, and NIMBY sentiments, and/or offsets are available, the utility could pursue additional emissions reduction ambition beyond their QTs. Other key milestones identified include whether a regional decarbonization transition (power sector or economy-wide) is occurring, the availability of CDR, electrification trends, and the costs and availability of fugitive SF₆ abatement technologies and strategies for reducing T&D line losses. In particular, tracking the emergence of sector-specific or economy-wide decarbonization will be important because of their impact on the size of the electricity market and the mix of generating technologies. The results above highlighted that Facilitating conditions with decarbonization transitions in the power sector or economy overall would enable the utility to achieve the CO₂ ATs. For SF₆, the ATs would be more attainable with Facilitating conditions and either no decarbonization transition or an electric sector decarbonization with More Limited options. All these factors shape the feasibility and timing for T&D emissions reductions. Tracking these trends informs the utility's evaluation and possible adjustment of its T&D targets.

Illustrative Results for Gas Utility Emissions Targets

This example illustrates the derivation of SMARTargets for gas utility emissions for a hypothetical company that purchases natural gas from producers and delivers the gas to consumers. The primary categories of GHG emissions for this utility are the following:

1. Direct fugitive CH₄ emissions from gas distribution (scope 1),
2. Upstream fugitive CH₄ emissions from gas production, processing, and transmission (scope 3), and
3. Downstream CO₂ emissions from end-use gas consumption (scope 3).

All of the gas utility's emissions categories are a function of the size and composition of the overall gas market. The gas market is determined by many factors that are generally not controlled by a gas utility. As such, it is a key uncertainty for the utility and essential to evaluate.

The illustrative example is based on a U.S. Midwestern region and the company sits within the region and is interested in understanding how the regional gas market and economy might evolve with decarbonization by 2050 and what might be the implications and risks. Using US-REGEN and fugitive CH₄ abatement cost data, we can constrain emissions and evaluate alternative potential low-emissions transitions for the regional gas market, the three categories of GHG emissions, and GHG abatement opportunities.

Below, we describe how this gas utility would apply each step of the SMARTargets Methodology.

Step 1: GHG Inventories and Base Year

Step 1 Outputs: Verified GHG inventory and emissions reductions base year.

The first step is for the company to assemble their GHG inventory and select a base year. The company will need GHG emissions accounts for each of the categories of emissions relevant to their business. The company will also need to decide on their emissions reduction base year and apply it consistently. Note that the company will need to have the inventories verified.

For this example, Table 45 summarizes the emissions for our illustrative utility. The primary source of emissions is downstream gas consumption CO₂, however, fugitive direct and indirect CH₄ emissions are also noteworthy. The utility could have other emission sources but, for this example, we assume they represent only a small fraction of the overall inventory and are immaterial.

The emissions trends for each category are shown in Figure 46. As illustrated, CH₄ emissions have shown a steady decline; however, since 2015, emissions have increased slightly and leveled off. In contrast, CO₂ emissions have been fairly flat with a slight upward trend and decrease in the last few years. An analysis of the emissions drivers reveals that these trends are primarily driven by steady gas consumption, efficiency improvements, and electrification with shifts away from gas.

Given the information available, we assume the utility decides to select 2015 as its base year. This decision is based on considering the accuracy of their inventories, the inventory years verified, and choosing a year that facilitates communications regarding the utility’s emissions trends.

Table 45. GHG inventory for key categories in gas utility illustrative example
Source: EPRI with data from EPA (2024), IEA (2025) and illustrative gas utility GHG inventories.

Key emissions categories	2015 (base year)	2021 (most recent)
	MMtonCO _{2e}	
Direct fugitive CH ₄ emissions from gas distribution	0.3	0.3
Upstream fugitive CH ₄ emissions from gas production, processing, and transmission	0.3	0.3
Downstream CO ₂ emissions from end-use gas consumption	4.7	3.6

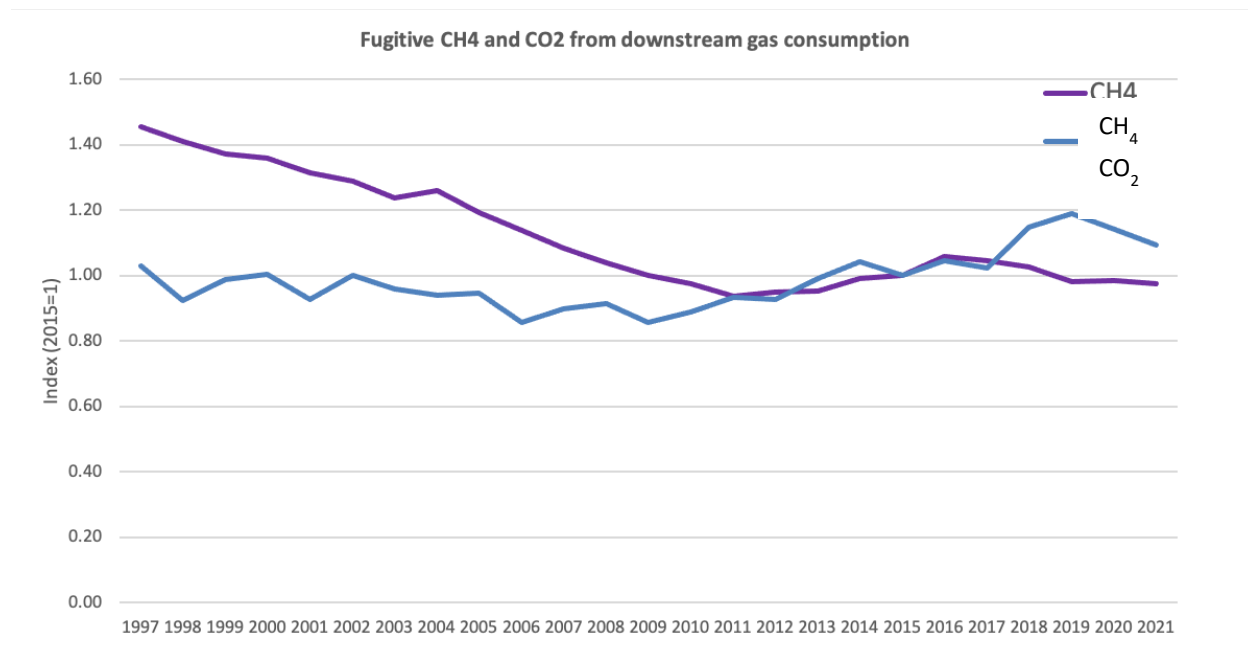


Figure 46. Emissions Trends for key categories in gas utility illustrative example

Step 2. Practical Issues and Climate Target Groupings

Step 2 Outputs: Decisions on practical issues, including emissions categories covered (by regulation or company targets), the categories for which targets are being set, and how these categories are grouped for target evaluation.

For this step, our illustrative gas utility works through the company target setting practical issues, as well as identifies their target groupings. Table 46 presents example responses to the set of practical issues questions for all three emissions categories, and below we discuss the implications. This step helps companies assess practical considerations that may influence their ability to set and achieve targets. The company's responses also inform their evaluation of targets (Step 4).

Table 46. Illustrative practical issues responses for the primary gas utility emissions categories

Company Target Setting Practical Considerations question	Response for gas distribution fugitive CH ₄ (Scope 1 Direct Emissions from Fugitive Sources)	Response for gas production, processing, and transmission fugitive CH ₄ (Scope 3 Category 3 Fuel- and energy-related activities not in scopes 1 and 2)	Response for gas combustion CO ₂ (Scope 3 Category 11 Use of sold products)
Is the emissions category adequately estimated and reported?	Yes, the emissions can be adequately characterized	There is uncertainty about the emissions due to uncertainty regarding upstream activity, operations, and technology practices	Yes, the emissions can be adequately characterized
Are the emissions material to the company's overall GHG inventory? (e.g., above 5% of total emissions or a threshold defined by the company)	Yes, the emissions from this source category are greater than 5% of emissions	Yes, the emissions from this source category are greater than 5% of emissions	Yes, the emissions from this source category are greater than 5% of emissions
Are the category's emissions regulated by a Paris Agreement aligned policy?	No, the emissions from this source category are not currently regulated	Yes, the emissions from this source category are currently regulated	Yes, the emissions from this source category are currently partly regulated
Does another company in the value chain already set Paris Agreement-aligned targets for this source? (applicable only to indirect emission sources)	n/a, these are direct emissions.	No, the source does not currently have a Paris Agreement-aligned target separate from emissions reduction compliance requirements	No, the source does not currently have a Paris Agreement-aligned target separate from emissions reduction compliance requirements
Does the company have control or influence over the emissions?	Yes, the utility has control over these emissions	Limited, these emissions come from the assets and operations of the provider companies	Limited, these emissions are produced by consumers based on their preferences and choices and available technologies

Based on the responses above, the company might want to deliberate on whether to set a target for each of the categories. Upstream fugitive CH₄ emissions are difficult to robustly quantify and may be regulated, such as by the U.S. Environmental Protection Agency. The company may still choose to evaluate and potentially set targets, but some of these issues will complicate the company's efforts to pursue the target, as well as society's interest in cost-effective decarbonization. For example, the lack of control of upstream fugitive CH₄ and downstream combustion CO₂ means that there is more uncertainty and risk for the company in achieving targets for these sources. More specifically, the company not only does not control the emitting technologies and activities, but they do not control the size and composition of the gas market, which as discussed below is the primary uncertainty the company needs to evaluate. Note, for downstream combustion CO₂, customer initiatives could be deployed, but their effectiveness may be limited.

Overall, in this example, the company has legitimate practical reasons for not setting targets for upstream fugitive CH₄ emissions. Recall that, for each emissions category, the company decides whether to:

1. Set a target (using the subsequent SMARTargets steps),
2. Not set a target because the emissions are already covered by a Paris Agreement aligned policy or another company's Paris Agreement aligned targets,
3. Not set a target with justification based on practical considerations that currently preclude target setting, or
4. Not set a target because the category is not relevant to their business.

For the sake of this illustration, we assume that the company chooses to set targets for all three categories despite the practical issues identified and they will therefore undertake subsequent steps, including the Step 4 analysis to evaluate the targets, transition opportunities and risks, and enabling conditions. However, the company should keep these issues in mind as it considers its strategies for achieving targets, including contingency plans for different possible future conditions.

Another utility might have different responses to some of the questions in Table 46 or reach a different conclusion based on the responses in the table. For instance, a different utility might conclude that existing regulations are resulting in Paris Agreement aligned reductions in fugitive CH₄ emissions and it is therefore practical and cost-effective to focus on compliance and not set targets.

In this step, the company also needs to decide on its target groupings. For this example, we assume the company pursues separate CH₄ and net CO₂ targets for each of the three emissions categories. The company's overall target setting and target grouping decisions are documented in Table 47.

Table 47. Emissions category targets

Table note: Emissions categories are those from Table Table 22. * Emissions not relevant to business or not material to inventory.

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by <u>future</u> Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company's Paris Agreement Aligned Targets (MtCO ₂ e)			
			(indicate if target for category, GHG, or CO ₂ e)					
Scope 1. Direct Emissions from Stationary Combustion	0.2							Not material*
Scope 1. Direct Emissions from Mobile Combustion	0.0 (< 0.1)							Not material*
Scope 1. Direct Emissions from Process Sources	n/a							n/a
Scope 1. Direct Emissions from Fugitive Sources	0.3	0.3	CH ₄					Fugitive methane emissions from gas distribution operations are material
Scope 2. Indirect Emissions from Purchased/Acquired Electricity	n/a							n/a
Scope 2. Indirect Emissions from Purchased/Acquired Steam	n/a							n/a

Table 47 (continued). Emissions category targets

Table note: Emissions categories are those from Table 22. * Emissions not relevant to business or not material to inventory.

Emissions category	GHG total (MtCO ₂ e or NA if not relevant to business)	Currently covered				To be covered by <u>future</u> Company SMARTargets (MtCO ₂ e)	Not Covered (MtCO ₂ e)	Justification (and, as relevant, plans for not covered emissions and future targets)
		By Our Current Company SMARTargets (MtCO ₂ e)	SMARTarget type	By Paris Agreement Aligned Regulation (MtCO ₂ e)	By Another Company’s Paris Agreement Aligned Targets (MtCO ₂ e)			
			(indicate if target for category, GHG, or CO ₂ e)					
Scope 2. Indirect Emissions from Purchased/Acquired Heating	n/a							n/a
Scope 2. Indirect Emissions from Purchased/Acquired Cooling	n/a							n/a
Scope 3. Fuel- and energy-related activities (not included in scope 1 or 2)	0.3	0.3	CH ₄					Fugitive methane emissions from upstream gas production are material
Scope 3. Use of sold products	3.6	3.6	CO ₂					CO ₂ emissions from use of natural gas by end user are material

Step 3. Aspirational Targets Aligned with International Goals

Step 3 Outputs: Aspirational targets that will be evaluated in the Step 4 transition risk analysis.

For this step, the utility will select generic Aspirational Targets (ATs) aligned with the international goals for their CH₄ and net CO₂ targets. The utility identifies intermediate and 2050 ATs using Table 48, which shows the CH₄ and net CO₂ ATs from Table 6 of the methodology document. The utility must select a 2030 or 2035 intermediate AT, as well as a 2050 AT. They are also welcome to select additional pre-2050 intermediate targets using the table. For this example, we assume the utility chooses to select 2030 and 2040 intermediate ATs and therefore their set of net CO₂ ATs are -56%, -88%, and net-zero (-100%) from 2015 for 2030, 2040, and 2050 respectively; and, their set of CH₄ ATs for both fugitive categories are -37%, -41%, and -47% from 2015 for 2030, 2040, and 2050 respectively.

Table 48. Generic Aspirational Targets (% reduction from 2015)

	CH ₄	Net CO ₂
2030	-37%	-56%
2035	-39%	-82%
2040	-41%	-88%
2045	-45%	-94%
2050	-47%	-100%

Step 4. Company Transition Risk Analysis of Opportunities and Risks

Step 4 Outputs: Company regional transition opportunities and risks, evaluation of aspirational and qualified targets and their enabling conditions, and transition strategies (low-carbon & risk management).

Given that the utility has decided to evaluate and pursue targets (Step 2), they will need to undertake transition risk analysis to identify their opportunities, risks, and risk management strategies for pursuing the Step 3 ATs and identifying qualified targets (QTs) and their associated conditions. The utility works through the seven phases associated with the transition risk analysis. The example analysis below informs all three of the company's target types.

Phase 1: Define Reference Conditions and Itemize Uncertainties

- *Phase inputs:* Previous company planning, strategy, analysis, and risk management experience
- *Phase outputs:* Scenario analysis reference conditions and list of uncertain factors to consider in scenario design
- *Who should be involved:* Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling

- *Documentation:* SMARTargets template summarizing reference conditions and uncertain transition factors for scenario design consideration

For this phase, the illustrative company works with their planning, policy, sustainability, risk management, strategy, and outreach staff, as well as any external experts on scenario analysis and modeling with which they choose to collaborate, to define their scenario analysis reference conditions and the uncertain factors that they wish to consider in their scenario design in Phase 2.

The key uncertainty for the gas utility CH₄ and CO₂ emissions categories is the size and composition of the future regional gas market. A gas market twice the size of today in 2050 will have twice the level of emissions, and a gas market half the size of today in 2050 will have half the level of emissions. Similarly, greater use of low-carbon gas alternatives to natural gas, such as renewable natural gas and synthetic gas made from CO₂, could lower the CO₂ emissions associated with gas consumption. Thus, the company will want to discuss potential reference conditions and identify uncertain factors that can influence the size and composition of the gas market.

Unsurprisingly, many of the conditions and uncertain factors relevant to the electricity generating utility's potential transitions in the previous example are also relevant for a gas utility. For example, for the reference conditions, our hypothetical company develops lists for announced energy infrastructure retirements and new builds that will be implemented as constraints in the modeling analysis. In addition, the company will also want to include local and federal policy drivers that should be a part of every transition future they evaluate.

The company staff, and any experts involved will also develop a list of uncertain factors that they believe are or could be important and that they want to evaluate in some way. In this case, we are using the same uncertain factors that were identified by the electricity generating utility since they are also factors relevant to power and economy-wide gas use transitions. We have, however, also added two additional uncertainties regarding gas system fugitive CH₄ abatement technology cost and performance improvement uncertainty and low-carbon gas cost improvement uncertainty:

- Electricity supply decarbonization options,
- Electricity supply technology cost and performance improvements,
- Future electricity demand (load),
- Low-carbon generation technology development and commercialization,
- Low-carbon generation technology permitting,
- Supply chain constraints related to low-carbon generation technologies,
- Not in my backyard (NIMBY) issues,
- Resource procurement,
- Support for new transmission infrastructure investments,
- New transmission infrastructure costs,

- New transmission siting and permitting,
- Fuel availability,
- Fuel pipeline capacity/augmentation,
- End-use technology cost and performance improvements,
- Gas system fugitive CH₄ abatement technology cost and performance improvements, and
- Low-carbon gas cost improvement.

Note that the SMARTargets Methodology scenario design for gas utility emissions already accounts for uncertainty about decarbonization transitions—reference (without decarbonization ambition), power sector only, and economy-wide. Like the electricity generating utility, these are relevant to the gas utility as significant factors influencing potential gas market transitions. At the end of Phase 1, the utility will document their reference conditions and itemize the identified uncertain factors in the SMARTargets Reporting Template.

Phase 2: Create Core Scenario Design – Define Plausible Extreme Alternative Future Conditions

- *Phase inputs:* Phase 1 scenario analysis reference conditions and list of uncertain factors to consider in scenario design.
- *Phase outputs:* The company's core scenario design for the required SMARTargets scenarios.
- *Who should be involved:* Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as possibly external experts on scenario analysis and modeling.
- *Documentation after iterations:* SMARTargets Reporting Template entries summarizing the core scenario design.

For this phase, the company continues working with the same team as in Phase 1 to define their core scenario design for the required SMARTargets scenarios. Table 49 presents the required core scenarios for gas utility emissions from the methodology. For evaluating the ATs for gas utility emissions, the required scenarios are the net-zero scenarios for electricity generation CO₂ targets and the reference scenarios. The reference scenarios help the gas utility evaluate the gas market and fugitive CH₄ and gas combustion CO₂ targets when electricity or economy-wide decarbonization are not pursued. This type of future is a part of the plausible transition space for a gas utility and therefore should be evaluated. The company will then want to evaluate a range of target levels for CH₄ and CO₂ for identifying their QTs after they review the results for the ATs and reference conditions.

To implement the Table 50 scenarios, the company needs to define the conditions—Fuller/More Limited decarbonization options and Facilitating/Challenging other conditions. Recall that the conditions should be intentionally more extreme, but plausible. In this example, we use the same condition definitions that were used for evaluating electricity generation targets. See the electricity generation target example for discussion of the three intermediate

steps associated with grouping and defining the Facilitating and Challenging non-climate-policy conditions.

The outcome of the second phase, combining uncertainty categories into sets of conditions and defining the relative conditions, is what is shown in Table 50. Table 50 is identical to Table 12 that was used for the electricity generation targets example. The set of uncertainties considered in both the Facilitating and Challenging conditions includes uncertainty regarding electricity generation cost improvements, electricity capacity additions, electricity transmission additions, fossil fuel prices, and electric technology cost improvements. Note that, Table 50 does not include fugitive CH₄ abatement technology or low-carbon gas cost and performance uncertainties. These are relevant uncertainties, however, current estimates of the cost of gas system fugitive CH₄ abatement opportunities and low-carbon gas costs do not typically provide quantitative uncertainty estimates. Thus, this uncertainty is not being considered in this example analysis due to the absence of data; but, these uncertainties could be incorporated in the future.

With the scenario conditions defined, the company is ready for Phase 3 and quantification of the alternative conditions. At the end of Phase 2, the company should document their definitions of Fuller/More Limited and Facilitating/Challenging using the SMARTargets Reporting Template.

Table 49. Required scenarios for evaluating targets for gas utility emissions

Target pathway type being evaluated	2050 targets for pathways	Electric sector transition	Economy-wide transition to net-zero in 2050	Decarbonization options	Other conditions
Aspirational Targets (CH ₄ & net CO ₂ ATs and QTs > ATs)	CH ₄ 47% & net CO ₂ 100%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging
Qualified Targets (CH ₄ & net CO ₂ QTs < ATs)	CH ₄ A, B, C% & net CO ₂ X, Y, Z%	Reference (without CO ₂ constraint)	No	N/A	Facilitating
		Reference (without CO ₂ constraint)	No	N/A	Challenging
		Transition to net-zero CO ₂ in 2050	No	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	No	More Limited	Challenging
		Transition to net-zero CO ₂ in 2050	Yes	Fuller	Facilitating
		Transition to net-zero CO ₂ in 2050	Yes	More Limited	Challenging

Table 50. Illustrative Fuller/More Limited and Facilitating/Challenging decarbonization condition definitions

Illustrative FULLER decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions) Advanced nuclear (additions) Fossil Fossil w/CCS CDR (biopower w/ CCS & direct air capture (DAC)) Storage Hydrogen Biopower carbon neutrality Other renewable fuels (w/ and w/o CCS) Low-carbon power imports Fossil capacity reserves permitted (within emissions constraint) Demand conservation and energy efficiency	Illustrative MORE LIMITED decarbonization option set Renewables (wind, solar, bio, hydro) Nuclear (extensions only) Storage Electrolytic Hydrogen only Other renewable fuels (w/ and w/o CCS) Fossil capacity reserve eligibility declines to zero in 2050 Demand conservation and energy efficiency
Illustrative FACILITATING other conditions Reference electricity load growth Faster electricity generation cost improvements Economic electricity capacity additions Economic electricity transmission additions Lower fossil fuel prices Faster electric technology cost improvements	Illustrative CHALLENGING other conditions Faster electricity load growth Slower electricity generation cost improvements Constrained electricity capacity additions Constrained electricity transmission additions Higher fossil fuel prices Slower electric technology cost improvements

Phase 3: Implement Scenario Design – Specify Assumptions for Scenario Modeling

- *Phase inputs:* The company’s Phase 2 core scenario design for the required SMARTargets scenarios.
- *Phase outputs:* Specification of core scenario design for the company’s modeling framework.
- *Who should be involved:* Staff related to planning, sustainability, policy, and strategy, as well as external experts on scenario analysis, modeling, and alternative assumptions.
- *Stakeholder engagement:* This is a good time to seek external stakeholder feedback.
- *Documentation after iterations:* SMARTargets Reporting Template entries summarizing the core scenario design specification.

For this phase, the company continues working with the same team as in Phases 1 and 2 to specify the core scenario design for their modeling framework. Phase 2 provides the company with everything they need conceptually and qualitatively in terms of the scenarios they want to run. Running the scenarios, however, requires quantification of the scenario design. This includes specifying decarbonization technology cost and performance assumptions, as well as what exactly is meant by “faster” and “slower” cost improvements, and “constrained” capacity and transmission additions. Since we are using the same condition definitions as used for the

electricity generation target example, we also use the same specifications (Table 51a and Table 51b are the same as Table 41 and Table 42).

Tables Table 51a and Table 51b present an illustrative specification for each of the assumptions associated with the Fuller/More Limited and Facilitating/Challenging conditions. Remember that the specification definitions are intentionally extreme, but plausible, alternative conditions. They are also grounded with justification provided for the specifications.

The utility will also need to define the emissions constraint pathways they will be imposing on the model. These constraints will constrain the model to producing no more than the emissions reductions specified (though the model could produce emissions below the constraints).

After completing Phase 3 is a good time for external stakeholder feedback on the scenario design and specifications. It is an opportunity to discuss setting SMARTargets and the planned transition risk analysis and preliminary future conditions the company is considering. At the end of Phase 3, the company should document their specifications for their Fuller and More Limited decarbonization options, their Facilitating and Challenging conditions, and their emissions constraints in the SMARTargets Reporting Template.

Table 51. Illustrative summary of energy technology assumptions

Decarbonization technology cost and performance	Comment/description	Reference
Renewables (wind, solar, bio, hydro), nuclear, advanced nuclear, coal, coal w/ CCS, gas, gas w/ CCS, CDR (biopower w/ CCS, DAC), storage, hydrogen	Engineering-based cost and performance specifications with gradual improvement over time	EPRI (2022b)
Biomass carbon neutrality	Carbon neutral assumption based on analysis of net land use and GHG changes associated with increasing biomass supplies	Blanford et al (2022)
Other renewable fuels	Renewable liquid and gas fuels, including renewable gasoline, diesel, and jet fuel, renewable natural gas, and synthetic natural gas produce with renewable energy	Blanford et al (2022)
Low-carbon power imports	Low-carbon electricity can be exported to other regions via intra-regional transmission	EPRI (2023)
Fossil capacity reserves permitted (within emissions constraint)	Fossil generation capacity can be used to meet capacity reserve requirements but declining emissions constraints over time apply to any generation from that capacity	EPRI (2022b)
Demand conservation and energy efficiency	End-use sectors can reduce emissions through technology and fuel substitution that increases efficiency, as well as reductions in overall use	Blanford et al (2022)

Table 52. Illustrative specifications for Facilitating and Challenging other condition assumptions

Other conditions	Facilitating	Challenging
Electricity load growth	Reference = load growth due to reference economic and population growth assumptions and electrification due to electric technology relative cost improvements	Faster = reference load growth conditions plus additional data center load growth based on recent trends resulting in 12% greater total load by 2030 (EPRI, 2024)
Electricity generation cost improvements	Faster = default cost projections and reference learning curve for battery cost improvements over time	Slower = higher cost projections by technology and slower learning curve for battery cost improvements over time
Electricity capacity additions	Economic = allows for all economically viable electric capacity to be built	Constrained = limits the building of economically viable electric capacity to historical build rates
Electricity transmission additions	Economic = allows for all economically viable interregional transmission to be built	Constrained = limits the building of economically viable interregional transmission based on historical transmission build rates
Fossil fuel prices*	Lower = No additional supply costs for fossil fuel production beyond the equilibrium cost determined by the energy economy model's balance of supply and demand for each fuel	Higher = Additional supply costs added to fossil fuel production modeling low resource availability or pipeline expansion
Electric technology cost improvements	Faster = Default cost reductions in EV & heat pump components, faster EV adoption reflective of less range anxiety and greater availability of new electric technologies	Slower = Higher component costs for EV & heat pumps, slower EV adoption representative of more range anxiety and less availability of new electric technologies

* The fossil fuel price assumptions in the table reflect the US-REGEN modeling structure. Other economy-wide frameworks might have fixed assumed (i.e., exogenous) energy prices. For instance, they might be based on lower/higher U.S. Department of Energy Annual Energy Outlook fuel price projections.

Phase 4: Identify Potential Transitions and Target Implications

- **Phase inputs:** Phase 3 specification of core scenario design for implementation in the company's modeling framework.
- **Phase outputs:** Preliminary transition analysis pathways and insights.
- **Who should be involved:** Staff and external experts running the model and processing outputs.
- **Documentation:** *None.* This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.

For this phase, the company's staff and external experts running the model and processing outputs will work together to develop preliminary transition analysis pathway results and insights. The team will be running the model with the scenario specification implementation

developed in Phase 3, and they will have to debug and likely revise the scenario specification implementation as they attempt to run each of the scenarios in the design. They will also want to review the results for non-intuitive, or otherwise problematic, outcomes that could be an artifact of an incorrect scenario specification or implementation. The team would want to revise and rerun scenarios in these cases as well.

In addition to running the scenarios, the team will develop preliminary graphics—figures and tables—and identify preliminary insights for review. In particular, the team will want to evaluate differences and similarities in transition results between scenarios. The differences between the scenarios are the most meaningful insights, not the specific values from individual scenarios, as you will learn something about the direction and order of magnitude of implications for the different conditions. For instance, gas utilities would be interested in the alternative potential transitions for the gas market size and composition, emissions, and the abatement costs for each emissions category. The modeling provides information regarding the alternative potential transitions, trade-offs, uncertainty, risks, enabling conditions, robust strategies, contingencies, and more. Below we present sample results for the company's scenario design. For this example, we are evaluating potential market transitions for a company in the midwestern U.S. These results represent potential alternative market and operational conditions for the company's system and assets.

The overall approach is, for each emissions category, to first estimate the future potential gas markets and the associated emissions under the different scenario conditions. The next step is to evaluate target setting. For both fugitive CH₄ emissions categories, we evaluate targets by applying marginal abatement cost curves to estimate costs for a range of emissions reduction levels for a given gas market. For gas consumption CO₂ emissions, we evaluate targets using the scenario modeling end-use decarbonization outcomes (emissions reductions and portfolios) from which costs can be estimated using the resulting CO₂ prices.

Figure 47 presents the model estimated total gas consumption transitions for all the required scenarios. Gas consumption includes gas used by the power sector, as well as the rest of the economy, including residential and industrial use. We observe the possibility of increasing or decreasing gas consumption, with natural gas still being consumed across the regional economy in 2050 in all the transitions. In 2050, we find the possibility of a regional gas market that is approximately 70% smaller than today or 80% larger, with the transitions from today rising, peaking, flat, or falling. We also find only a modest potential role for low-carbon gas in this region. The smallest gas market by 2050 occurs in the economy-wide decarbonization scenarios, and the largest occurs in the reference scenarios without power sector or economy-wide decarbonization. We also find that the Facilitating conditions, with assumed lower gas prices, result in a larger gas market compared to the comparable Challenging future. Finally, the Fuller versus More Limited decarbonization conditions have their greatest impact when there is an economy-wide decarbonization policy. In this case, the absence of carbon dioxide removal in the More Limited context is contributing to a larger reduction in the size of the future gas market. The key uncertainty for the gas market is whether there is an economy-wide, an electric sector, or no decarbonization transition.

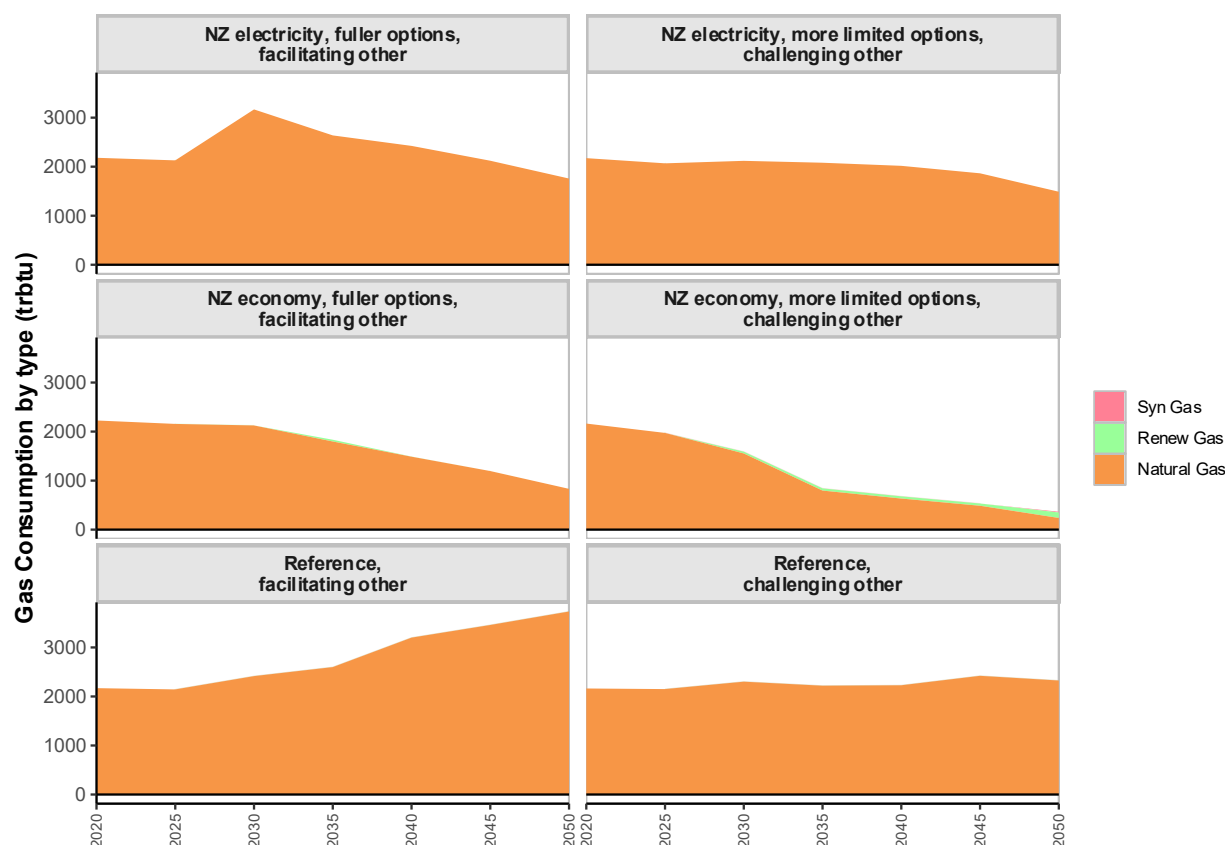


Figure 47. Total regional gas consumption transitions for the required scenarios

Source: EPRI analysis

Figure note: Illustrative results for a Midwest region of the U.S. and for the illustrative scenario design and specification. Results will vary by region of the world and scenario design and specification.

Figure 48 presents the 2030 and 2050 CH₄ and CO₂ emissions implications of the gas transitions in percentage change terms (relative to 2015). The figure illustrates that the gas market transitions could result in a fugitive CH₄ emissions decrease of 80% or increase of 70%. Gas consumption CO₂, on the other hand, could decrease by as much as 80% or increase by as much as 95%. The gas consumption CO₂ reduction percentage range differs from the CH₄ ranges because the CH₄ emissions are primarily a function of gas volumes, while the CO₂ emissions are a function of gas volumes and shifts in energy technologies and the overall make-up of the energy system. In Figure 48, we also observe that the 2050 47% methane reduction AT is met in the two economy-wide decarbonization scenarios. Therefore, additional mitigation effort is not required in 2050 in these futures, but additional abatement effort is required in the other futures, as well as in 2030 since none of the scenarios reach the 37% reduction AT. The gas market transition to 2050 is not linear in any of the futures (Figure 47). The CH₄ transitions follow these profiles and, therefore, additional abatement is required along the economy-wide transitions, as well as the other transitions. For CO₂, additional abatement is required under all the future conditions to meet the 2030, 2040, and 2050 AT targets.

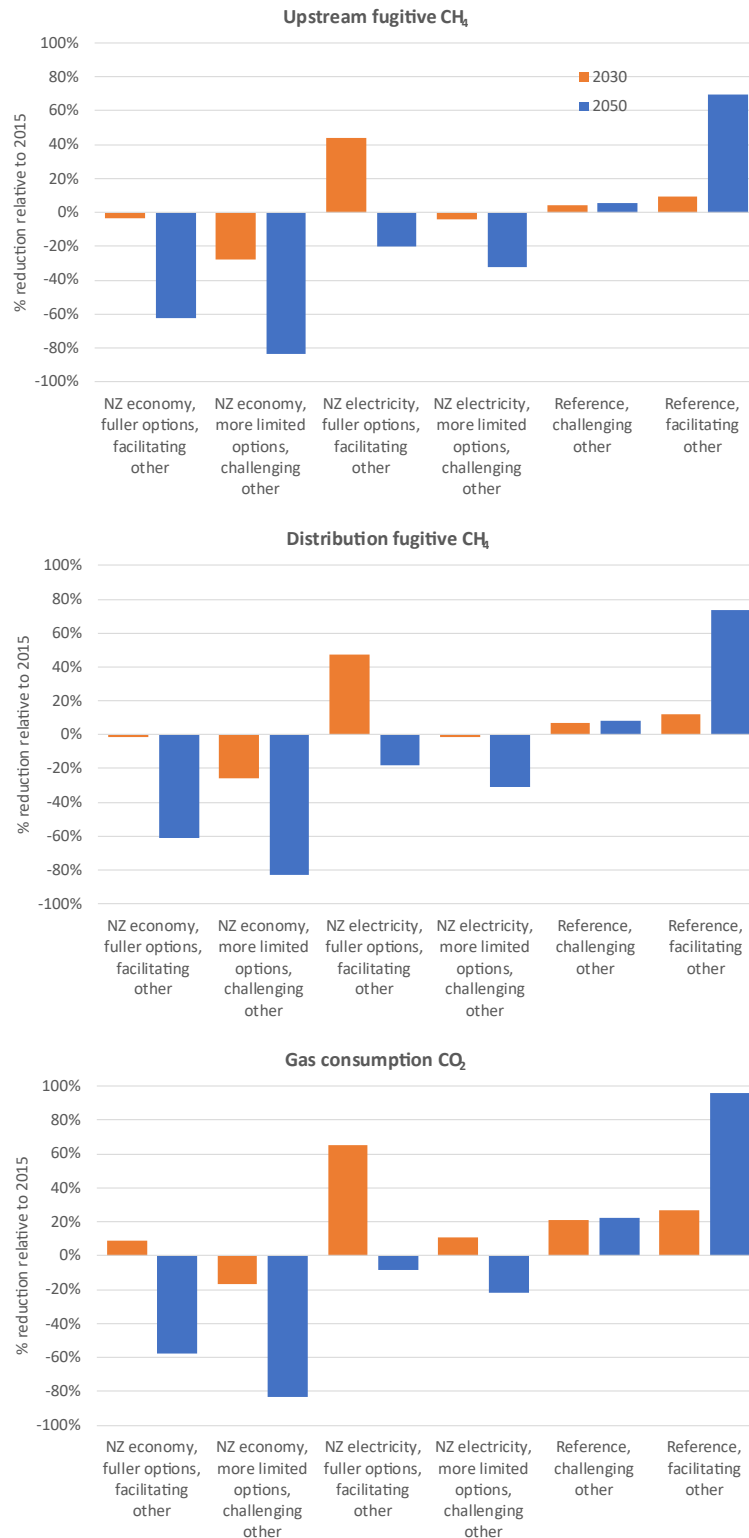


Figure 48. 2030 and 2050 percent reductions in regional upstream fugitive CH₄, distribution fugitive CH₄, and gas consumption CO₂ for each required scenario

Source: EPRI analysis

There are a variety of abatement measures that can be applied for fugitive CH₄ from gas production, processing, transmission, and distribution (EPRI, 2013; Table 53). All seventeen measures shown in Table 53 can be applied to upstream fugitive CH₄, and eleven can be applied to distribution fugitive CH₄. Furthermore, estimates of abatement costs for the set of measures can be used to evaluate the emissions reduction opportunities. Figure 49 is an example of a marginal abatement cost curve for reducing gas distribution fugitive CH₄ emissions in our example region. The curve has been adjusted for negative cost options, which as noted in EPRI (2013) should be treated as omitted costs.

Table 53. Natural gas fugitive CH₄ emissions abatement measures

Source: EPRI (2013)

1	Installing Catalytic Converters on Gas Engines and Turbines - Vented and Combusted Exhaust.
2	Reducing Methane Emissions from Compressor Rod Packing Systems
3	Directed Inspection and Maintenance
4	Replacing Wet Seals with Dry Seals in Centrifugal Compressors
5	Enhanced Inspection and Maintenance
6	Electronic monitoring at Large Surface Facilities
7	Installing Electronic Starters on Production Field Compressors
8	Fuel Gas Retrofit for BD valve - Take Recip. Compressors Offline
9	Installing Flash Tank Separators on Dehydrators (at production, transmission, and storage sites without flash tanks)
10	Convert Gas Pneumatic Controls to Instrument Air
11	Inspection and Maintenance at Transmission and Storage Facilities
12	Using a Portable Evacuation Compressor for Pipeline Blowdowns
13	Installing Plunger Lift Systems in Gas Wells
14	Reduced Emission Completions from Hydraulically Fractured Oil and Natural Gas Wells
15	Optimize Glycol Circulation Rates in Dehydrators (at production sites with/without flash tanks and gas assisted pumps)
16	Replacing High-bleed Pneumatic Devices in the Natural Gas Industry
17	Installing Surge Vessels for Capturing Blowdown Vents

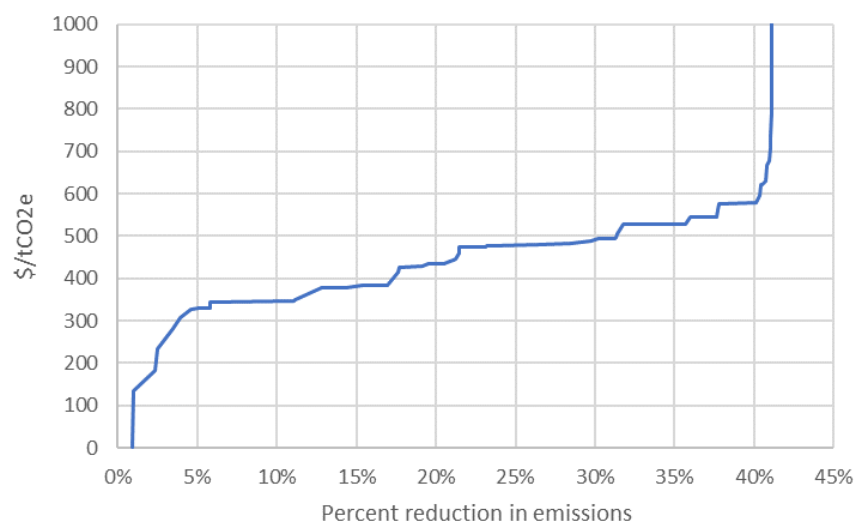


Figure 49. Marginal abatement cost curve in 2030 for regional gas distribution fugitive CH₄ emissions

Source: Developed from EPRI (2013)

The marginal abatement cost estimates are then applied to the alternative CH₄ transition results (Figure 48) to evaluate the additional abatement costs for achieving the AT levels, as well as to evaluate the costs of different reduction levels. Figure 50 presents net present value cost results for achieving the CH₄ ATs of 37%, 41%, and 47% reductions by 2030, 2040, and 2050 respectively under each of the scenarios. The costs are for the additional abatement required in each scenario, where, as shown, some scenarios produce declining emissions simply due to the decline in total gas consumption (Figure 47 and Figure 48). CO₂ abatement costs are derived using the additional CO₂ abatement required in each scenario for achieving the net CO₂ ATs of 56%, 88%, and 100% reductions by 2030, 2040, and 2050 respectively and the implied CO₂ prices associated with achieving the decarbonization policies under the varying conditions modeled.²⁴

The uncertainty in the gas market (Figure 47) is resulting in significant uncertainty in the cost of achieving the targets. The net present value cost of achieving 47% reductions in upstream fugitive CH₄ by 2050 ranges from \$2.3 billion to \$868.1 billion. The cost of achieving 47% reductions in distribution fugitive CH₄ by 2050 ranges from \$19 million to \$95.8 billion; and, the net present value cost of achieving 100% reductions in gas consumption CO₂ by 2050 ranges from \$7.3 billion to \$46.5 billion. The lowest cost outcomes are associated with the transitions with the smallest gas market, which occur in the economy-wide decarbonization transitions, with More Limited CO₂ decarbonization options, and with higher gas prices (Challenging). The costs are highest when the gas market is largest, which is without a decarbonization policy and when gas prices are lower (Facilitating). The extremely costly upstream fugitive CH₄ outcomes of \$636 and \$868 billion are due to the early bump up in the size of the gas market in the power sector decarbonization transition with Fuller decarbonization options and low gas prices in Figure 47 or the growing gas market in the no policy future with low gas prices. In both cases, the limits of the fugitive CH₄ abatement measures are being reached, which is forcing use of the most expensive options. When gas consumption is increasing, as is the case in these futures, the company is less likely to be able to meet the 47% CH₄ reduction 2050 target due to the high costs. Even \$10-20 billion in abatement costs might be a challenge, especially when we consider the total abatement costs across the three emissions categories. This threshold is exceeded in other future conditions as well, such as in the power sector only decarbonization transitions. Since the results in Figure 50 are for the region, the company would scale them for their system, utilizing the implied growth rates and/or their market share, to help them consider their company-specific cost thresholds.

The results thus far highlight that the enabling conditions for the CH₄ targets are those conditions that reduce the size of the gas market (economy-wide policy, higher gas prices, and more limited decarbonization options), as well as cost improvements or additional options for fugitive CH₄ abatement that help reduce the risk of high-cost outcomes. A smaller gas market, of course, has broader business implications that the utility would want to consider. Note that Facilitating conditions can result in higher costs for fugitive CH₄ abatement, but lower costs for gas consumption CO₂ abatement (not shown). This is because the low gas prices result in a

²⁴ In 2050, the implied CO₂ prices range from \$130 to \$1,330/tCO₂. The lower price is associated with Fuller-Facilitating conditions, while the higher price is associated with More Limited-Challenging conditions.

larger gas market, but they also make gas a more cost-effective decarbonization strategy, while the other Facilitating conditions, such as slower load growth, lower technology costs, and reduced additions barriers, also facilitate a less expensive transition to net-zero for CO₂.

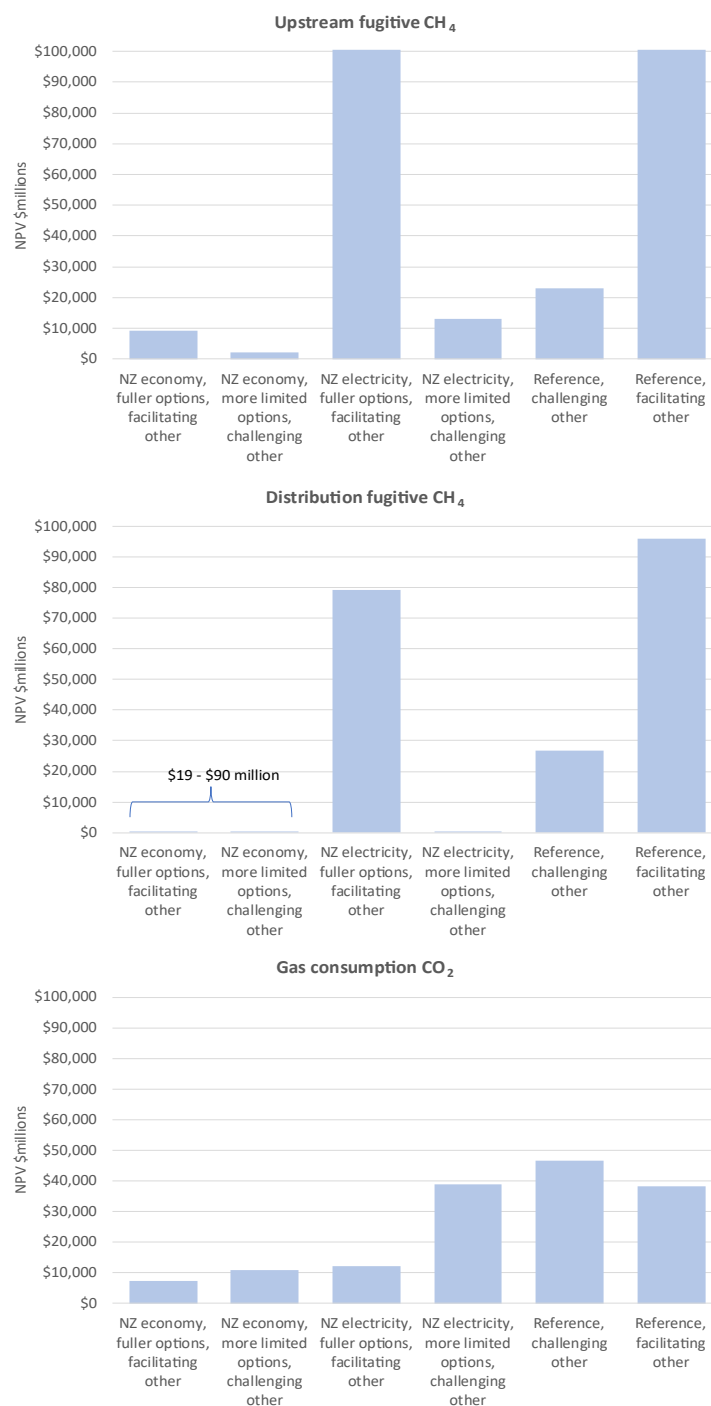


Figure 50. Net present value additional abatement costs for reaching each of the fugitive CH₄ and gas combustion CO₂ ATs for each required scenario

Source: EPRI analysis

The company will also need to evaluate the level of abatement possible if future conditions are less conducive to CH₄ and CO₂ abatement. Figure 51 presents incremental net present value abatement cost results for each of the fugitive CH₄ emissions categories for reducing emissions by 20%, 30%, 40%, or 47% by 2050 and reducing gas consumption CO₂ emissions by 30%, 50%, 80%, or 100% by 2050. Note that, the upstream fugitive CH₄ abatement cost curves for power sector decarbonization with Fuller options and Facilitating conditions, as well as for Reference-Facilitating conditions are separated. This is because the costs under these conditions are an order of magnitude higher, and therefore they are presented on a different y-axis scale.

For abating the emissions from all three categories, we find that costs increase with the reduction level; however, the levels and rates of increase depend on the emissions category and the conditions. We also see notable cost uncertainty for all three categories and at every abatement level, with the variation in costs across conditions varying by category. The different future conditions clearly affect the abatement implications. For example, for the same cost of 20% reductions in upstream fugitive CH₄ with reference conditions and high gas prices, 35% reductions could be achieved under power sector decarbonization conditions, and 47% reductions could be achieved with economy-wide decarbonization conditions with Fuller options and Facilitating conditions. Furthermore, for a quarter of the cost, 47% reductions could be achieved with economy-wide decarbonization but More Limited and Challenging conditions where CDR is unavailable and gas prices are higher.

Futures without a decarbonization transition and low gas prices (Reference-Facilitating) or with power sector decarbonization and CDR and low gas prices (NZ Power-Fuller-Facilitating) are particular challenging conditions for upstream fugitive CH₄ abatement, with the costs of 47% reductions high, and even 20% reductions expensive. They are also the most challenging futures for distribution fugitive CH₄ abatement. These results illustrate that reducing gas fugitive CH₄ emissions depends heavily on conditions that are not fully in the gas utility's control, such as economy-wide or electric sector decarbonization transitions. Also, as noted earlier, factors favorable to cost-effective CO₂ reductions, such as CDR availability and low gas prices, can increase the gas market size and make CH₄ reductions more challenging.

For gas consumption CO₂, we clearly see the cost uncertainty of 100% reductions by 2050 under the different conditions, with a net present value regional cost range of \$7.3 billion to \$46.5 billion. The upper end represents the possibility of over six times the costs. If the more expensive conditions emerge, such as reference and high gas price conditions, the company may have to reduce their emissions reduction objective. For example, if \$10 billion is viewed as a threshold, abatement of 30%, 90%, or 100% reductions could be supported if there is respectively electric sector decarbonization with More Limited options and high gas prices, electric sector decarbonization with Fuller decarbonization options and low gas prices, or economy-wide decarbonization with More Limited decarbonization options and high gas prices. The company can estimate their costs from the regional results by considering their emissions share. With this information, they can consider their company-specific cost thresholds and the corresponding emissions reduction rates from the regional results.

Overall, the results from this step provide broad and transparent transition perspectives that are a useful complement to company system planning. The results provide insights regarding potential transitions in the gas market, as well as other aspects of the energy system that are relevant to the company's system and service territory transition opportunities and CH₄ and CO₂ emissions reduction strategies.

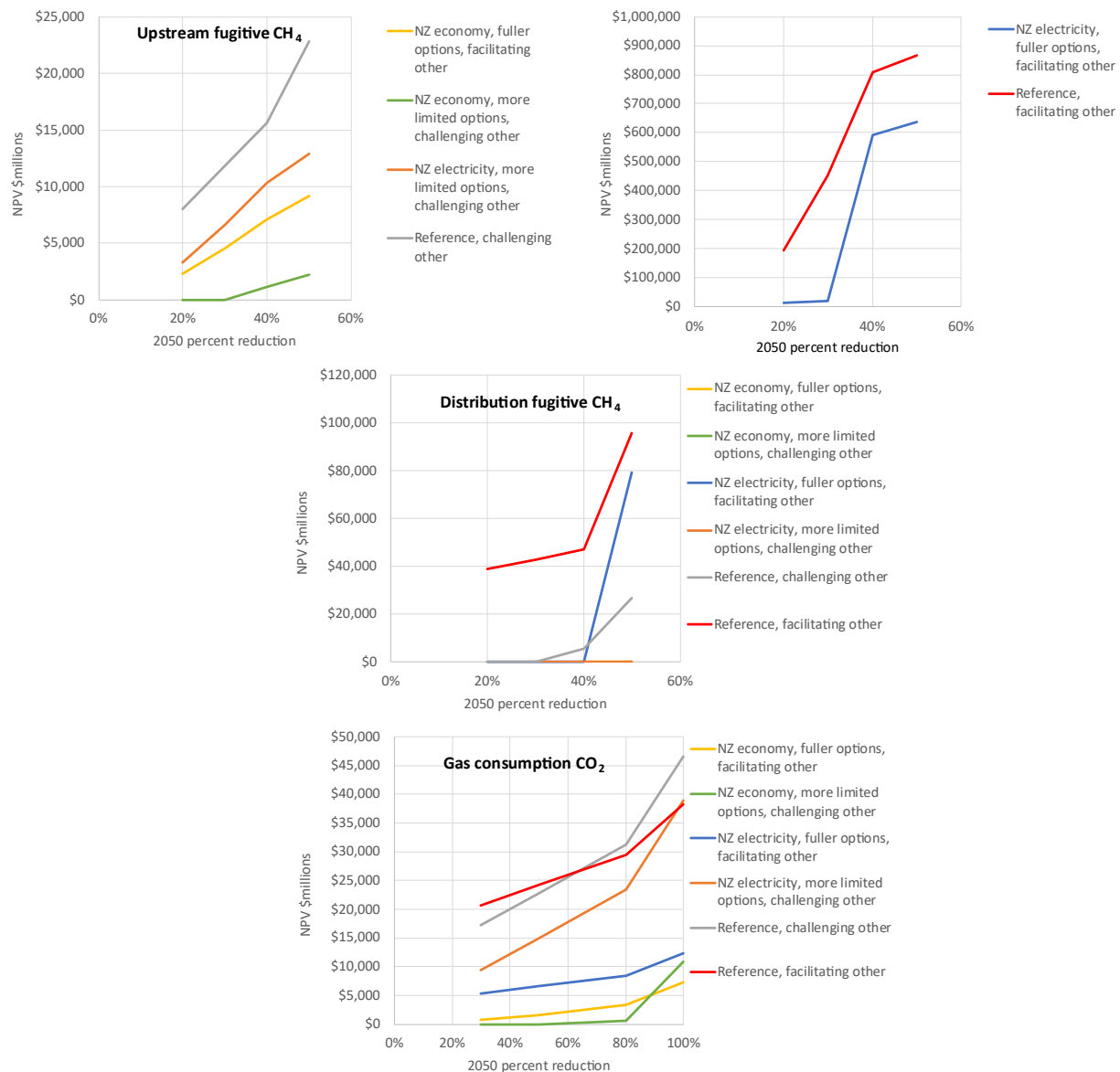


Figure 51. Incremental net present value costs through 2050 for increasing 2050 emissions reduction targets under the different future conditions for upstream gas production fugitive CH₄ (top), gas distribution fugitive CH₄ (middle), and gas consumption CO₂ (bottom)

Source: EPRI analysis

Figure notes: Abatement cost results for two of the upstream fugitive CH₄ abatement conditions are separate due to their significantly higher costs and the need for an alternative y-axis range.

The set of results also provide milestone and contingency insights. For instance, the results suggest the need to evaluate in a few years which future is emerging to determine if the strategy needs to be adjusted. If, for example, no decarbonization policy is emerging for the regional power sector or economy, and gas prices are low, the company might have to reduce their emissions reduction plans or seek reasonably priced offsets for emissions too costly to directly abate. Similarly, if net-zero transitions are underway in the power sector or economy, the availability of CDR and low-carbon fuels options will matter. Both technology options will impact the fugitive CH₄ and gas consumption CO₂ emissions levels and abatement effort required. Other milestones might include end-use technology adoption, fugitive CH₄ abatement technology costs, low carbon fuel costs, and energy demand trends. The milestones related to the conditions that influence the gas market size and composition will be priorities.

It is important to keep in mind that transition opportunities and risks can vary dramatically between regions, with different possibilities, sensitivity to future conditions, and risks. See EPRI (2023) for examples of differences in regional gas markets, decarbonization opportunities, and risks within the U.S.

Outside of noting that they completed this phase, there is no other documentation requirement since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete. The company will, of course, be generating draft results and insights from this phase. However, those results are deliberative and documenting them can be confusing and misleading to audiences that are non-technical and that have not been intimately engaged in the development process.

Phase 5: Review Potential Transitions and Target Implications – Based on Decision-Making Needs

- *Phase inputs: Phase 4 preliminary transition analysis pathways and insights.*
- *Phase outputs:*
 1. *Preliminary insights regarding potential transitions, targets, and risks,*
 2. *Revisions to core scenario design, model implementation, and output reporting, and*
 3. *Additional recommended and sensitivity scenarios and specifications the company would like to evaluate.*
- *Who should be involved: Company decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.*

For this phase, the company's planning, policy, sustainability, risk management, strategy, and outreach, staff, as well as external experts, will want to engage the company's decision-makers—directors and executives. The main objective here is to critically evaluate and discuss the preliminary results and insights and identify those they find reliable and meaningful. The additional objectives here are to identify any revisions to core scenario design, model

implementation, and output reporting they might want to make, and to identify additional scenarios and specifications that the company would like to evaluate. The last two items emerge from the company's critique and discussion of the draft results and insights.

Given the need to identify enabling conditions, this is the time for the company to think concretely about trade-offs, risks, and risk management from their perspective. The Phase 4 discussion above highlighted related insights from the perspective of our example company and analysis, noting specific risks and enabling conditions that identify and justify the company's QTs. Using the results from Phase 4, the company identifies trade-offs, risks, and risk management strategies. For instance:

- *Risks identified:* The primary risk is the size of the gas market, which is not only a risk in terms of the emissions abatement challenge, but also an overall revenue and operations risk for a gas utility. Related to gas market risk, there is also abatement cost and target attainability risk, where we find that there is the possibility of significantly higher costs associated with achieving the ATs in some plausible future conditions, especially for 47% reductions in fugitive CH₄ emissions by 2050.
- *Risk sources:* The gas market size risk is due to decarbonization transition, decarbonization option, and other uncertainties, with the size of the gas market responsive to whether there is a decarbonization transition and if it is in the power sector or economy-wide, whether the decarbonization options are fuller, and include carbon dioxide removal in particular, or more constrained, and whether gas prices are low or high and other conditions are facilitating or challenging the decarbonization transitions. Emissions abatement cost risk is a function of the gas market size uncertainty, as well as the cost and availability of fugitive CH₄ abatement options, and the cost-effectiveness of low carbon gas fuel options. The riskiest fugitive CH₄ abatement cost conditions are when the gas market is growing, which is projected to occur when there is no decarbonization transition, gas prices are low, CDR is available, and fugitive CH₄ abatement options are limited. The riskiest gas consumption CO₂ abatement cost conditions are when the gas market is flat or growing, which is projected to occur when there is no decarbonization transition, gas prices are high or low, and low-carbon technologies are more costly with slower cost improvement and barriers delaying new builds.
- *Risk management strategies:* Looking across the scenarios, we can identify enabling conditions and contingencies. The former are conditions that allow for a low-emissions transition with more manageable trade-offs. In this case, for fugitive CH₄ emissions, the conditions that result in a smaller gas market help reduce the level of CH₄ emissions and the challenge of abating CH₄. These include economy-wide decarbonization, higher gas prices, and more limited decarbonization options. In addition, research and development that lowers the cost and increases the availability of gas system fugitive CH₄ abatement technologies would help manage the abatement cost risk.

For gas combustion CO₂ abatement, economy-wide decarbonization, lower gas prices, and more decarbonization options help manage the cost risk. Note that, some of the conditions conducive to gas consumption CO₂ abatement are at odds with those conducive to gas system fugitive CH₄ abatement. Research and development is also a risk management

strategy for gas consumption CO₂ abatement, where improving the cost-effectiveness and availability of low-carbon gas could help manage the abatement cost risk.

As for contingency strategies, if the company finds itself in a growing gas market future, CH₄ abatement costs may become prohibitive, and they may need to lower their emissions reduction target. For CO₂ abatement, the company would want to adjust the portfolio of CO₂ reduction strategies depending on the conditions. This would imply shifting the mix of strategies between sourcing low-carbon gas, energy efficiency improvements, electrification, offsets, and emissions allowances (if available).

Given the set of results and insights, this company might specify QTs for 2050 of 20% for both fugitive CH₄ categories and 30% for gas combustion CO₂. The former being defined by what the company could support under Reference and low gas price conditions, where additional reductions could be pursued if any of the other conditions emerge where the gas market is not growing. Even 20% reductions would be difficult without offsets in futures resulting in a growing gas market (Reference-Facilitating and power sector net-zero with Fuller-Facilitating conditions). The CO₂ QT is defined by what the company could support under Reference conditions, where additional ambition could be pursued under all other conditions.

This phase is also the time when the company would want to consider the likelihoods of the different futures and potential hedging strategies they might pursue. For example, if the company believes that the likelihood of an economy-wide policy is low and Challenging conditions are more likely than Facilitating, they might want to invest more in R&D for gas system CH₄ abatement options, as well as explore and communicate what emissions reductions are possible if they cannot bring the abatement costs down.

After developing preliminary results and insights, the company would want to identify whether revisions to the scenario design, model implementation, and/or output reporting. In terms of identifying potential revisions to the scenario design and implementation in particular, the group discussion should consider whether they want to tweak anything. This could include revising what is included in each of the Fuller and More Limited decarbonization options sets, expanding Facilitating/Challenging to include additional uncertainties (e.g., CH₄ marginal abatement cost uncertainty), making the specifications more extreme to further differentiate them, or simply adjusting a specification such as the annual additions constraints for new electricity generating capacity or transmission. An additional option would be to add low-carbon gas cost uncertainty.

The company's discussion on potential sensitivity scenarios should look for meaningful adjustments to one or two assumptions. For example, based on the results above, this company might want to consider running More Limited with CDR scenarios (power sector and economy-wide net-zero) that add CDR to the More Limited decarbonization options set. They might also want to consider running an alternative power sector emissions target (e.g., 50% by 2050) to reflect the power sector's potential reductions without power sector decarbonization enabling conditions (see the potential QTs in the electricity generation supply emissions target setting example in this appendix).

Like Phase 4, outside of noting that they completed this step, there is no documentation requirement for the company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete.

Phase 6: Revise and Iterate as Needed – Revising Scenario Design, Implementation, and Model Results

- *Phase inputs:*
 1. *Phase 5 revisions to core scenario design, model implementation, and output reporting, and*
 2. *Phase 5 sensitivity scenarios and specifications the company would like to evaluate.*
- *Phase outputs:*
 1. *Revised insights regarding potential transitions, targets, and risks, and*
 2. *As needed, additional revisions to scenario design, model implementation, and transition pathways.*
- *Who should be involved: Staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Stakeholder engagement: This is a good time to seek external stakeholder feedback.*
- *Documentation: None. This is an intermediate phase. Phase 7 documents the final modeling results and insights once iterations are complete.*

For this phase, the company's planning, policy, sustainability, risk management, strategy, and outreach, staff, as well as external experts, will work together to revise the insights regarding potential transitions, targets, and risks, and, as needed, implement any needed revisions to the scenario design (core and sensitivity scenarios), model implementation, and transition pathways.

Phase 6 is also a good time for external stakeholder feedback. After internal iterations are complete, the company will have final drafts of their scenario design, results, insights, and strategies to share and discuss. This gives the company an opportunity for education and dialogue, as well as an opportunity to solicit feedback on their potential transitions, challenges, uncertainties, risks, enabling conditions, and collaboration opportunities.

For this phase, the company continues to iterate on the scenario design, specification, results, and insights until they have agreement with their decision-makers. Like Phase 5, outside of noting that they completed this phase, there is no documentation requirement for the company since it is an intermediate phase. SMARTargets requires documenting the final modeling results and insights in Phase 7 once iterations are complete.

Phase 7: Communicate Results – Targets, Transitions, Risks, Enabling Conditions, Risk Management

- *Phase inputs:*
 1. *Phase 6 final scenario design (core and additional scenarios), model implementation, and transition pathways, and*
 2. *Phase 6 final insights regarding potential transitions, targets, risks, enabling conditions, and risk management.*
- *Phase outputs:*
 1. *Documented scenario analysis, outputs, and insights,*
 2. *Aspirational Targets (ATs), strategies, and risk management, and*
 3. *Qualified Targets and their qualifying conditions (QTs) and strategies.*
- *Who should be involved: The company's decision-makers (e.g., directors, executives), staff related to planning, policy, sustainability, risk management, strategy, and outreach, as well as any external experts on scenario analysis and modeling with which you are collaborating.*
- *Documentation: SMARTargets Reporting Template summarizing the following:*
 1. *The company's ATs and QTs with corresponding conditions,*
 2. *The company's enabling conditions, strategy, risks, and risk management,*
 3. *The company's transition risk analysis (modeling, scenario design and specification, potential transitions, insights), and*
 4. *The implementation process.*

This phase is the critical communications of all the company has done for the Step 4 analysis. For this phase, the company will want to work with their decision-makers and company planning, policy, sustainability, risk management, strategy, and outreach teams. They will also want to engage their external experts to help with documenting the methodology and effectively, rigorously, and defensibly communicating the results. The experts can also help the company communicate the relevant science to educate readers and ground the methods.

At the end of Phase 7, the company will use the SMARTargets Reporting Template to finalize the documentation for their Step 4 transition risk analysis (Phases 1 through 7). This includes the summary tables of key output indicators, as well as graphics to help communicate the company's modeling results and insights. Additionally, the company will report the identified enabling conditions for ATs, QTs and corresponding conditions, and the risks and opportunities identified. Finally, the company will indicate their risk management strategies along with their milestones and contingency plans.

The company's strategy will consist of more than targets. It will also include transition plans as well as strategies for pursuing enabling conditions and managing risks, where the latter might entail research and development plans, collaboration with customers or other sectors, and policy dialogue. Step 4 has a minimum set of required outputs (Table 20 in the methodology document). Additional variables may be added based on company need.

Step 5. Strategy and Qualified Target Alignment with International Goals

Step 5 Outputs: International goal alignment assessments of qualified targets and strategies.

In this step, the company will evaluate *qualitative* alignment of their transition strategy with the international goals, as well as *quantitative* alignment of their QTs.

Assessing Qualitative Alignment of the Transition Strategy

The company will assess two types of qualitative alignment of their strategy with international goals: (1) alignment with the types of transitions robustly observed in global emissions pathways consistent with the Paris Agreement temperature goals, and (2) alignment with the provisions in the Paris Agreement associated with achieving the temperature goals.

As discussed in the methodology document, the following are robust types of transitions found in all global 1.5°C and 2°C emissions pathways. The following are changes found consistently in well-below 2°C pathways regardless of the model and assumptions:

- Increasing low-carbon technology deployments,
- Increasing reliance on electricity,
- Transitions away from fossil fuels but continued use,
- Use of carbon dioxide removal,
- Improvements in energy conservation and efficiency,
- Differences in regional low-carbon transitions, and
- For any climate goal, substantial uncertainty in the levels, rates of change, and composition (e.g., technologies, energy, activity) of the transition of the global economy.

Methane reductions are also a robust characteristic of the global pathways, with reductions in CH₄ producing activity levels and the utilization of CH₄ abatement measures. The transition analysis results can be assessed for consistency with these robust Paris Agreement aligned transition characteristics. The assessment, however, is different from that for electricity generation emissions strategies since the utility is needing to consider alternative gas markets conditions that are primarily out of their control. The assessment here should therefore focus on the strategy elements the utility can control and influence. For example, in all cases the company strategy is reducing fugitive CH₄ and gas consumption CO₂ emissions by utilizing fugitive CH₄ and gas consumption CO₂ abatement technologies. The set of strategies also includes increasing deployment of low-carbon technologies and energy conservation and efficiency and the use of carbon dioxide removal. They also exhibit significant uncertainty in the transitions and of course, by design, recognize the unique regional transition opportunities.

Furthermore, the transitions and insights align with Paris Agreement's provisions recognizing differences in national decarbonization opportunities, multiple societal priorities, transparency, and the value of innovation and cooperation (in this case across sectors via the economy-wide scenarios).

Assessing Alignment of the Qualified Targets

The company will also assess quantitative alignment of their QTs through comparison to the relevant global emissions pathway ranges consistent with the Paris Agreement objective of limiting global average temperature to “well-below” 2°C. Table 50 shows this comparison. Based on the Step 4 transition risk analysis, we assume this utility identified the following fugitive CH₄ and gas consumption CO₂ sets of QTs:

1. Fugitive CH₄ QTs: 20% reductions by 2050 (with 16% and 17% reductions in 2030 and 2040), and
2. Gas consumption CO₂ QTs: 30% reductions by 2050 (with 10% and 20% reductions in 2030 and 2040).

The conditions associated with the fugitive CH₄ QTs include a growing regional gas market due to low gas prices, no decarbonization transition in the power sector or economy-wide or power sector only decarbonization with CDR available, and non-CDR offsets being unavailable. The conditions associated with the gas consumption CO₂ QTs include a flat or growing regional gas market due to no regional decarbonization transition in the power sector or economy-wide and non-CDR offsets being unavailable. The QTs were identified by primarily considering the transition implications for affordability and the companies threshold for manageable costs. Other factors could also be considered, such as new infrastructure requirements, operational reliability, and the likelihood of the future conditions.

Based on the Table 50 comparison, this utility would conclude that their fugitive CH₄ QTs are quantitatively Paris Agreement aligned. However, the comparison for the gas consumption QTs is less definitive. The 2030 QT falls within the well-below 2°C range, but the 2040 and 2050 QTs do not. As discussed in the methodology document, the QTs should not be judged negatively based on this comparison. This outcome is simply an indication of the company’s challenges. The QTs should be evaluated in terms of their challenges and the strategies for pursuing them and enabling conditions for additional ambition.

The high-level insight from the Table 50 comparison is that QTs within the ranges are aligned with global futures consistent with limiting warming to well-below 2°C. Furthermore, if all companies have QTs within the range, the sum will also be aligned with well-below 2°C. However, it is important to keep in mind that this comparison is not conclusive because the global pathways do not capture the transition opportunities and challenges for individual companies, or differences across companies. In addition, the global pathway ranges do not reflect all the sub-global and local transitions found to be aligned with a global temperature. Thus, QTs for some companies are likely to fall outside the global pathway ranges due to their transition opportunities and challenges. The QTs represent the highest possible ambition for a company under different conditions while balancing trade-offs and risks that the company would be transparently identifying and communicating. In this example, the company would communicate what conditions are determining their QTs. The 30% by 2050 reduction in CO₂ was defined by a regional cost threshold of \$20 billion, which corresponds to a company-specific threshold of \$5 billion (25% of the cost based on the company’s market share). The QT

is associated with Reference conditions without decarbonization transitions elsewhere in the regional economy. The company could also communicate how additional reductions would be possible under the other conditions.

Note that, this utility might also communicate fugitive CH₄ QTs greater than ATs. This occurs when there is a net-zero decarbonization transition economy-wide and the regional gas market shrinks.

Table 54. Qualified Target Paris Agreement alignment assessment

Qualified Target set	Key qualifying conditions	GHG	Qualified Target 2050		Well-below 2°C ranges	Within range
Fugitive CH ₄	Growing regional gas market due to low gas prices, no decarbonization transition in power or economy-wide or power sector only decarbonization with CDR available, and non-CDR offsets unavailable	CH ₄	2030	-16%	13% to -60%	Yes
			2040	-17%	-5% to -65%	Yes
			2050	-20%	-4% to -78%	Yes
Gas consumption CO ₂	Flat or growing regional gas market due to no regional decarbonization transition in power sector or economy-wide and non-CDR offsets unavailable	Net CO ₂	2030	-10%	10% to -100%	Yes
			2040	-20%	-24% to -121%	No
			2050	-30%	-43% to -126%	No

The company can also visually communicate overall alignment with the Paris Agreement (PA). A graphic like Figure 52 can be created and is an effective means to defensibly and quantitatively communicate ambition, as well as alignment of the company targets and transition risk analysis pathways with the Paris Agreement temperature goal. The company will want to include a narrative with Figure 52 discussing the information shown and describing SMARTargets' comprehensive Paris Agreement alignment—quantitative, qualitative, and in the company-specific evaluation of two reduction types (ATs and QTs) that recognizes differences in transition opportunities, challenges, risks, and enabling conditions.

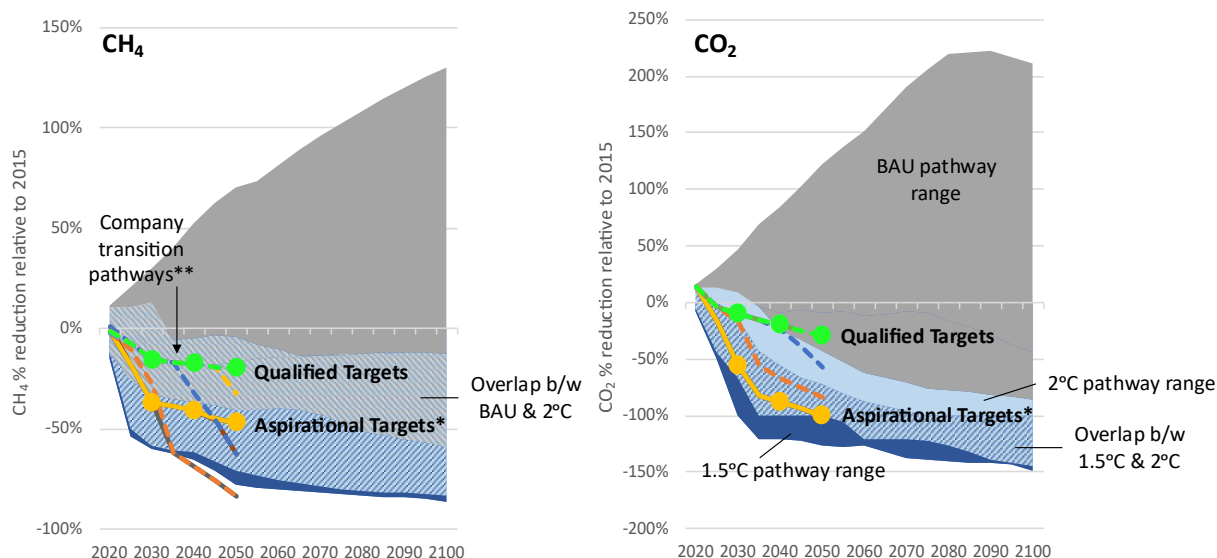


Figure 52. Alignment of illustrative company fugitive gas CH₄ and gas consumption CO₂ Aspirational and Qualified Targets with the global emissions pathways consistent with the Paris Agreement temperature goals

Source: EPRI

Figure notes: Aspirational Targets denoted with yellow dots with solid line transition pathways. Qualified Targets denoted with green dots with dashed line transition pathways. Developed from EPRI (forthcoming) analysis of the IPCC's Sixth Assessment Report, IEA Net Zero by 2050, and other recent global emissions pathways, and the EPRI illustrative US-REGEN analysis completed for this example. See EPRI (forthcoming) and EPRI (2020 and 2018) for detailed assessments of recent and previous global emissions pathway modeling and results, including discussion of global pathway limitations and the importance of using full ranges due to the nature of the IPCC data. Solid and dotted lines indicate AT and QT transition pathways respectively.

* Enabling conditions may be needed to achieve the Aspirational Targets. See the company transition risk analysis and narrative that identifies the enabling conditions and Qualified Targets reductions and their qualifying conditions.

** Shown are the CH₄ and CO₂ transitions after the additional abatement required for achieving the Aspirational and Qualified Targets identified from the Step 4 transition risk analysis.

Step 6. Documentation and Communication

Step 6 Output: SMARTargets documentation.

In this step, the utility will document and communicate their SMARTargets implementation using the SMARTargets Reporting Template. This includes documenting implementation choices with justification, transition analysis insights, targets, strategy, and milestones. Through the template, the following information will be assembled:

- General company information and stakeholder engagement,
- GHG Inventories and base year,
- Practical issues and climate target groupings,
- Company transition risk analysis of opportunities and risks,

- Alignment assessment,
- Documentation and communication of targets, strategies and indicators, and
- Implementation and monitoring.

The set of information represents a clear, comprehensive, and transparent package of information about the process and the outcomes. Overall, in applying the methodology, the utility will produce the following standardized and comparable results that reflect their implementation and climate strategy:

- Paris Agreement-aligned targets and enabling conditions, including Aspirational Targets (AT) and Qualified Targets (QT) and their associated conditions,
- Insights into potential low-carbon transitions,
- Assessment of risks, risk management strategies, and overall transition strategy, and
- Documentation of methodology implementation, including:
 - GHG inventory by required emissions category,
 - Responses to practical considerations and target-setting decisions for each category,
 - Transition risk analysis components:
 - Modeling approach
 - Reference conditions and key uncertainties
 - Core scenario definitions and specifications
 - Criteria for identifying risks and enabling conditions

In addition, the company is encouraged to holistically communicate their strategy. This goes beyond emissions targets and includes leadership strategies to help create the enabling conditions for achieving greater emissions reductions and economy-wide decarbonization, such as research and development (R&D), educational outreach (stakeholder and policy), and collaboration and coordination with customers and supply chain partners. For this illustrative utility, the analysis has identified that risk management is important given the significant uncertainty and lack of control over the size and composition of the gas market. Among other things, for fugitive CH₄, this could include research and development that lowers the cost and increases the availability of gas system fugitive CH₄ abatement technologies. For their largest source of emissions, gas combustion CO₂, abatement cost risk management might include encouraging the availability of additional decarbonization options and research and development that improves the cost-effectiveness and availability of low-carbon gas. Recall, some of the conditions conducive for gas consumption CO₂ abatement are at odds with those conducive for gas system fugitive CH₄ abatement. For all three categories of emissions, offsets (CDR and non-CDR) would help the utility achieve the targets, increase ambition, and manage trade-offs.

Finally, the company would also want to consider communicating on the SMARTargets Methodology itself, including its scientific foundations. This can involve referencing the publicly available methodology, the SMARTargets Scientific Foundations, and other materials and

emphasizing how the approach is science-based and aligned with the Paris Agreement (see Justification for the Methodology section).

Step 7. Validation and Verification

Step 7 Outputs: Validated SMARTargets and verified progress.

In this step, the utility will validate their SMARTargets and verify progress. After completing the SMARTargets Reporting Template in Step 6, the utility will be able to initiate validation activities. Validation assesses whether the methodology has been properly implemented and whether the target and strategy—by nature forward-looking statements—are supported by sound assumptions, data, and methods. The primary input for validation is the SMARTargets Reporting Template.

The utility will begin with internal validation, leveraging internal audit processes aligned with recognized standards. This step ensures the methodology has been correctly applied and that the resulting targets and strategies are scientifically grounded and aligned with the Paris Agreement. Internal validation also builds confidence by thoroughly documenting the rationale and process behind the targets—information that is essential for both third-party validation and external communications.

Internal validation is all that is possible in the near-term. Accredited, independent third-party validation services for corporate GHG targets do not currently exist. Third-party validation is recommended once it becomes available. It provides an independent, authoritative assessment of conformity with the SMARTargets Methodology and validation criteria.

Following validation, the utility will communicate the results to the public, including regulators, investors, and other interested stakeholders.

In addition to validating, the utility is required to verify progress periodically. The utility will evaluate actual emissions reductions against its targets using the most recent GHG emissions inventory. The same third-party provider that verifies the GHG inventory could also perform this progress verification, thereby ensuring consistency and credibility with reporting. This verification relies on GHG inventories, which are based on observable data and are therefore subject to assurance-level assessments.

Step 8. Monitoring and Adjusting

Step 8 Outputs: Periodic milestone assessment and, as needed, strategy adjustment and/or SMARTargets refresh.

Under this step, the company will annually monitor their milestones and adjust their strategy as needed. The utility will need to monitor conditions with respect to their milestones to determine if strategy adjustments are needed and/or if updated SMARTargets analysis, targets, and strategies are needed.

The transition risk analysis (Step 4) provides milestone and contingency insights as well as insights regarding transition opportunities and risks. The company should articulate meaningful and observable milestones, and potential responses, based on their analysis, with the milestones clearly communicated in the Reporting Template. For instance, the Step 4 results above suggest the need to evaluate which future is emerging. If, for example, regional decarbonization transitions are emerging in the power sector or economy-wide, and/or offsets are available, the utility could pursue additional emissions reduction ambition beyond their QTs. Other key milestones identified include gas prices, the availability of CDR, electrification trends, and the costs and availability of fugitive CH₄ abatement technologies and low-carbon gas.

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